

PRELIMINARY TRIM AND STABILITY BOOKLET OF PONTOON (YARD NO. CHISTI-036,037)&038

PRELIMINARY



APPROVED

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**INDIAN REGISTER OF SHIPPING
MUMBAI**

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GENERAL PARTICULARS

Name of the Vessel	PONTOON
Client	K.B. SHIPPING & CO. (JAMNAGAR)
Builder	CHISTI MARINE SERVICES
Yard No.	CHISTI 036,037 & 038
Type of Vessel	PONTOON
Class Notation	HSU, PONTOON, INDIAN COASTAL SERVICE DESCRIPTIVE NOTE "FOR CRANE OPERATION" "DECK STRENGTHENED FOR 10T/M ² "

Main Particulars

LOA	50.000 M
LBP	50.000 M
Breadth (Mld.)	18.000 M
Depth (Mld.)	03.000 M
Draft (Mld.)	2.200 M
Draft (Ext.)	2.212 M
Displacement (Ext.)	1893.06 T
Lightship	405.97 T
Deadweight	1487.09 T

Stability Criteria

REGULATION

Intact Stability Criteria in accordance **IS CODE 2008**

- 1) The area under a righting lever curve up to the angle of maximum righting lever should not be less than 0.080 meter-radian.
- 2) The static angle of heel due to a uniformly distributed wind load of 0.54 kPa (wind speed 30 m/s) should not exceed an angle corresponding to half freeboard for the relevant loading condition, where the lever of wind heeling moment is measured from the centroid of the windage area to half the draught.
- 3) The minimum range of stability should be:
For $L \leq 100\text{m}$: 20°
For $L \geq 150\text{m}$: 15°
For intermediate length by interpolation

Intact stability criteria during cargo lifting

The following intact stability criteria are to be complied with:

- $\theta_C \leq 15^\circ$
- $GZC \leq 0,6 GZ_{MAX}$
- $A1 \geq 0.4 ATOT$

where:

θ_C : Heeling angle of equilibrium, corresponding to the first intersection between heeling and righting arms (see Fig 1)

GZC , GZ_{MAX} : Defined in Fig 1

$A1$: Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θ_C to the heeling angle equal to the lesser of:

- heeling angle θ_R of loss of stability, corresponding to the second intersection between heeling and righting arms (see Fig 1)
- heeling angle θ_F , corresponding to flooding of unprotected openings

$ATOT$: Total area, in m.rad, below the righting lever curve.

In the above formula, the heeling arm, corresponding to the cargo lifting, is to be obtained, in m, from the following formula:

$$b = (Pd - Zz) / \Delta$$

where:

P : Cargo lifting mass, in t

d : Transversal distance, in m, of lifting cargo to the longitudinal plane (see Fig 1)

Z : Mass, in t, of ballast used for righting the pontoon, if applicable (see Fig 1)

z : Transversal distance, in m, of the centre of gravity of Z to the longitudinal plane (see Fig 1)

Δ : Displacement, in t, at the loading condition

The above check is to be carried out considering the most unfavourable situations of cargo lifting combined with the lesser initial metacentric height GM . The vertical position of the centre of gravity of cargo lifting is to be assumed in correspondence of the suspension point.

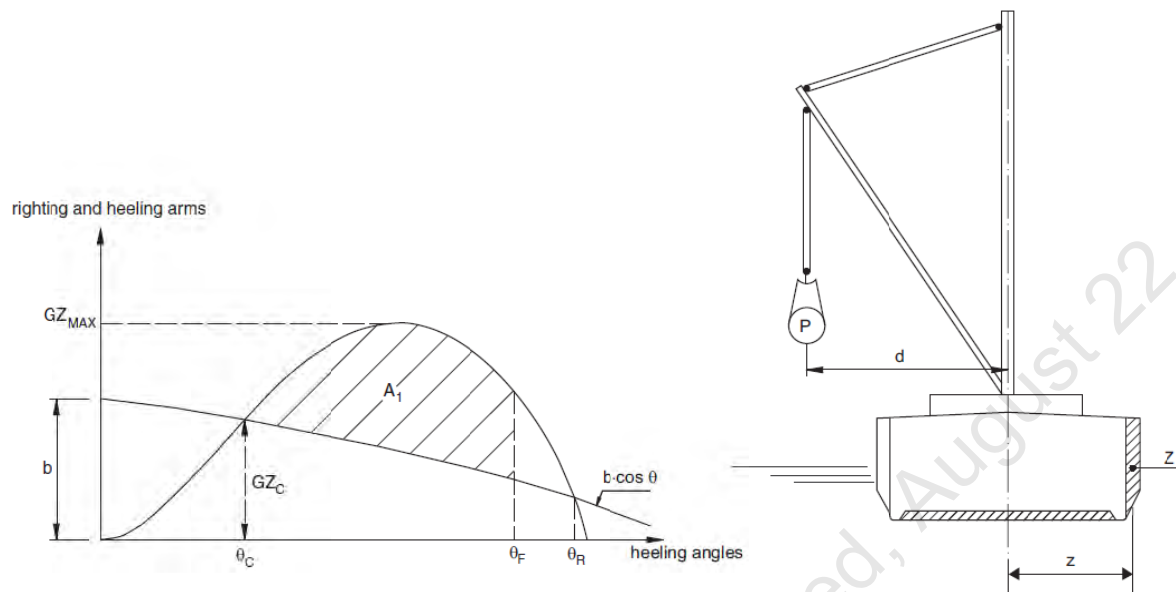


FIGURE 1: CARGO LIFTING

Intact stability criteria in the event of sudden loss of cargo during lifting

This additional requirement is compulsory when counterweights or ballasting of the ship are necessary or when deemed necessary by the Society taking into account the ship dimensions and the weights lifted.

The case of a hypothetical loss of cargo during lifting due to a break of the lifting cable is to be considered.

In this case, the following intact stability criteria are to be compiled with

- $A_2 / A_1 \geq 1$
- $\theta_2 - \theta_3 \geq 20^\circ$

where:

A_1 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θ_1 to the heeling angle θ_C (see Fig 2)

A_2 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θ_C to the heeling angle θ_2 (see Fig 2)

A_3 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θ_C to the heeling angle θ_3 (see Fig 2)

θ_1 : Heeling angle of equilibrium during lifting (see Fig 2)

θ_2 : Heeling angle corresponding to the lesser of θ_R and θ_F

θ_C : Heeling angle of equilibrium, corresponding to the first intersection between heeling and righting arms (see Fig 2)

θ_3 : Maximum heeling angle due to roll, at which $A_3 = A_1$, to be taken not greater than 30° (angle in correspondence of which the loaded cargo on deck is assumed to shift (see Fig 2)

θ_R : Heeling angle of loss of stability, corresponding to the second intersection between heeling and righting arms (see Fig 2).

θ_F : Heeling angle at which progressive flooding may occur (see Fig 2)

In the above formulae, the heeling arm, induced on the ship by the cargo loss, is to be obtained, in m, from the following formula

$$b = Zz \cos\theta/\Delta$$

HEELING AND RIGHTING ARMS

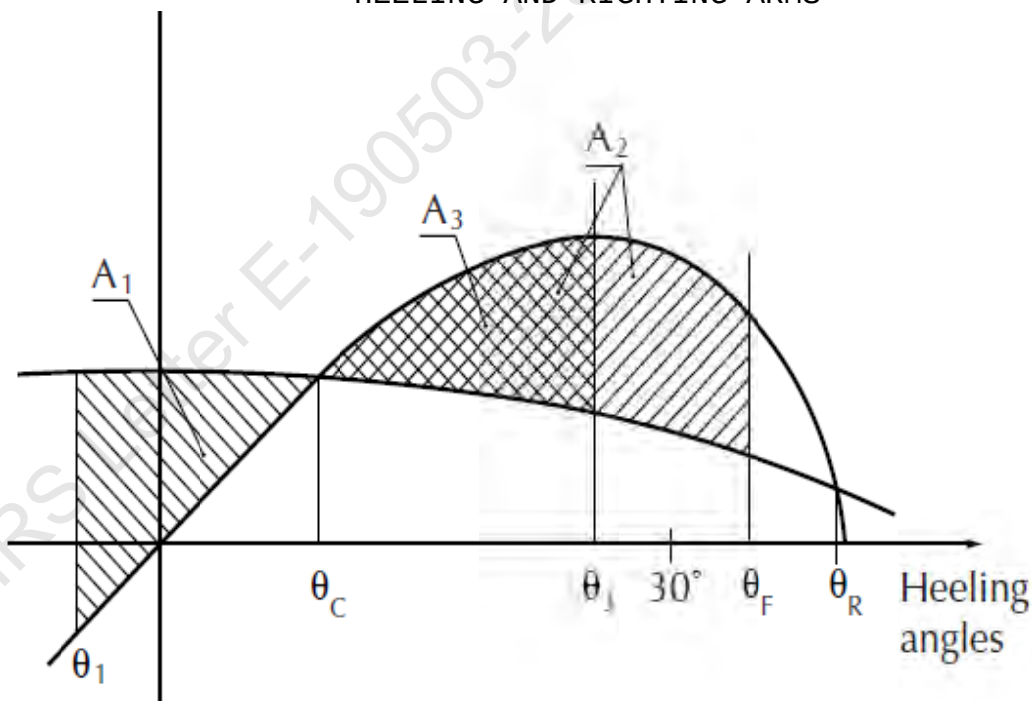


FIGURE 2: CARGO LOSS

Notes to the Master

General precautions against capsizing

1. Compliance with the required minimum stability criteria does not insure immunity against capsizing, regardless of the circumstances, or absolve the Master from his responsibilities. Masters should therefore exercise prudence and good seamanship, having regard to the season of the year, weather forecasts, the navigational zone, and should take appropriate action as to speed and course warranted by the prevailing circumstances.
2. Loading conditions shown in this booklet are typical of those which may be encountered in service and are worked out to be used for guidance only. Further the booklet contains sufficient data to enable the ship's staff to compute the ship's stability in various loading conditions.
3. Excessive trim by forward is to be avoided in all cases as this may lead to difficulties of maneuverability and course keeping as also infringe the minimum Bow-height requirements.
4. All longitudinal levers, namely LCB, LCG, and LCF are computed with reference to the A.P of Ship.
5. In this booklet hydrostatic particulars and cross curves are given at following trims.

-1.00m, -0.50m, 0.00m, 0.50m, 1.00m

These trims are the expected operational trims.

Notes on Use of Cross Curves of Stability

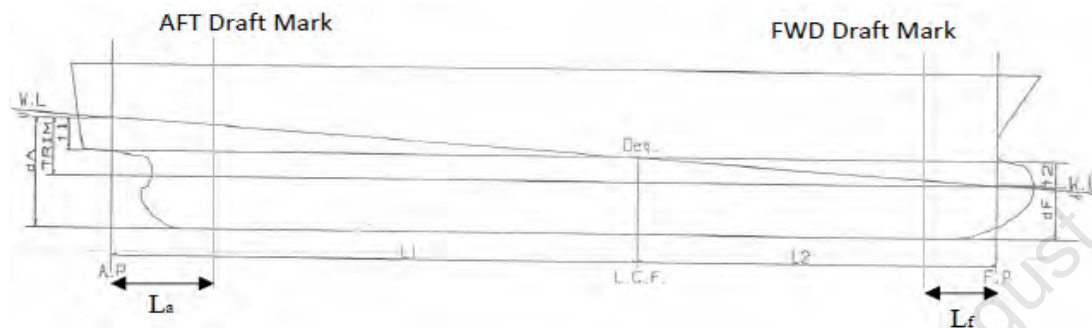
The purpose of the Cross-Curves of Stability is to enable Statical Stability curves to be drawn for the ship in any loading condition. Intact stability is then determined on the basis of this Curve. WORKED OUT EXAMPLE demonstrates the use of Cross Curves.

Note:

1. During unmanned tow minimum forward draft of 1.94m to be maintained at all times for compliance with slamming requirements or else it is recommended that unmanned tow of vessel should be carried out in fair weather and against favorable weather forecast.

2. Deck strength is examined considering Deck loading of 25 T/m² in way of crane operating area & 10 T/m² in way of other deck area as marked in Structural Plan. The selected crane's ground pressure as well as Deck cargo shall not exceed above considered values.

Draft and Trim Calculation



The following data are required for ascertaining trim and draft:

- a) Longitudinal Centre of gravity of the ship calculated as follows:

$$LCG = \frac{\text{Summation of longitudinal weight moments}}{\text{Displacement}}$$

- b) Longitudinal centre of buoyancy, LCB
 c) Displacement, Δ
 d) Moment to change trim, MCT1cm
 e) Longitudinal centre of floatation, LCF

LCB, LCF and MCT1cm are taken from the hydrostatic tables (trim = 0 m) corresponding to the actual displacement. The trim is calculated according to the following formula:

$$\text{Trim} = \frac{\Delta * (LCB - LCG)}{MCT1cm * 100}$$

The Extreme drafts at AP (TAP) and at FP (TFP) are calculated according to the following formulae:

$$T_{AP} = \frac{LCF * \text{Trim}}{LBP} + T (\text{extreme})$$

$$T_{FP} = T_{AP} - \text{Trim}$$

Where T is the Extreme draft taken from the hydrostatic tables (at trim = 0) corresponding to the actual displacement.

The draft at the Aft and Forward reading marks are as follows:

$$T_a = T_{AP} - \frac{\text{Trim} * L_a}{LBP}$$

$$T_f = T_{FP} + \frac{\text{Trim} * L_f}{LBP}$$

Where,

T_a = Draft at Aft Draft Mark

T_f = Draft at Fwd Draft Mark

L_a = Distance of Aft Draft Mark from AP and it is consider positive when Aft Draft Mark is Fwd of AP

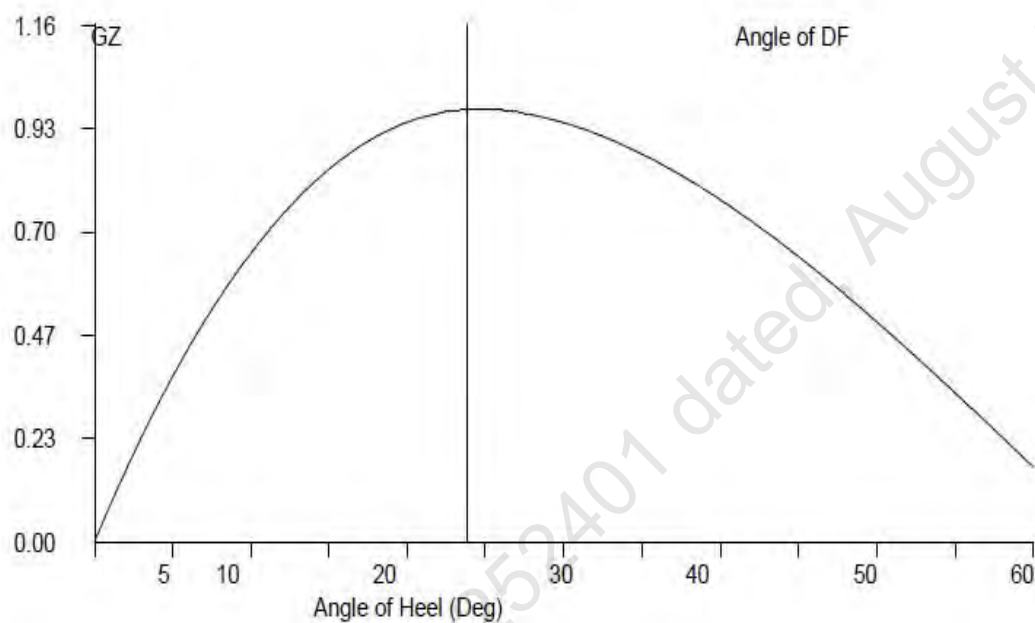
L_f = Distance of Fwd Draft Mark from FP and it is consider positive when Fwd Draft Mark is Aft of FP

GZ Calculation:

- 1) $GZ = KN - KGo \sin (\theta)$
- 2) Area under 0-10 = $10 * 0.01745 * [GZ0 + GZ10] / 2$
- 3) Area under 10-20 = $10 * 0.01745 * [GZ10 + GZ20] / 2$
- 4) Area under 20-30 = $10 * 0.01745 * [GZ20 + GZ30] / 2$
- 5) Area under 30-40 = $10 * 0.01745 * [GZ30 + GZ40] / 2$
- 6) Area under 40-50 = $10 * 0.01745 * [GZ40 + GZ50] / 2$
- 7) Area under 50-60 = $10 * 0.01745 * [GZ50 + GZ60] / 2$
- 8) Area upto 40 deg / Angle of DF = sum of 2+3+4+5
- 9) Max Gz = Read from Gz Curve

Plot GZ values against the angles of heel as above. At 57.3 degree erect an ordinate equal to GM fluid on the same scale as GZ ordinates. Join the point to the origin. This is the Tangent to the GZ Curve at the Origin. Find the

Maxima of GZ curve, ordinates under GZ curve upto 30 degrees, 40 degrees/angle of flooding & between 30 degrees to 40 degrees/angle of flooding. If maximum GZ occurs at an angle less than 30 degrees then area upto this angle must be calculated.



Worked Out Examples**1 Loading the Vessel**

List out the weights, LCG, VCG, TCG and work out longitudinal moment and vertical moment for each load and finally sum up the entire loads to get Displacement, LCG and VCG of the vessel.

For this worked out Example,

Condition 1: Lightship + Crane + Ballast

Displacement	1040.870	Tonnes
LCG	24.220	M Fwd of AP
VCG (KG)	3.975	M above Base Line
TCG	0.000	M Off centerline

2 Free Surface Correction

A tank which is partially filled, adversely affects the stability of the vessel. This is known as Free Surface Effect and is expressed as 'Loss of GM' or 'Virtual Rise in KG' and is calculated as follows:

$$\text{Free Surface Correction (M)} = \frac{\text{Sum of Free Surface Moments of all Tanks}}{\text{Displacement of Vessel}}$$

$$\text{Free Surface moment for a Tank} = \frac{\text{Moment of Inertia (M}^4\text{) for a Tank} \times \text{Density of Liquid in Tank}}{\text{Density of Liquid in Tank}}$$

$$\text{KGo (Fluid)} = \text{KG (Solid)} + \text{Free Surface Correction.}$$

Hydrostatic Particulars

The trim of a particular loading condition is obtained by

$$\text{Trim} = (\text{LCB} - \text{LCG}) \times \text{displacement} / (\text{MCT} \times 100) + \text{Trim}^*$$

Where,

Trim* = Trim at which hydrostatic particulars have been computed.

Condition 1 : Lightship + Crane + Ballast

TANK NAME	VOL (Cu.M.)	SpGr	Weight (MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT	FSM
				METER	T-M	METER	T-M	METER	T-M	T-M
07_WBTK.P	48.39	1.025	49.60	47.660	2363.94	-5.971	-296.16	1.880	93.25	72.32
06_WBTK.C	132.36	1.025	135.67	42.251	5732.19	0.000	0.00	1.499	203.37	135.61
07_WBTK.S	48.39	1.025	49.60	47.660	2363.94	5.971	296.16	1.880	93.25	72.32
TOTAL TANK			234.87	44.536	10460.07	0.000	0.00	1.660	389.87	280.25

FIXED WEIGHT			Weight (MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
LIGHT SHIP			406.00	25.000	10,150.00	0.000	0.00	3.000	1,218.00
CRANE			400.00	11.500	4,600.00	0.000	0.00	6.325	2,530.00
TOTAL FIXED WEIGHT			806.00	18.300	14,750.00	0.000	0.00	4.650	3,748.00

ITEM			Weight (MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
TOTAL TANK			234.870	44.536	10460.065	0.000	0.000	1.660	389.865
TOTAL FIXED WEIGHT			806.000	18.300	14750.000	0.000	0.000	4.650	3748.000
TOTAL WEIGHT			1040.870	24.220	25210.065	0.000	0.000	3.975	4137.865

Condition 1 : Lightship + Crane + Ballast

Free Surface Correction for VCG = FSM/Displacement

	0.269	m
VCGc	4.245	m

From the MaxVCG table,

Displacement	1040.87	T
Allowable Max VCG	14.946	m

VCGc < Maxvcg

PASS

Note : The Master should always check the obtained VCG value against Maximum limiting VCG curve for intact
Refer to the Hydrostatic Table for even trim & Even Heel Condition of Stability Booklet.

Displacement =	1040.870	T
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Draft ,Tm @ 25.0f =	1.269	m
LCB =	25.000	m
LCF =	25.000	m
MCT =	32.722	T-m/cm
KMT =	23.341	m

The vessels trim is to be determined as follow,

Trim=(LCB-LCG)xDisplacement / (MCTX100)

-0.248 m, (+) is aft trim, (-) is fwd trim

Determining the Forward perpendicular,Midship and aft Perpendicular Draft

LBP = 50m

Draft @ 25.00 (MS),Tmid	1.269	m
-------------------------	-------	---

Draft @ 50.0f (FP),Tfwd	=Tm - Trim/2
	1.145 m

Draft@0(AP),Taft	=Tm + Trim/2
	1.393 m

Condition 1 : Lightship + Crane + Ballast

From HYDROSTATICS

Hydrostatic Particulars			Drafts & Trims	
MCT	32.722	T-m	Draft (Mean)	1.269 m
LCB	25.000	m	Draft AP (USK)	1.393 m
			Draft FP (USK)	1.145 m
Metacentric Height				
KMt	23.341	m		
KG	3.975	m		
KGO	4.245	m		
GMt	19.096	m	Trim	0.248 m

The condition shown above is first worked using Even Keel hydrostatic particulars Program then interpolates the hydrostatic particulars from the values available for the displacement

From HYDROSTATICS

Hydrostatic Particulars			Drafts & Trims	
MCT	31.910	T-m	Draft (Mean)	1.269 m
LCB	24.203	m	Draft AP (USK)	1.396 m
			Draft FP (USK)	1.141 m
Metacentric Height				
KMt	23.320	m		
KG	3.975	m		
KGO	4.245	m		
GMt	19.075	m	Trim	0.255 m

Trim is calculated as follows:

$$\text{Trim} = (\text{LCB} - \text{LCG}) \times \text{displacement} / (\text{MCT} \times 100) + \text{Trim}^*$$

Where Trim* is trim at which hydrostatics were calculated.

Condition 1 : Lightship + Crane + Ballast

Intact Stability (Primary Criterion)

Minimum Required

Displacement	1040.87	T
KG0	4.245	M
GMt	19.075	M

0.150 M

Residual Arm Curve

Angle (Deg)	0	15	25	35	45	55
KN (M)	0.000	4.805	5.227	5.128	4.788	4.274
GZ (M)	-0.021	3.706	3.433	2.693	1.787	0.797

	Computed values	Minimum Required
Residual Area from abs 0 deg to MAXRA	0.7180 m-rad	0.080 m-rad
Angle from Equilibrium to RAZero or flood	LARGE m-rad	20.00 Deg.
Angle from Equilibrium to 50% DK Imm Angle	5.23 m-rad	0.00 Deg.

Plot GZ values against the angles of heel as above.

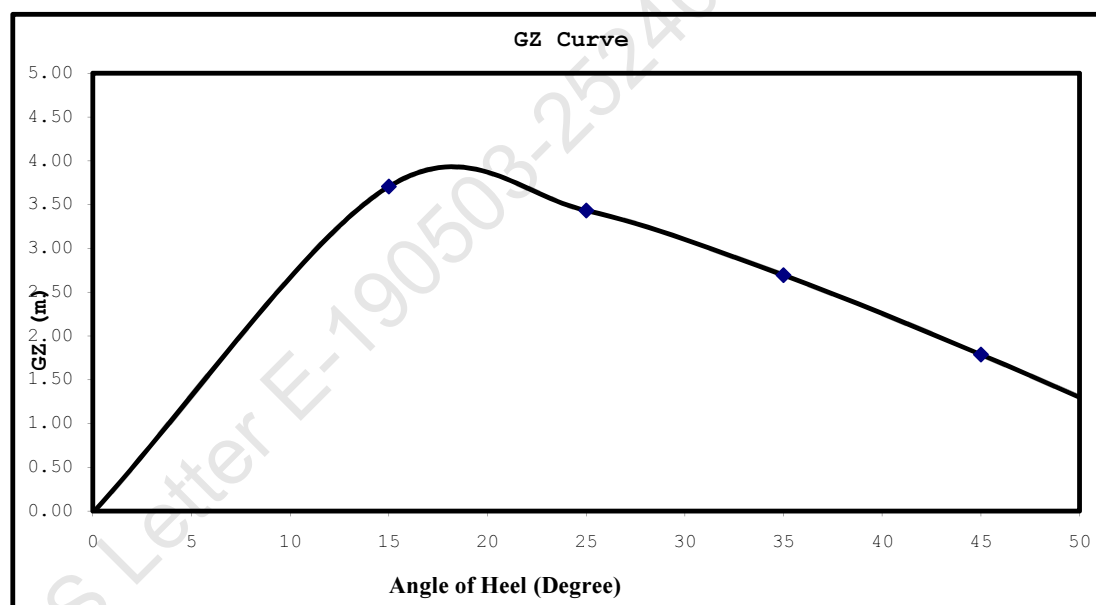
At 57.3 degree erect an ordinate equal to GM fluid on the same scale as GZ ordinates. Join the point to the origin.

This is the Tangent to the GZ Curve at the Origin. Find the Maxima of GZ

curve, ordinates under GZ curve upto 30 degrees, 40 degrees/angle of flooding

If maximum GZ occurs at an angle less than 30 degrees

then area upto this angle must be calculated.



Worked Out Example Contd.

Mean Draft	1.269	M
LCF from AP	24.767	M
LCB from AP	24.203	M
MCT	31.910	T-M/Cm
Trim	0.255	M
Draft (AP)	1.396	M
Draft (FP)	1.141	M
KG	3.975	M
KGo	4.245	M
GoMt	19.075	M

LOADLINE DRAFT: Check that Mean Draft (Midship) does not exceed the Summer Draft (Ext) for the corresponding season, as per the assigned loadline.

BLANK CALCULATION SHEET

[illegible][illegible][illegible]

	m
VCGc	m

Displacement	T
Allowable Max VCG	m

Note : The Master should always check the obtained VCG value against Maximum limiting VCG curve for intact
Refer to the Hydrostatic Table for even trim & Even Heel Condition of Stability Booklet.

Displacement =	T
Draft ,Tm @ 25.0f =	m
LCB =	m
LCF =	m
MCT =	T-m/cm
KMT =	m

$$\text{Trim} = (\text{LCB} - \text{LCG}) \times \text{Displacement} / (\text{MCTX} \times 100)$$

m, (+) is aft trim, (-) is fwd trim

LBP = 50m

Draft @ 25.00 (MS), Tmid m

Draft @ 50.0f (FP), Tfwd = Tm - Trim/2 m

Draft @ 0 (AP), Taft = Tm + Trim/2 m

Definitions, Symbols and Sign Conventions:

AP	Fr.0
FP	Fr.100
Midship	Fr.50
Frame Spacing	500mm throughout

List of Symbols

Symbol	Unit	Description
LOA	M	Length Overall
LBP	M	Length between perpendiculars
B (MLD)	M	Moulded Breadth
D (MLD)	M	Moulded Depth
TPCm	T/Cm	Tonnes per 1 Cm immersion
MCTCm	T-M/Cm	Moment to change Trim by 1 Cm
Draft AP (USK)	M	Draft at AP wrt Keel line
Draft FP (USK)	M	Draft at FP wrt Keel line
Draft (Mean)	M	Draft wrt Keel line at Midship
Draft (Mean)	M	[Draft AP(USK) + Draft FP(USK)] x 0.5
VCG, KG	M	Vertical Center of Gravity above Base Line before correcting for free surface
KGo	M	Vertical Center of Gravity above Base Line after correcting for free surface
KB	M	Vertical Center of Buoyancy from Base Line
KMt	M	Transverse Metacenter above Base Line
GoMt	M	Transverse Metacenter above KGo
GZ	M	Righting Arm
FSM	T-M	Free Surface Moment
sp.gr.		Specific Gravity
LCG	M	Longitudinal Center of Gravity
LCB	M	Longitudinal Center of Buoyancy
LCF	M	Longitudinal Center of Floatation

Sign Convention

LCB, LCG, LCF	Fwd of AP is denoted by f and aft of AP is denoted by a
TCG	port is negative and stbd is positive
Trim by Aft	Positive
Trim by Ford	Negative

Metric Conversion Table

<u>MULTIPLY BY</u>	<u>TO CONVERT FROM</u>	<u>TO OBTAIN</u>	
0.03937	millimeters	Inches	25.400
0.39370	Centimeters	Inches	2.5400
3.28080	Meters	Feet	0.3050
2.20460	kilograms	Pounds	0.4540
0.00098	kilogram's	tons(2240 lb.)	1016.1
0.98420	Metric tonnes (i.e. tonnes of 1000 kg.)	tons(2240 lb.)	1.0160
2.49980	Metric tonnes per centimeter of immersion	tons per inch (immersion)	0.4000
8.20140	moment to change trim one centimeter one (tonne meter units)	moment to change trim inch(foot ton units)	0.1220
187.976	Meter Radians	Feet degrees	0.0053
	<u>TO OBTAIN</u>	<u>TO CONVERT FROM</u>	<u>MULTIPLY</u>

Relation between Weight and volume

1000mm cubed	1 Cubic Centimeters
1 Cubic Centimeter of Fresh Water (S.G.1.0)	1 Gram
1000 cubic centimeter of fresh Water (S.G.1.0)	1 Kilogram (1000 Gm.)
1 Cubic Meter of fresh water (S.G.1.0)	1 Tonne (1000 Kg.)
1 Cubic Meter of fresh water (S.G.1.0)	1 tonne
1 Cubic Meter of Salt water (S.G. 1.025)	1.025 Tonnes
1 Tonne of Salt Water (S.G.1.025)	0.975 Cubic Meters

Angle of Downflooding

1. The angle of flooding is the angle of immersion of the sill or coaming of the opening on or above the freeboard deck through which progressive downflooding can occur as the vessel heels.
2. If this opening has a weathertight closing device which cannot be kept closed and secured at all times while the vessel is underway (e.g. engine room air supply trunk), or has no weathertight closing device, progressive downflooding will occur when the sill of the opening immerses. This angle is to be marked on the GZ curve if it comes within the apparent positive range of the GZ curve, and the GZ curve terminates at this angle: i.e., the actual angle of flooding becomes the range of the GZ curve.
3. If an opening is fitted with a weathertight closing device which can be kept closed and secured at all times while the vessel is underway, progressive downflooding can only occur through the opening if its closing device is not closed and secured. The angle of immersion of the lower sill of this opening is a 'potential' angle of flooding. If the closing appliance is not closed and secured, downflooding will occur when the lower sill of this opening reaches the water level.
If the lower sill of the opening through which downflooding can occur immerses at 40 degree or smaller angles of heel, its closing appliances must be kept closed and secured at all times when the vessel is underway or at sea where this opening is used for access in the normal working of the vessel, an alternate access should be used in heavy weather. In moderate weather, the opening may be used for access, but the closing appliances must be closed and secured again immediately after use.

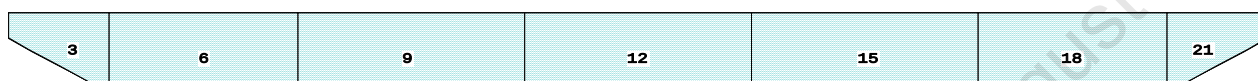
As there is no any unprotected opening exists above freeboard deck, hence the downflooding angle is considered at 50 deg.

01-08-20 21:16:44
GHS 17.22

Cybermarine Technologies PTE. Ltd.
PONTOON

CAPACITY PLAN AND TABLE

Profile View



Plan View

1	4	7	10	13	16	19
2	5	8	11	14	17	20
3	6	9	12	15	18	21

Tanks

1 01_WBTK.P	4 02_WBTK.P	8 03_WBTK.C	12 04_WBTK.S	16 06_WBTK.P	20 07_WBTK.C
2 01_WBTK.C	5 02_WBTK.C	9 03_WBTK.S	13 05_WBTK.P	17 06_WBTK.C	21 07_WBTK.S
3 01_WBTK.S	6 02_WBTK.S	10 04_WBTK.P	14 05_WBTK.C	18 06_WBTK.S	
	7 03_WBTK.P	11 04_WBTK.C	15 05_WBTK.S	19 07_WBTK.P	

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GHS 17.22 PONTOON

TANK STATUS							
Trim: zero, Heel: zero							
Part-----	Cu.M.----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
01_WBTK.P	48.4	1.025	49.60	2.340f	5.971p	1.880	72.32*
01_WBTK.C	48.9	1.025	50.11	2.336f	0.000	1.872	72.33*
01_WBTK.S	48.4	1.025	49.60	2.340f	5.971s	1.880	72.32*
02_WBTK.P	131.4	1.025	134.72	7.749f	5.980p	1.509	135.61*
02_WBTK.C	132.4	1.025	135.67	7.749f	0.000	1.499	135.61*
02_WBTK.S	131.4	1.025	134.72	7.749f	5.980s	1.509	135.61*
03_WBTK.P	157.7	1.025	161.60	16.000f	5.980p	1.509	162.73*
03_WBTK.C	158.8	1.025	162.73	16.000f	0.000	1.500	162.73*
03_WBTK.S	157.7	1.025	161.60	16.000f	5.980s	1.509	162.73*
04_WBTK.P	157.7	1.025	161.60	25.000f	5.980p	1.509	162.73*
04_WBTK.C	158.8	1.025	162.73	25.000f	0.000	1.500	162.73*
04_WBTK.S	157.7	1.025	161.60	25.000f	5.980s	1.509	162.73*
05_WBTK.P	157.7	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.C	158.8	1.025	162.73	34.000f	0.000	1.500	162.73*
05_WBTK.S	157.7	1.025	161.60	34.000f	5.980s	1.509	162.73*
06_WBTK.P	131.4	1.025	134.72	42.251f	5.980p	1.509	135.61*
06_WBTK.C	132.4	1.025	135.67	42.251f	0.000	1.499	135.61*
06_WBTK.S	131.4	1.025	134.72	42.251f	5.980s	1.509	135.61*
07_WBTK.P	48.4	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	48.9	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	48.4	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			2,566.63	25.000f	0.000	1.549	2712.14
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

HYDROSTATIC PARTICULARS

31-07-20 12:18:46 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONTOON

HYDROSTATIC PROPERTIES

Trim: Fwd 1.000/50.000, No Heel, Fixed VCG = 0.000

Draft@	Displacement	Buoyancy-Ctr.		Weight/		Moment/			
25.000f	---Weight(MT)---	---LCB---	---VCB---	---cm---	---LCF---	cm trim	---KML---	---KMT	
0.112	133.29	37.689f	0.175	5.02	33.490f	6.52	244.36	96.881	
0.212	188.30	36.107f	0.211	5.98	31.124f	10.90	289.46	82.456	
0.312	252.92	34.523f	0.246	6.94	28.751f	16.92	334.36	71.772	
0.412	327.16	32.938f	0.281	7.91	26.375f	24.82	379.30	63.564	
0.512	407.76	31.583f	0.319	8.13	26.034f	26.65	326.70	53.012	
0.612	489.54	30.651f	0.361	8.23	25.975f	27.51	280.91	45.226	
0.712	572.35	29.971f	0.406	8.33	25.923f	28.38	247.92	39.491	
0.812	656.10	29.452f	0.452	8.42	25.886f	29.26	222.97	35.068	
0.912	740.70	29.044f	0.499	8.50	25.870f	30.12	203.29	31.535	
1.012	826.02	28.717f	0.548	8.57	25.880f	30.81	186.46	28.540	
1.112	912.01	28.450f	0.597	8.63	25.878f	31.57	173.02	26.147	
1.212	998.69	28.226f	0.646	8.70	25.875f	32.35	161.94	24.171	
1.312	1,086.07	28.037f	0.696	8.78	25.872f	33.17	152.69	22.519	
1.412	1,174.15	27.875f	0.746	8.85	25.869f	34.03	144.90	21.124	
1.512	1,262.91	27.735f	0.797	8.91	25.870f	34.71	137.41	19.834	
1.612	1,352.12	27.609f	0.848	8.94	25.783f	35.14	129.91	18.700	
1.712	1,441.67	27.493f	0.899	8.97	25.694f	35.58	123.36	17.712	
1.812	1,531.58	27.384f	0.950	9.01	25.603f	36.03	117.61	16.848	
1.912	1,621.83	27.283f	1.000	9.05	25.508f	36.51	112.52	16.088	
2.012	1,712.42	27.187f	1.051	9.08	25.436f	36.88	107.68	15.374	
2.112	1,803.32	27.097f	1.102	9.11	25.351f	37.30	103.41	14.750	
2.212	1,894.54	27.011f	1.153	9.14	25.263f	37.74	99.58	14.190	
2.312	1,986.08	26.928f	1.203	9.17	25.174f	38.19	96.11	13.689	
2.412	2,077.95	26.849f	1.254	9.21	25.081f	38.65	92.98	13.238	
2.512	2,170.15	26.772f	1.305	9.23	25.013f	39.01	89.86	12.803	
2.612	2,258.72	26.664f	1.353	8.40	22.760f	29.59	65.48	11.408	
2.712	2,338.10	26.490f	1.395	7.48	20.257f	21.09	45.09	10.043	
2.812	2,408.27	26.272f	1.431	6.56	17.755f	14.46	30.01	8.792	
2.912	2,469.20	26.032f	1.462	5.63	15.252f	9.46	19.15	7.632	
3.012	2,520.92	25.786f	1.488	4.71	12.750f	5.86	11.62	6.543	

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.

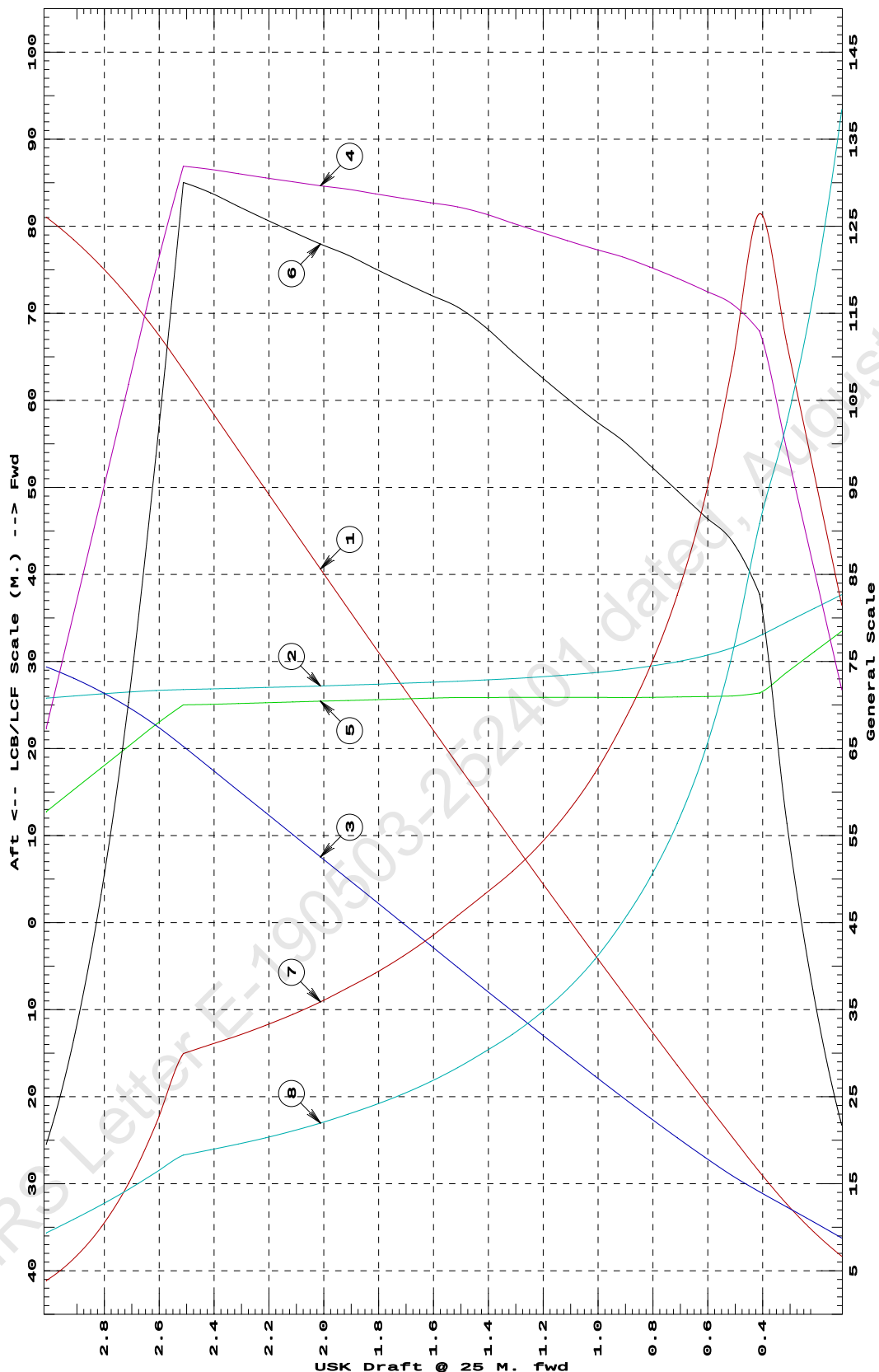
Trim is per 50.00m.

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Cybermarine Technologies PTE. Ltd.
PONTOON

HYDROSTATIC PROPERTIES at 1 M. FWD TRIM



Specific Gravity = 1.025 Assumed KG = 0.00 M.
Trim is per 50 M. "K" = BPL

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GHS 17.22 PONTOON

HYDROSTATIC PROPERTIES

Trim: Fwd 0.500/50.000, No Heel, Fixed VCG = 0.000

Draft@	Displacement	Buoyancy-Ctr.		Weight/		Moment/			
25.000f	---Weight(MT)---	LCB----	VCB-----	cm-----	LCF----	cm trim----	KML-----	KMT	
0.112	94.61	35.657f	0.098	5.88	30.439f	10.72	566.32	155.843	
0.212	162.65	32.417f	0.132	7.74	25.617f	24.15	742.47	120.816	
0.312	240.79	30.208f	0.173	7.89	25.602f	25.06	520.34	85.097	
0.412	320.44	29.059f	0.218	8.04	25.566f	25.94	404.70	66.389	
0.512	401.45	28.349f	0.266	8.16	25.509f	26.81	333.87	54.610	
0.612	483.61	27.862f	0.315	8.27	25.457f	27.70	286.38	46.395	
0.712	566.72	27.508f	0.365	8.35	25.449f	28.52	251.60	40.187	
0.812	650.57	27.242f	0.415	8.42	25.440f	29.21	224.49	35.343	
0.912	735.11	27.034f	0.466	8.49	25.439f	29.96	203.76	31.632	
1.012	820.35	26.868f	0.517	8.56	25.438f	30.74	187.34	28.689	
1.112	906.31	26.733f	0.567	8.63	25.436f	31.56	174.08	26.309	
1.212	992.99	26.620f	0.619	8.71	25.451f	32.35	162.89	24.309	
1.312	1,080.33	26.524f	0.670	8.77	25.439f	33.07	153.05	22.576	
1.412	1,168.34	26.443f	0.722	8.83	25.438f	33.86	144.89	21.132	
1.512	1,257.02	26.372f	0.773	8.90	25.437f	34.68	137.92	19.898	
1.612	1,346.39	26.310f	0.825	8.97	25.435f	35.53	131.95	18.836	
1.712	1,436.46	26.255f	0.877	9.04	25.448f	36.37	126.58	17.889	
1.812	1,527.11	26.206f	0.929	9.08	25.389f	36.90	120.81	16.991	
1.912	1,618.11	26.157f	0.981	9.12	25.304f	37.34	115.37	16.198	
2.012	1,709.44	26.109f	1.033	9.15	25.217f	37.79	110.53	15.496	
2.112	1,801.11	26.062f	1.085	9.19	25.127f	38.26	106.21	14.872	
2.212	1,893.13	26.014f	1.137	9.22	25.034f	38.75	102.33	14.317	
2.312	1,985.42	25.968f	1.189	9.23	25.012f	38.91	97.97	13.759	
2.412	2,077.73	25.925f	1.240	9.23	25.009f	38.96	93.75	13.255	
2.512	2,170.06	25.886f	1.292	9.23	25.006f	39.02	89.90	12.799	
2.612	2,262.39	25.850f	1.343	9.23	25.004f	39.08	86.36	12.384	
2.712	2,354.73	25.817f	1.394	9.24	25.001f	39.14	83.10	12.006	
2.812	2,445.61	25.772f	1.445	8.50	23.000f	30.67	62.71	10.846	
2.912	2,521.33	25.617f	1.485	6.65	18.000f	15.11	29.97	8.622	
3.012	2,578.59	25.395f	1.516	4.80	13.000f	6.19	12.01	6.555	

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.

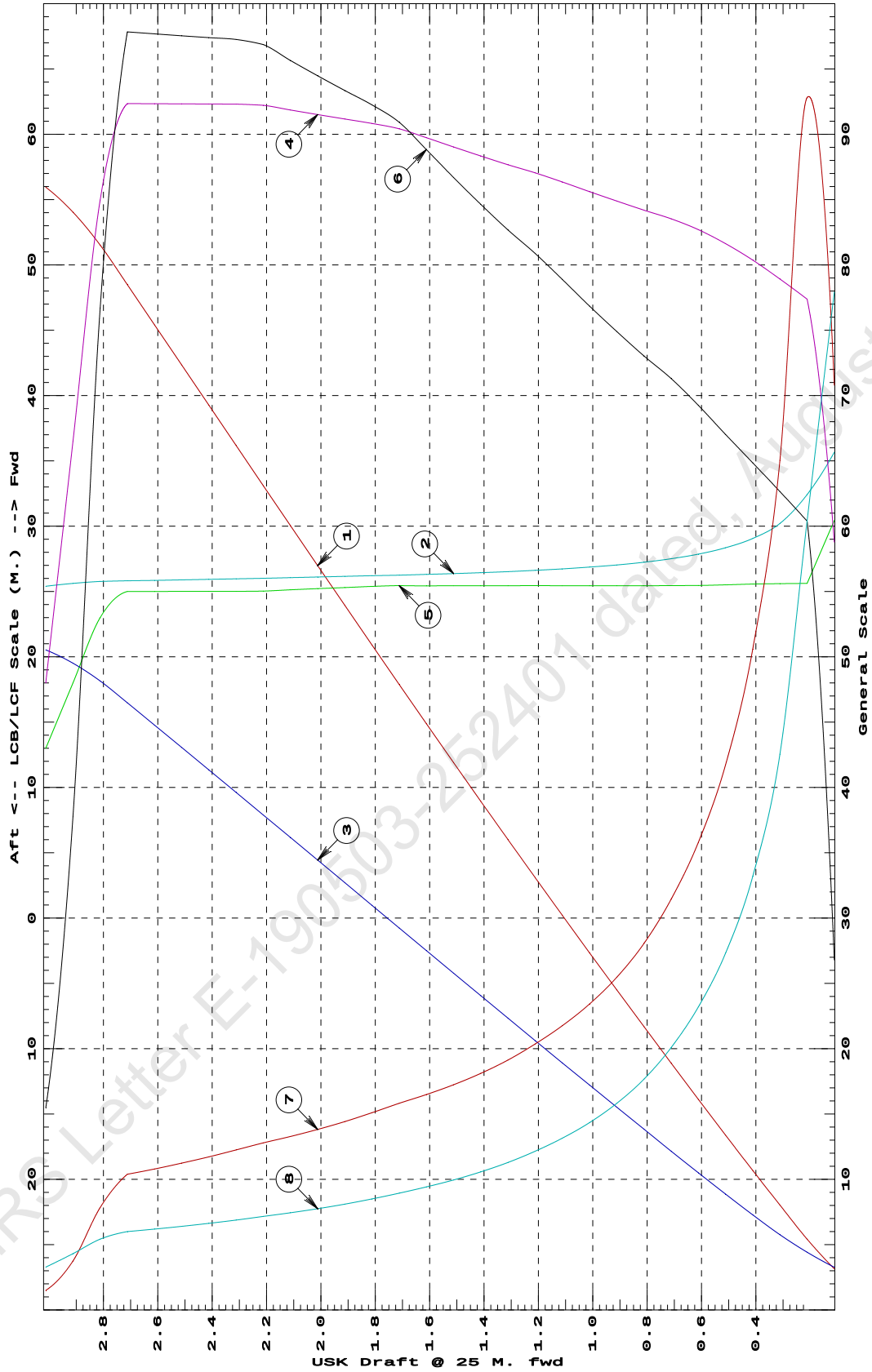
Trim is per 50.00m.

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Cybermarine Technologies PTE. Ltd.
PONTOON

HYDROSTATIC PROPERTIES at 0.5 M. FWD TRIM



- | | | |
|------------------------|-------------------------|-----------------------------|
| ① Displacement 1=30 MT | ④ Immersion 1=1.1 MT/cm | ⑥ Moment/Trim 1=.4 M.-MT/cm |
| ② LCB (use top scale) | ④ WPA 1=9.76 Sq.M. | ⑦ KML 1=8 M. |
| ③ VCB (KB) 1=.03 M. | ⑤ LCF (use top scale) | ⑧ KMT 1=2 M. |

Specific Gravity = 1.025 Assumed KG = 0.00 M.
Trim is per 50 M. "K" = BPL

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GHS 17.22 PONTOON

HYDROSTATIC PROPERTIES
No Trim, No Heel, Fixed VCG = 0.000

Draft@	Displacement	Buoyancy-Ctr.		Weight/		Moment/		
25.000f	---Weight(MT)---	LCB----	VCB-----	cm-----	LCF----	cm trim----	KML-----	KMT
0.112	83.79	25.000f	0.044	7.57	25.000f	23.17	1382.58	223.643
0.212	160.29	25.000f	0.095	7.73	25.000f	24.11	752.15	122.246
0.312	238.38	25.000f	0.145	7.89	25.000f	25.08	526.13	85.934
0.412	318.09	25.000f	0.196	8.06	25.000f	26.09	410.04	67.317
0.512	399.40	25.000f	0.248	8.20	25.000f	26.98	337.80	55.685
0.612	481.78	25.000f	0.300	8.27	25.000f	27.70	287.43	46.641
0.712	564.87	25.000f	0.351	8.35	25.000f	28.44	251.70	40.223
0.812	648.69	25.000f	0.403	8.42	25.000f	29.21	225.14	35.449
0.912	733.24	25.000f	0.454	8.50	25.000f	30.02	204.72	31.774
1.012	818.51	25.000f	0.506	8.56	25.000f	30.72	187.65	28.736
1.112	904.44	25.000f	0.558	8.63	25.000f	31.47	173.99	26.304
1.212	991.06	25.000f	0.610	8.70	25.000f	32.26	162.74	24.297
1.312	1,078.37	25.000f	0.661	8.77	25.000f	33.07	153.35	22.621
1.412	1,166.40	25.000f	0.713	8.84	25.000f	33.93	145.46	21.208
1.512	1,255.11	25.000f	0.765	8.90	25.000f	34.66	138.07	19.915
1.612	1,344.46	25.000f	0.818	8.97	25.000f	35.45	131.83	18.822
1.712	1,434.46	25.000f	0.870	9.03	25.000f	36.27	126.41	17.872
1.812	1,525.13	25.000f	0.922	9.10	25.000f	37.12	121.69	17.042
1.912	1,616.48	25.000f	0.974	9.17	25.000f	38.01	117.57	16.315
2.012	1,708.51	25.000f	1.027	9.23	25.000f	38.71	113.30	15.610
2.112	1,800.77	25.000f	1.079	9.23	25.000f	38.78	107.66	14.924
2.212	1,893.06	25.000f	1.132	9.23	25.000f	38.84	102.58	14.309
2.312	1,985.36	25.000f	1.184	9.23	25.000f	38.90	97.98	13.756
2.412	2,077.68	25.000f	1.236	9.23	25.000f	38.97	93.78	13.257
2.512	2,170.03	25.000f	1.287	9.24	25.000f	39.04	89.95	12.804
2.612	2,262.38	25.000f	1.339	9.24	25.000f	39.09	86.38	12.385
2.712	2,354.73	25.000f	1.390	9.24	25.000f	39.14	83.10	12.003
2.812	2,447.09	25.000f	1.442	9.24	25.000f	39.19	80.07	11.654
2.912	2,539.44	25.000f	1.493	9.24	25.000f	39.24	77.26	11.334
3.012	2,631.79	25.000f	1.544	9.24	25.000f	39.29	74.65	11.040

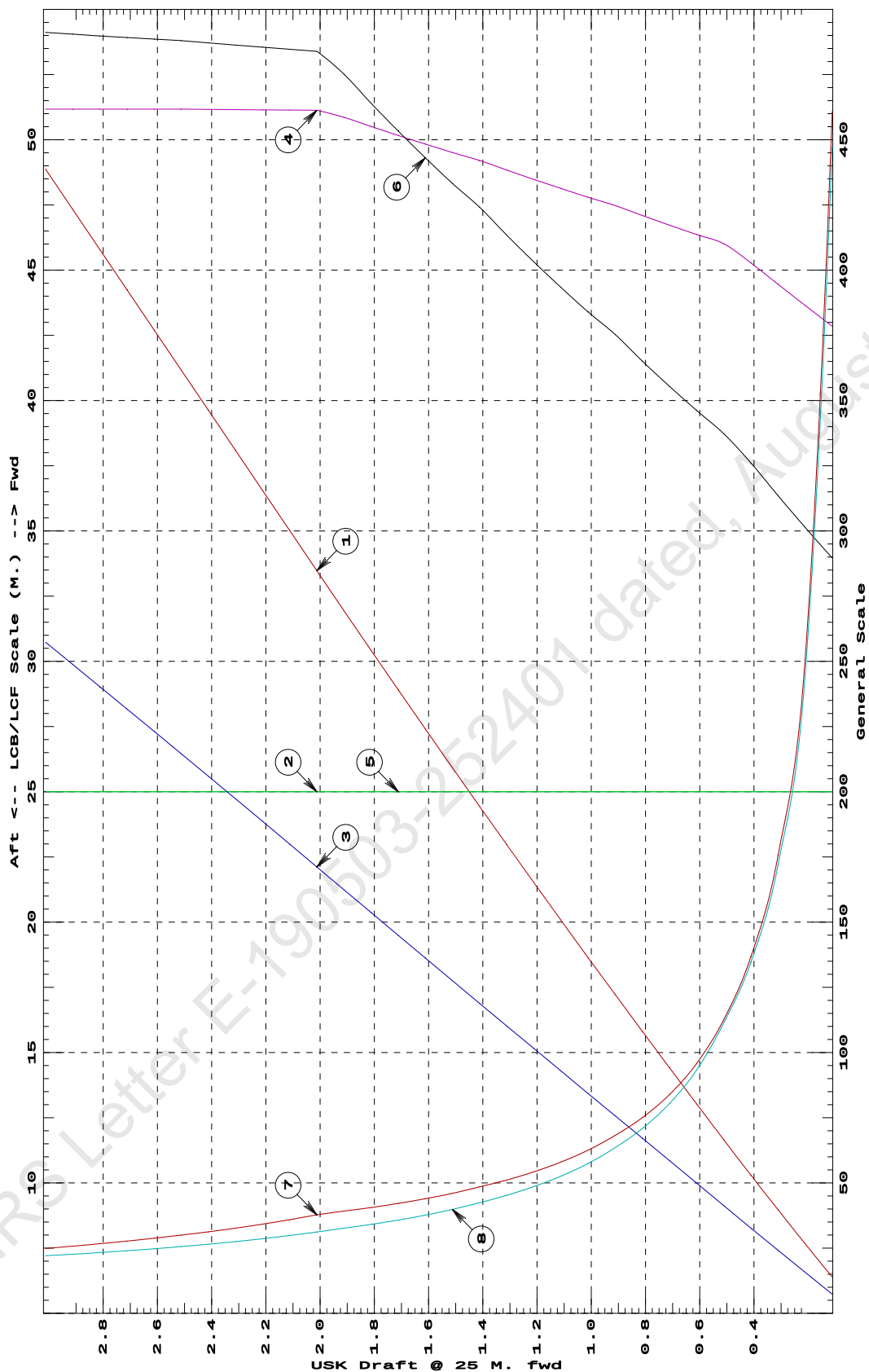
Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

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Cybermarine Technologies PTE. Ltd.
PONTOON

HYDROSTATIC PROPERTIES at LEVEL TRIM



- ① Displacement 1=6 MT
- ② LCB (use top scale)
- ③ VCB (KB) 1=0.006 M.
- ④ Immersion 1=0.02 MT/cm
- ④ WPA 1=1.95 Sq.M.
- ⑤ LCF (use top scale)
- ⑥ Moment/Trim 1=0.08 M.-MT/cm
- ⑦ KML 1=3 M.
- ⑧ KMT 1=1.5 M.

Specific Gravity = 1.025 Assumed KG = 0.00 M.
Trim is per 50 M. "K" = BPL

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GHS 17.22 PONTOON

HYDROSTATIC PROPERTIES

Trim: Aft 0.500/50.000, No Heel, Fixed VCG = 0.000

Draft@	Displacement	Buoyancy-Ctr.		Weight/		Moment/			
25.000f	---Weight(MT)---	LCB----	VCB-----	cm-----	LCF----	cm trim----	KML-----	KMT	
0.112	94.61	14.343f	0.098	5.88	19.561f	10.72	566.32	155.836	
0.212	162.65	17.583f	0.132	7.74	24.383f	24.15	742.47	120.811	
0.312	240.79	19.792f	0.173	7.89	24.398f	25.06	520.34	85.098	
0.412	320.44	20.941f	0.218	8.04	24.434f	25.94	404.70	66.390	
0.512	401.45	21.651f	0.266	8.16	24.491f	26.81	333.87	54.611	
0.612	483.61	22.138f	0.315	8.27	24.543f	27.70	286.38	46.395	
0.712	566.72	22.492f	0.365	8.35	24.551f	28.52	251.60	40.186	
0.812	650.57	22.758f	0.415	8.42	24.560f	29.21	224.49	35.343	
0.912	735.11	22.966f	0.466	8.49	24.561f	29.96	203.76	31.632	
1.012	820.35	23.132f	0.517	8.56	24.562f	30.74	187.34	28.689	
1.112	906.31	23.267f	0.567	8.63	24.564f	31.56	174.08	26.309	
1.212	992.99	23.380f	0.619	8.71	24.549f	32.35	162.89	24.308	
1.312	1,080.33	23.476f	0.670	8.77	24.561f	33.07	153.05	22.576	
1.412	1,168.34	23.557f	0.722	8.83	24.562f	33.86	144.89	21.132	
1.512	1,257.02	23.628f	0.773	8.90	24.563f	34.68	137.92	19.898	
1.612	1,346.39	23.690f	0.825	8.97	24.565f	35.53	131.95	18.836	
1.712	1,436.46	23.745f	0.877	9.04	24.552f	36.37	126.58	17.889	
1.812	1,527.11	23.794f	0.929	9.08	24.611f	36.90	120.81	16.991	
1.912	1,618.10	23.843f	0.981	9.12	24.696f	37.34	115.37	16.198	
2.012	1,709.44	23.891f	1.033	9.15	24.783f	37.79	110.53	15.496	
2.112	1,801.11	23.938f	1.085	9.19	24.873f	38.26	106.21	14.872	
2.212	1,893.13	23.986f	1.137	9.22	24.966f	38.75	102.33	14.317	
2.312	1,985.42	24.032f	1.189	9.23	24.988f	38.91	97.97	13.758	
2.412	2,077.73	24.075f	1.240	9.23	24.991f	38.96	93.75	13.255	
2.512	2,170.06	24.114f	1.292	9.23	24.994f	39.02	89.90	12.799	
2.612	2,262.39	24.150f	1.343	9.23	24.996f	39.08	86.36	12.384	
2.712	2,354.73	24.183f	1.394	9.24	24.999f	39.14	83.10	12.006	
2.812	2,445.61	24.228f	1.445	8.50	27.000f	30.67	62.71	10.846	
2.912	2,521.33	24.383f	1.485	6.65	32.000f	15.11	29.97	8.622	
3.012	2,578.59	24.605f	1.516	4.80	37.000f	6.19	12.01	6.555	

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.

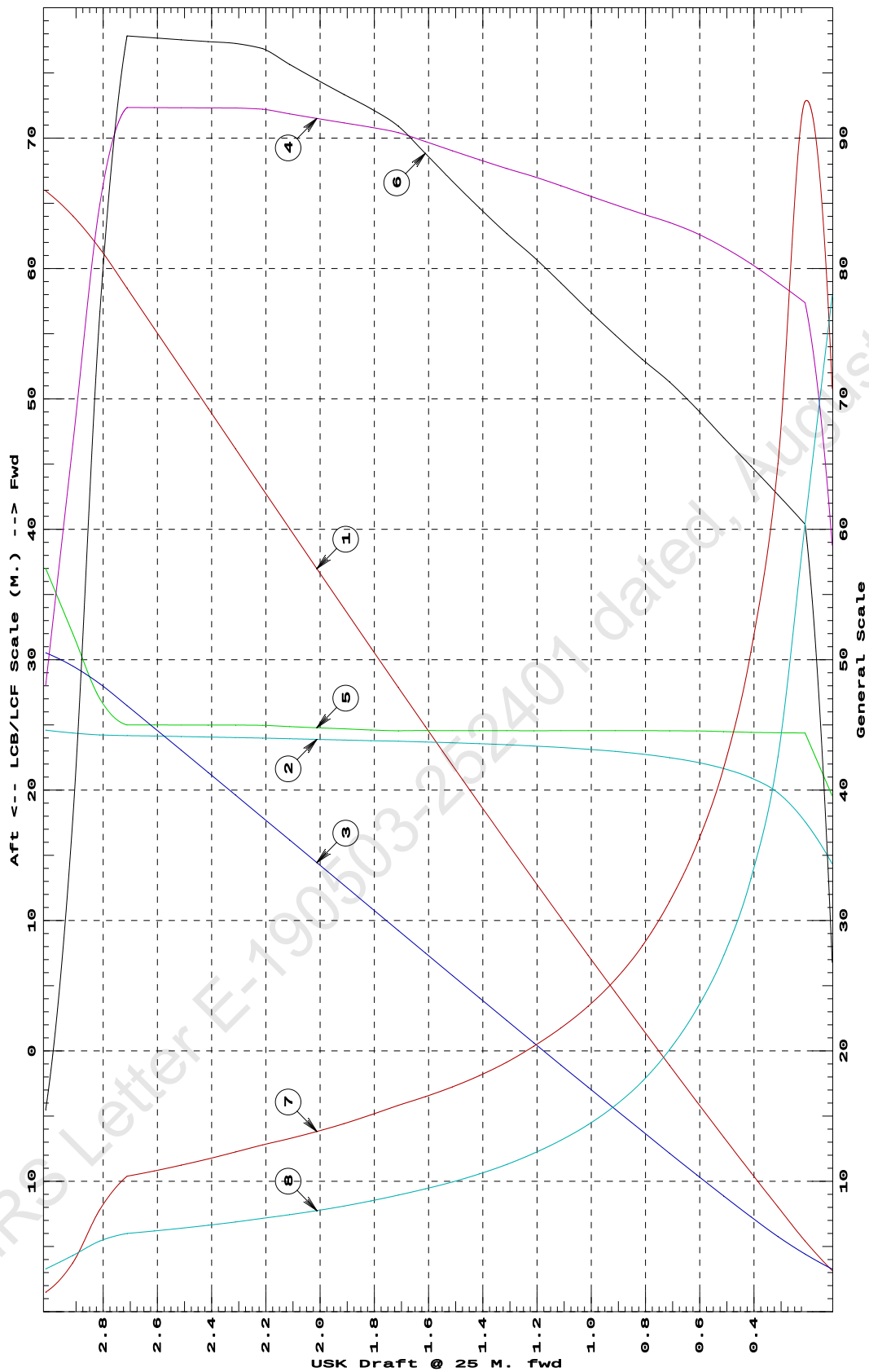
Trim is per 50.00m.

Draft is from USK.

31-07-20 12:18:46
GHS 17.22

Cybermarine Technologies PTE. Ltd.
PONTOON

HYDROSTATIC PROPERTIES at 0.5 M. AFT TRIM



- ① Displacement 1=30 MT
② LCB (use top scale)
③ VCB (KB) 1=.03 M.
④ Immersion 1=.1 MT/cm
④ WPA 1=9.76 Sq.M.
⑤ LCF (use top scale)
⑥ Moment/Trim 1=.4 M.-MT/cm
⑦ KML 1=8 M.
⑧ KMT 1=2 M.

Specific Gravity = 1.025 Assumed KG = 0.00 M.
Trim is per 50 M. "K" = BPL

31-07-20 12:18:46 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONTOON

HYDROSTATIC PROPERTIES

Trim: Aft 1.000/50.000, No Heel, Fixed VCG = 0.000

Draft@ 25.000f	Displacement ---Weight(MT)	Buoyancy-Ctr. ----LCB-----VCB	Weight/ -----cm	Moment/ -----LCF-----cm trim	KML	KMT
0.112	133.29	12.311f 0.175	5.02 16.510f	6.52 244.36	96.880	
0.212	188.30	13.893f 0.211	5.98 18.876f	10.90 289.46	82.454	
0.312	252.92	15.477f 0.246	6.94 21.249f	16.92 334.36	71.771	
0.412	327.16	17.062f 0.281	7.91 23.625f	24.82 379.31	63.562	
0.512	407.76	18.417f 0.319	8.13 23.966f	26.65 326.70	53.012	
0.612	489.54	19.349f 0.361	8.23 24.026f	27.51 280.91	45.226	
0.712	572.35	20.029f 0.406	8.33 24.077f	28.38 247.92	39.491	
0.812	656.10	20.548f 0.452	8.42 24.114f	29.26 222.97	35.069	
0.912	740.70	20.956f 0.499	8.50 24.130f	30.12 203.29	31.535	
1.012	826.02	21.283f 0.548	8.57 24.120f	30.81 186.46	28.540	
1.112	912.01	21.550f 0.597	8.63 24.122f	31.57 173.02	26.147	
1.212	998.69	21.774f 0.646	8.70 24.125f	32.35 161.94	24.171	
1.312	1,086.07	21.963f 0.696	8.78 24.128f	33.17 152.69	22.519	
1.412	1,174.15	22.125f 0.746	8.85 24.131f	34.03 144.90	21.125	
1.512	1,262.91	22.265f 0.797	8.91 24.130f	34.71 137.41	19.834	
1.612	1,352.12	22.391f 0.848	8.94 24.217f	35.14 129.91	18.700	
1.712	1,441.67	22.507f 0.899	8.97 24.306f	35.58 123.36	17.712	
1.812	1,531.58	22.616f 0.950	9.01 24.397f	36.03 117.61	16.848	
1.912	1,621.83	22.717f 1.000	9.05 24.492f	36.51 112.52	16.088	
2.012	1,712.42	22.813f 1.051	9.08 24.564f	36.88 107.68	15.374	
2.112	1,803.32	22.903f 1.102	9.11 24.649f	37.30 103.41	14.749	
2.212	1,894.54	22.989f 1.153	9.14 24.737f	37.74 99.58	14.190	
2.312	1,986.08	23.072f 1.203	9.17 24.826f	38.19 96.11	13.689	
2.412	2,077.95	23.151f 1.254	9.21 24.919f	38.65 92.98	13.238	
2.512	2,170.15	23.228f 1.305	9.23 24.987f	39.01 89.86	12.803	
2.612	2,258.72	23.336f 1.353	8.40 27.240f	29.59 65.48	11.408	
2.712	2,338.10	23.510f 1.395	7.48 29.743f	21.09 45.09	10.042	
2.812	2,408.27	23.728f 1.431	6.56 32.245f	14.46 30.01	8.792	
2.912	2,469.21	23.968f 1.462	5.63 34.748f	9.46 19.15	7.632	
3.012	2,520.92	24.214f 1.488	4.71 37.250f	5.86 11.62	6.543	

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.

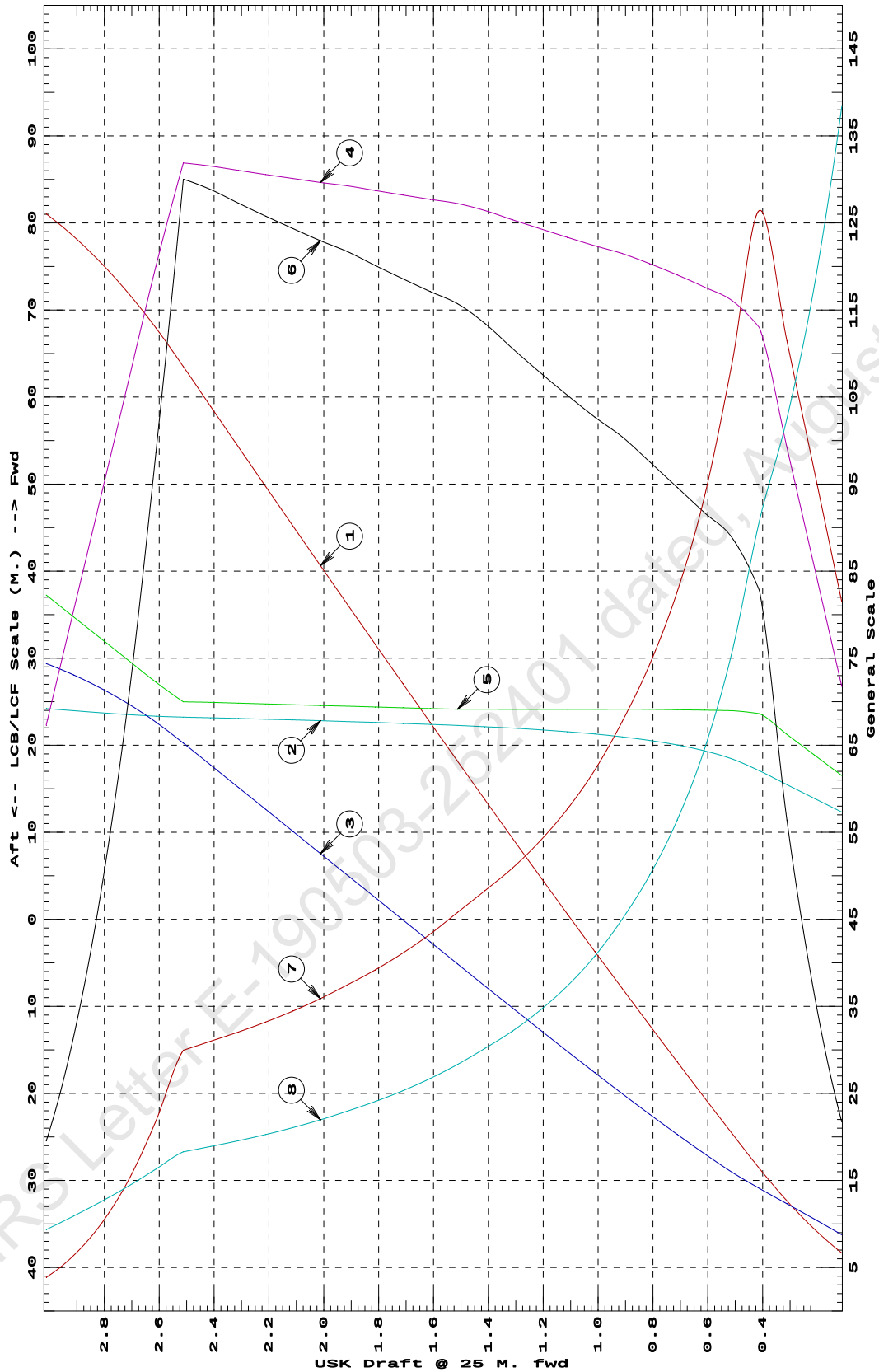
Trim is per 50.00m.

Draft is from USK.

31-07-20 12:18:46
GHS 17.22

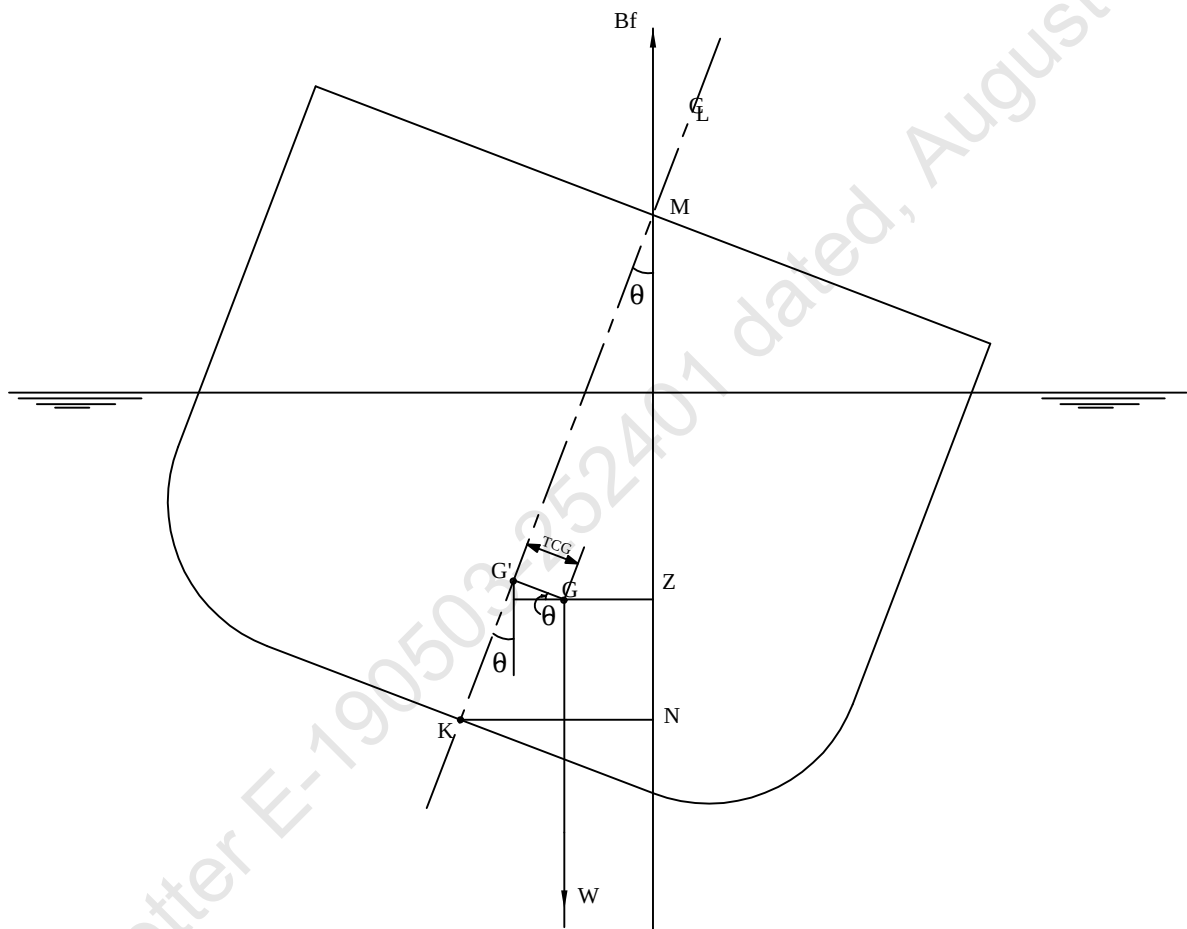
Cybermarine Technologies PTE. Ltd.
PONTOON

HYDROSTATIC PROPERTIES at 1 M. AFT TRIM



Specific Gravity = 1.025 Assumed KG = 0.00 M.
Trim is per 50 M. "K" = BPL

KN CONCEPT



$$\underline{GZ = KN - KG' \sin \theta - TCG \cos \theta}$$

01-08-20 21:16:24
GHS 17.22

Cybermarine Technologies PTE. Ltd.
PONTOON

VOLUME USED FOR KN COMPUTATION

Condition Graphic

Profile View



Plan View



CROSS CURVE TABLES

31-07-20 12:20:24 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONTOON

CROSS CURVES OF STABILITY

Showing righting arms in heel at VCG = 0.00

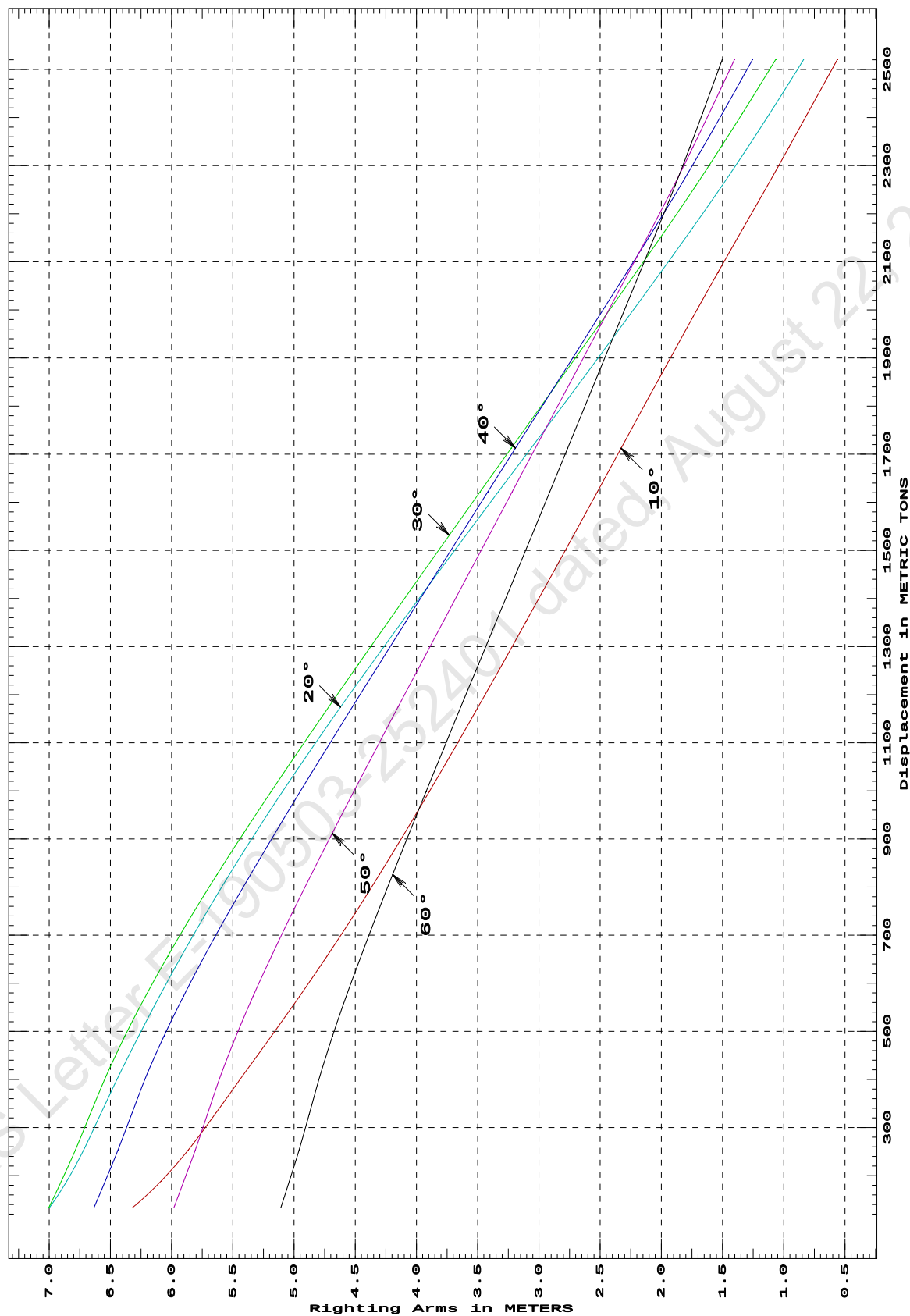
Trim: Fwd 1.000/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees					
	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
133.29	6.320s	6.999s	7.003s	6.634s	5.981s	5.107s
188.30	6.087s	6.857s	6.900s	6.542s	5.901s	5.038s
252.92	5.865s	6.719s	6.786s	6.438s	5.806s	4.959s
327.16	5.646s	6.581s	6.666s	6.327s	5.707s	4.876s
407.76	5.419s	6.431s	6.534s	6.205s	5.600s	4.788s
489.54	5.186s	6.270s	6.382s	6.063s	5.474s	4.686s
572.35	4.956s	6.100s	6.214s	5.904s	5.335s	4.574s
656.10	4.731s	5.920s	6.031s	5.732s	5.184s	4.453s
740.70	4.513s	5.729s	5.836s	5.548s	5.025s	4.326s
826.02	4.303s	5.528s	5.629s	5.355s	4.859s	4.194s
912.01	4.097s	5.316s	5.412s	5.155s	4.687s	4.059s
998.69	3.895s	5.094s	5.186s	4.949s	4.512s	3.921s
1,086.07	3.695s	4.861s	4.955s	4.739s	4.333s	3.781s
1,174.15	3.497s	4.620s	4.717s	4.524s	4.151s	3.638s
1,262.91	3.300s	4.371s	4.475s	4.306s	3.966s	3.494s
1,352.12	3.105s	4.116s	4.229s	4.086s	3.781s	3.349s
1,441.67	2.910s	3.858s	3.982s	3.864s	3.594s	3.203s
1,531.58	2.716s	3.596s	3.732s	3.641s	3.406s	3.057s
1,621.83	2.523s	3.331s	3.480s	3.417s	3.218s	2.911s
1,712.42	2.329s	3.065s	3.227s	3.192s	3.029s	2.765s
1,803.32	2.134s	2.798s	2.972s	2.965s	2.840s	2.618s
1,894.54	1.938s	2.532s	2.717s	2.738s	2.649s	2.471s
1,986.08	1.740s	2.268s	2.462s	2.511s	2.459s	2.323s
2,077.95	1.538s	2.006s	2.207s	2.283s	2.268s	2.175s
2,170.15	1.332s	1.746s	1.954s	2.056s	2.077s	2.027s
2,258.72	1.132s	1.503s	1.718s	1.843s	1.897s	1.888s
2,338.10	0.958s	1.294s	1.514s	1.660s	1.743s	1.767s
2,408.27	0.806s	1.115s	1.338s	1.502s	1.610s	1.664s
2,469.20	0.674s	0.963s	1.190s	1.368s	1.497s	1.576s
2,520.92	0.561s	0.836s	1.066s	1.255s	1.402s	1.502s
Distances in METERS.---Specific Gravity = 1.025.-----						

31-07-20 12:20:24
GHS 17.22

Cybermarine Technologies PTE. Ltd.
PONTOON

CROSS CURVES OF STABILITY - Stbd Heel
at 1 M. FWD TRIM (initial)



Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL

31-07-20 12:20:24 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONTOON

CROSS CURVES OF STABILITY

Showing righting arms in heel at VCG = 0.00

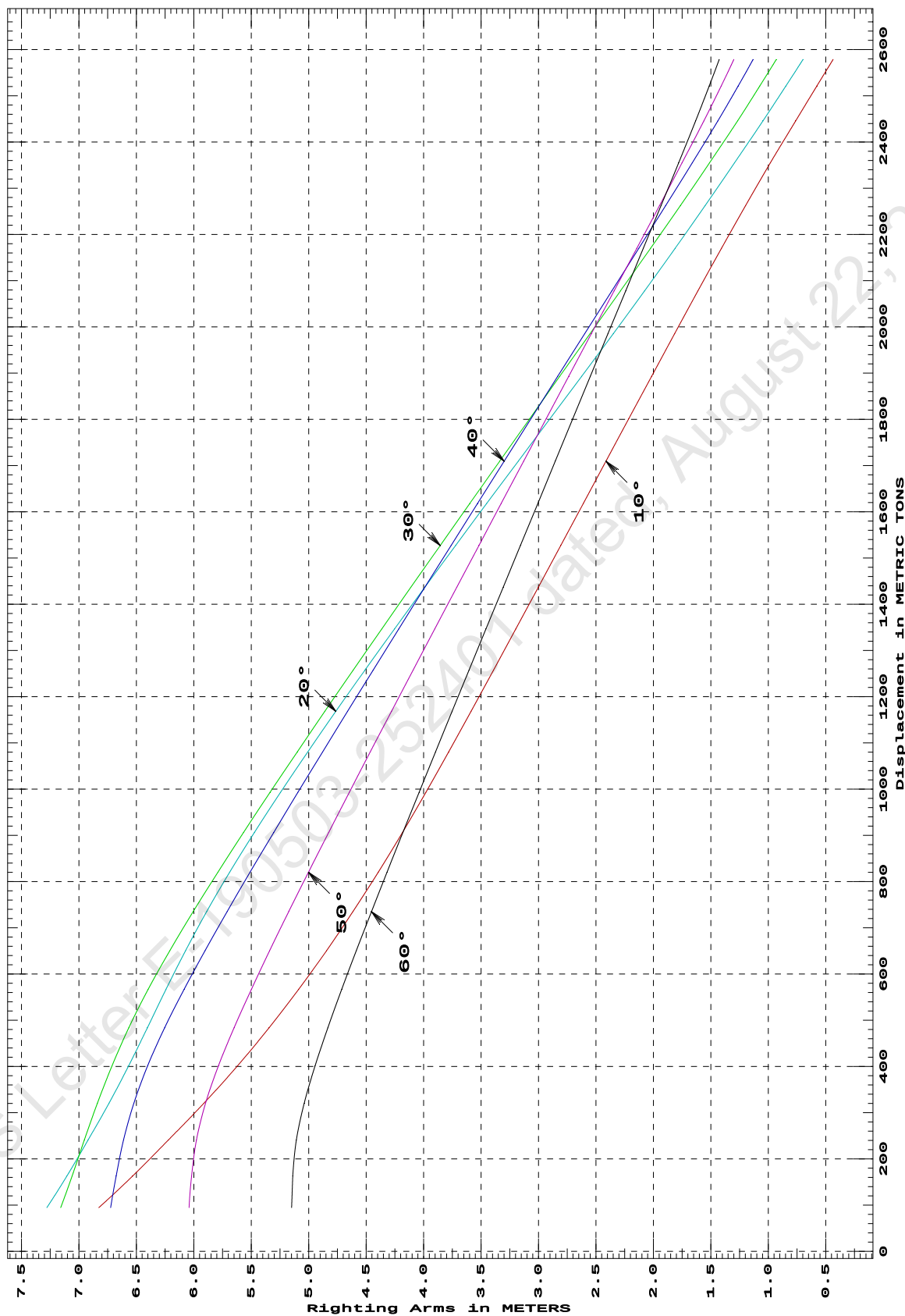
Trim: Fwd 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees					
	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
94.61	6.827s	7.278s	7.158s	6.723s	6.041s	5.149s
162.65	6.533s	7.105s	7.059s	6.676s	6.020s	5.139s
240.79	6.221s	6.920s	6.953s	6.612s	5.977s	5.107s
320.44	5.912s	6.738s	6.839s	6.522s	5.897s	5.039s
401.45	5.619s	6.567s	6.712s	6.401s	5.785s	4.944s
483.61	5.346s	6.404s	6.566s	6.254s	5.649s	4.833s
566.72	5.088s	6.242s	6.397s	6.085s	5.499s	4.712s
650.57	4.845s	6.071s	6.209s	5.902s	5.340s	4.585s
735.11	4.615s	5.887s	6.005s	5.710s	5.173s	4.454s
820.35	4.395s	5.688s	5.790s	5.510s	5.002s	4.318s
906.31	4.184s	5.473s	5.567s	5.305s	4.826s	4.180s
992.99	3.980s	5.246s	5.336s	5.094s	4.647s	4.039s
1,080.33	3.782s	5.009s	5.100s	4.880s	4.465s	3.896s
1,168.34	3.587s	4.764s	4.859s	4.662s	4.280s	3.752s
1,257.02	3.393s	4.512s	4.614s	4.441s	4.093s	3.605s
1,346.39	3.198s	4.253s	4.364s	4.217s	3.903s	3.457s
1,436.46	3.003s	3.990s	4.111s	3.990s	3.712s	3.308s
1,527.11	2.807s	3.722s	3.856s	3.761s	3.519s	3.157s
1,618.11	2.610s	3.451s	3.598s	3.530s	3.325s	3.006s
1,709.44	2.413s	3.177s	3.338s	3.299s	3.130s	2.854s
1,801.11	2.215s	2.901s	3.077s	3.066s	2.934s	2.702s
1,893.13	2.016s	2.624s	2.815s	2.833s	2.738s	2.549s
1,985.42	1.816s	2.349s	2.551s	2.598s	2.540s	2.396s
2,077.73	1.613s	2.077s	2.287s	2.363s	2.343s	2.242s
2,170.06	1.408s	1.811s	2.025s	2.128s	2.145s	2.088s
2,262.39	1.199s	1.551s	1.766s	1.894s	1.947s	1.934s
2,354.73	0.983s	1.295s	1.512s	1.662s	1.751s	1.781s
2,445.61	0.763s	1.048s	1.268s	1.440s	1.562s	1.633s
2,521.33	0.578s	0.848s	1.074s	1.263s	1.411s	1.515s
2,578.59	0.436s	0.699s	0.931s	1.133s	1.301s	1.428s
Distances in METERS.---Specific Gravity = 1.025.-----						

31-07-20 12:20:24
GHS 17.22

Cybermarine Technologies PTE. Ltd.
PONTOON

CROSS CURVES OF STABILITY - Stbd Heel
at 0.5 M. FWD TRIM (initial)



Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL

31-07-20 12:20:24 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONTOON

CROSS CURVES OF STABILITY

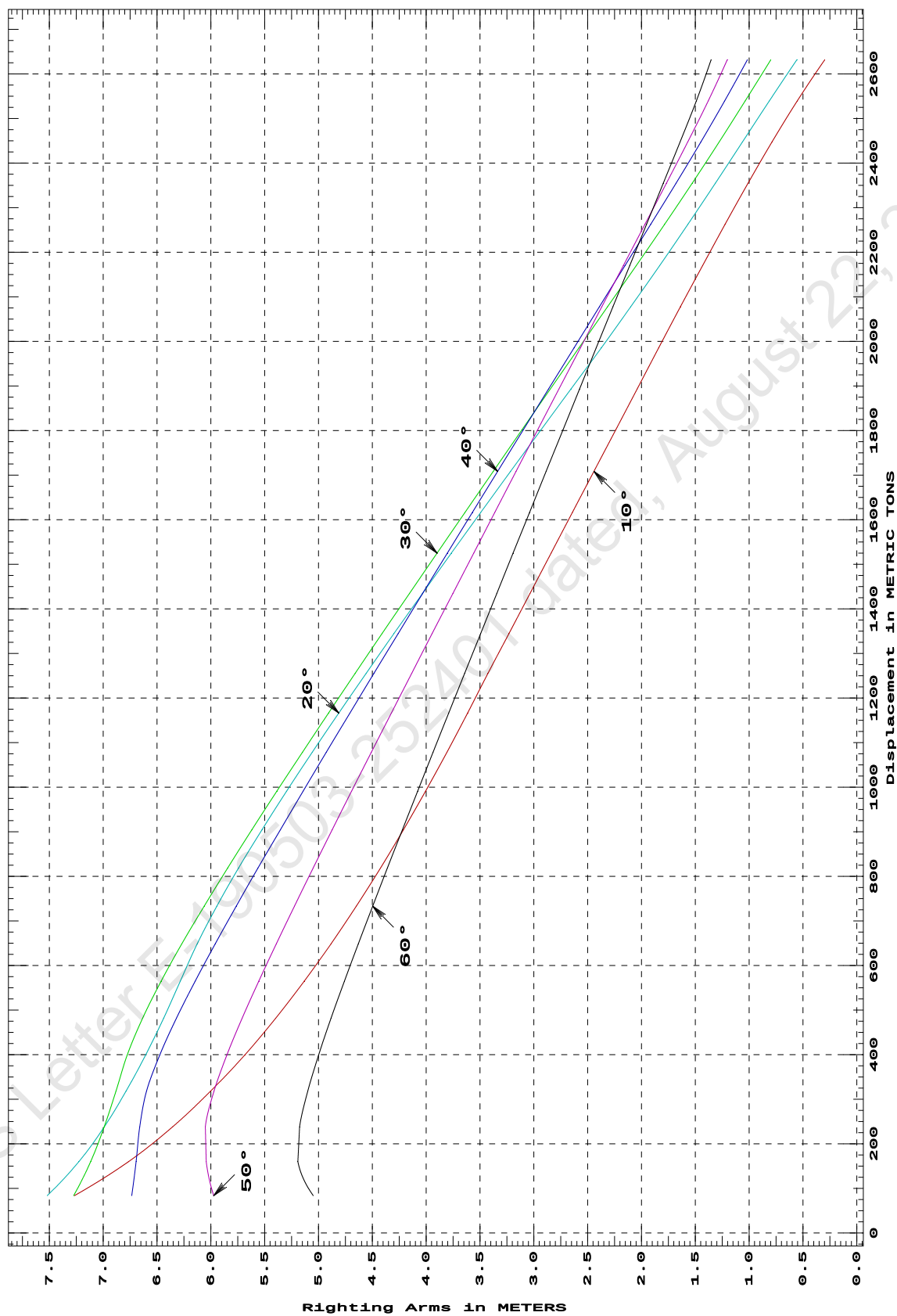
Showing righting arms in heel at VCG = 0.00

Trim: zero at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees					
	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
83.79	7.268s	7.519s	7.274s	6.735s	5.973s	5.049s
160.29	6.763s	7.226s	7.116s	6.694s	6.043s	5.192s
238.38	6.354s	6.990s	6.990s	6.660s	6.051s	5.176s
318.09	6.003s	6.788s	6.881s	6.597s	5.971s	5.100s
399.40	5.685s	6.606s	6.776s	6.474s	5.851s	4.999s
481.78	5.397s	6.438s	6.634s	6.319s	5.709s	4.883s
564.87	5.132s	6.283s	6.462s	6.146s	5.555s	4.760s
648.69	4.884s	6.127s	6.269s	5.960s	5.392s	4.630s
733.24	4.648s	5.945s	6.061s	5.764s	5.224s	4.497s
818.51	4.425s	5.743s	5.844s	5.562s	5.050s	4.360s
904.44	4.211s	5.525s	5.619s	5.355s	4.873s	4.221s
991.06	4.006s	5.297s	5.387s	5.143s	4.693s	4.080s
1,078.37	3.807s	5.058s	5.149s	4.927s	4.509s	3.936s
1,166.40	3.615s	4.812s	4.907s	4.708s	4.324s	3.791s
1,255.11	3.427s	4.559s	4.660s	4.486s	4.135s	3.643s
1,344.46	3.231s	4.300s	4.410s	4.261s	3.945s	3.495s
1,434.46	3.036s	4.035s	4.156s	4.033s	3.753s	3.345s
1,525.13	2.840s	3.767s	3.899s	3.803s	3.559s	3.193s
1,616.48	2.642s	3.493s	3.640s	3.571s	3.363s	3.040s
1,708.51	2.442s	3.216s	3.377s	3.336s	3.165s	2.886s
1,800.77	2.241s	2.937s	3.113s	3.100s	2.966s	2.731s
1,893.06	2.041s	2.656s	2.848s	2.864s	2.767s	2.576s
1,985.36	1.840s	2.374s	2.582s	2.627s	2.568s	2.421s
2,077.68	1.638s	2.100s	2.316s	2.390s	2.369s	2.265s
2,170.03	1.433s	1.832s	2.049s	2.153s	2.169s	2.110s
2,262.38	1.225s	1.570s	1.786s	1.916s	1.969s	1.954s
2,354.73	1.010s	1.314s	1.529s	1.679s	1.769s	1.798s
2,447.09	0.789s	1.063s	1.280s	1.449s	1.571s	1.642s
2,539.44	0.554s	0.813s	1.038s	1.228s	1.380s	1.491s
2,631.79	0.298s	0.558s	0.799s	1.016s	1.204s	1.354s
Distances in METERS.---Specific Gravity = 1.025.-----						

31-07-20 12:20:24
GHS 17.22Cybermarine Technologies PTE. Ltd.
PONTOON

CROSS CURVES OF STABILITY - Stbd Heel
at LEVEL TRIM (initial)



Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL

31-07-20 12:20:24 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONTOON

CROSS CURVES OF STABILITY

Showing righting arms in heel at VCG = 0.00

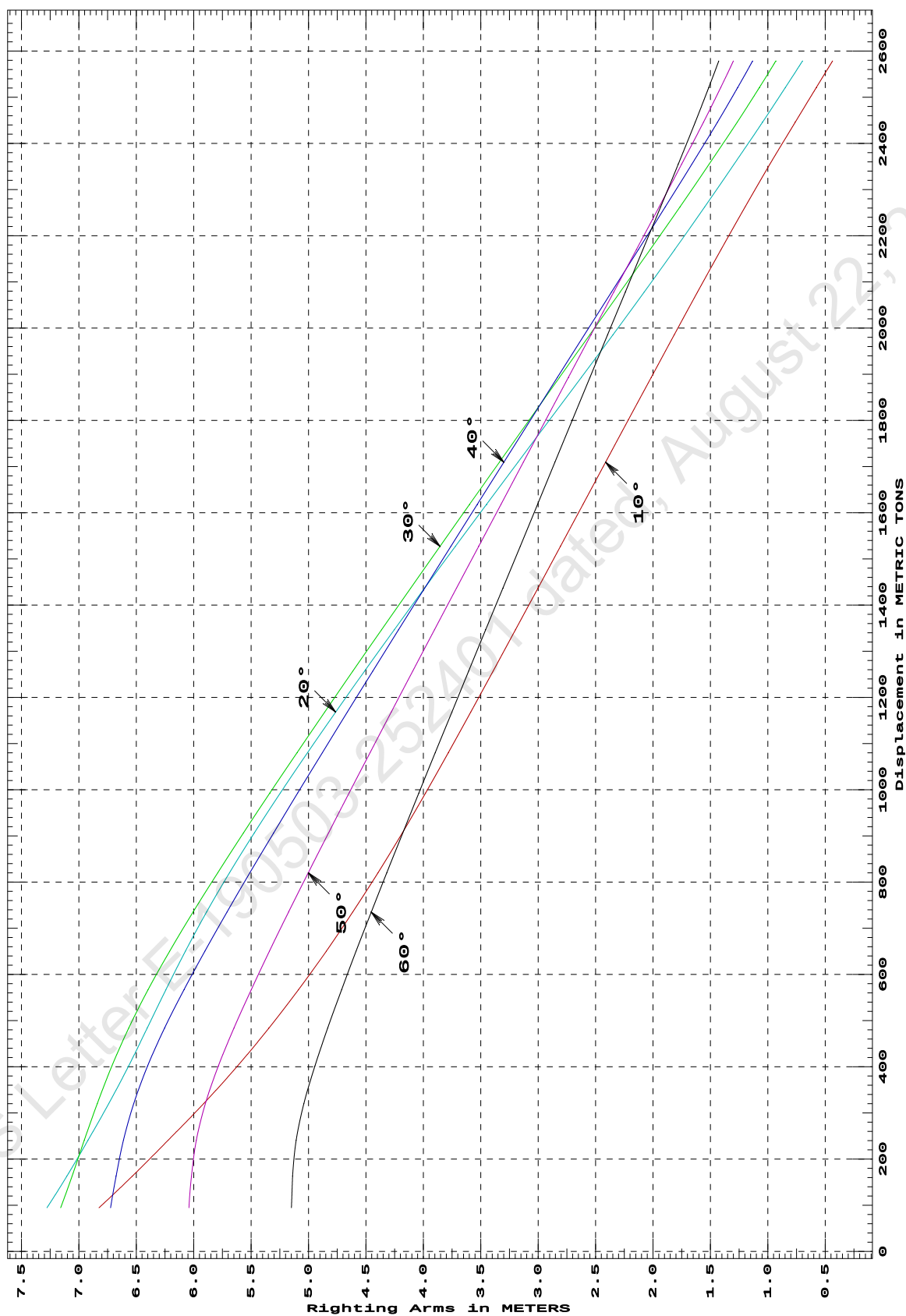
Trim: Aft 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees					
	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
94.61	6.824s	7.277s	7.158s	6.723s	6.041s	5.149s
162.65	6.535s	7.105s	7.059s	6.676s	6.020s	5.139s
240.79	6.222s	6.920s	6.953s	6.612s	5.977s	5.107s
320.44	5.912s	6.738s	6.839s	6.522s	5.897s	5.040s
401.45	5.619s	6.567s	6.712s	6.401s	5.785s	4.944s
483.61	5.346s	6.405s	6.566s	6.254s	5.649s	4.833s
566.72	5.088s	6.242s	6.397s	6.085s	5.499s	4.712s
650.57	4.845s	6.071s	6.209s	5.902s	5.340s	4.585s
735.11	4.615s	5.887s	6.005s	5.710s	5.173s	4.454s
820.35	4.395s	5.688s	5.790s	5.510s	5.002s	4.318s
906.31	4.184s	5.473s	5.567s	5.305s	4.826s	4.180s
992.99	3.980s	5.246s	5.336s	5.094s	4.647s	4.039s
1,080.33	3.782s	5.009s	5.100s	4.880s	4.465s	3.896s
1,168.34	3.587s	4.764s	4.859s	4.662s	4.280s	3.752s
1,257.02	3.393s	4.511s	4.614s	4.441s	4.093s	3.605s
1,346.39	3.198s	4.253s	4.364s	4.216s	3.903s	3.457s
1,436.46	3.003s	3.990s	4.111s	3.990s	3.712s	3.308s
1,527.11	2.807s	3.722s	3.855s	3.761s	3.519s	3.157s
1,618.10	2.610s	3.451s	3.598s	3.530s	3.325s	3.006s
1,709.44	2.413s	3.177s	3.338s	3.299s	3.130s	2.854s
1,801.11	2.215s	2.901s	3.077s	3.066s	2.934s	2.702s
1,893.13	2.016s	2.624s	2.815s	2.833s	2.738s	2.549s
1,985.42	1.816s	2.349s	2.551s	2.598s	2.540s	2.396s
2,077.73	1.613s	2.077s	2.287s	2.363s	2.343s	2.242s
2,170.06	1.408s	1.811s	2.025s	2.128s	2.145s	2.088s
2,262.39	1.198s	1.551s	1.766s	1.894s	1.947s	1.934s
2,354.73	0.983s	1.295s	1.512s	1.662s	1.751s	1.781s
2,445.61	0.763s	1.048s	1.268s	1.440s	1.562s	1.633s
2,521.33	0.578s	0.848s	1.074s	1.263s	1.411s	1.515s
2,578.59	0.436s	0.698s	0.931s	1.133s	1.301s	1.428s
Distances in METERS.---Specific Gravity = 1.025.-----						

31-07-20 12:20:24
GHS 17.22

Cybermarine Technologies PTE. Ltd.
PONTOON

CROSS CURVES OF STABILITY - Stbd Heel
at 0.5 M. AFT TRIM (initial)



Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL

31-07-20 12:20:24 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONTOON

CROSS CURVES OF STABILITY

Showing righting arms in heel at VCG = 0.00

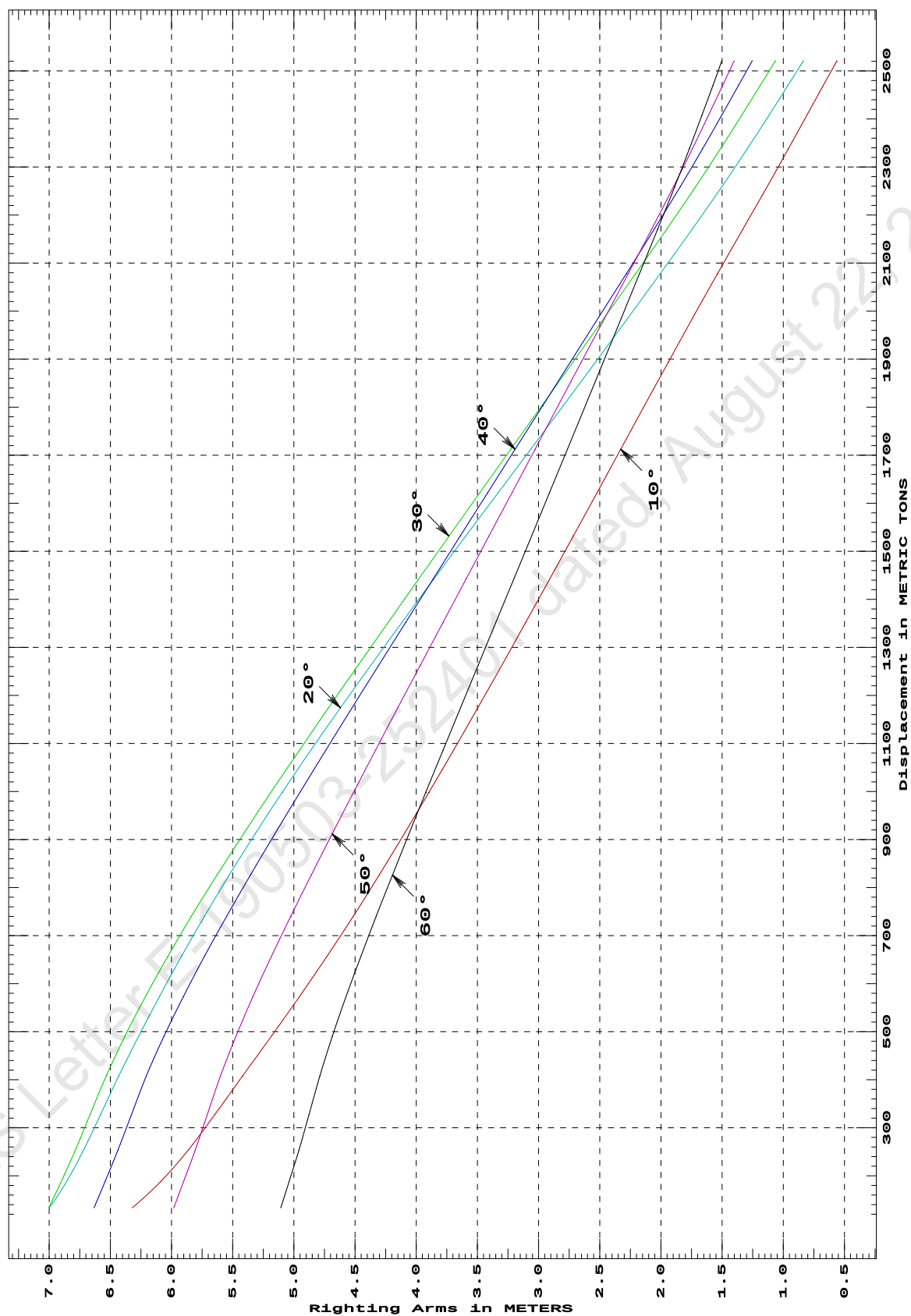
Trim: Aft 1.000/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees					
	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
133.29	6.322s	6.999s	7.004s	6.633s	5.981s	5.107s
188.30	6.087s	6.857s	6.900s	6.542s	5.901s	5.038s
252.92	5.864s	6.719s	6.786s	6.438s	5.807s	4.959s
327.16	5.646s	6.580s	6.666s	6.327s	5.708s	4.876s
407.76	5.419s	6.431s	6.533s	6.205s	5.600s	4.787s
489.54	5.186s	6.270s	6.382s	6.063s	5.474s	4.686s
572.35	4.956s	6.100s	6.214s	5.904s	5.335s	4.574s
656.10	4.731s	5.920s	6.031s	5.732s	5.184s	4.453s
740.70	4.513s	5.729s	5.836s	5.548s	5.025s	4.326s
826.02	4.303s	5.528s	5.629s	5.355s	4.858s	4.194s
912.01	4.097s	5.316s	5.411s	5.155s	4.687s	4.059s
998.69	3.895s	5.093s	5.186s	4.949s	4.511s	3.921s
1,086.07	3.695s	4.861s	4.954s	4.738s	4.333s	3.780s
1,174.15	3.497s	4.619s	4.717s	4.524s	4.151s	3.638s
1,262.91	3.300s	4.370s	4.474s	4.306s	3.966s	3.494s
1,352.12	3.105s	4.116s	4.229s	4.085s	3.780s	3.348s
1,441.67	2.910s	3.858s	3.981s	3.863s	3.593s	3.203s
1,531.58	2.716s	3.596s	3.732s	3.640s	3.406s	3.057s
1,621.83	2.523s	3.330s	3.480s	3.416s	3.218s	2.911s
1,712.42	2.329s	3.065s	3.227s	3.192s	3.029s	2.764s
1,803.32	2.134s	2.798s	2.973s	2.966s	2.840s	2.618s
1,894.54	1.938s	2.532s	2.717s	2.738s	2.650s	2.471s
1,986.08	1.740s	2.268s	2.462s	2.511s	2.458s	2.323s
2,077.95	1.538s	2.006s	2.207s	2.283s	2.268s	2.175s
2,170.15	1.332s	1.746s	1.954s	2.056s	2.077s	2.027s
2,258.72	1.132s	1.503s	1.718s	1.843s	1.898s	1.888s
2,338.10	0.958s	1.294s	1.513s	1.660s	1.743s	1.768s
2,408.27	0.806s	1.115s	1.339s	1.502s	1.610s	1.664s
2,469.21	0.674s	0.963s	1.189s	1.367s	1.497s	1.576s
2,520.92	0.561s	0.836s	1.065s	1.254s	1.402s	1.503s
Distances in METERS.---Specific Gravity = 1.025.-----						

31-07-20 12:20:24
GHS 17.22

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PONTOON

**CROSS CURVES OF STABILITY - Stbd Heel
at 1 M. AFT TRIM (initial)**



Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL

DEADWEIGHT TABLE

Draft	Disp	Dwt	
M	T	T	
0.612	481.78	75.81	
0.712	564.87	158.90	
0.812	648.69	242.72	
0.912	733.24	327.27	
1.012	818.51	412.54	
1.112	904.44	498.47	
1.212	991.06	585.09	
1.312	1078.37	672.40	
1.412	1166.40	760.43	
1.512	1255.11	849.14	
1.612	1344.46	938.49	
1.712	1434.46	1028.49	
1.812	1525.13	1119.16	
1.912	1616.48	1210.51	
2.012	1708.51	1302.54	
2.112	1800.77	1394.80	
2.212	1893.06	1487.09	Full Load Displacement

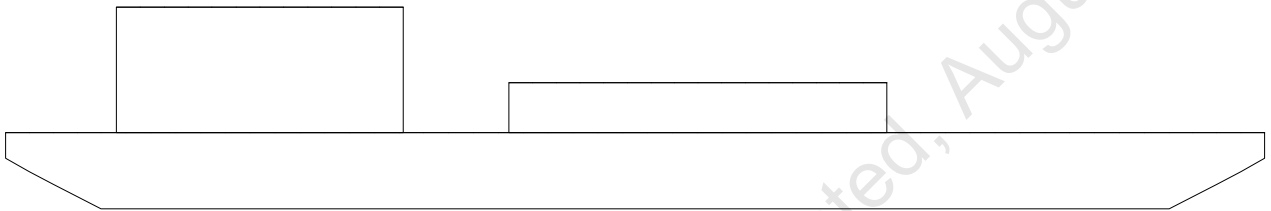
01-08-20 22:26:06 Cybermarine Technologies PTE. Ltd.
GHS 17.22 PONTOON

USK draft refers to the line:

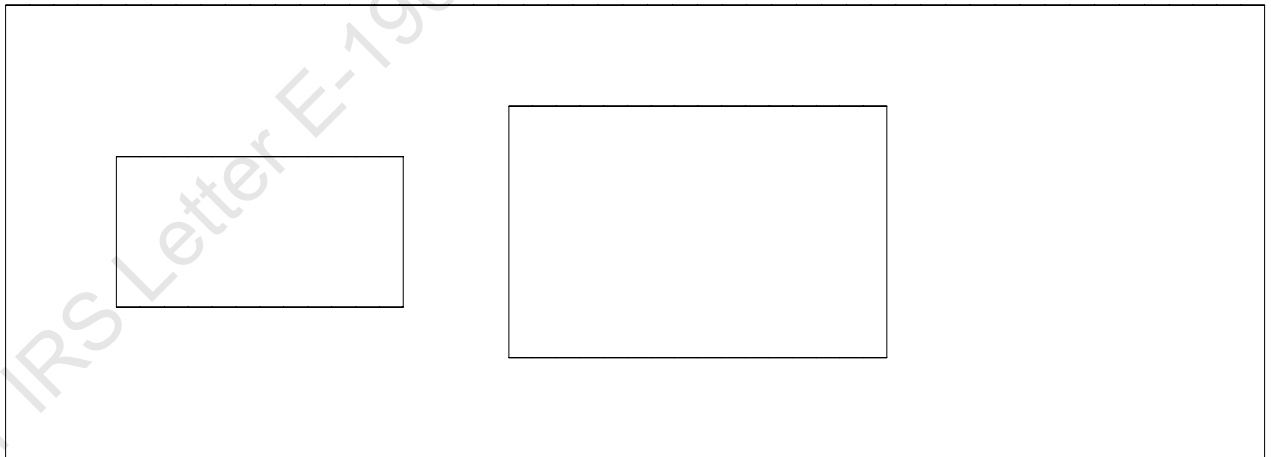
0.012 below baseline @ 50.000f and 0.012 below baseline @ 0.000

Condition Graphic

Profile View



Plan View



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MAXIMUM VCG vs. DISPLACEMENT

Heeling moment is present from: wind

Trim = Fwd 1.000/50.000 at zero heel (trim righting arm held at zero)

Displacement		--- Margins ---			
METRIC TONS	Max VCG	LIM1	LIM2	LIM3	
400.00	18.489	122%	0d	7d	
600.00	17.349	115%	0d	6d	
800.00	16.087	112%	0d	5d	
1,000.00	14.631	73%	0d	4d	
1,200.00	13.079	51%	0d	3d	
1,400.00	11.429	30%	0d	3d	
1,600.00	9.770	12%	0d	2d	
1,800.00	7.778	0%	1d	1d	
2,000.00	5.433	0%	5d	1d	

Heeling moment is present from: wind

Trim = Fwd 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement		--- Margins ---			
METRIC TONS	Max VCG	LIM1	LIM2	LIM3	
400.00	18.792	138%	0d	10d	
600.00	17.834	103%	0d	7d	
800.00	16.573	83%	0d	6d	
1,000.00	15.128	70%	0d	5d	
1,200.00	13.510	57%	0d	4d	
1,400.00	11.824	35%	0d	3d	
1,600.00	10.137	7%	0d	3d	
1,800.00	7.959	0%	2d	2d	
2,000.00	5.618	3%	6d	1d	

Heeling moment is present from: wind

Trim = zero at zero heel (trim righting arm held at zero)

Displacement		--- Margins ---			
METRIC TONS	Max VCG	LIM1	LIM2	LIM3	
400.00	18.899	137%	0d	14d	
600.00	17.990	117%	0d	10d	
800.00	16.745	80%	0d	7d	
1,000.00	15.270	69%	0d	6d	
1,200.00	13.683	59%	0d	5d	
1,400.00	11.943	37%	0d	4d	
1,600.00	10.176	8%	0d	3d	
1,800.00	8.007	0%	2d	3d	
2,000.00	5.806	0%	5d	2d	

Heeling moment is present from: wind

Trim = Aft 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement		--- Margins ---			
METRIC TONS	Max VCG	LIM1	LIM2	LIM3	
400.00	18.793	138%	0d	10d	
600.00	17.835	103%	0d	7d	
800.00	16.573	83%	0d	6d	
1,000.00	15.129	70%	0d	5d	
1,200.00	13.510	57%	0d	4d	

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GHS 17.22 PONTOON

1,400.00	11.825	35%	0d	3d
1,600.00	10.136	7%	0d	3d
1,800.00	7.958	0%	2d	2d
2,000.00	5.599	4%	6d	1d

Heeling moment is present from: wind
Trim = Aft 1.000/50.000 at zero heel (trim righting arm held at zero)

Displacement	--- Margins ---			
METRIC TONS	Max VCG	LIM1	LIM2	LIM3
400.00	18.491	119%	0d	7d
600.00	17.392	114%	0d	6d
800.00	16.095	112%	0d	5d
1,000.00	14.632	73%	0d	4d
1,200.00	13.079	51%	0d	3d
1,400.00	11.430	31%	0d	3d
1,600.00	9.769	12%	0d	2d
1,800.00	7.781	0%	1d	1d
2,000.00	5.434	0%	5d	1d

Distances in METERS.---Specific Gravity = 1.025.---d = degrees.

LIM-----IS CODE 2008 CRITERION-----Min/Max
(1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad
(2) Angle from Equilibrium to RAzero or Flood > 20.00 deg
(3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg

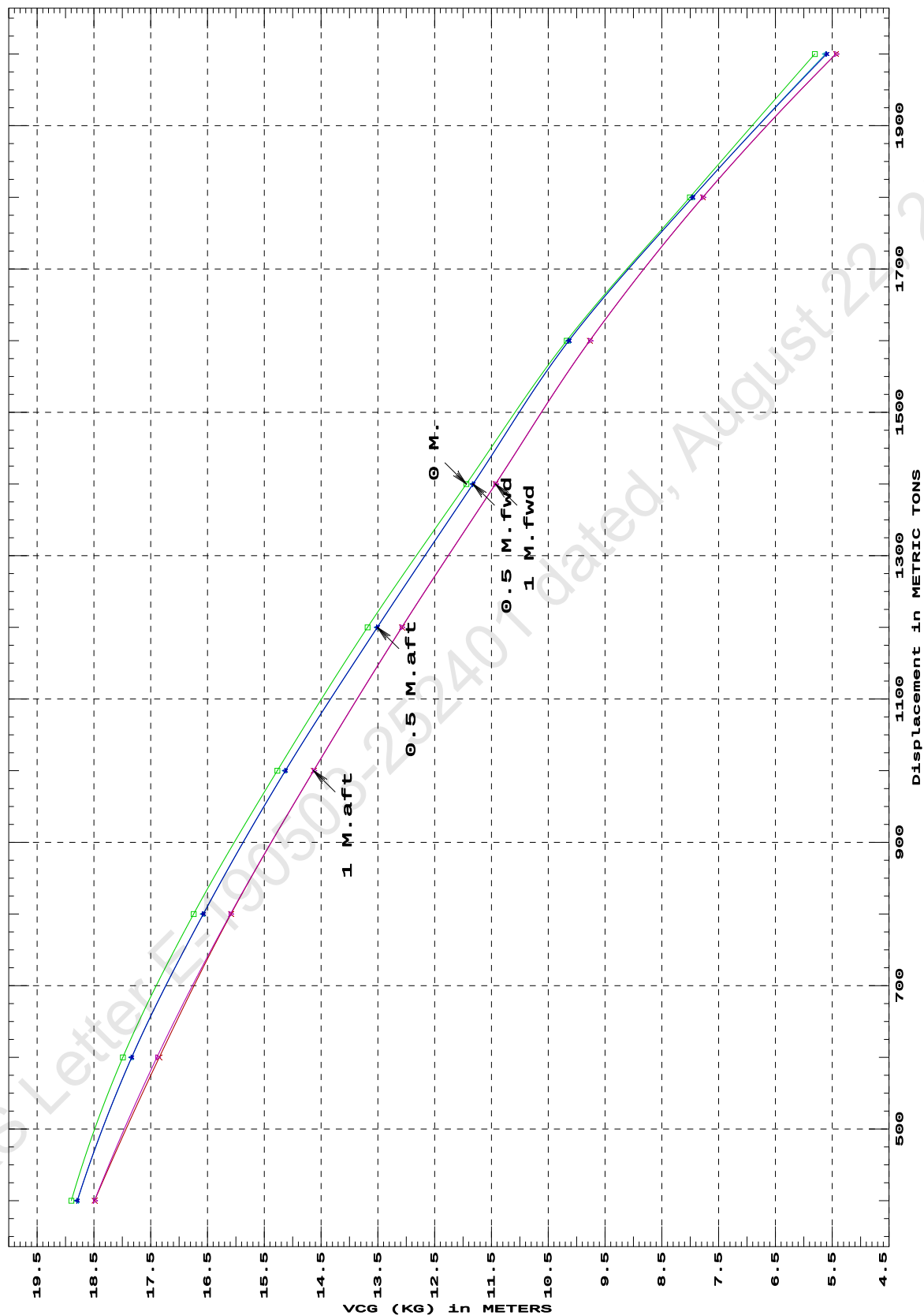
MAX VCG DOES NOT COVER CRANE OPERATIONS/LIFTING OF WEIGHTS AND IS VALID ONLY FOR UNMANNED TOW, WITH CRANE IN STOWED POSITION.

MAX DECK CARGO HEIGHT = 2.00M ABOVE MAIN DECK

01-08-20 22:26:06
GHS 17.22

Cybermarine Technologies PTE. Ltd.
PONTOON

IS CODE 2008 criterion MAXIMUM VCG (KG)
at various trims (initial)



Specific Gravity = 1.025
Trim is per 50 M.
"K" = BPL

LOADING CONDITIONS

SUMMARY OF CONDITIONS				
CONDITION NO.	1	2	3	4
CONDITION	Condition 00 : Lightship Condition	Condition 1 : Lightship + Crane + Ballast	Condition 2 : Lightship + Crane + Deck Cargo + Ballast / Maximum Draft Condition	Condition 3 : Lightship + Deck Cargo / Maximum Deck Cargo Condition
TOTAL TANK LOAD (T)	0.000	234.870	234.870	0.000
DWT CONST. (T)	0.000	400.000	1252.000	1487.090
DEADWEIGHT (T)	0.000	634.870	1486.870	1487.090
LIGHT WEIGHT (T)	405.970	405.970	405.970	405.970
DISPLACEMENT (T)	405.760	1040.840	1892.840	1893.060
FWD DRAFT (M)	0.520	1.141	2.210	2.212
AFT DRAFT (M)	0.520	1.396	2.213	2.212
MEAN DRAFT (M)	0.520	1.268	2.212	2.212
TRIM (M)	0.000	0.255a	0.000	0.000
L.C.G. (FROM AP) (M)	25.000f	24.220f	24.994f	25.000f
L.C.B. (FROM AP) (M)	25.000f	24.203f	24.994f	25.000f
M.C.T. (T-CM)	26.790	31.910	37.330	37.410
KG (M)	3.000	3.975	3.987	3.786
GM _T (M)	51.854	19.075	10.176	10.523
MAX. GZ (M)	5.543	3.733	1.357	1.446
Note: For details of deadweight constant refer respective loading condition				

SUMMARY OF CONDITIONS				
CONDITION NO.	5	6	7	8
CONDITION	Condition 4 : Operating Condition of 400T Crane Crane Lifting 40T @ 30m Radius aft	Condition 5 : Operating Condition of 400T Crane Crane Lifting 40T @ 30m Radius stbd	Condition 6 : Operating Condition of 400T Crane Crane Lifting 40T @ 30m Radius fwd	Condition 7 : Operating Condition of 350T Crane Crane Lifting 40T @ 35m Radius aft
TOTAL TANK LOAD (T)	284.980	284.980	284.980	422.400
DWT CONST. (T)	440.000	440.000	440.000	390.000
DEADWEIGHT (T)	724.980	724.980	724.980	812.400
LIGHT WEIGHT (T)	406.000	406.000	406.000	405.970
DISPLACEMENT (T)	1130.980	1130.980	1130.980	1218.370
FWD DRAFT (M)	1.145	1.339	1.529	1.243
AFT DRAFT (M)	1.594	1.405	1.213	1.695
MEAN DRAFT (M)	1.370	1.372	1.371	1.469
TRIM (M)	0.449a	0.067a	0.316f	0.452a
L.C.G. (FROM AP) (M)	23.713f	24.809f	25.906f	23.762f
L.C.B. (FROM AP) (M)	23.674f	24.803f	25.933f	23.732f
M.C.T. (T-CM)	32.400	32.460	32.410	33.320
KG (M)	4.965	4.965	4.965	4.075
GM _T (M)	16.440	16.471	16.441	15.953
MAX. GZ (M)	3.186	3.221	3.204	3.173
Note: For details of deadweight constant refer respective loading condition				

SUMMARY OF CONDITIONS				
CONDITION NO.	9	10	11	12
CONDITION	Condition 8 : Operating Condition of 350T Crane Crane Lifting 40T @ 35m Radius stbd	Condition 9 : Operating Condition of 350T Crane Crane Lifting 40T @ 35m Radius fwd	Condition 10 : Operating Condition of 250T Crane Crane Lifting 36.6T @ 18m Radius aft	Condition 11 : Operating Condition of 250T Crane Crane Lifting 36.6T @ 18m Radius stbd
TOTAL TANK LOAD (T)	422.400	422.400	422.400	422.400
DWT CONST. (T)	390.000	390.000	286.600	286.600
DEADWEIGHT (T)	812.400	812.400	709.000	709.000
LIGHT WEIGHT (T)	405.970	405.970	405.970	405.970
DISPLACEMENT (T)	1218.370	1218.370	1114.970	1114.970
FWD DRAFT (M)	1.454	1.663	1.273	1.376
AFT DRAFT (M)	1.487	1.276	1.434	1.332
MEAN DRAFT (M)	1.471	1.469	1.353	1.354
TRIM (M)	0.032a	0.387f	0.161a	0.044f
L.C.G. (FROM AP) (M)	24.911f	26.060f	24.536f	25.127f
L.C.B. (FROM AP) (M)	24.909f	26.086f	24.521f	25.132f
M.C.T. (T-CM)	33.400	33.330	32.230	32.230
KG (M)	4.075	4.075	5.390	5.390
GM _T (M)	15.982	15.953	16.199	16.193
MAX. GZ (M)	3.209	3.182	3.119	3.122
Note: For details of deadweight constant refer respective loading condition				

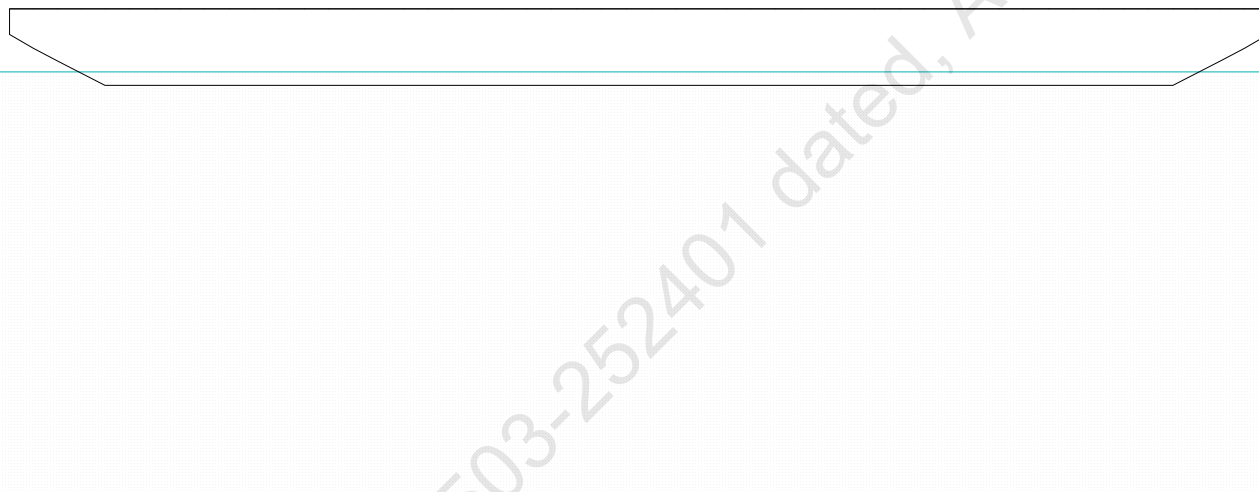
SUMMARY OF CONDITIONS				
CONDITION NO.	13	14	15	16
CONDITION	Condition 12 : Operating Condition of 250T Crane Crane Lifting 36.6T @ 18m Radius fwd	Condition 13 : Operating Condition of 150T Crane Crane Lifting 51.3T @ 12m Radius aft	Condition 14 : Operating Condition of 150T Crane Crane Lifting 51.3T @ 12m Radius stbd	Condition 15 : Operating Condition of 150T Crane Crane Lifting 51.3T @ 12m Radius fwd
TOTAL TANK LOAD (T)	422.400	323.200	323.200	323.200
DWT CONST. (T)	286.600	201.300	201.300	201.300
DEADWEIGHT (T)	709.000	524.500	524.500	524.500
LIGHT WEIGHT (T)	405.970	405.970	405.970	405.970
DISPLACEMENT (T)	1114.970	930.470	930.470	930.470
FWD DRAFT (M)	1.477	0.942	1.042	1.142
AFT DRAFT (M)	1.229	1.340	1.241	1.142
MEAN DRAFT (M)	1.353	1.141	1.142	1.142
TRIM (M)	0.248f	0.398a	0.199a	0.000
L.C.G. (FROM AP) (M)	25.718f	23.679f	24.340f	25.002f
L.C.B. (FROM AP) (M)	25.741f	23.646f	24.325f	25.001f
M.C.T. (T-CM)	32.190	30.960	30.880	30.870
KG (M)	5.390	4.503	4.503	4.503
GM _T (M)	16.172	20.875	20.815	20.806
MAX. GZ (M)	3.111	3.816	3.839	3.847
Note: For details of deadweight constant refer respective loading condition				

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PONTOON**Condition 00 : Lightship****Lightship condition**

Condition Graphic - Draft: 0.520 Trim: zero Heel: zero

Profile View



Plan View



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PONTOON

Condition 00 : Lightship
Lightship condition

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 0.520 @ 50.00f, 0.520 @ 25.00f, 0.520 @ 0.00
Trim: zero, Heel: zero

Part-----	Weight (MT)----	LCG-----	TCG-----	VCG-----	FSM-----
WEIGHT	405.97	25.000f	0.000	3.000	
Load-----	SpGr-----	Weight (MT)----	LCG-----	TCG-----	VCG-----
Total Tanks----->		0.00			0.00
		Displ (MT)----	LCB-----	TCB-----	VCB-----
HULL	1.025	405.76	25.000f	0.000	0.252

Righting Arms:		0.000	0.000		
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----BMT
Total Waterplane---->	1.025	800.8	25.000f	0.000	332.92 54.603
		MT/cm-----	m.-MT/cm-----	GML-----	GMT
		8.21	26.79	330.18	51.854

Distances in METERS.-----Moments in m.-MT.

SUMMARY OF LOADING
0.0 Cu.M. (0%) SALT WATER

DRAFT AT FP (M) = 0.520
DRAFT AT MS (M) = 0.520
DRAFT AT AP (M) = 0.520
DRAFT AT FWD MARK (M) = 0.520
DRAFT AT AFT MARK (M) = 0.520

HYDROSTATIC PROPERTIES
No Trim, No Heel, VCG = 3.000

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/
Draft----	Weight (MT)----	LCB-----	VCB-----	cm-----LCF-----cm trim-----GML-----GMT
0.520	405.76	25.000f	0.252	8.21 25.000f 26.79 330.17 51.854

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

HEELING MOMENT specification
Lateral Plane Method

Wind pressure toward starboard
USK draft: 0.520 @ 50.00f, 0.520 @ 25.00f, 0.520 @ 0.00
Trim: zero, Heel: zero

Part-----	LPA*SF-----	HCP-----	Arm-----	Pressure-----	Moment-----
HULL	121.1	1.277	1.534	0.05500	10.21

Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT

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PONTOON**Condition 00 : Lightship****Lightship condition**

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

LCG = 25.000f TCG = 0.000 VCG = 3.000

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---Trim---Heel---	Weight (MT)---	in Trim--in Heel---	Area----	Angle
0.508 0.00 0.00	405.81	0.000 -0.025	0.0000	13.835
0.508 0.00 0.03s	405.82	0.000 0.000	-0.0000	13.808
0.459 0.00 5.03s	406.05	0.000 3.927	0.1713	8.808
0.174 0.00 10.03s	406.15	0.000 5.118	0.5769	3.808
-0.133 0.00 13.83s	405.97	0.000 5.427	0.9302	0.000
-0.239 0.00 15.03s	405.97	0.000 5.477	1.0436	-1.192
-0.639 0.00 19.24s	405.76	0.000 5.543	1.4510	-5.406
-0.716 0.00 20.03s	406.23	0.000 5.540	1.5271	-6.192
-1.234 0.00 25.03s	405.99	0.000 5.445	2.0064	-11.192
-1.772 0.00 30.03s	405.92	0.000 5.240	2.4734	-16.192
-2.301 0.00 35.03s	406.14	0.000 4.916	2.9175	-21.192
-2.813 0.00 40.03s	405.93	0.000 4.507	3.3293	-26.192
-3.304 0.00 45.03s	405.83	0.000 4.035	3.7024	-31.192
-3.767 0.00 50.03s	406.10	0.000 3.513	4.0322	-36.192
-4.204 0.00 55.03s	406.09	0.000 2.954	4.3146	-41.192
-4.609 0.00 60.03s	405.86	0.000 2.364	4.5469	-46.192
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.				

Note: No tank loads are present.

Note: The Residual Righting Arms shown above are in excess of the
wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 10.21 (constant)

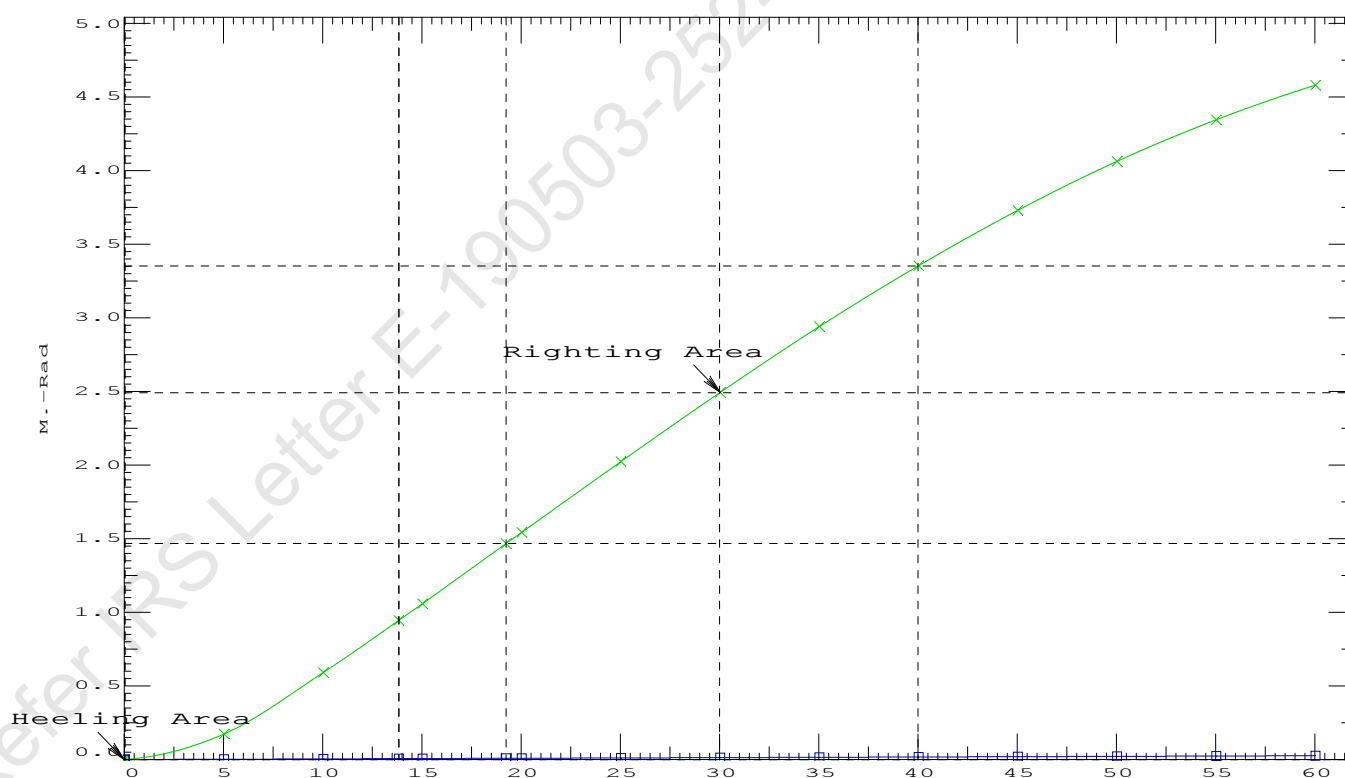
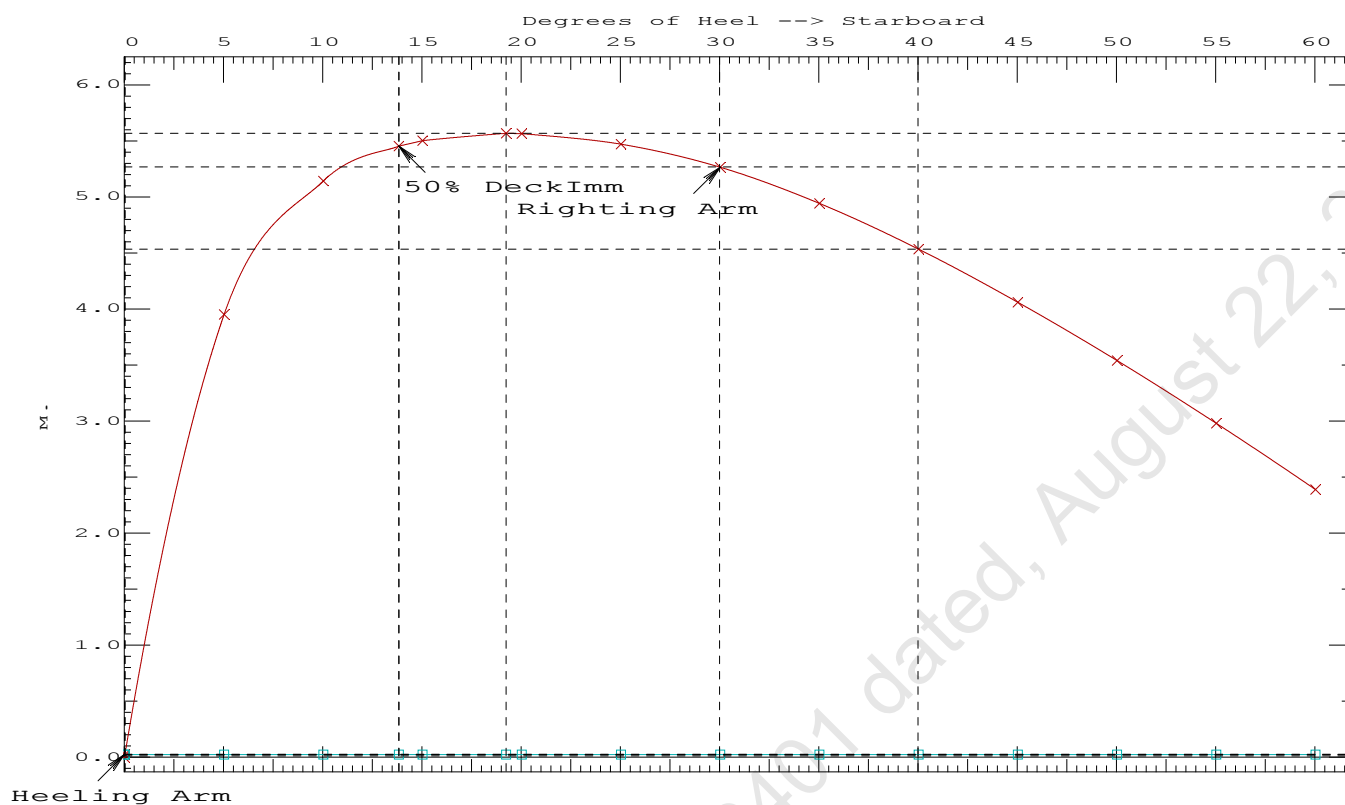
LIM-----	IS CODE 2008 PART B CHAPTER-2 SEC.2----	Min/Max-----	Attained
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	1.4510 P
(2) Angle from Equilibrium to RAzero or Flood	>	20.00 deg	LARGE
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	13.81 P

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PONTOON

Condition 00 : Lightship

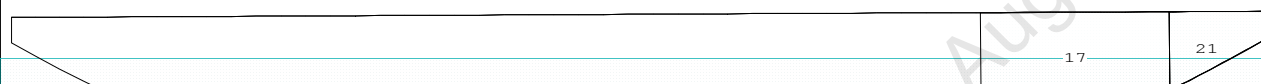
Lightship condition



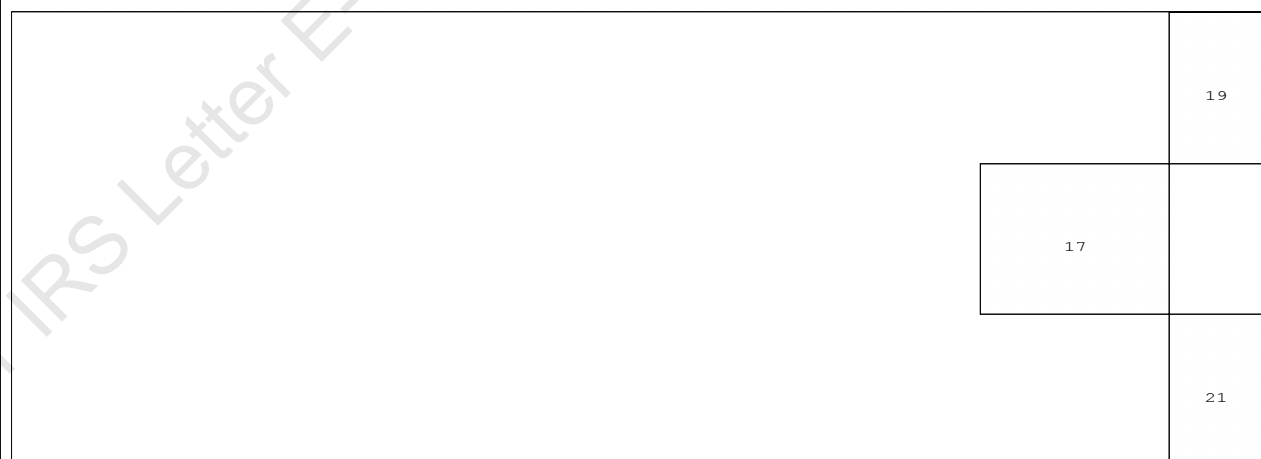
PONTOON

Condition Graphic - Draft: 1.141 @ 50.000f, 1.396 @ 0.000 Heel: zero

Profile View



Plan View



Rank	Material	Volume (m³)	Weight (kg)	Material	Volume (m³)	Weight (kg)	Material	Volume (m³)	Weight (kg)	Material	Volume (m³)	Weight (kg)	
1	17-06-WBTK-G-100%	19	07	WBTK-P-100%	19	07	WBTK-P-100%	19	07	WBTK-P-100%	19	07	WBTK-P-100%
2	17-06-WBTK-G-100%	21	07	WBTK-S-100%	21	07	WBTK-S-100%	21	07	WBTK-S-100%	21	07	WBTK-S-100%

PONTOON

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

Trim: Aft 0.255/50.000, Heel: zero

Righting Arms:		0.001	0.000		
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----BMT
Total Waterplane----	> 1.025	852.6	24.767f	0.000	156.64 22.410
		MT/cm-----	m.-MT/cm-----	GML-----	GMT
		8.74	31.91	153.31	19.075
Distances in METERS.-----		Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

229.1 Cu.M. (9%) SALT WATER

400.00 MT Crane

DRAFT AT FP	(M)	=	1.141
DRAFT AT MS	(M)	=	1.268
DRAFT AT AP	(M)	=	1.396
DRAFT AT FWD MARK	(M)	=	1.161
DRAFT AT AFT MARK	(M)	=	1.375

HYDROSTATIC PROPERTIES

Trim: Aft 0.255/50.000, No Heel, VCG = 3.975

LCF	Displacement	Buoyancy-Ctr.		Weight/		Moment/			
Draft----	Weight (MT)----	LCB-----	VCB-----	cm-----	LCF----	cm trim----	GML-----	GMT-----	
1.270	1,040.79	24.203f	0.641	8.74	24.767f	31.91	153.31	19.075	
Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.									
Trim is per 50.00m.									

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 280.2 m.-MT

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PONTOON**Condition 01 : Lightship + Crane + Ballast**

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 1.141 @ 50.00f, 1.268 @ 25.00f, 1.396 @ 0.00

Trim: Aft 0.255/50.000, Heel: zero

Part	LPA*Sf	HCP	Arm	Pressure	Moment
HULL	86.6	0.887	1.510	0.05500	7.19
CRANE	57.0	4.168	4.791	0.05500	15.02
Total wind heeling moment to starboard-->					22.21
Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT					

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.220f TCG = 0.000 VCG = 3.975

Free Surface Adjustment: 0.269

Adjusted CG: LCG = 24.222f TCG = 0.000 VCG = 4.245

Origin Depth	Degrees of Trim	Displacement Heel	Weight (MT)	Residual Arms in Trim	Res. 50% DeckImm in Heel	Area	Angle
1.384	0.29a	0.00	1,040.8	0.000	-0.021	0.0000	5.281
1.384	0.29a	0.05s	1,040.8	0.000	0.000	-0.0000	5.226
1.369	0.29a	5.05s	1,040.8	0.000	1.672	0.0729	0.226
1.367	0.29a	5.28s	1,040.8	0.000	1.746	0.0797	0.000
1.319	0.32a	10.05s	1,040.8	0.000	3.140	0.2846	-4.774
1.192	0.45a	15.05s	1,040.8	0.000	3.706	0.5899	-9.774
1.141	0.51a	17.02s	1,040.8	0.000	3.733	0.7180	-11.742
1.061	0.60a	20.05s	1,040.8	0.000	3.675	0.9148	-14.774
0.925	0.76a	25.05s	1,040.8	0.000	3.433	1.2274	-19.774
0.785	0.92a	30.05s	1,040.8	0.000	3.092	1.5128	-24.774
0.642	1.08a	35.05s	1,040.8	0.000	2.693	1.7657	-29.774
0.496	1.23a	40.05s	1,040.8	0.000	2.254	1.9818	-34.774
0.350	1.39a	45.05s	1,040.8	0.000	1.787	2.1583	-39.774
0.202	1.53a	50.05s	1,040.8	0.000	1.299	2.2931	-44.774
0.054	1.67a	55.05s	1,040.8	0.000	0.797	2.3846	-49.774
-0.096	1.79a	60.05s	1,040.8	0.000	0.286	2.4320	-54.774
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.							

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 280.2 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 22.21 (constant)

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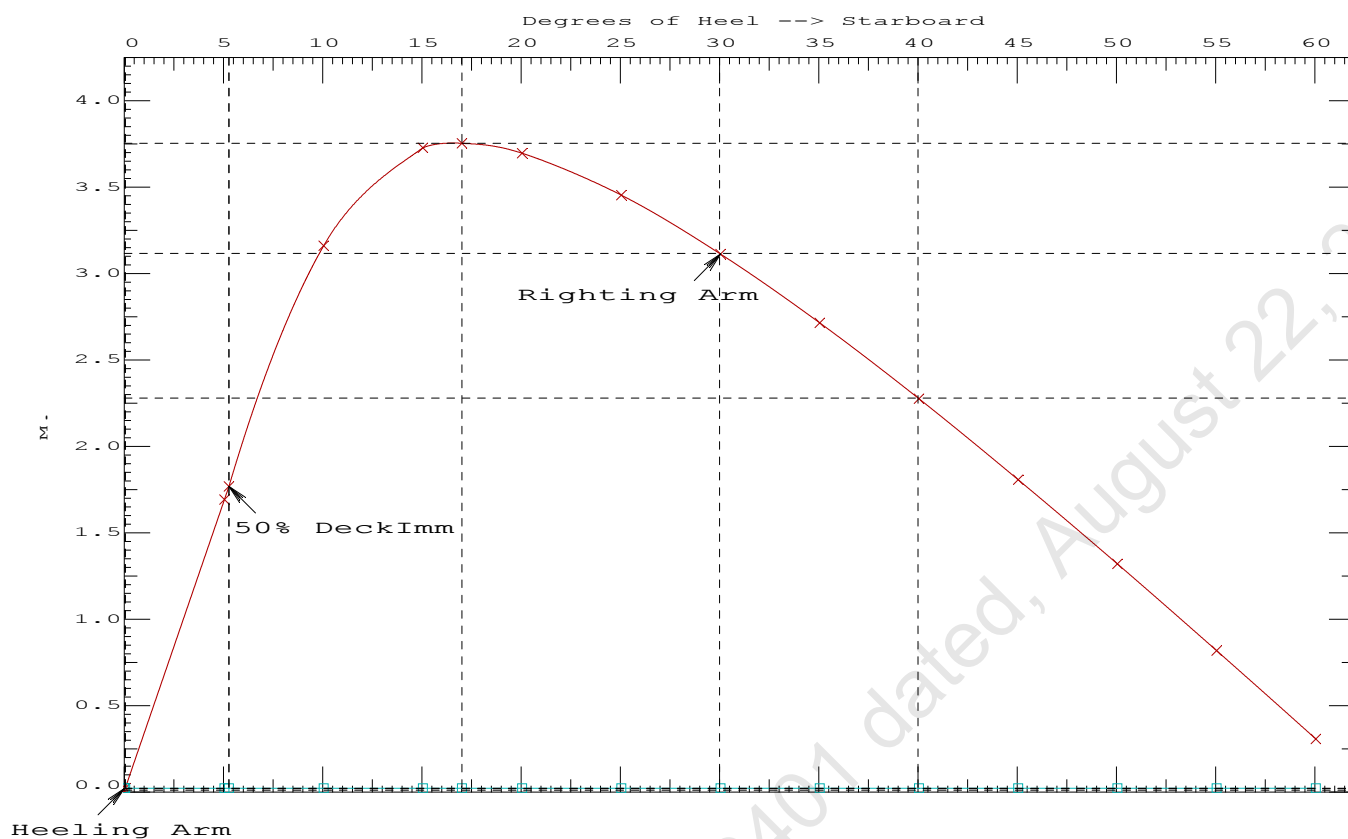
PONTOON**Condition 01 : Lightship + Crane + Ballast**

LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max-----Attained
(1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad 0.7180 P
(2) Angle from Equilibrium to RAzero or Flood > 20.00 deg LARGE
(3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg 5.23 P

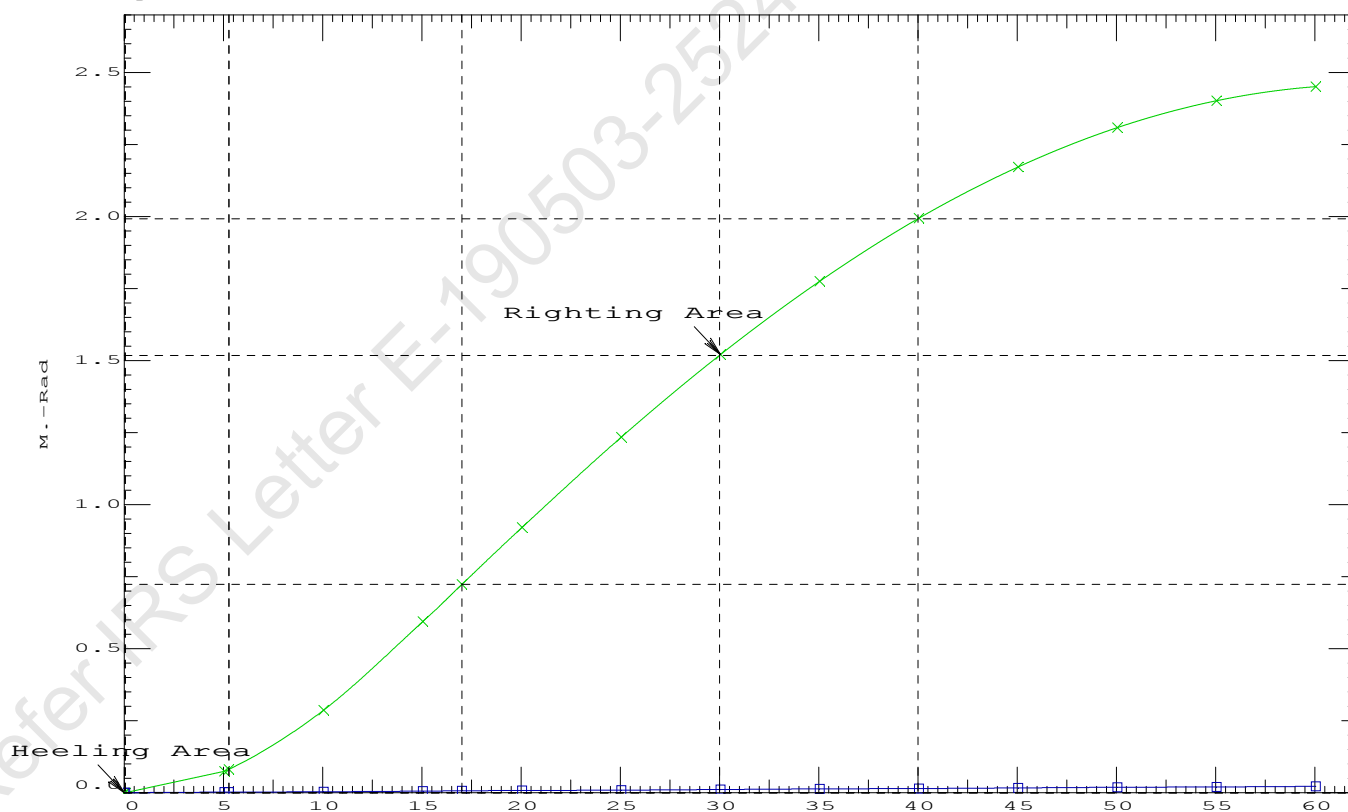
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PONTOON

Condition 01 : Lightship + Crane + Ballast

Heeling Arm



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PONTOON

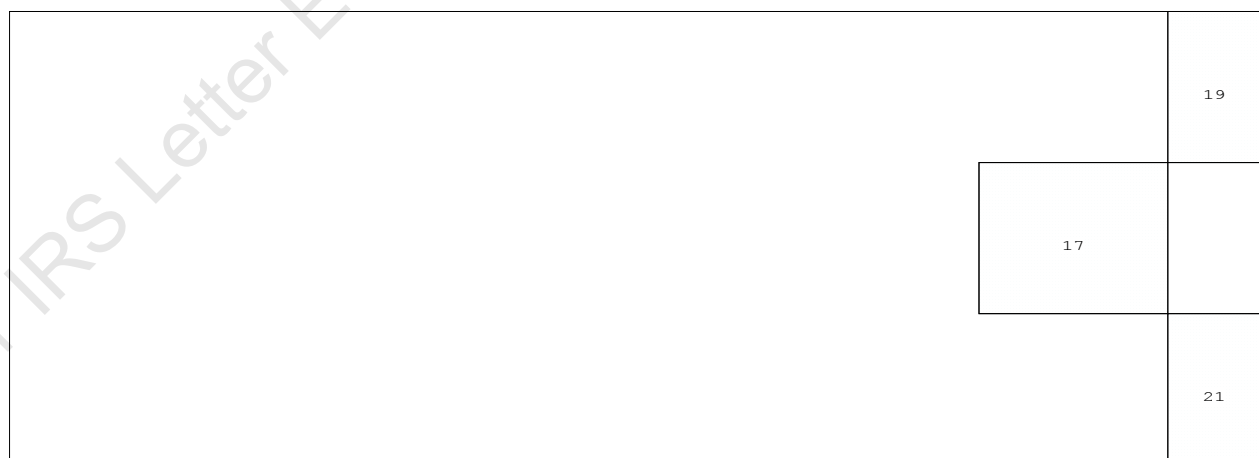
Condition 02 : Lightship + Crane + Deck Cargo + Ballast
Maximum Draft Condition

Condition Graphic - Draft: 2.210 @ 50.000f, 2.213 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

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PONTOON

Condition 02 : Lightship + Crane + Deck Cargo + Ballast
Maximum Draft Condition

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 2.210 @ 50.00f, 2.212 @ 25.00f, 2.213 @ 0.00
Trim: 0.000/50.000, Heel: zero

Part-----	Weight (MT)	LCG	TCG	VCG	
LIGHT SHIP	405.97	25.000f	0.000	3.000	
Crane	400.00	11.500f	0.000	6.325	
Deck Cargo	852.00	25.940f	0.000	4.000	
Total Fixed----->	1,657.97	22.226f	0.000	4.316	
Load-----	SpGr-----	Weight (MT)	LCG	TCG	VCG
06_WBTK.C	1.000	1.025	135.67	42.251f	0.000
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s
Total Tanks----->		234.87	44.536f	0.000	1.660
Total Weight----->		1,892.84	24.994f	0.000	3.987
		Displ (MT)	LCB	TCB	VCB
HULL	1.025	1,892.84	24.994f	0.000	1.132

Righting Arms:		0.000	0.000		
Part-----	SpGr-----	WPA	LCF	TCF	BML
Total Waterplane---->	1.025	900.4	25.000f	0.000	101.46
		MT/cm	m.	MT/cm	GML
		9.23		37.33	98.61
Distances in METERS.-----					10.176
					Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

229.1 Cu.M. (9%) SALT WATER

400.00 MT Crane

852.00 MT Deck Cargo

DRAFT AT FP (M) = 2.210
DRAFT AT MS (M) = 2.212
DRAFT AT AP (M) = 2.213
DRAFT AT FWD MARK (M) = 2.211
DRAFT AT AFT MARK (M) = 2.213

HYDROSTATIC PROPERTIES

Trim: 0.000/50.000, No Heel, VCG = 3.987

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/	
Draft---	Weight (MT)	LCB	VCB	cm	trim---
2.212	1,892.84	24.994f	1.132	9.23	25.000f
				37.33	98.61
Distances in METERS.-----					10.176
					Specific Gravity = 1.025
					Moment in m.-MT.
					Trim is per 50.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 280.2 m.-MT

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PONTOON

Condition 02 : Lightship + Crane + Deck Cargo + Ballast
Maximum Draft Condition

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 2.210 @ 50.00f, 2.212 @ 25.00f, 2.213 @ 0.00

Trim: 0.000/50.000, Heel: zero

Part-----	LPA*Sf-----	HCP-----	Arm----	Pressure-----	Moment
HULL	40.5	0.405	1.478	0.05500	3.29
CRANE	57.0	3.299	4.372	0.05500	13.71
DECKCARGO.C	29.4	1.800	2.873	0.05500	4.65
Total wind heeling moment to starboard----->					21.65
Distances in METERS.-----Pressure in MT/Sqm.-----					Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.994f TCG = 0.000 VCG = 3.987

Free Surface Adjustment: 0.148

Adjusted CG: LCG = 24.994f TCG = 0.000 VCG = 4.135

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---Trim---	Heel---	Weight (MT)---	in Trim--in Heel---	Area---Angle
2.201 0.00a 0.00	1,892.8	0.000 -0.011	0.0000 2.572	
2.201 0.00a 0.06s	1,892.8	0.000 0.000	-0.0000 2.509	
2.198 0.00a 2.57s	1,892.8	0.000 0.450	0.0098 -0.000	
2.190 0.00a 5.06s	1,892.8	0.000 0.890	0.0390 -2.491	
2.286 0.00a 10.06s	1,892.8	0.000 1.314	0.1403 -7.491	
2.427 0.01a 13.33s	1,892.8	0.000 1.357	0.2174 -10.762	
2.514 0.01a 15.06s	1,892.8	0.000 1.345	0.2582 -12.491	
2.785 0.01a 20.06s	1,892.8	0.000 1.229	0.3722 -17.491	
3.047 0.01a 25.06s	1,892.8	0.000 1.021	0.4710 -22.491	
3.286 0.01a 30.06s	1,892.8	0.000 0.766	0.5494 -27.491	
3.501 0.01a 35.06s	1,892.8	0.000 0.486	0.6042 -32.491	
3.688 0.02a 40.06s	1,892.8	0.000 0.192	0.6339 -37.491	
3.794 0.02a 43.25s	1,892.8	0.000 0.000	0.6392 -40.680	
3.848 0.02a 45.06s	1,892.8	0.000 -0.110	0.6375 -42.491	
3.978 0.02a 50.06s	1,892.8	0.000 -0.415	0.6146 -47.491	
4.078 0.02a 55.06s	1,892.8	0.000 -0.719	0.5652 -52.491	
4.149 0.03a 60.06s	1,892.8	0.000 -1.019	0.4893 -57.491	
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.				

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 280.2 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.65 (constant)

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PONTOON**Condition 02 : Lightship + Crane + Deck Cargo + Ballast****Maximum Draft Condition**

LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max-----Attained

(1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad 0.2174 P

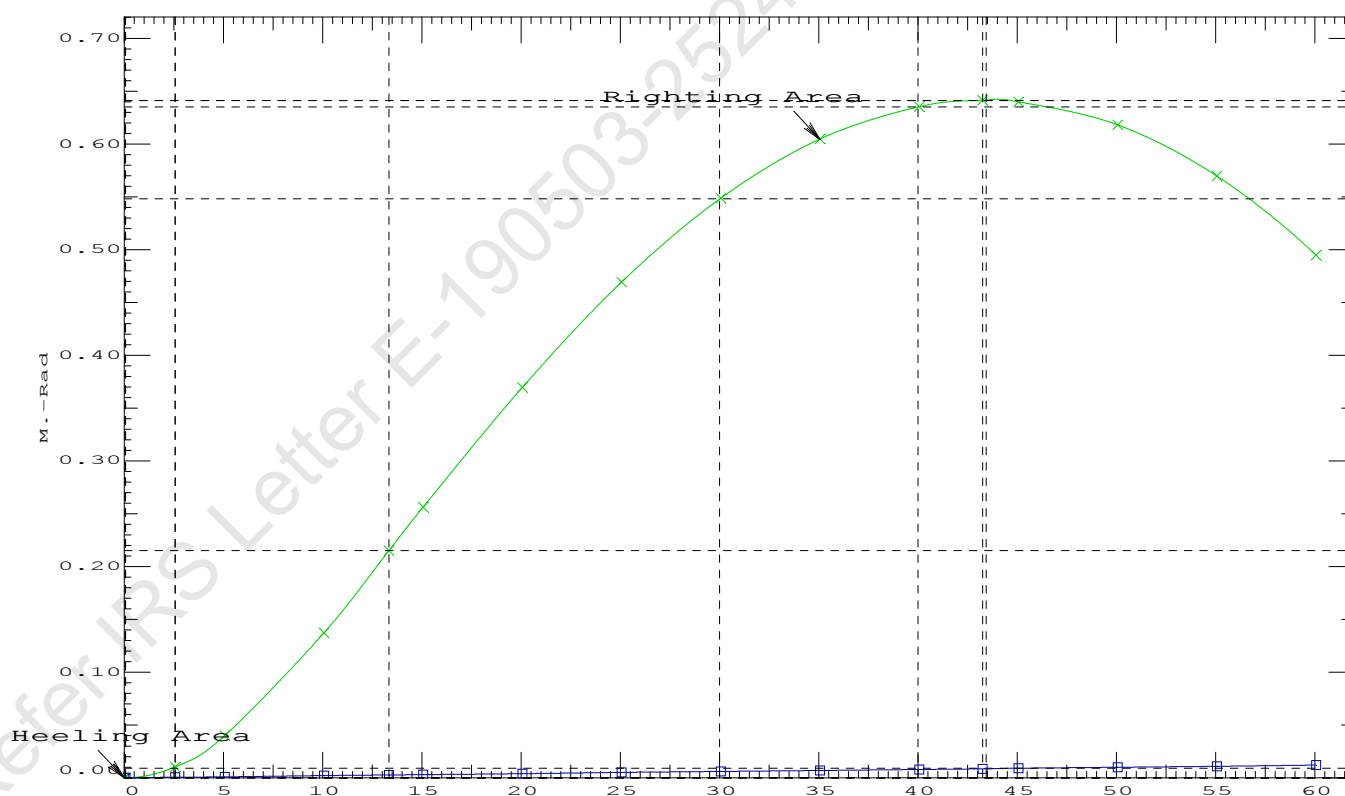
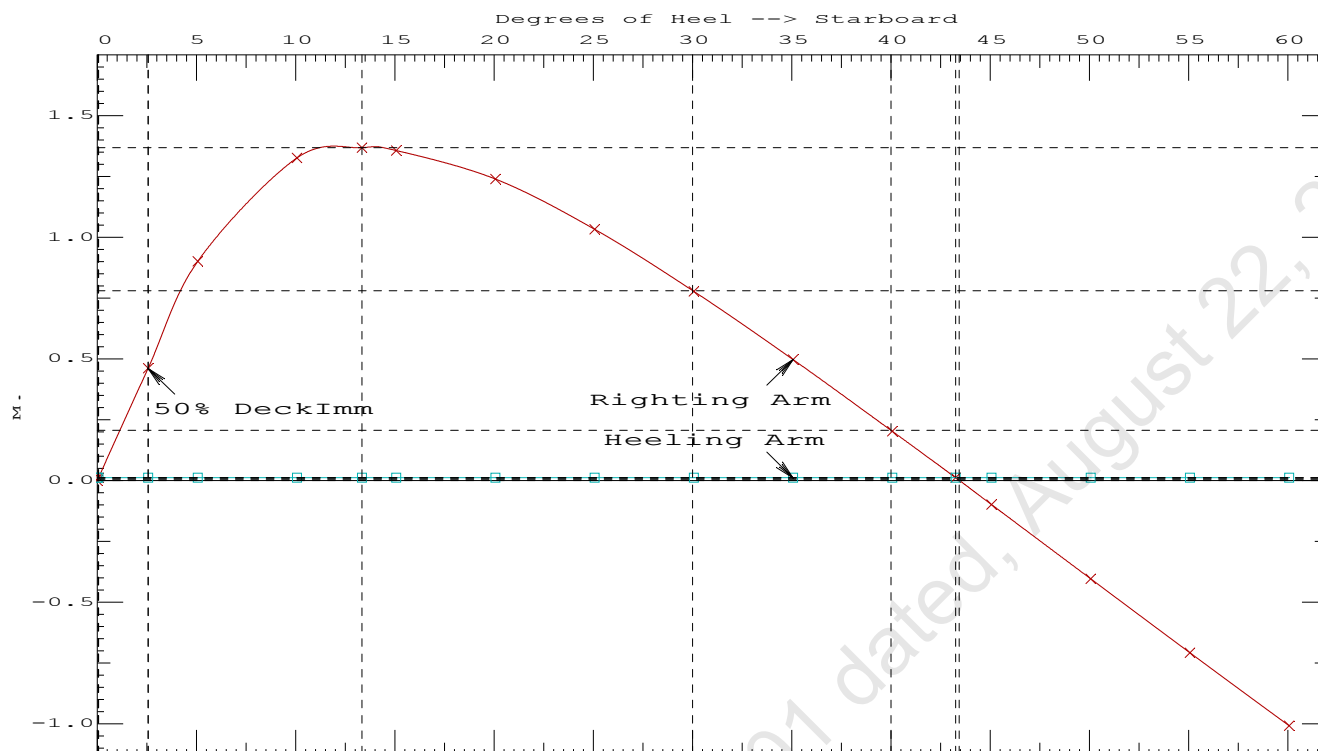
(2) Angle from Equilibrium to RAzero or Flood > 20.00 deg 43.19 P

(3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg 2.51 P

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PONTOON

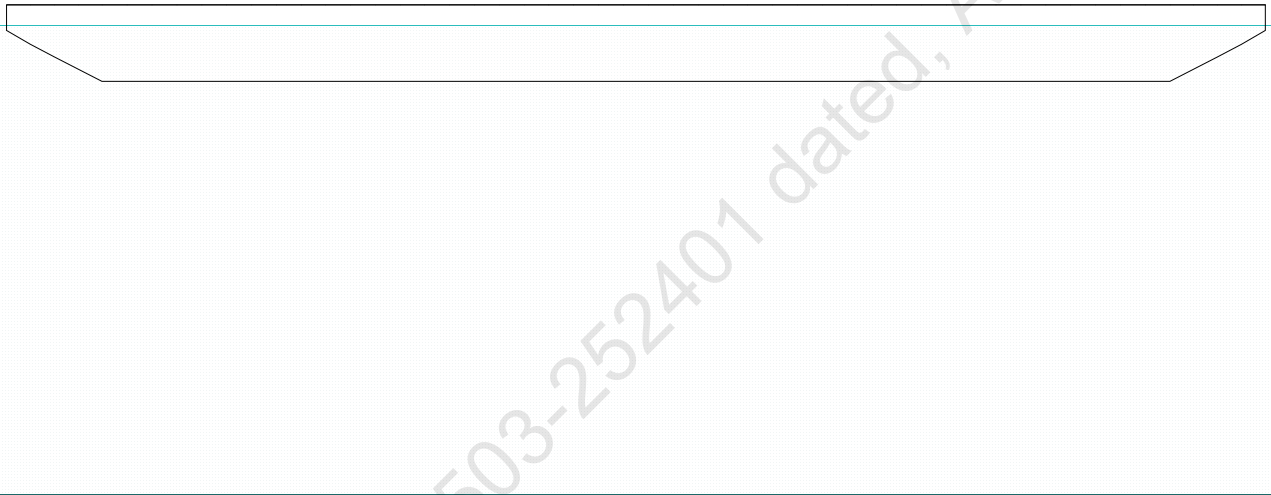
Condition 02 : Lightship + Crane + Deck Cargo + BallastMaximum Draft Condition

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PONTOON**Condition 03 : Lightship + Deck Cargo****Maximum Deck Cargo Condition**

Condition Graphic - Draft: 2.212 Trim: zero Heel: zero

Profile View

The profile view shows a cross-section of the pontoon hull. It features a flat top deck, a vertical side wall, and a flat bottom. The hull is symmetrical about a central vertical axis. The draft is indicated as 2.212. The trim is zero, and the heel is zero.

Plan View

The plan view shows the top-down perspective of the pontoon hull. It is a rectangular shape with a central vertical axis of symmetry. The width and length of the hull are indicated by the dimensions of the rectangle.

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PONTOON

Condition 03 : Lightship + Deck Cargo
Maximum Deck Cargo Condition

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.212 @ 50.00f, 2.212 @ 25.00f, 2.212 @ 0.00

Trim: zero, Heel: zero

Part-----	Weight (MT)----	LCG-----	TCG-----	VCG-----	FSM-----
LIGHT SHIP	405.97	25.000f	0.000	3.000	
Deck Cargo	1,487.09	25.000f	0.000	4.000	
Total Weight----->	1,893.06	25.000f	0.000	3.786	
Load-----	SpGr-----	Weight (MT)---- <td>LCG-----</td> <td>TCG-----</td> <td>VCG-----</td>	LCG-----	TCG-----	VCG-----
Total Tanks----->		0.00			0.00
		Displ (MT)---- <td>LCB-----</td> <td>TCB-----</td> <td>VCB-----</td>	LCB-----	TCB-----	VCB-----
HULL	1.025	1,893.06	25.000f	0.000	1.132

Righting Arms: 0.000 0.000

Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025	900.4	25.000f	0.000	101.45	13.177
		MT/cm-----	m.-MT/cm-----		GML-----	GMT-----
		9.23		37.41	98.80	10.523

Distances in METERS.-----Moments in m.-MT.

SUMMARY OF LOADING

0.0 Cu.M. (0%) SALT WATER

1,487.09 MT Deck Cargo

DRAFT AT FP (M) = 2.212
DRAFT AT MS (M) = 2.212
DRAFT AT AP (M) = 2.212
DRAFT AT FWD MARK (M) = 2.212
DRAFT AT AFT MARK (M) = 2.212

HYDROSTATIC PROPERTIES

No Trim, No Heel, VCG = 3.786

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/
Draft----	Weight (MT)----	LCB-----	VCB-----	cm-----
2.212	1,893.06	25.000f	1.132	9.23
				25.000f
				37.41
				98.80
				10.523

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 2.212 @ 50.00f, 2.212 @ 25.00f, 2.212 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA*SF-----	HCP-----	Arm-----	Pressure-----	Moment-----
HULL	40.5	0.405	1.478	0.05500	3.29
DECKCARGO.C	29.4	1.800	2.873	0.05500	4.65
Total wind heeling moment to starboard----->					7.94
Distances in METERS.-----	Pressure in MT/Sqm.-----				Moment in m.-MT

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PONTOON

Condition 03 : Lightship + Deck CargoMaximum Deck Cargo Condition

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

LCG = 25.000f TCG = 0.000 VCG = 3.786

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---Trim---Heel---Weight (MT)---in Trim--in Heel---> Area---Angle				
2.200 0.00 0.00	1,893.1	0.000	-0.004	0.0000 2.576
2.200 0.00 0.02s	1,893.1	0.000	0.000	-0.0000 2.554
2.197 0.00 2.58s	1,893.3	0.000	0.473	0.0105 0.000
2.189 0.00 5.02s	1,893.1	0.000	0.921	0.0403 -2.446
2.283 0.00 10.02s	1,893.1	0.000	1.381	0.1459 -7.446
2.464 0.00 14.12s	1,893.1	0.000	1.446	0.2487 -11.547
2.509 0.00 15.02s	1,893.1	0.000	1.443	0.2714 -12.446
2.779 0.00 20.02s	1,893.0	0.000	1.356	0.3952 -17.446
3.041 0.00 25.02s	1,893.1	0.000	1.177	0.5064 -22.446
3.280 0.00 30.02s	1,893.1	0.000	0.950	0.5996 -27.446
3.494 0.00 35.02s	1,893.1	0.000	0.695	0.6715 -32.446
3.681 0.00 40.02s	1,893.1	0.000	0.425	0.7205 -37.446
3.840 0.00 45.02s	1,893.1	0.000	0.146	0.7455 -42.446
3.910 0.00 47.60s	1,893.1	0.000	0.000	0.7488 -45.024
3.969 0.00 50.02s	1,893.1	0.000	-0.138	0.7459 -47.446
4.069 0.00 55.02s	1,893.1	0.000	-0.424	0.7214 -52.446
4.136 0.00f 60.02s	1,893.1	0.000	-0.708	0.6720 -57.446
Distances in METERS.----Specific Gravity = 1.025.-----Area in m.-Rad.				

Note: No tank loads are present.

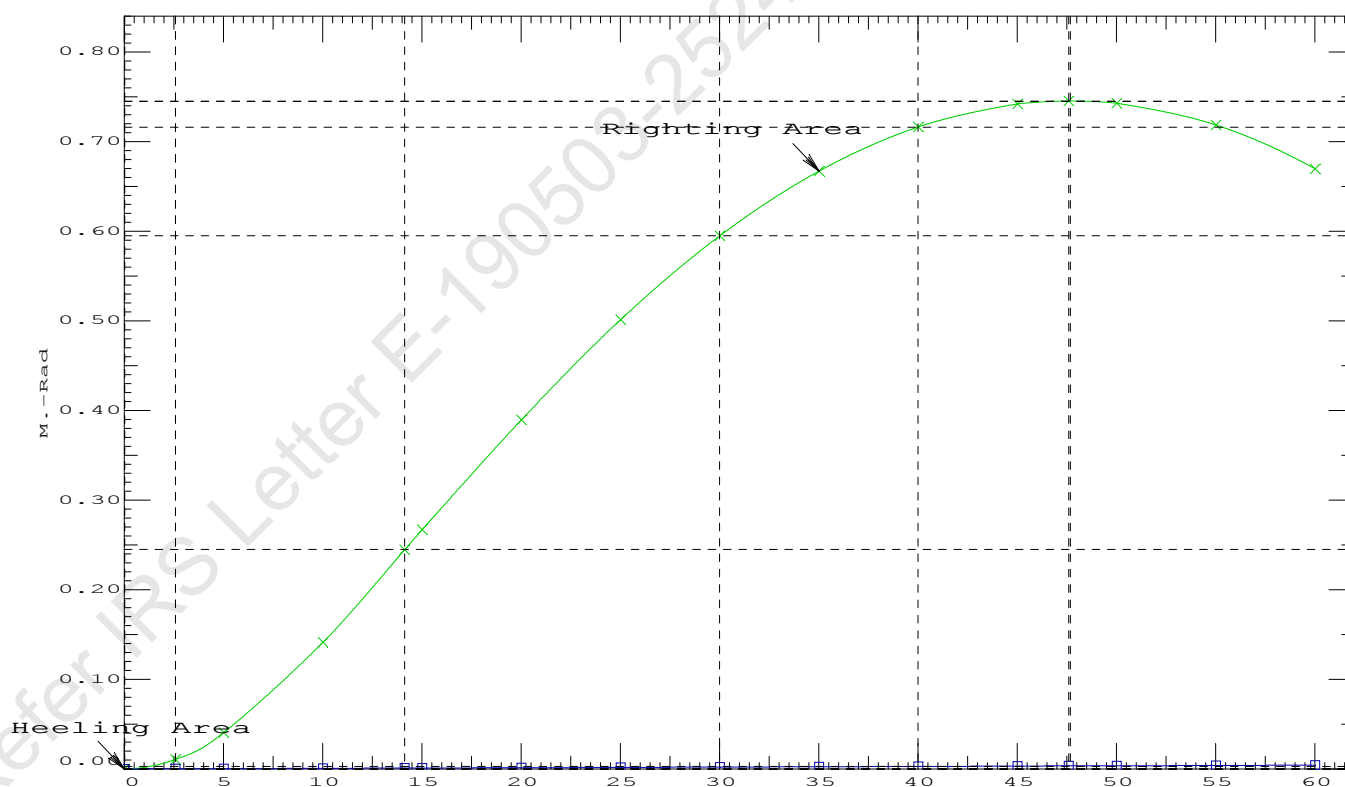
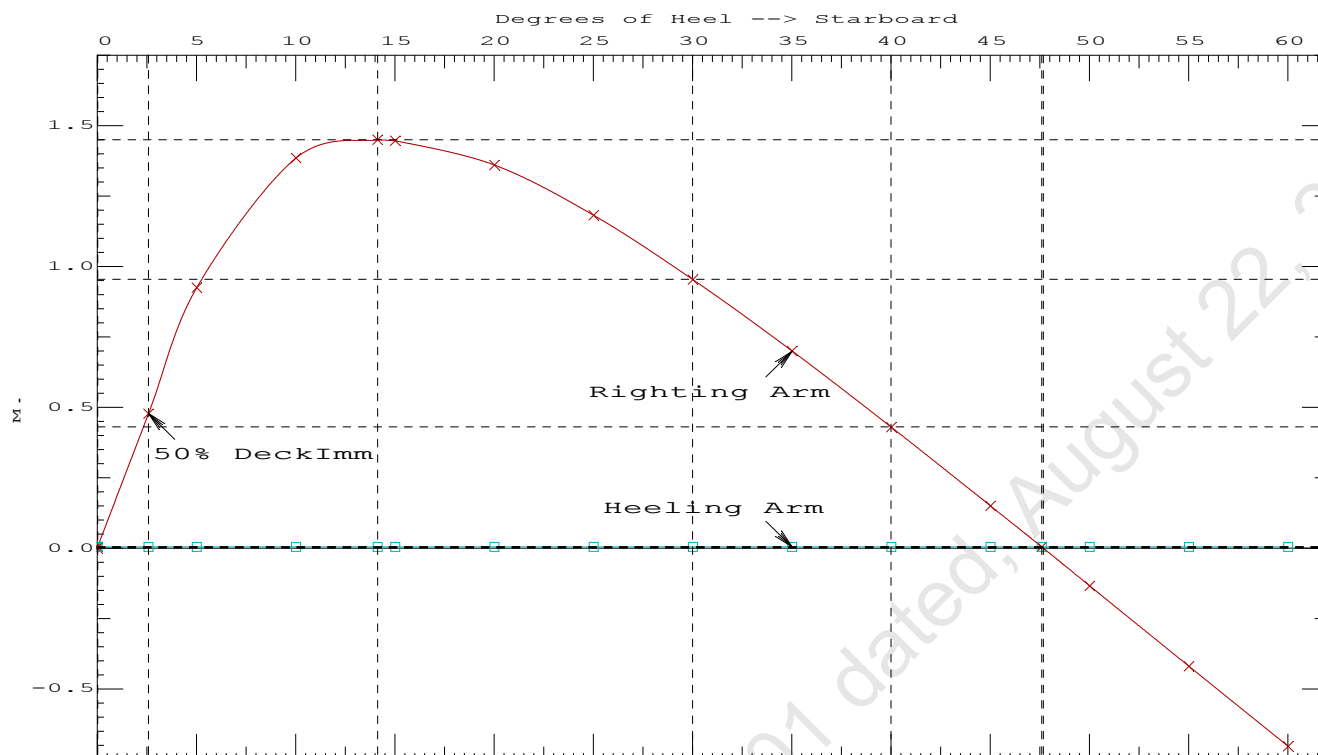
Note: The Residual Righting Arms shown above are in excess of the
wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 7.94 (constant)

LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2-----Min/Max-----Attained			
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	0.2487 P
(2) Angle from Equilibrium to RAzero or Flood	>	20.00 deg	47.58 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	2.55 P

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PONTOON

Condition 03 : Lightship + Deck CargoMaximum Deck Cargo Condition

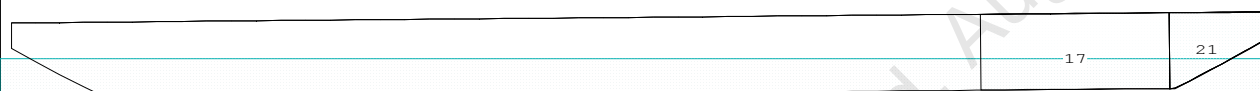
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PONTOON

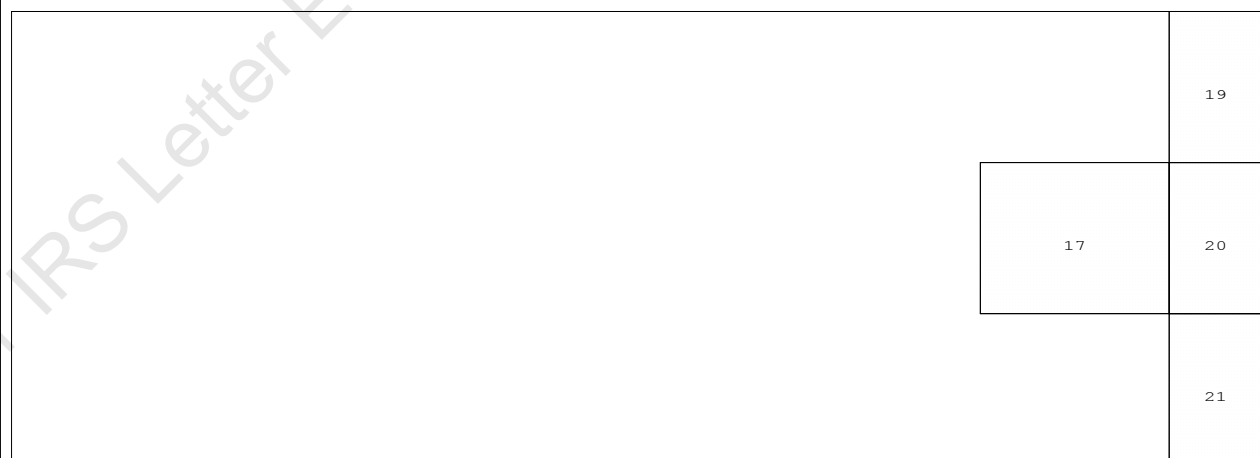
Condition 04 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Aft

Condition Graphic - Draft: 1.145 @ 50.000f, 1.594 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

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PONTOON

Condition 04 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Aft

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.145 @ 50.00f, 1.370 @ 25.00f, 1.594 @ 0.00

Trim: Aft 0.449/50.000, Heel: zero

Part-----			Weight (MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			406.00	25.000f	0.000	3.000	
Crane			400.00	11.500f	0.000	6.325	
Crane Load			40.00	19.500a 18.500a	0.000	34.602	
Total Fixed----->			846.00	16.513f	0.000	6.066	
	Load-----	SpGr-----	Weight (MT)----	LCG-----	TCG-----	VCG-----	FSM-----
06_WBTK.C	1.000	1.025	135.67	42.251f	0.000	1.499	135.61*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			284.98	45.086f	0.000	1.697	352.57
Total Weight----->			1,130.98	23.713f	0.000	4.965	
			Displ (MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,130.87	23.674f	0.000	0.698	

Righting Arms: 0.000 0.000

Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025	859.1	24.606f	0.000	147.51	20.708
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
		8.81	32.40	143.24	16.440	

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

278.0 Cu.M. (11%) SALT WATER

400.00 MT Crane

40.00 MT Crane Load

DRAFT AT FP (M) = 1.145
DRAFT AT MS (M) = 1.370
DRAFT AT AP (M) = 1.594
DRAFT AT FWD MARK (M) = 1.181
DRAFT AT AFT MARK (M) = 1.559

HYDROSTATIC PROPERTIES

Trim: Aft 0.449/50.000, No Heel, VCG = 4.965

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/
Draft----	Weight (MT)----	LCB-----	VCB-----	cm trim-----
1.373	1,130.87	23.674f	0.698	8.81
				24.606f
				32.40
				143.24
				16.440

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 352.6 m.-MT

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PONTOON

Condition 04 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Aft

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 1.145 @ 50.00f, 1.370 @ 25.00f, 1.594 @ 0.00

Trim: Aft 0.449/50.000, Heel: zero

Part-----	LPA*Sf-----	HCP-----	Arm----	Pressure-----	Moment
HULL	81.7	0.838	1.513	0.05500	6.80
CRANE	57.0	4.008	4.683	0.05500	14.68
Total wind heeling moment to starboard----->					21.48
Distances in METERS.-----Pressure in MT/Sqm.-----					Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 23.713f TCG = 0.000 VCG = 4.965

Free Surface Adjustment: 0.312

Adjusted CG: LCG = 23.716f TCG = 0.000 VCG = 5.277

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---Trim---Heel---	Weight (MT)---	in Trim--in Heel---	Area----	Angle
1.583 0.51a 0.00	1,131.0	0.000 -0.019	0.0000	4.566
1.583 0.51a 0.06s	1,131.0	0.000 0.000	-0.0000	4.503
1.570 0.51a 4.57s	1,131.0	0.000 1.302	0.0512	0.000
1.568 0.52a 5.06s	1,131.0	0.000 1.445	0.0631	-0.497
1.534 0.56a 10.06s	1,130.9	0.000 2.752	0.2472	-5.497
1.511 0.82a 15.06s	1,131.0	0.000 3.185	0.5132	-10.497
1.509 0.84a 15.42s	1,131.0	0.000 3.186	0.5327	-10.849
1.488 1.10a 20.06s	1,131.0	0.000 3.048	0.7888	-15.497
1.460 1.40a 25.06s	1,131.0	0.000 2.719	1.0418	-20.497
1.427 1.69a 30.06s	1,131.0	0.000 2.304	1.2616	-25.497
1.390 1.99a 35.06s	1,131.0	0.000 1.838	1.4427	-30.497
1.347 2.28a 40.06s	1,131.0	0.000 1.342	1.5817	-35.497
1.297 2.57a 45.06s	1,131.0	0.000 0.826	1.6764	-40.497
1.240 2.84a 50.06s	1,131.0	0.000 0.297	1.7255	-45.497
1.204 2.98a 52.84s	1,131.0	0.000 0.000	1.7327	-48.278
1.173 3.08a 55.06s	1,131.0	0.000 -0.238	1.7281	-50.497
1.095 3.30a 60.06s	1,131.0	0.000 -0.773	1.6840	-55.497
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.				

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 352.6 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.48 (constant)

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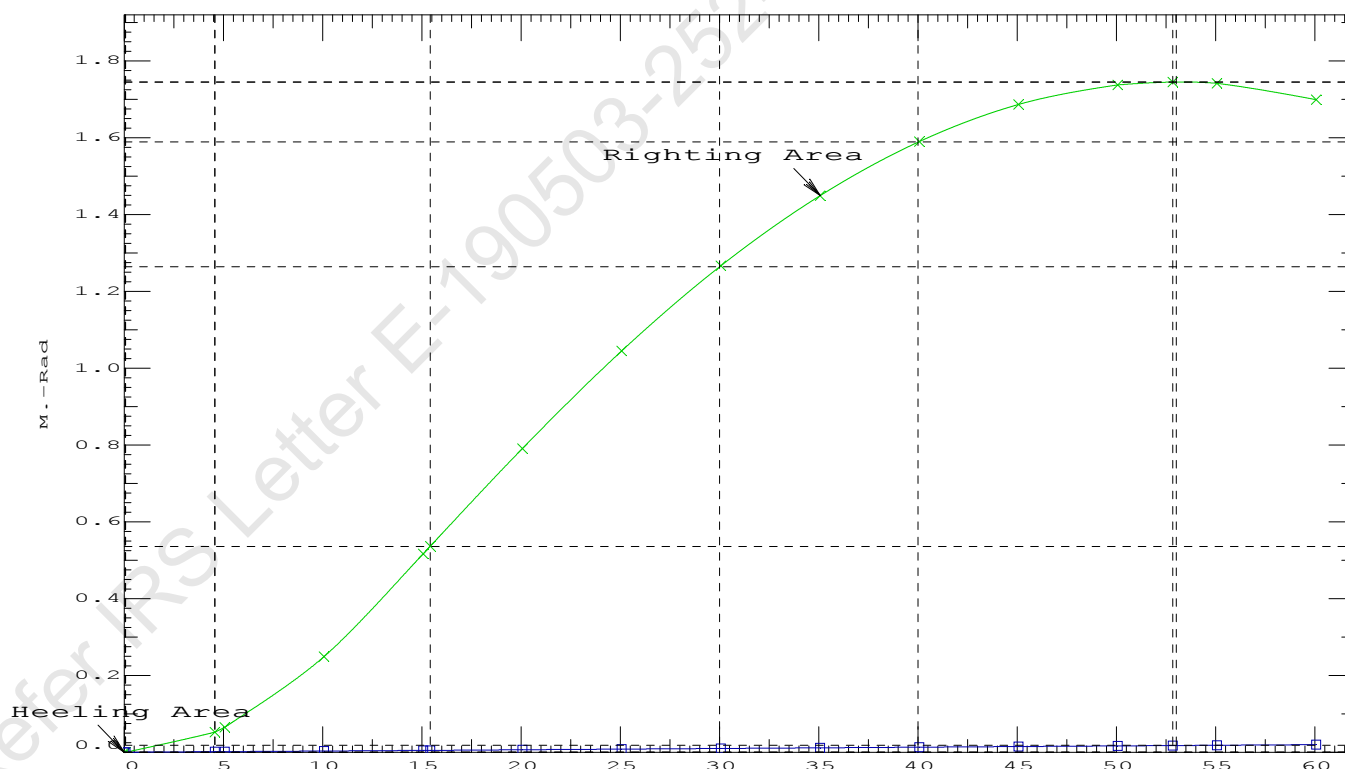
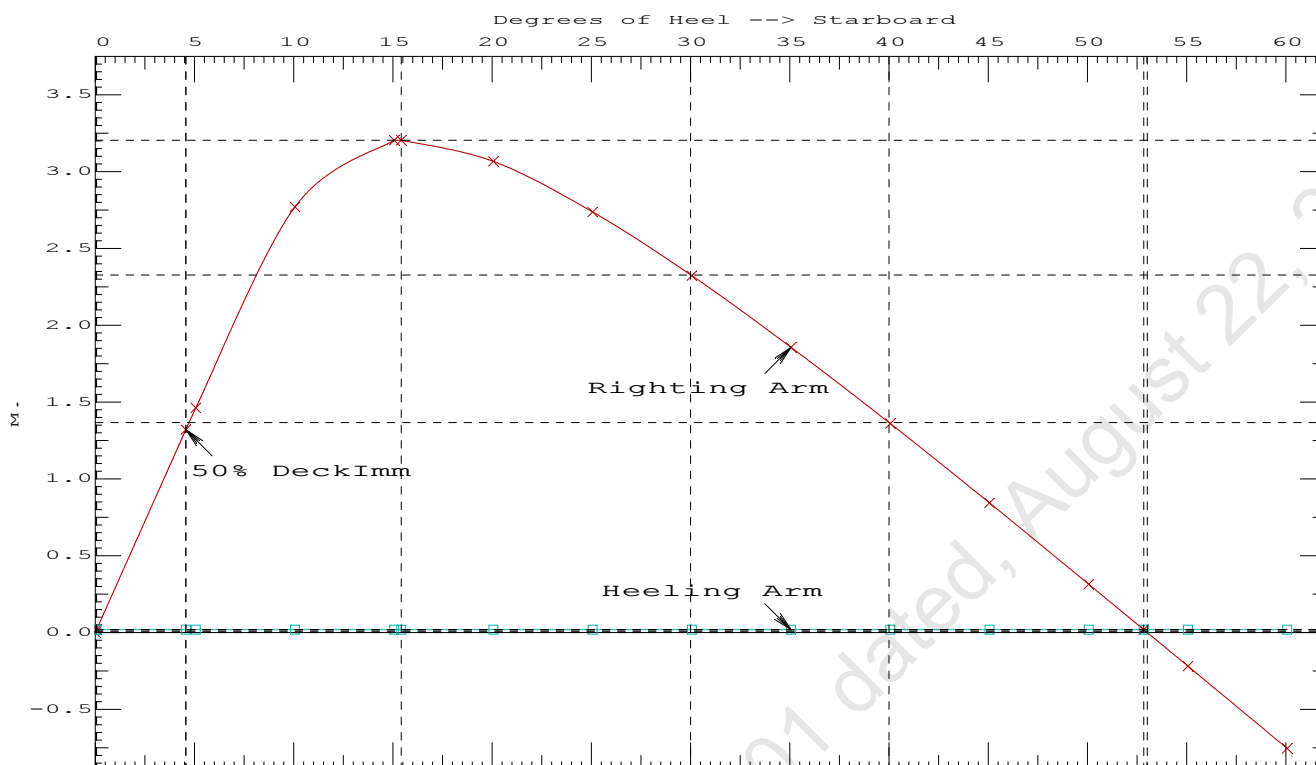
PONTOON**Condition 04 :Operating Condition of 400T Crane**
Crane Lifting 40.0T @ 30m Radius Aft

LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max-----Attained
(1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad 0.5327 P
(2) Angle from Equilibrium to RAzero or Flood > 20.00 deg 52.78 P
(3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg 4.50 P

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PONTOON

Condition 04 :Operating Condition of 400T CraneCrane Lifting 40.0T @ 30m Radius Aft

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PONTOON

Condition 04 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Aft

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 23.713f TCG = 0.000 VCG = 4.965
Free Surface Adjustment: 0.312
Adjusted CG: LCG = 23.716f TCG = 0.000 VCG = 5.277

Origin Depth	Degrees of Trim	Displacement Heel	Weight (MT)	Residual Arms in Trim	Res. in Heel	Area
1.583	0.51a	0.01s	1,131.0	0.000	0.000	0.0000
1.568	0.52a	5.01s	1,131.0	0.000	1.445	0.0631
1.534	0.56a	10.01s	1,130.9	0.000	2.756	0.2474
1.511	0.81a	15.01s	1,131.0	0.000	3.198	0.5142
1.509	0.84a	15.43s	1,131.0	0.000	3.200	0.5372
1.488	1.10a	20.01s	1,131.0	0.000	3.065	0.7912
1.460	1.39a	25.01s	1,131.0	0.000	2.738	1.0457
1.428	1.69a	30.01s	1,131.0	0.000	2.322	1.2671
1.390	1.98a	35.01s	1,131.0	0.000	1.857	1.4499
1.347	2.28a	40.01s	1,131.0	0.000	1.361	1.5905
1.298	2.56a	45.01s	1,131.0	0.000	0.845	1.6870
1.240	2.83a	50.01s	1,131.0	0.000	0.317	1.7377
1.202	2.98a	52.98s	1,131.0	0.000	0.000	1.7459
1.174	3.08a	55.01s	1,131.0	0.000	-0.218	1.7421
1.096	3.30a	60.01s	1,131.0	0.000	-0.753	1.6997

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 352.6 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

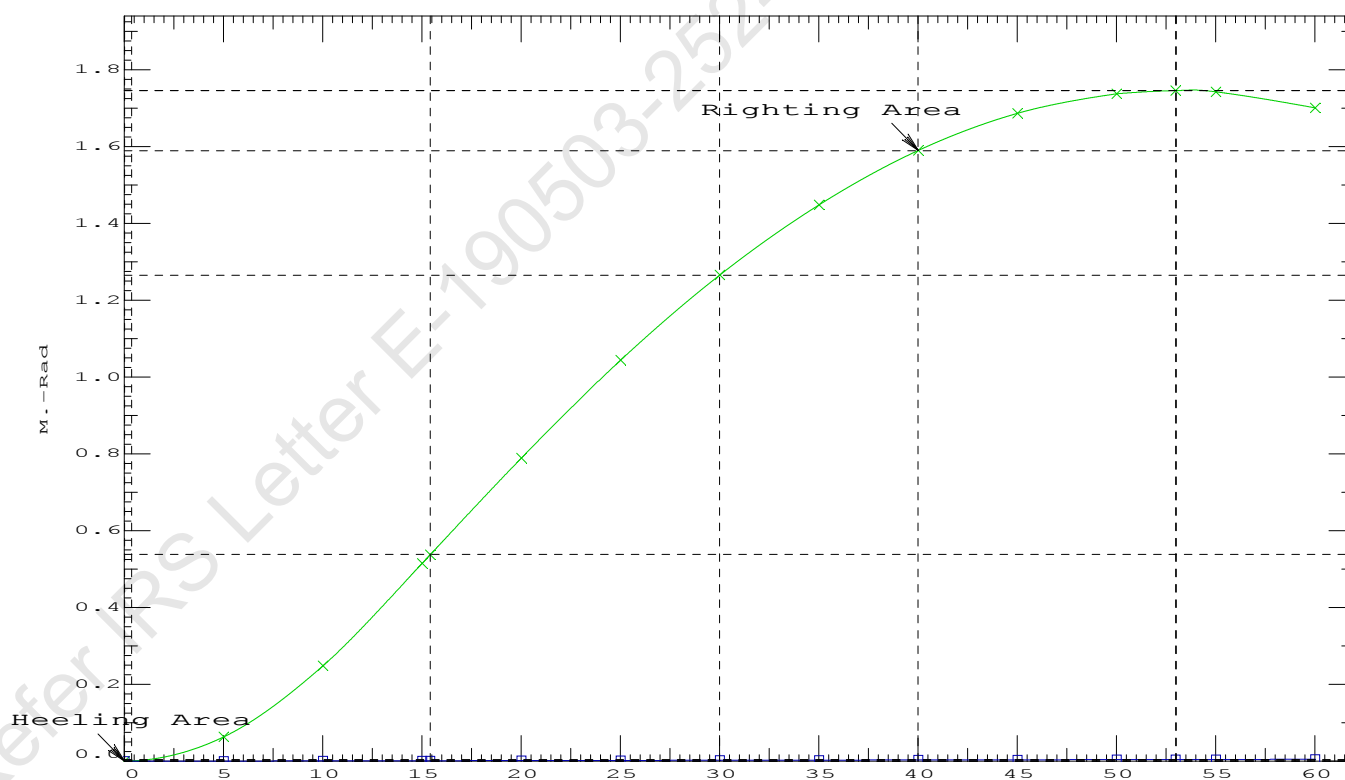
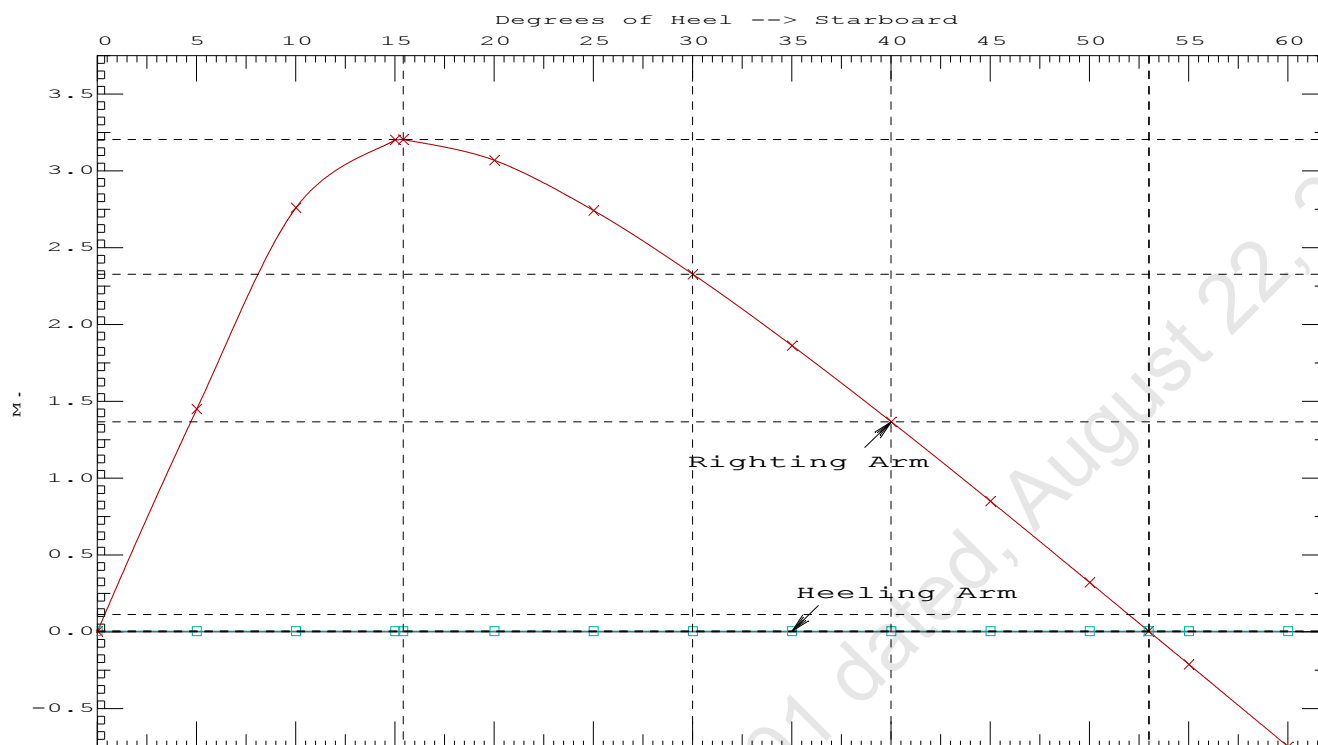
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM	STABILITY CRITERIA FOR CRANE OPERAT	Min/Max	Attained
(1)	Absolute Angle at Equilibrium	< 15.00 deg	0.01 P
(2)	Rise in Abs. RA from Equilibrium to MaxRA	> 66.7 %	72391.6 P
(3)	Abs Ratio from Equilibrium to RAzero or Flood	> 1.667	428.216 P

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PONTOON

Condition 04 :Operating Condition of 400T CraneCrane Lifting 40.0T @ 30m Radius Aft

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Condition 04 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Aft

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.146 @ 50.00f, 1.370 @ 25.00f, 1.595 @ 0.00

Trim: Aft 0.449/50.000, Heel: 0.00 deg.

Part-----			Weight (MT)	-----LCG-----	TCG-----	VCG-----	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			400.00	11.500f	0.000	6.325	
Crane Load			40.00	18.500a	0.000	34.602	
Total Fixed----->			845.97	16.513f	0.000	6.066	
	Load-----	SpGr-----	Weight (MT)	-----LCG-----	TCG-----	VCG-----	FSM-----
06_WBTK.C	1.000	1.025	135.67	42.251f	0.000	1.499	135.61*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			284.98	45.086f	0.000	1.697	352.57
Total Weight----->			1,130.95	23.713f	0.000	4.965	
			Displ (MT)	-----LCB-----	TCB-----	VCB-----	
HULL		1.025	1,130.95	23.674f	0.001s	0.698	

	Righting Arms:			0.000	0.001s		
	External Arms:			0.000	0.001		
	Residual Righting Arms:			0.000	0.000		
Part-----	SpGr-----		WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		857.5	24.582f	0.014s	146.66	20.634
			MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
			8.79		32.21	142.40	16.367
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

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PONTOON

After hook load drop

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.376 @ 50.00f, 1.326 @ 25.00f, 1.276 @ 0.00

Trim: Fwd 0.100/50.000, Heel: 0.00 deg.

Part-----			Weight (MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			400.00	11.500f	0.000	6.325	
Total Fixed----->			805.97	18.300f	0.000	4.650	
	Load----	SpGr-----	Weight (MT)----	LCG-----	TCG-----	VCG-----	FSM-----
06_WBTK.C	1.000	1.025	135.67	42.251f	0.000	1.499	135.61*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			284.98	45.086f	0.000	1.697	352.57
Total Weight----->			1,090.95	25.297f	0.000	3.879	
			Displ (MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,090.96	25.303f	0.005s	0.669	

	Righting Arms:			0.000	0.005s		
	External Arms:			0.000	0.005s		
	Residual Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->		1.025	847.9	25.033f	0.057s	147.05	21.176
			MT/cm				
			8.69				

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: INFINITY; t3: 30.0; t: 15.00
limmarg: INFINITY; t3: 15.0; t: 7.50
limmarg: INFINITY; t3: 7.5; t: 3.75
limmarg: INFINITY; t3: 3.8; t: 1.88
limmarg: INFINITY; t3: 1.9; t: 0.94
limmarg: INFINITY; t3: 1.0; t: 0.47
limmarg: INFINITY; t3: 0.5; t: 0.23
limmarg: INFINITY; t3: 0.3; t: 0.12
limmarg: INFINITY; t3: 0.2; t: 0.06
limmarg: INFINITY; t3: 0.1; t: 0.03
limmarg: INFINITY; t3: 0.1; t: 0.01

Theta3 is 0.07 degrees, as demonstrated below.

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PONTOON

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.300f TCG = 0.000 VCG = 4.650

Origin Depth	Degrees of Trim	Heel	Displacement Weight (MT)	Residual Arms in Trim	in Heel	Res. Area
1.264	0.12f	0.00s	1,091.0	0.000	0.000	0.0000
1.264	0.12f	0.07s	1,091.1	0.000	0.021	0.0000
1.250	0.11f	5.00s	1,091.0	0.000	1.624	0.0709
1.188	0.13f	10.00s	1,090.9	0.000	3.101	0.2781
1.008	0.18f	15.00s	1,091.0	0.000	3.698	0.5812
0.916	0.21f	17.36s	1,091.0	0.000	3.733	0.7344
0.811	0.24f	20.00s	1,091.0	0.000	3.690	0.9064
0.606	0.30f	25.00s	1,091.0	0.000	3.478	1.2216
0.396	0.37f	30.00s	1,091.0	0.000	3.169	1.5123

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 805.97 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 0.1 > 1.000 LARGE

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PONTOON

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

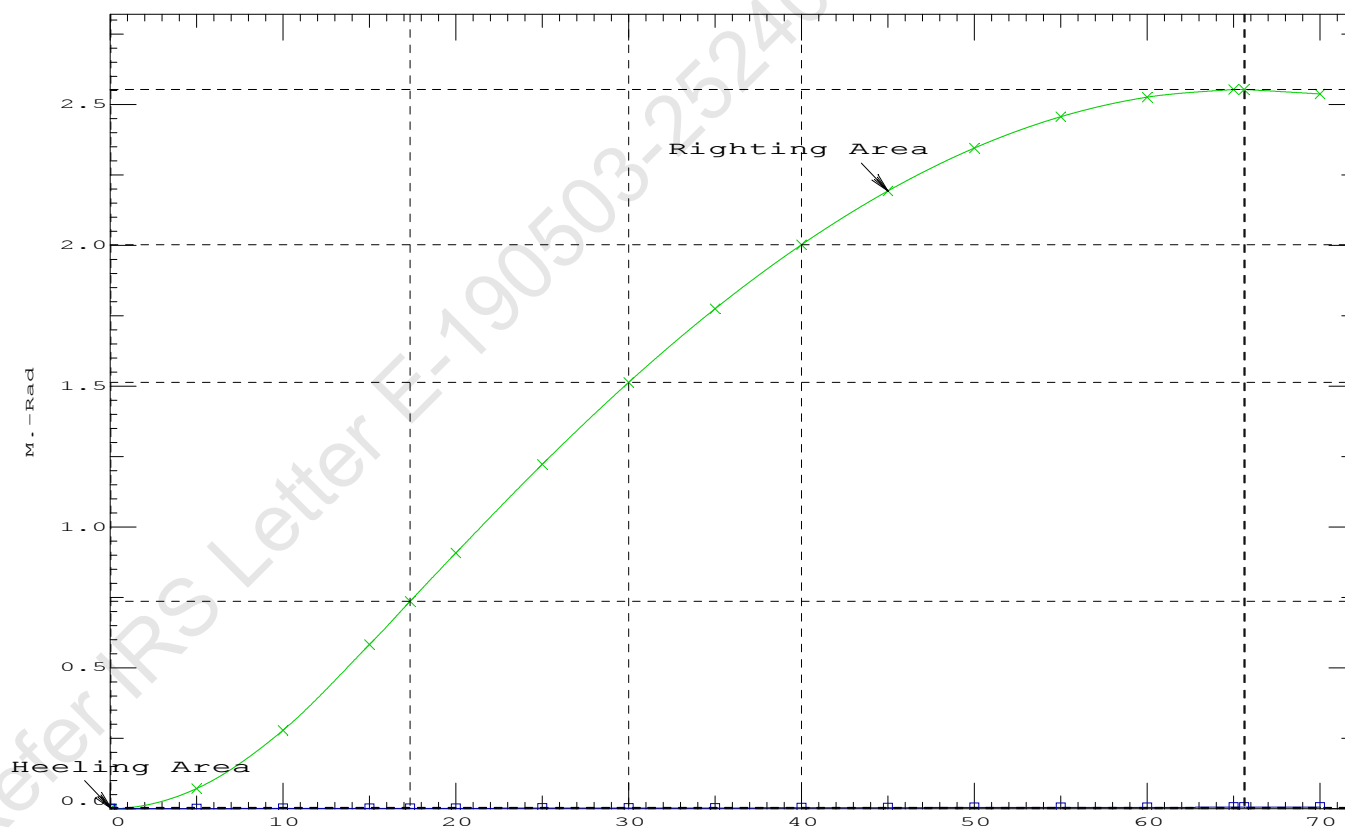
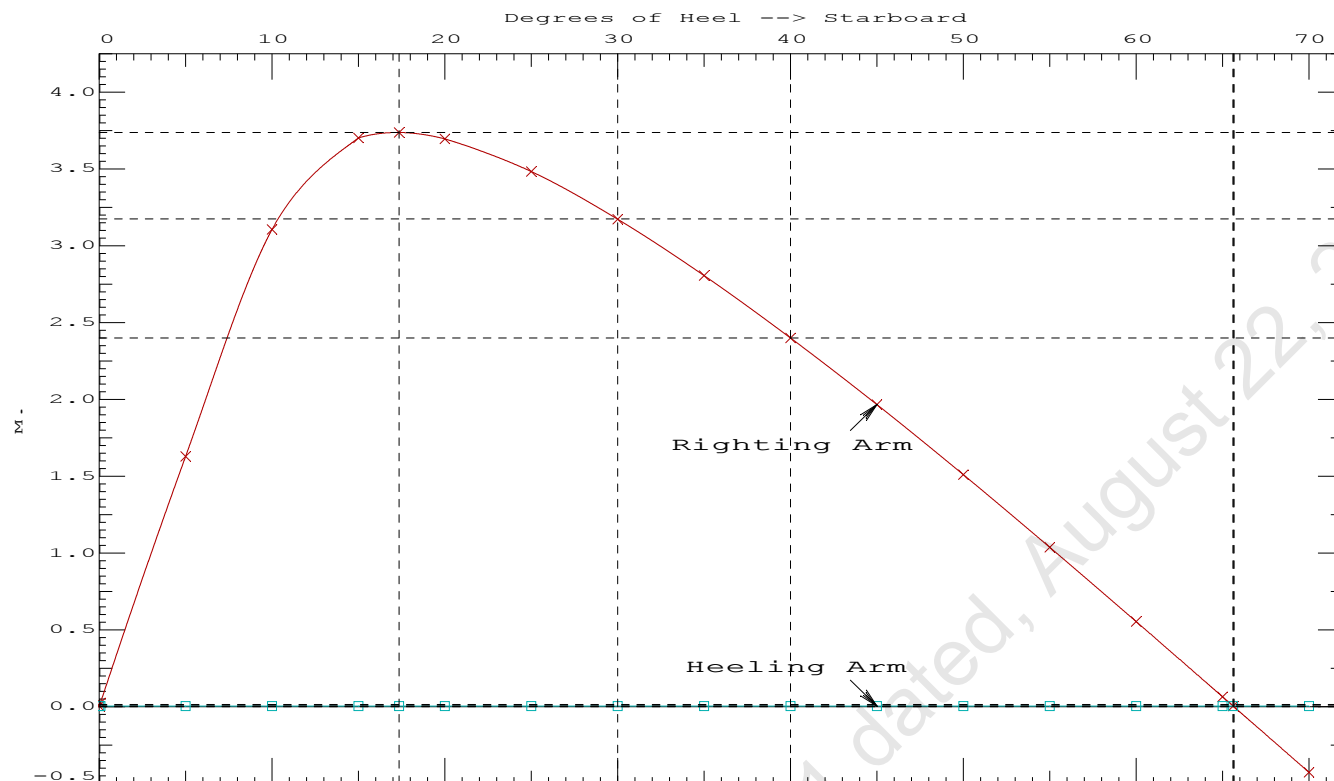
Fixed CG: LCG = 18.300f TCG = 0.000 VCG = 4.650

Origin	Degrees of		Displacement	Residual Arms		Res.
Depth---	Trim---	Heel---	Weight (MT)---	in Trim--	in Heel---	Area
1.264	0.12f	0.00s	1,091.0	0.000	0.000	0.0000
1.264	0.12f	0.07s	1,091.1	0.000	0.021	0.0000
1.250	0.11f	5.00s	1,091.0	0.000	1.624	0.0709
1.188	0.13f	10.00s	1,090.9	0.000	3.101	0.2781
1.008	0.18f	15.00s	1,091.0	0.000	3.698	0.5812
0.916	0.21f	17.36s	1,091.0	0.000	3.733	0.7344
0.811	0.24f	20.00s	1,091.0	0.000	3.690	0.9064
0.606	0.30f	25.00s	1,091.0	0.000	3.478	1.2216
0.396	0.37f	30.00s	1,091.0	0.000	3.169	1.5123
0.182	0.43f	35.00s	1,091.0	0.000	2.802	1.7733
-0.035	0.49f	40.00s	1,091.0	0.000	2.396	2.0004
-0.253	0.55f	45.00s	1,091.0	0.000	1.961	2.1907
-0.469	0.61f	50.00s	1,091.0	0.000	1.505	2.3421
-0.682	0.66f	55.00s	1,091.0	0.000	1.033	2.4530
-0.890	0.71f	60.00s	1,091.0	0.000	0.550	2.5221
-1.091	0.76f	65.00s	1,091.0	0.000	0.060	2.5487
-1.115	0.76f	65.62s	1,091.0	0.000	0.000	2.5491
-1.282	0.79f	70.00s	1,091.0	0.000	-0.431	2.5326
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.						

Note: The Center of Gravity shown above is for the Fixed Weight of 805.97 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAzero > 1.000 LARGE
(2) Angle from abs 0.1 deg to RAzero > 20.00 deg 65.55 P

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PONTOON

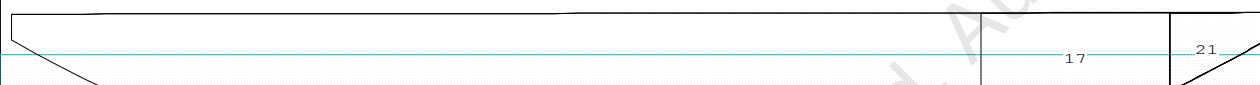
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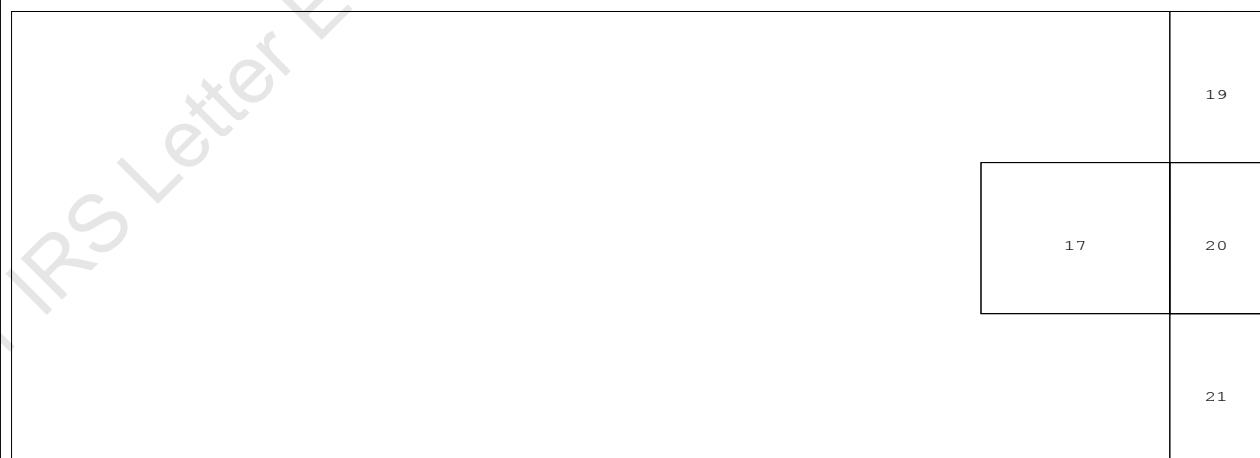
Condition 05 : Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Stbd

Condition Graphic - Draft: 1.339 @ 50.000f, 1.405 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

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PONTOON

Condition 05 : Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Stbd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.339 @ 50.00f, 1.372 @ 25.00f, 1.405 @ 0.00

Trim: Aft 0.067/50.000, Heel: zero

Part-----			Weight (MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			406.00	25.000f	0.000	3.000	
Crane			400.00	11.500f	0.000	6.325	
Crane Load			40.00	11.500f	0.000	34.602	
Total Fixed----->			846.00	17.979f	0.000	6.066	
	Load-----	SpGr-----	Weight (MT)----	LCG-----	TCG-----	VCG-----	FSM-----
06_WBTK.C	1.000	1.025	135.67	42.251f	0.000	1.499	135.61*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			284.98	45.086f	0.000	1.697	352.57
Total Weight----->			1,130.98	24.809f	0.000	4.965	
			Displ (MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,130.98	24.803f	0.000	0.693	

Righting Arms: 0.000 0.000

Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025	859.7	24.937f	0.000	147.78	20.744
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
		8.81	32.46	143.51	16.471	

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

278.0 Cu.M. (11%) SALT WATER

400.00 MT Crane

40.00 MT Crane Load

DRAFT AT FP (M) = 1.339
DRAFT AT MS (M) = 1.372
DRAFT AT AP (M) = 1.405
DRAFT AT FWD MARK (M) = 1.344
DRAFT AT AFT MARK (M) = 1.400

HYDROSTATIC PROPERTIES

Trim: Aft 0.067/50.000, No Heel, VCG = 4.965

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/
Draft----	Weight (MT)----	LCB-----	VCB-----	cm-----
1.372	1,130.98	24.803f	0.693	8.81
				24.937f
				32.46
				143.51
				16.471

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 352.6 m.-MT

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PONTOON

Condition 05 : Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Stbd

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 1.339 @ 50.00f, 1.372 @ 25.00f, 1.405 @ 0.00

Trim: Aft 0.067/50.000, Heel: zero

Part	LPA*Sf	HCP	Arm	Pressure	Moment
HULL	81.7	0.832	1.503	0.05500	6.76
CRANE	57.0	4.120	4.792	0.05500	15.02
Total wind heeling moment to starboard-->					21.78
Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT					

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.809f TCG = 0.000 VCG = 4.965

Free Surface Adjustment: 0.312

Adjusted CG: LCG = 24.810f TCG = 0.000 VCG = 5.277

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---Trim---Heel---	Weight (MT)---	in Trim--in Heel---	Area---Angle	
1.393 0.08a 0.00	1,131.0	0.000 -0.019	0.0000	5.248
1.393 0.08a 0.05s	1,131.0	0.000 0.000	-0.0000	5.197
1.378 0.08a 5.05s	1,131.0	0.000 1.442	0.0629	0.197
1.377 0.08a 5.25s	1,131.0	0.000 1.499	0.0680	0.000
1.328 0.08a 10.05s	1,131.0	0.000 2.766	0.2475	-4.803
1.208 0.12a 15.05s	1,131.0	0.000 3.220	0.5162	-9.803
1.198 0.12a 15.43s	1,131.0	0.000 3.221	0.5373	-10.178
1.079 0.16a 20.05s	1,131.0	0.000 3.084	0.7950	-14.803
0.942 0.21a 25.05s	1,131.0	0.000 2.757	1.0513	-19.803
0.800 0.25a 30.05s	1,131.0	0.000 2.342	1.2744	-24.803
0.652 0.29a 35.05s	1,131.0	0.000 1.876	1.4588	-29.803
0.500 0.34a 40.05s	1,131.0	0.000 1.379	1.6011	-34.803
0.345 0.38a 45.05s	1,131.0	0.000 0.862	1.6990	-39.803
0.187 0.42a 50.05s	1,131.0	0.000 0.331	1.7511	-44.803
0.090 0.44a 53.13s	1,131.0	0.000 0.000	1.7600	-47.882
0.028 0.46a 55.05s	1,131.0	0.000 -0.207	1.7566	-49.803
-0.131 0.49a 60.05s	1,131.0	0.000 -0.746	1.7150	-54.803
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.				

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 352.6 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.78 (constant)

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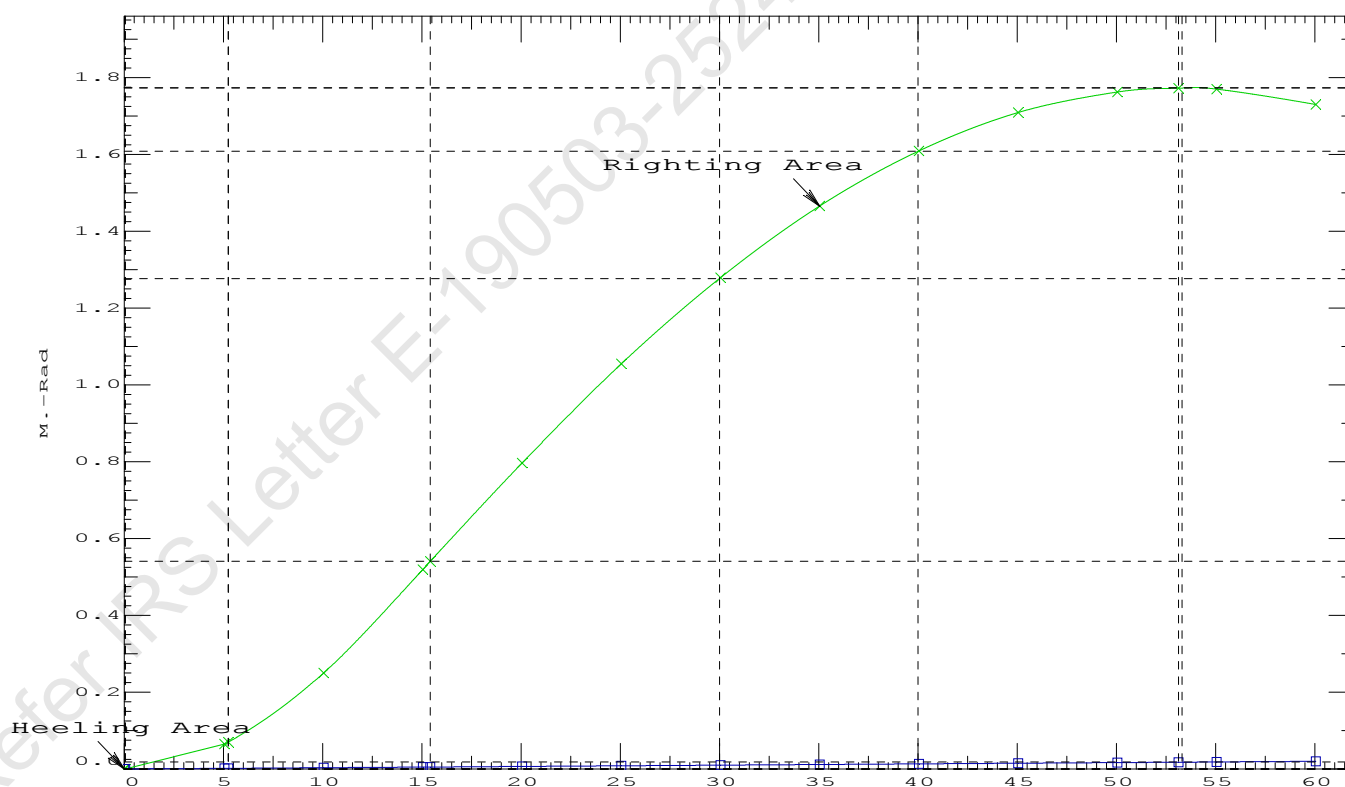
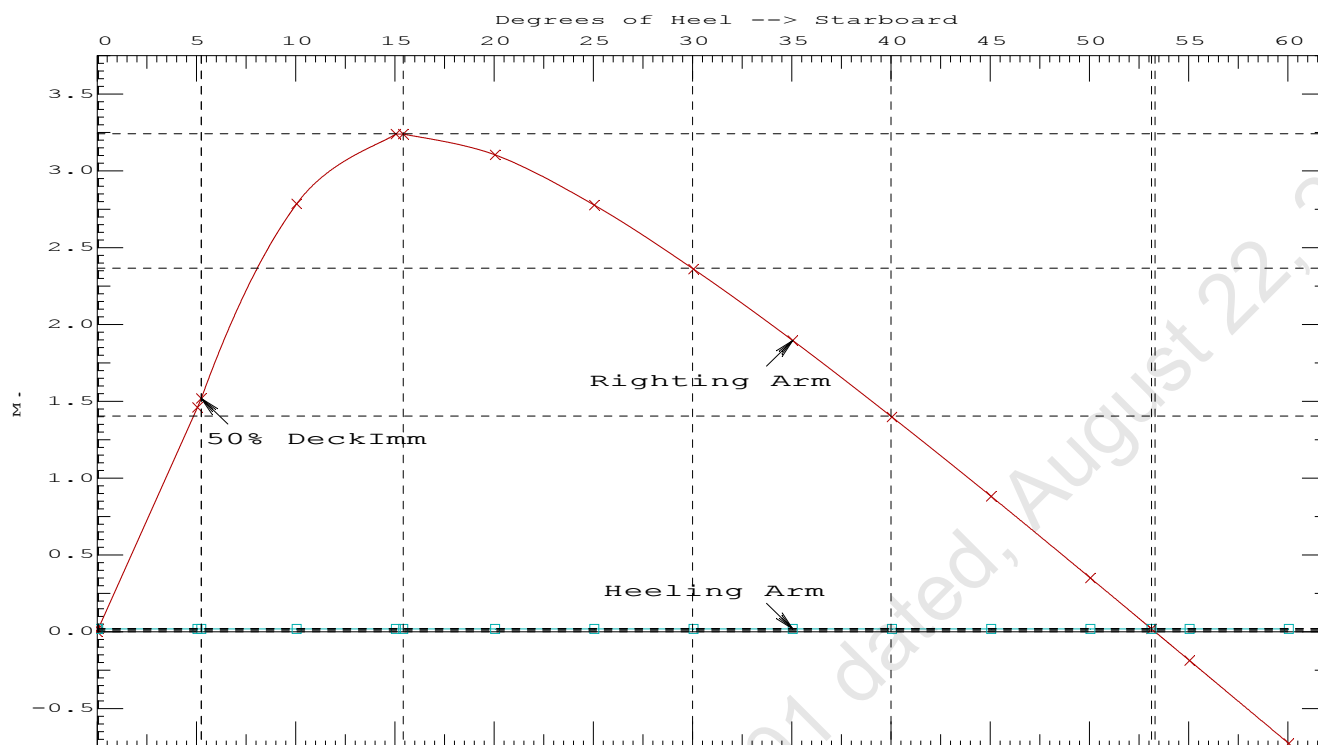
PONTOON**Condition 05 : Operating Condition of 400T Crane**
Crane Lifting 40.0T @ 30m Radius Stbd

LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max-----Attained
(1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad 0.5373 P
(2) Angle from Equilibrium to RAzero or Flood > 20.00 deg 53.08 P
(3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg 5.20 P

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PONTOON

Condition 05 : Operating Condition of 400T CraneCrane Lifting 40.0T @ 30m Radius Stbd

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PONTOON

Condition 05 : Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.809f TCG = 0.000 VCG = 4.965
Free Surface Adjustment: 0.312
Adjusted CG: LCG = 24.810f TCG = 0.020p VCG = 5.277

Origin	Degrees of	Displacement	Residual Arms	Res.
Depth---Trim---Heel---	Weight (MT)---	in Trim--in Heel---	Area	
1.385 0.08a 3.59s	1,131.0	0.000 0.000	0.0000	
1.350 0.08a 8.59s	1,131.0	0.000 1.397	0.0610	
1.244 0.11a 13.59s	1,131.0	0.000 2.161	0.2252	
1.193 0.13a 15.65s	1,131.0	0.000 2.198	0.3038	
1.117 0.15a 18.59s	1,131.0	0.000 2.128	0.4160	
0.983 0.19a 23.59s	1,131.0	0.000 1.840	0.5924	
0.842 0.24a 28.59s	1,131.0	0.000 1.445	0.7365	
0.695 0.28a 33.59s	1,131.0	0.000 0.991	0.8432	
0.545 0.33a 38.59s	1,131.0	0.000 0.501	0.9085	
0.394 0.37a 43.47s	1,131.0	0.000 0.000	0.9299	
0.390 0.37a 43.59s	1,131.0	0.000 -0.013	0.9299	
0.234 0.41a 48.59s	1,131.0	0.000 -0.542	0.9058	
0.075 0.45a 53.59s	1,131.0	0.000 -1.080	0.8351	
-0.084 0.48a 58.59s	1,131.0	0.000 -1.620	0.7173	
-0.244 0.51a 63.59s	1,131.0	0.000 -2.159	0.5524	

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 352.6 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1200.00

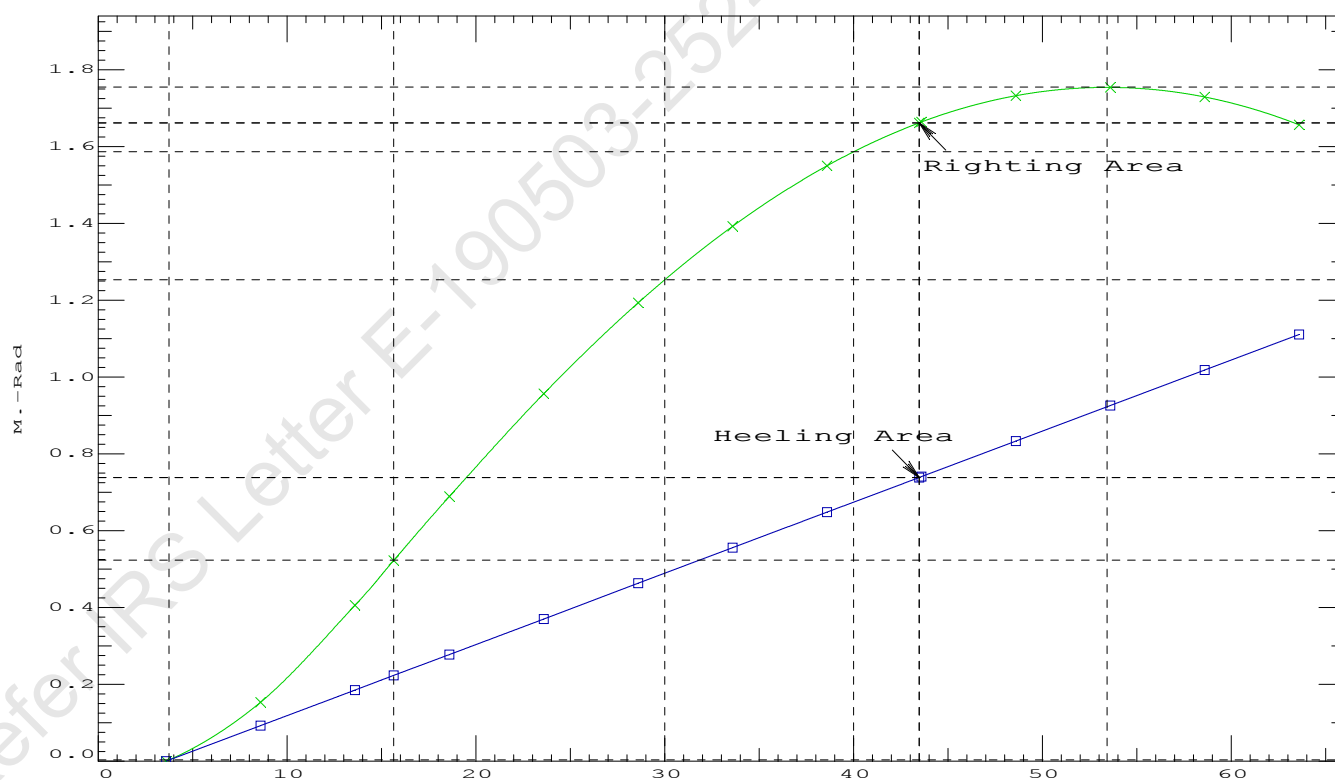
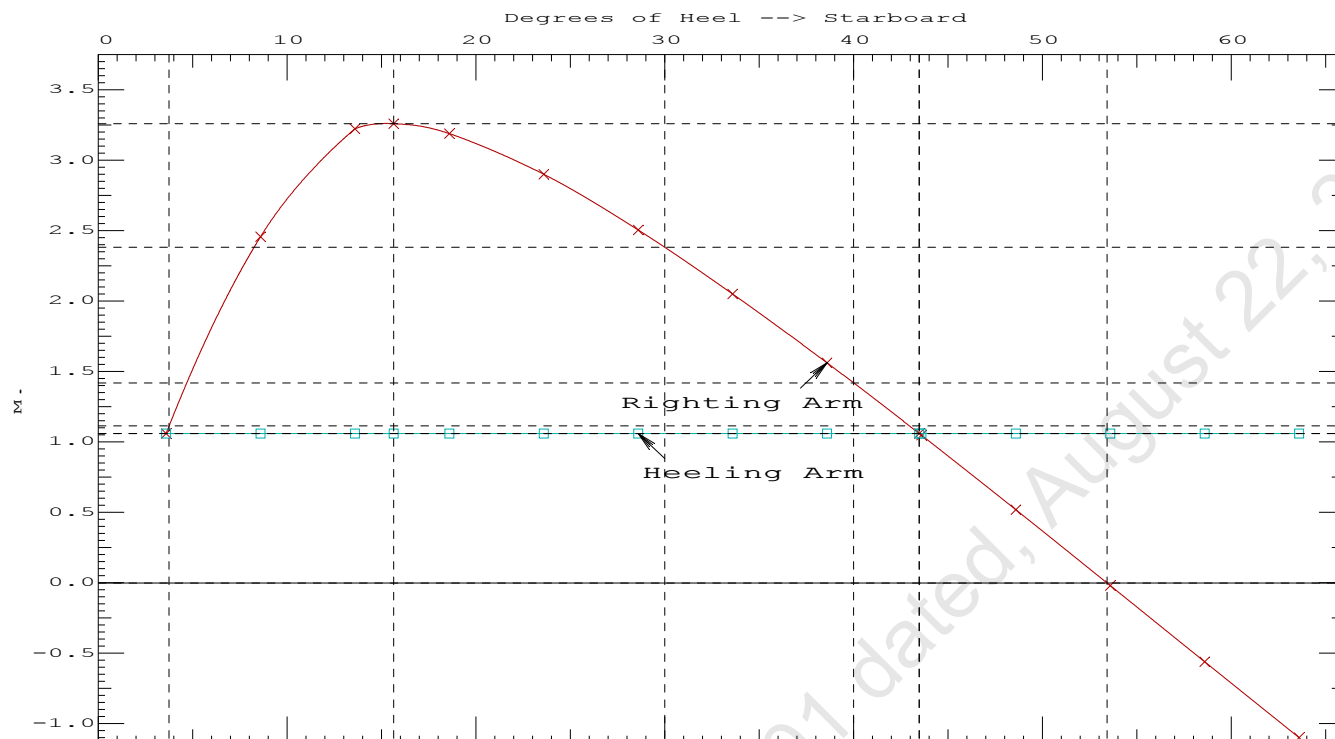
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	3.59 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	207.2 P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667	2.259 P

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PONTOON

Condition 05 : Operating Condition of 400T CraneCrane Lifting 40.0T @ 30m Radius Stbd

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PONTOON

Condition 05 :Operating Condition of 400T CraneCrane Lifting 40.0T @ 30m Radius Stbd

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.116 @ 50.00f, 1.990 @ 25.00f, 1.864 @ 0.00

Trim: Fwd 0.251/50.000, Heel: Stbd 3.67 deg.

Part-----			Weight (MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			400.00	11.500f	0.000	6.325	
Crane Load			40.00	11.500f	0.000	34.602	
Deck Cargo			560.00	27.000f	0.000	4.000	
Total Fixed----->			1,405.97	21.572f	0.000	5.243	
	Load-----	SpGr-----	Weight (MT)----	LCG-----	TCG-----	VCG-----	FSM-----
06_WBTK.C	1.000	1.025	135.67	42.251f	0.000	1.499	135.61*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			284.98	45.086f	0.000	1.697	352.57
Total Weight----->			1,690.95	25.535f	0.000	4.646	
			Displ (MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,690.98	25.553f	0.942s	1.049	

Righting Arms:	0.000	0.710s
External Arms:	0.000	0.710s
Residual Righting Arms:	0.000	0.000s
Part-----	SpGr-----	WPA-----
Total Waterplane---->	1.025	890.9
		MT/cm
		9.13

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 3.67

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PONTOON

After hook load drop

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.145 @ 50.00f, 1.949 @ 25.00f, 1.752 @ 0.00

Trim: Fwd 0.393/50.000, Heel: 0.00 deg.

Part	Weight (MT)	LCG	TCG	VCG	FSM		
LIGHT SHIP	405.97	25.000f	0.000	3.000			
Crane	400.00	11.500f	0.000	6.325			
Deck Cargo	560.00	27.000f	0.000	4.000			
Total Fixed	1,365.97	21.867f	0.000	4.384			
Load	SpGr	Weight (MT)	LCG	TCG	VCG	FSM	
06_WBTK.C	1.000	1.025	135.67	42.251f	0.000	1.499	135.61*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks			284.98	45.086f	0.000	1.697	352.57
Total Weight			1,650.95	25.875f	0.000	3.920	
			Displ (MT)	LCB	TCB	VCB	
HULL	1.025		1,650.95	25.898f	0.001	0.998	
Righting Arms:			0.000	0.001s			
External Arms:			0.000	0.001			
Residual Righting Arms:			0.000	0.000			
Part	SpGr	WPA	LCF	TCF	BML	BMT	
Total Waterplane	1.025	890.0	25.288f	0.017s	112.32	14.696	
		MT/cm	m.	MT/cm	GML	GMT	
		9.12		36.12	109.40	11.774	
Distances in METERS.						Moments in m.-MT.	

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: 3331.54; t3: 30.0; t: 15.0
limmarg: 1318.14; t3: 15.0; t: 6.0
limmarg: 485.61; t3: 9.0; t: 2.4
limmarg: 223.05; t3: 6.6; t: 1.0
limmarg: 132.93; t3: 5.6; t: 0.4
limmarg: 100.88; t3: 5.2; t: 0.2
limmarg: 85.64; t3: 5.0; t: 0.1
limmarg: 78.41; t3: 4.9; t: 0.0
limmarg: 78.41; t3: 4.9; t: 0.0

Theta3 is 4.90 degrees, as demonstrated below.

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PONTOON**RESIDUAL RIGHTING ARMS vs HEEL ANGLE**

Fixed CG: LCG = 21.867f TCG = 0.000 VCG = 4.384

Origin Depth	Degrees of Trim	Heel	Displacement Weight (MT)	Residual Arms in Trim	in Heel	Res. Area
1.728	0.46f	3.67s	1,651.0	0.000	-0.774	0.0000
1.740	0.45f	0.00	1,651.0	0.000	0.000	-0.0248
1.722	0.46f	4.90p	1,651.0	0.000	1.033	0.0194
1.721	0.47f	5.00p	1,651.0	0.000	1.054	0.0212
1.680	0.59f	10.00p	1,650.9	0.000	1.869	0.1505
1.717	0.81f	15.00p	1,651.0	0.000	2.098	0.3265
1.723	0.85f	15.76p	1,651.0	0.000	2.101	0.3543
1.741	1.09f	20.00p	1,651.0	0.000	2.025	0.5083
1.745	1.37f	25.00p	1,651.0	0.000	1.821	0.6775
1.730	1.66f	30.00p	1,650.9	0.000	1.557	0.8253

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1365.97 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Res. Area Ratio from abs -3.7 deg to abs 4.9 > 1.000 1.784 P

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GHS 17.22

PONTOON**RESIDUAL RIGHTING ARMS vs HEEL ANGLE**

Fixed CG: LCG = 21.867f TCG = 0.000 VCG = 4.384

Origin	Degrees of	Displacement	Residual Arms	Res.
Depth---Trim---Heel---Weight (MT)---in Trim--in Heel---> Area				
1.728	0.46f	3.67s	1,651.0	0.000 -0.774 0.0000
1.740	0.45f	0.00	1,651.0	0.000 0.000 -0.0248
1.722	0.46f	4.90p	1,651.0	0.000 1.033 0.0194
1.721	0.47f	5.00p	1,651.0	0.000 1.054 0.0212
1.680	0.59f	10.00p	1,650.9	0.000 1.869 0.1505
1.717	0.81f	15.00p	1,651.0	0.000 2.098 0.3265
1.723	0.85f	15.76p	1,651.0	0.000 2.101 0.3543
1.741	1.09f	20.00p	1,651.0	0.000 2.025 0.5083
1.745	1.37f	25.00p	1,651.0	0.000 1.821 0.6775
1.730	1.66f	30.00p	1,650.9	0.000 1.557 0.8253
1.696	1.95f	35.00p	1,650.9	0.000 1.259 0.9484
1.644	2.24f	40.00p	1,650.9	0.000 0.939 1.0444
1.576	2.52f	45.00p	1,650.9	0.000 0.606 1.1119
1.494	2.78f	50.00p	1,650.9	0.000 0.264 1.1500
1.424	2.97f	53.82p	1,651.0	0.000 0.000 1.1588
1.401	3.02f	55.00p	1,651.0	0.000 -0.082 1.1580

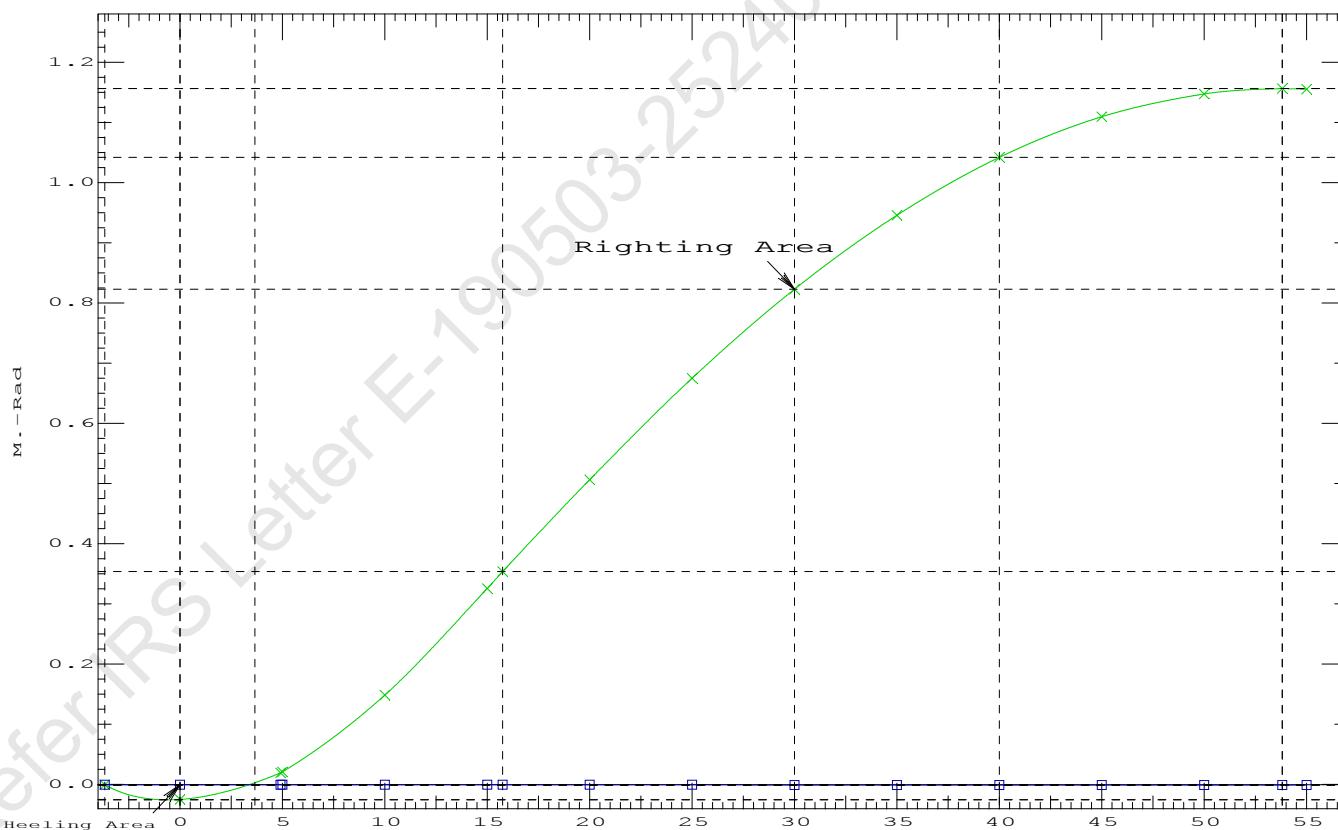
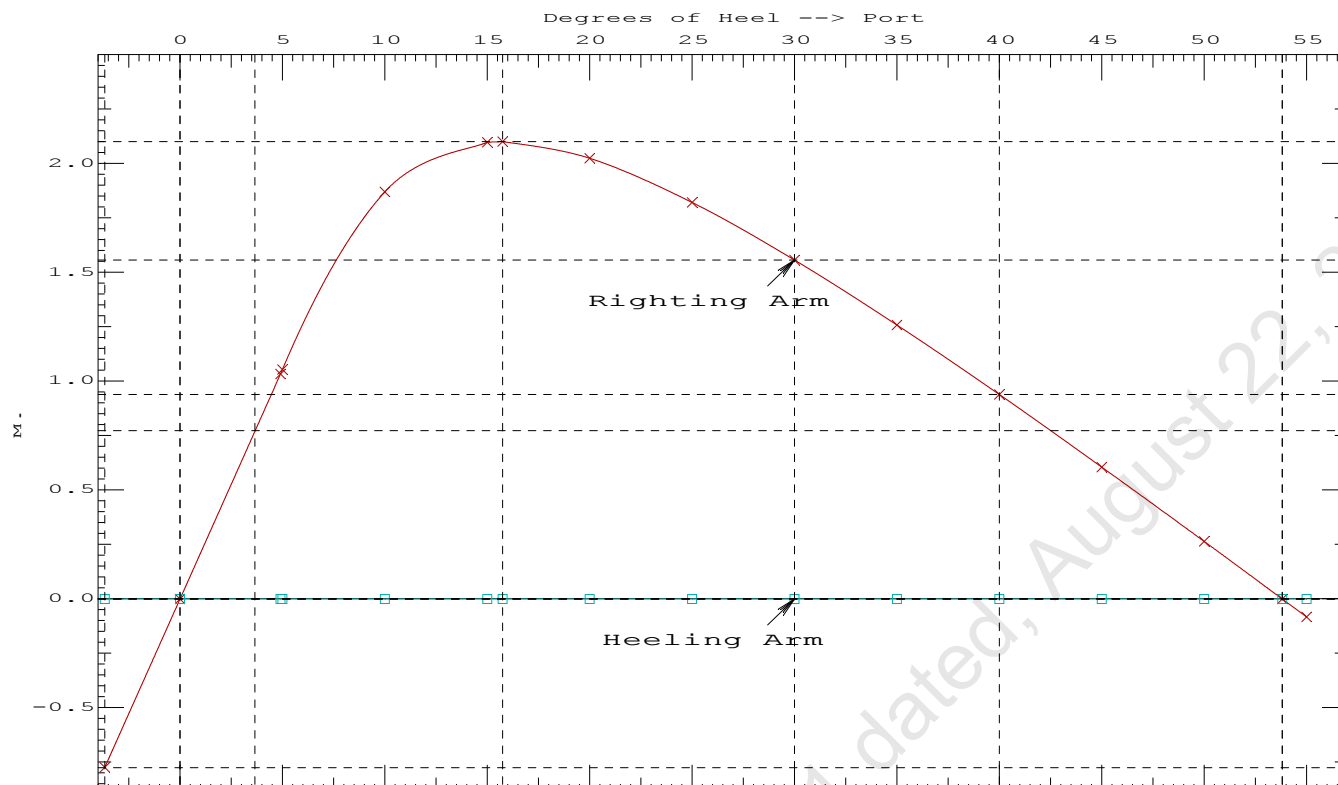
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1365.97 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Res. Area Ratio from abs -3.7 deg to RAzero > 1.000 47.780 P
(2) Angle from abs 4.9 deg to RAzero > 20.00 deg 48.92 P

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GHS 17.22 PONTOON



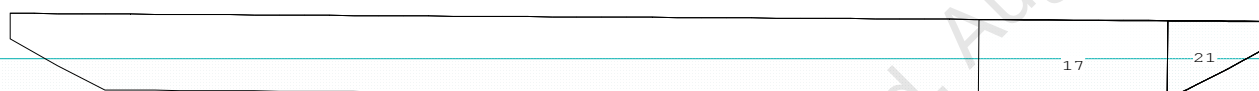
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PONTOON

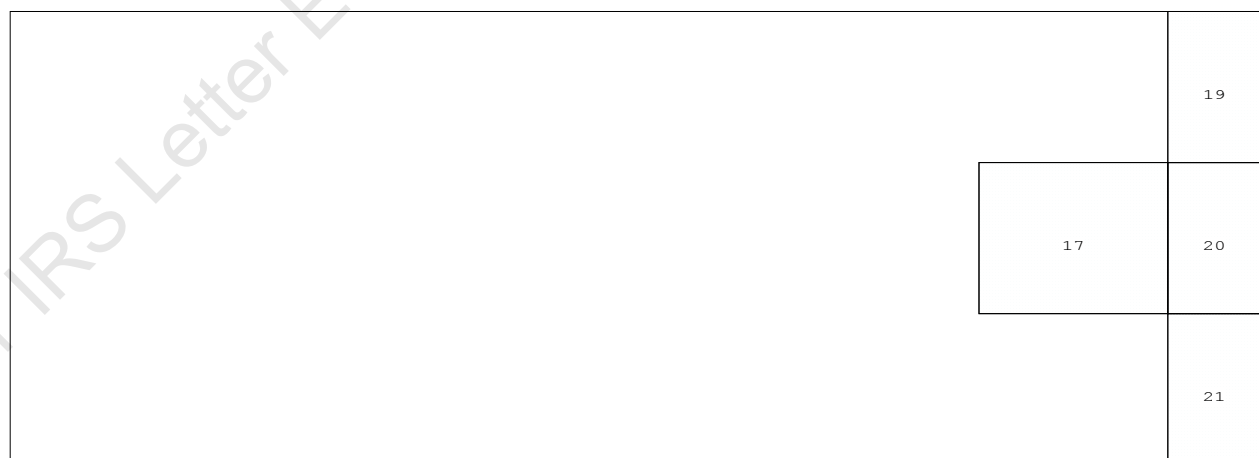
Condition 06 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Fwd

Condition Graphic - Draft: 1.529 @ 50.000f, 1.213 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

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PONTOON

Condition 06 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Fwd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.529 @ 50.00f, 1.371 @ 25.00f, 1.213 @ 0.00

Trim: Fwd 0.316/50.000, Heel: zero

Part-----	Weight (MT)	LCG-----	TCG-----	VCG-----	FSM-----
LIGHT SHIP	406.00	25.000f	0.000	3.000	
Crane	400.00	11.500f	0.000	6.325	
Crane Load	40.00	42.500f	0.000	34.602	
Total Fixed----->	846.00	19.444f	0.000	6.066	
Load-----	SpGr-----	Weight (MT)	LCG-----	TCG-----	VCG-----
06_WBTK.C	1.000	1.025	135.67	42.251f	0.000
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s
Total Tanks----->		284.98	45.086f	0.000	1.697
Total Weight----->		1,130.98	25.906f	0.000	4.965
		Displ (MT)	LCB-----	TCB-----	VCB-----
HULL	1.025	1,130.92	25.933f	0.000	0.695

Righting Arms: 0.000 0.000

Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025	859.2	25.278f	0.000	147.54	20.712
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
		8.81	32.41	143.27	16.441	

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

278.0 Cu.M. (11%) SALT WATER

400.00 MT Crane

40.00 MT Crane Load

DRAFT AT FP (M) = 1.529
DRAFT AT MS (M) = 1.371
DRAFT AT AP (M) = 1.213
DRAFT AT FWD MARK (M) = 1.504
DRAFT AT AFT MARK (M) = 1.238

HYDROSTATIC PROPERTIES

Trim: Fwd 0.316/50.000, No Heel, VCG = 4.965

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/
Draft----	Weight (MT)	LCB-----	VCB-----	cm-----
1.373	1,130.92	25.933f	0.695	8.81
				25.278f
				32.41
				143.27
				16.441

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 352.6 m.-MT

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PONTOON

Condition 06 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Fwd

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 1.529 @ 50.00f, 1.371 @ 25.00f, 1.213 @ 0.00

Trim: Fwd 0.316/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	81.7	0.835	1.507	0.05500	6.78
CRANE	57.0	4.235	4.908	0.05500	15.39
Total wind heeling moment to starboard-->					22.16
Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT					

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.906f TCG = 0.000 VCG = 4.965

Free Surface Adjustment: 0.312

Adjusted CG: LCG = 25.904f TCG = 0.000 VCG = 5.277

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---Trim---Heel---	Weight (MT)---	in Trim--in Heel---	Area---Angle	
1.201 0.36f 0.00	1,130.9	0.000 -0.019	0.0000	4.792
1.201 0.36f 0.06s	1,131.0	0.000 0.000	-0.0000	4.728
1.187 0.36f 4.79s	1,131.0	0.000 1.367	0.0564	0.000
1.186 0.36f 5.06s	1,131.0	0.000 1.445	0.0631	-0.272
1.117 0.39f 10.06s	1,131.0	0.000 2.761	0.2475	-5.272
0.903 0.58f 15.06s	1,131.0	0.000 3.202	0.5149	-10.272
0.887 0.59f 15.42s	1,131.0	0.000 3.204	0.5349	-10.629
0.668 0.78f 20.06s	1,131.0	0.000 3.066	0.7922	-15.272
0.423 0.98f 25.06s	1,131.0	0.000 2.738	1.0467	-20.272
0.170 1.19f 30.06s	1,131.0	0.000 2.322	1.2682	-25.272
-0.088 1.40f 35.06s	1,131.0	0.000 1.857	1.4509	-30.272
-0.349 1.61f 40.06s	1,131.0	0.000 1.360	1.5915	-35.272
-0.610 1.81f 45.06s	1,131.0	0.000 0.843	1.6878	-40.272
-0.868 2.00f 50.06s	1,131.0	0.000 0.313	1.7383	-45.272
-1.016 2.10f 52.99s	1,131.0	0.000 0.000	1.7463	-48.195
-1.119 2.17f 55.06s	1,131.0	0.000 -0.223	1.7423	-50.272
-1.360 2.33f 60.06s	1,131.0	0.000 -0.760	1.6994	-55.272
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.				

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 352.6 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 22.16 (constant)

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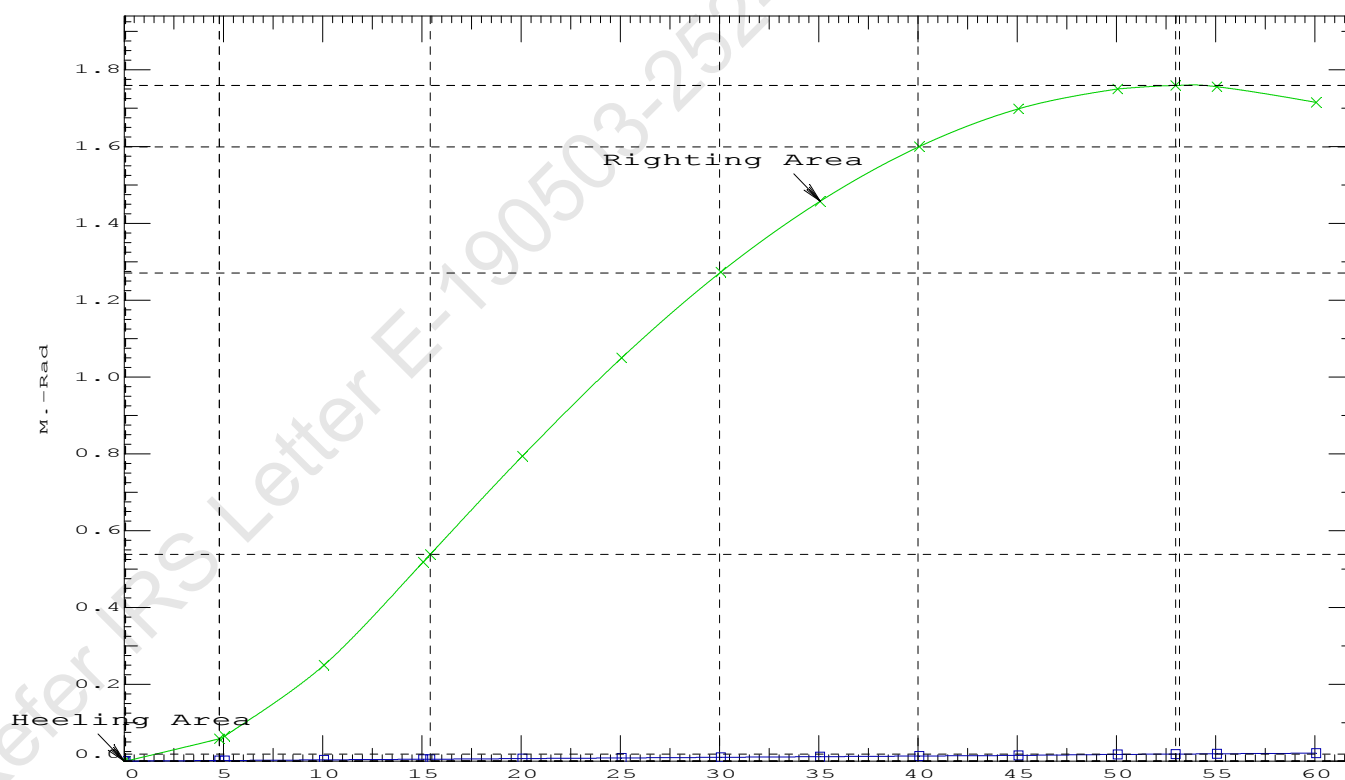
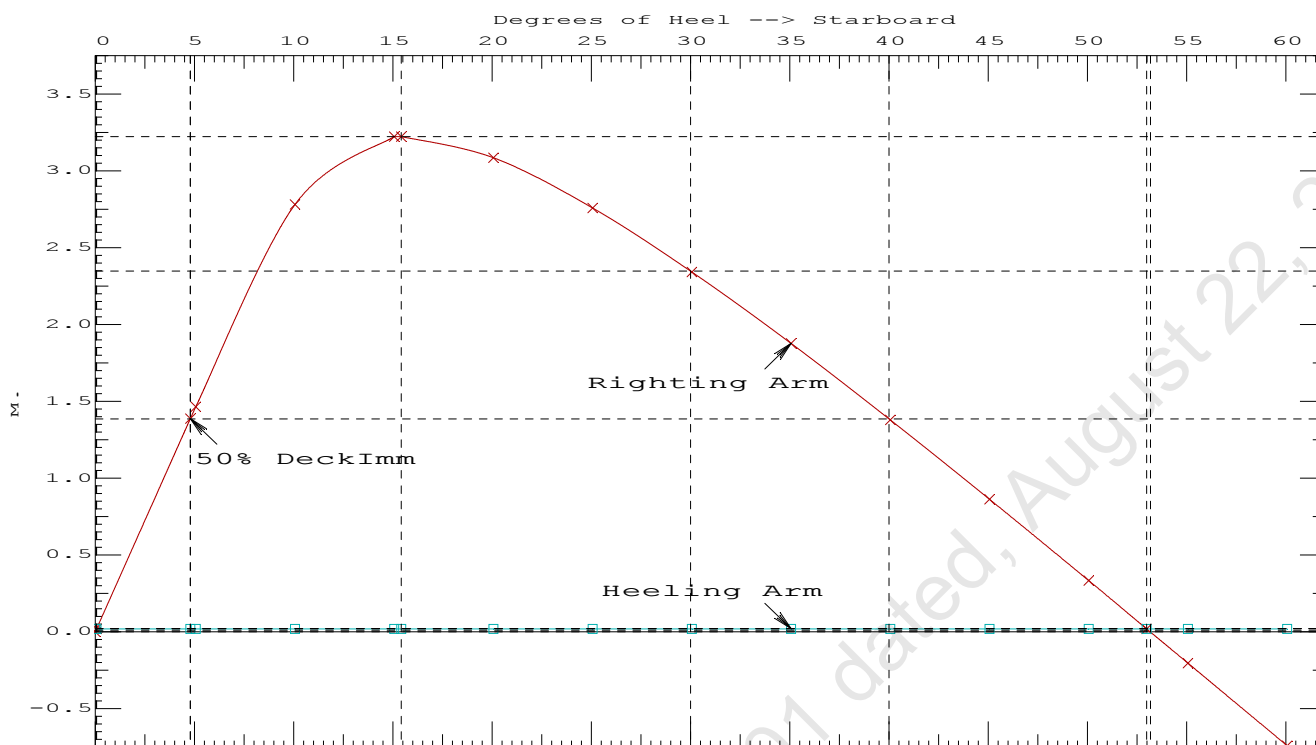
PONTOON**Condition 06 :Operating Condition of 400T Crane**
Crane Lifting 40.0T @ 30m Radius Fwd

LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max-----Attained
(1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad 0.5349 P
(2) Angle from Equilibrium to RAzero or Flood > 20.00 deg 52.92 P
(3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg 4.73 P

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PONTOON

Condition 06 :Operating Condition of 400T CraneCrane Lifting 40.0T @ 30m Radius Fwd

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PONTOON

Condition 06 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Fwd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.906f TCG = 0.000 VCG = 4.965
Free Surface Adjustment: 0.312
Adjusted CG: LCG = 25.904f TCG = 0.000 VCG = 5.277

Origin Depth	Degrees of Trim	Displacement Heel	Residual Arms Weight (MT)	Res. in Trim	Res. in Heel	Area
---	---	---	---	---	---	---
1.201	0.36f	0.00	1,131.0	0.000	0.000	0.0000
1.186	0.36f	5.00s	1,130.9	0.000	1.446	0.0631
1.119	0.39f	10.00s	1,131.0	0.000	2.767	0.2478
0.906	0.57f	15.00s	1,131.0	0.000	3.220	0.5163
0.886	0.59f	15.43s	1,131.0	0.000	3.222	0.5406
0.671	0.77f	20.00s	1,131.0	0.000	3.087	0.7952
0.426	0.98f	25.00s	1,131.0	0.000	2.761	1.0528
0.174	1.19f	30.00s	1,131.0	0.000	2.346	1.2763
-0.084	1.40f	35.00s	1,131.0	0.000	1.881	1.4611
-0.346	1.60f	40.00s	1,131.0	0.000	1.385	1.6039
-0.607	1.80f	45.00s	1,131.0	0.000	0.868	1.7023
-0.865	1.99f	50.00s	1,131.0	0.000	0.338	1.7550
-1.025	2.11f	53.16s	1,131.0	0.000	0.000	1.7644
-1.116	2.17f	55.00s	1,131.0	0.000	-0.198	1.7612
-1.357	2.32f	60.00s	1,131.0	0.000	-0.735	1.7205

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 352.6 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

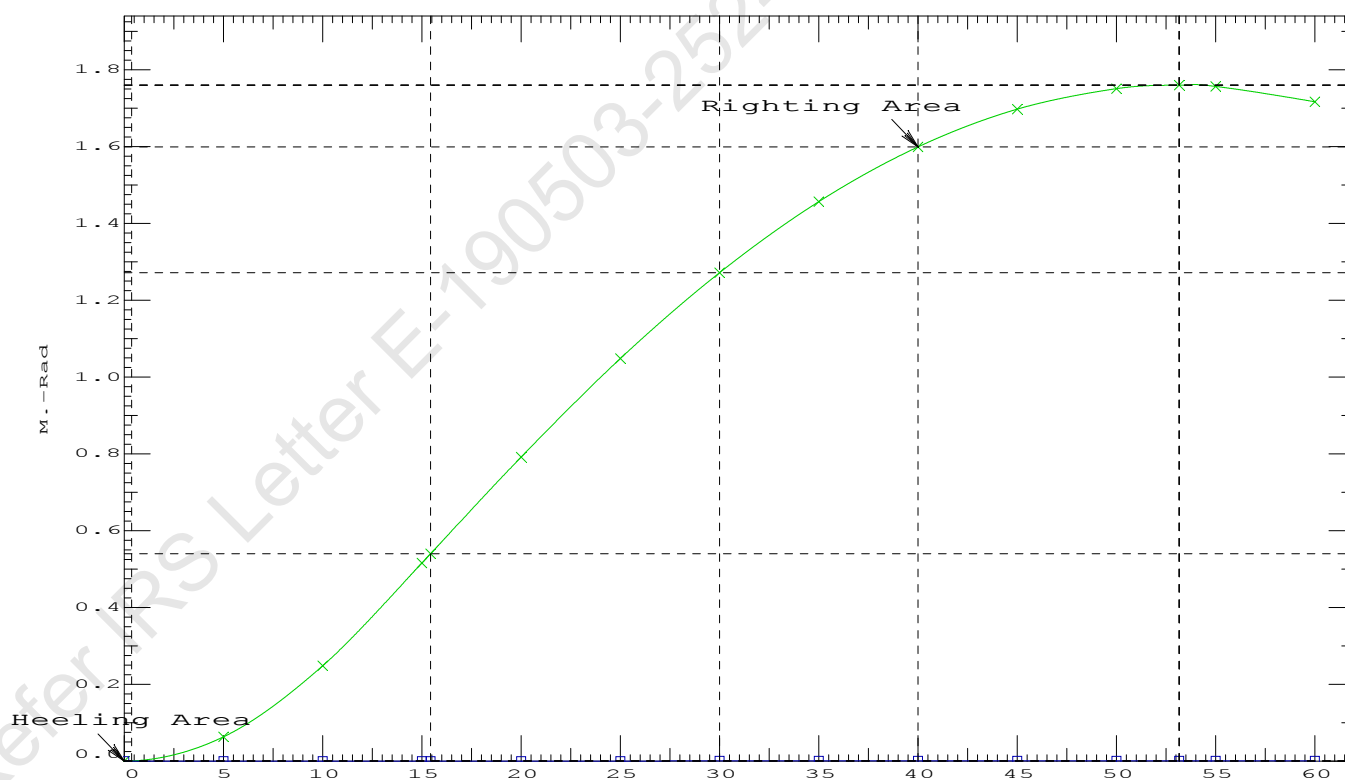
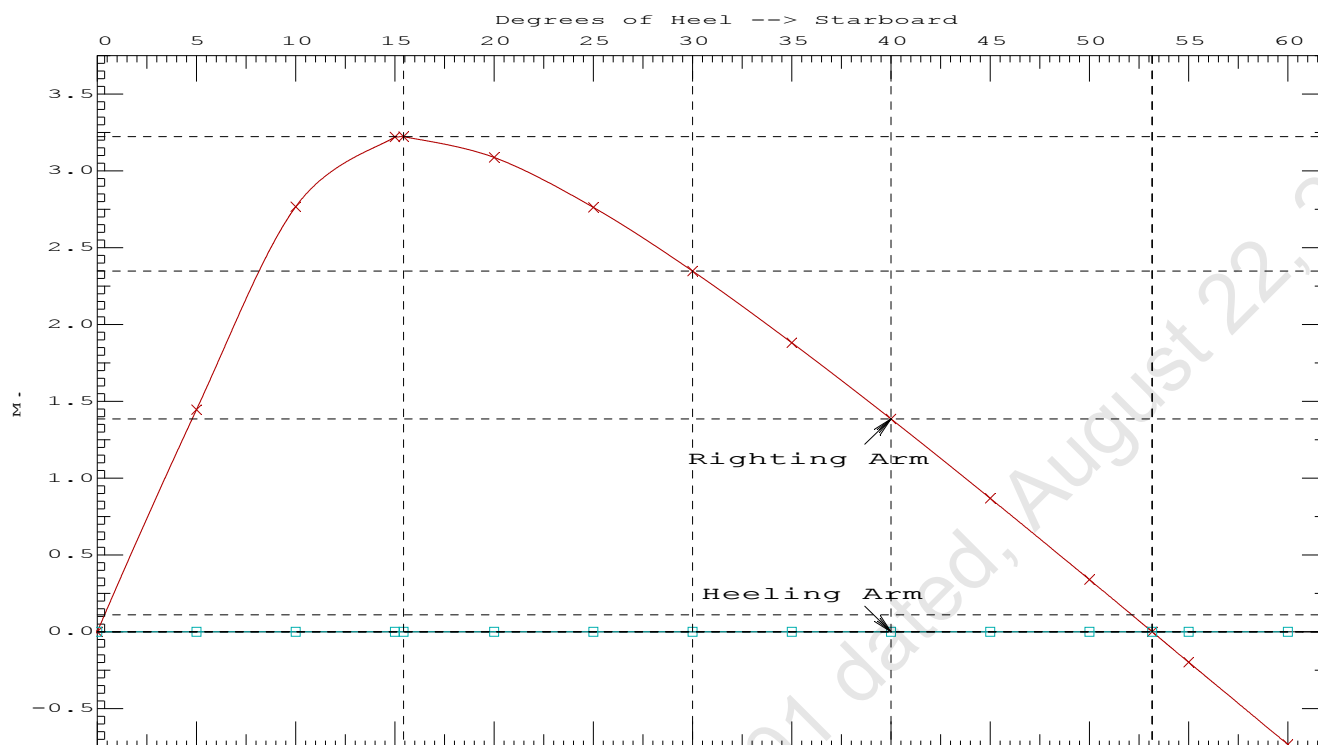
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM	STABILITY CRITERIA FOR CRANE OPERAT	Min/Max	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	0.00 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	364429 P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667	2151.77 P

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PONTOON

Condition 06 :Operating Condition of 400T CraneCrane Lifting 40.0T @ 30m Radius Fwd

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GHS 17.22 PONTOON

Condition 06 :Operating Condition of 400T Crane
Crane Lifting 40.0T @ 30m Radius Fwd

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.529 @ 50.00f, 1.371 @ 25.00f, 1.213 @ 0.00

Trim: Fwd 0.316/50.000, Heel: Stbd 0.01 deg.

Part-----			Weight (MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			400.00	11.500f	0.000	6.325	
Crane Load			40.00	42.500f	0.000	34.602	
Total Fixed----->			845.97	19.444f	0.000	6.066	
	Load-----	SpGr-----	Weight (MT)----	LCG-----	TCG-----	VCG-----	FSM-----
06_WBTK.C	1.000	1.025	135.67	42.251f	0.000	1.499	135.61*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.C	1.000	1.025	50.11	47.663f	0.000	1.872	72.33*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			284.98	45.086f	0.000	1.697	352.57
Total Weight----->			1,130.95	25.906f	0.000	4.965	
			Displ (MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,130.95	25.932f	0.005s	0.695	

	Righting Arms:			0.000	0.004s		
	External Arms:			0.000	0.004s		
	Residual Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		857.0	25.336f	0.017s	146.44	20.633
			MT/cm				
			8.78				

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.01

PONTOON

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.377 @ 50.00f, 1.326 @ 25.00f, 1.276 @ 0.00
Trim: Fwd 0.101/50.000, Heel: 0.00 deg.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

```
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limmarg: 1120152.75; t3: 1; t: 9.84
limmarg: ; t3: -9; t: 8.86
limmarg: ; t3: 0; t: 7.97
limmarg: 78413896.00; t3: 8; t: 7.17
limmarg: 876633.81; t3: 1; t: 6.46
limmarg: ; t3: -5; t: 5.81
limmarg: 835590.25; t3: 1; t: 5.23
limmarg: ; t3: -4; t: 4.71
limmarg: 645314.31; t3: 1; t: 4.24
limmarg: ; t3: -3; t: 3.81
limmarg: 835590.25; t3: 1; t: 3.43
limmarg: ; t3: -2; t: 3.09
limmarg: 1500798.25; t3: 1; t: 2.78
limmarg: ; t3: -2; t: 2.50
limmarg: 325069.03; t3: 1; t: 2.25
limmarg: ; t3: -1; t: 2.03
limmarg: 1341808.13; t3: 1; t: 1.82
limmarg: ; t3: -1; t: 1.64
limmarg: 526809.75; t3: 1; t: 1.48
limmarg: ; t3: 0; t: 1.33
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GHS 17.22 PONTOON

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limmarg: ; t3: 0; t: 0.38
limmarg: 190801.91; t3: 0; t: 0.34
limmarg: ; t3: 0; t: 0.30
limmarg: 121051.44; t3: 0; t: 0.27
limmarg: ; t3: 0; t: 0.25
limmarg: 85497.51; t3: 0; t: 0.22
limmarg: ; t3: 0; t: 0.20
limmarg: 56103.97; t3: 0; t: 0.18
limmarg: ; t3: 0; t: 0.16
limmarg: 37020.79; t3: 0; t: 0.15
limmarg: ; t3: 0; t: 0.13
limmarg: 25290.12; t3: 0; t: 0.12
limmarg: ; t3: 0; t: 0.11
limmarg: 18700.75; t3: 0; t: 0.10
limmarg: ; t3: 0; t: 0.09
limmarg: 13094.23; t3: 0; t: 0.08
limmarg: ; t3: 0; t: 0.07
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limmarg: ; t3: 0; t: 0.05
limmarg: 4828.56; t3: 0; t: 0.04
limmarg: ; t3: 0; t: 0.04
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limmarg: ; t3: 0; t: 0.03
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limmarg: ; t3: 0; t: 0.02
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Theta3 is -0.02 degrees, as demonstrated below.

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.300f TCG = 0.000 VCG = 4.650

Origin Depth	Degrees of Trim	Displacement Heel	Weight (MT)	Residual Arms in Trim	Res. in Heel	Area
1.264	0.11f	0.01s	1,091.0	0.000	-0.003	0.0000
1.264	0.12f	0.00s	1,091.0	0.000	0.000	-0.0000
1.250	0.11f	5.00p	1,091.0	0.000	1.631	0.0712
1.188	0.13f	10.00p	1,091.0	0.000	3.109	0.2791
1.008	0.18f	15.00p	1,091.0	0.000	3.707	0.5829
0.916	0.21f	17.36p	1,091.0	0.000	3.742	0.7370
0.811	0.24f	20.00p	1,091.0	0.000	3.700	0.9089

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0.607	0.30f	25.00p	1,091.0	0.000	3.487	1.2249
0.397	0.37f	30.00p	1,091.0	0.000	3.178	1.5164

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 805.97 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

LIM-----	STABILITY CRITERION-----	Min/Max-----	Attained
(1) Residual Area Ratio from abs 0 deg to abs 0	>	1.000	1.000 P

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GHS 17.22

PONTOON

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.300f TCG = 0.000 VCG = 4.650

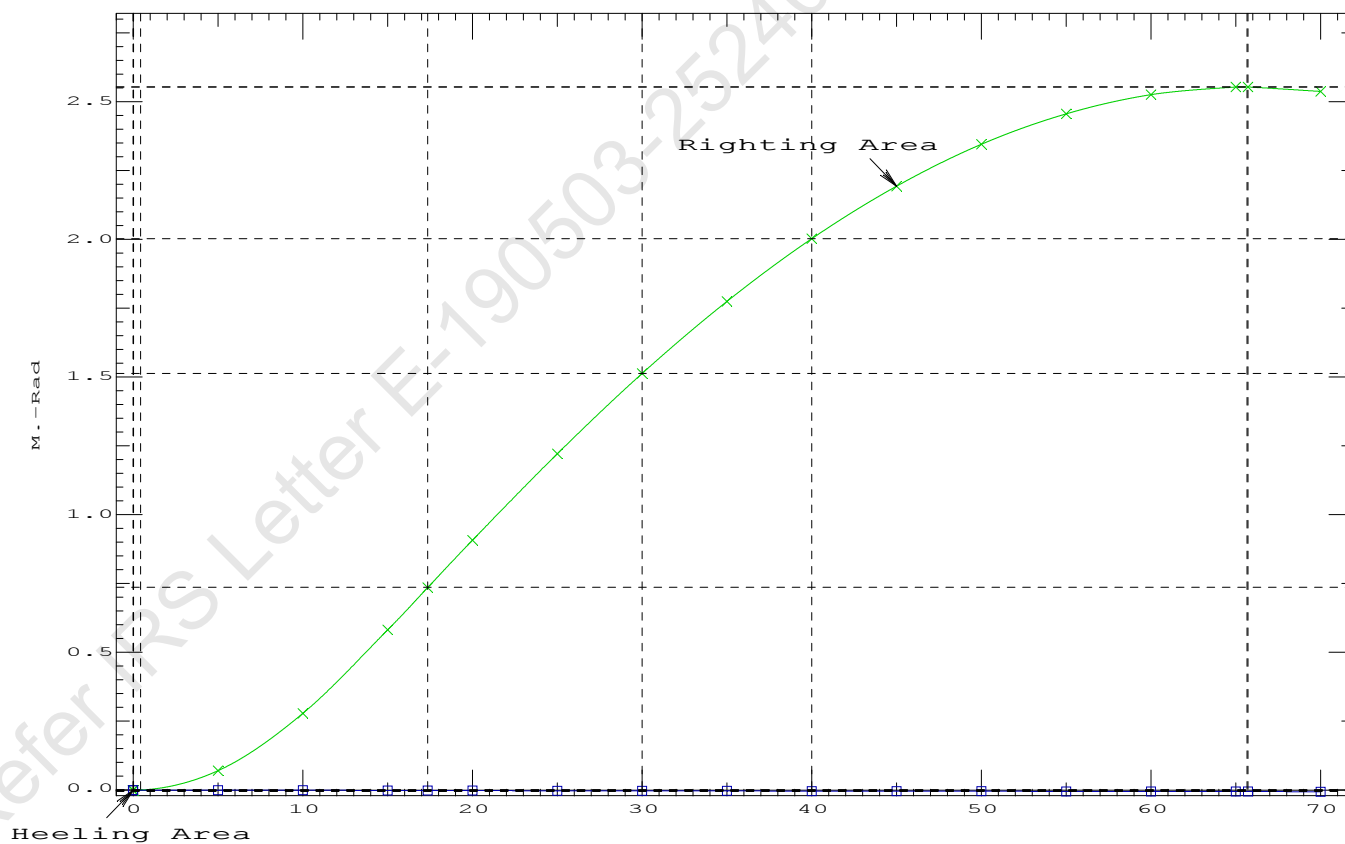
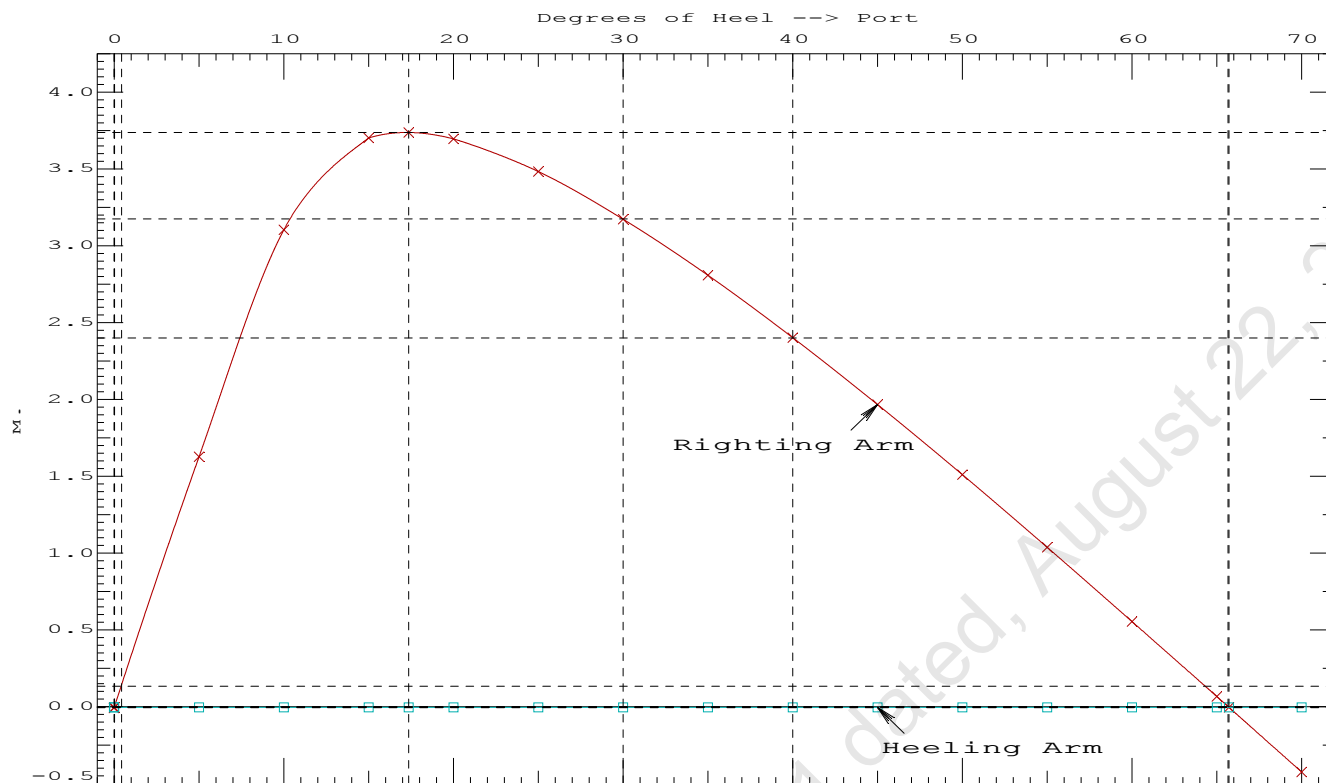
Origin	Degrees of	Displacement	Residual Arms	Res.
Depth---Trim---Heel---	Weight (MT)---	in Trim--in Heel---	Area	
1.264 0.11f 0.01s	1,091.0	0.000 -0.003	0.0000	
1.264 0.12f 0.00s	1,091.0	0.000 0.000	-0.0000	
1.250 0.11f 5.00p	1,091.0	0.000 1.631	0.0712	
1.188 0.13f 10.00p	1,091.0	0.000 3.109	0.2791	
1.008 0.18f 15.00p	1,091.0	0.000 3.707	0.5829	
0.916 0.21f 17.36p	1,091.0	0.000 3.742	0.7370	
0.811 0.24f 20.00p	1,091.0	0.000 3.700	0.9089	
0.607 0.30f 25.00p	1,091.0	0.000 3.487	1.2249	
0.397 0.37f 30.00p	1,091.0	0.000 3.178	1.5164	
0.182 0.43f 35.00p	1,091.0	0.000 2.812	1.7782	
-0.035 0.49f 40.00p	1,091.0	0.000 2.406	2.0062	
-0.252 0.55f 45.00p	1,091.0	0.000 1.971	2.1974	
-0.469 0.61f 50.00p	1,091.0	0.000 1.515	2.3497	
-0.682 0.66f 55.00p	1,091.0	0.000 1.043	2.4614	
-0.890 0.71f 60.00p	1,091.0	0.000 0.560	2.5314	
-1.090 0.76f 65.00p	1,091.0	0.000 0.070	2.5589	
-1.119 0.76f 65.71p	1,091.0	0.000 0.000	2.5593	
-1.282 0.79f 70.00p	1,091.0	0.000 -0.421	2.5436	

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 805.97 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAzero > 1.000 LARGE
(2) Angle from abs 0 deg to RAzero > 20.00 deg 65.73 P

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PONTOON

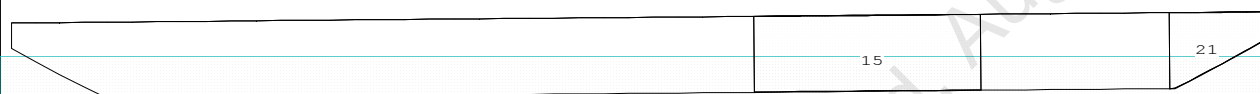
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PONTOON

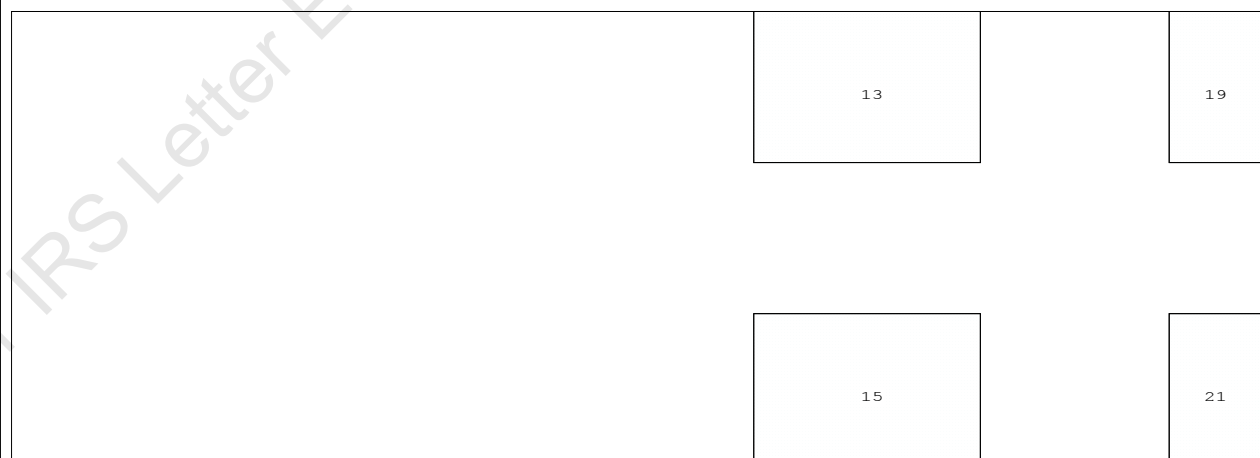
Condition 07 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Aft

Condition Graphic - Draft: 1.243 @ 50.000f, 1.695 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

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PONTOON

Condition 07 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Aft

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.243 @ 50.00f, 1.469 @ 25.00f, 1.695 @ 0.00

Trim: Aft 0.452/50.000, Heel: zero

Part-----			Weight (MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			350.00	11.500f	0.000	6.325	
Crane Load			40.00	23.500a	0.000	21.478	
Total Fixed----->			795.97	16.627f	0.000	5.391	
	Load----	SpGr-----	Weight (MT)----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			422.40	37.208f	0.000	1.596	470.09
Total Weight----->			1,218.37	23.762f	0.000	4.075	
			Displ (MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,218.25	23.732f	0.000	0.750	

Righting Arms: 0.000 0.000

Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025	865.7	24.605f	0.000	140.09	19.278
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
		8.87	33.32	136.77	15.953	

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

412.1 Cu.M. (16%) SALT WATER

350.00 MT Crane

40.00 MT Crane Load

DRAFT AT FP (M) = 1.243
DRAFT AT MS (M) = 1.469
DRAFT AT AP (M) = 1.695
DRAFT AT FWD MARK (M) = 1.279
DRAFT AT AFT MARK (M) = 1.659

HYDROSTATIC PROPERTIES

Trim: Aft 0.452/50.000, No Heel, VCG = 4.075

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/				
Draft----	Weight (MT)----	LCB-----	VCB-----	cm-----	LCF-----	cm trim----	GML-----	GMT-----
1.472	1,218.25	23.732f	0.750	8.87	24.605f	33.32	136.77	15.953
Distances in METERS.-----				Specific Gravity = 1.025.-----Moment in m.-MT.				
				Trim is per 50.00m.				

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 470.1 m.-MT

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GHS 17.22

PONTOON

Condition 07 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Aft

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 1.243 @ 50.00f, 1.469 @ 25.00f, 1.695 @ 0.00

Trim: Aft 0.452/50.000, Heel: zero

Part-----	LPA*Sf-----	HCP-----	Arm----	Pressure-----	Moment
HULL	76.9	0.788	1.510	0.05500	6.39
CRANE	57.0	3.908	4.630	0.05500	14.52
Total wind heeling moment to starboard----->					20.91
Distances in METERS.-----Pressure in MT/Sqm.-----					Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 23.762f TCG = 0.000 VCG = 4.075

Free Surface Adjustment: 0.386

Adjusted CG: LCG = 23.766f TCG = 0.000 VCG = 4.461

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---Trim---Heel---	Weight (MT)---	in Trim--in Heel---	Area----	Angle
1.683 0.52a 0.00	1,218.4	0.000 -0.017	0.0000	4.235
1.683 0.52a 0.06s	1,218.4	0.000 0.000	-0.0000	4.178
1.673 0.52a 4.23s	1,218.4	0.000 1.171	0.0427	0.000
1.670 0.52a 5.06s	1,218.4	0.000 1.400	0.0611	-0.822
1.642 0.57a 10.06s	1,218.3	0.000 2.702	0.2408	-5.822
1.675 0.84a 15.06s	1,218.4	0.000 3.166	0.5038	-10.822
1.682 0.90a 16.09s	1,218.4	0.000 3.173	0.5609	-11.855
1.704 1.13a 20.06s	1,218.4	0.000 3.083	0.7793	-15.822
1.727 1.43a 25.06s	1,218.4	0.000 2.820	1.0391	-20.822
1.742 1.73a 30.06s	1,218.4	0.000 2.474	1.2707	-25.822
1.750 2.03a 35.06s	1,218.4	0.000 2.079	1.4697	-30.822
1.750 2.33a 40.06s	1,218.4	0.000 1.652	1.6327	-35.822
1.742 2.62a 45.06s	1,218.4	0.000 1.204	1.7575	-40.822
1.722 2.90a 50.06s	1,218.4	0.000 0.741	1.8425	-45.822
1.690 3.15a 55.06s	1,218.4	0.000 0.269	1.8866	-50.822
1.666 3.29a 57.88s	1,218.4	0.000 0.000	1.8932	-53.644
1.644 3.38a 60.06s	1,218.4	0.000 -0.208	1.8892	-55.822
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.				

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 20.91 (constant)

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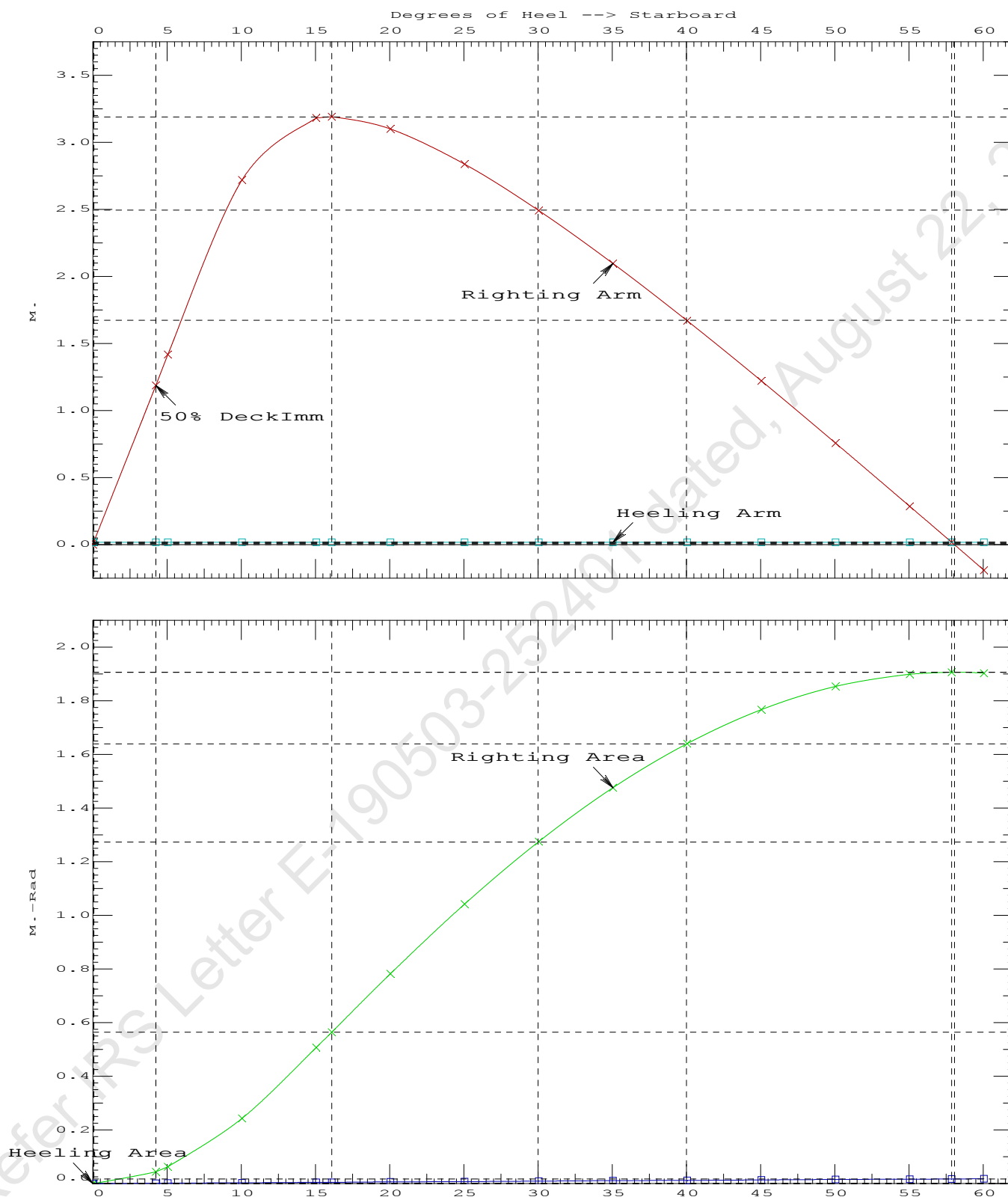
PONTOON**Condition 07 : Operating Condition of 350T Crane**
Crane Lifting 40.0T @ 35m Radius Aft

LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max-----Attained
(1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad 0.5609 P
(2) Angle from Equilibrium to RAzero or Flood > 20.00 deg 57.82 P
(3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg 4.18 P

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PONTOON

Condition 07 : Operating Condition of 350T CraneCrane Lifting 40.0T @ 35m Radius Aft

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PONTOON

Condition 07 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Aft

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 23.762f TCG = 0.000 VCG = 4.075
Free Surface Adjustment: 0.386
Adjusted CG: LCG = 23.766f TCG = 0.000 VCG = 4.461

Origin Depth	Degrees of Trim	Displacement Heel	Weight (MT)	Residual Arms in Trim	Res. in Heel	Area
1.683	0.52a	0.00	1,218.4	0.000	0.000	0.0000
1.670	0.52a	5.00s	1,218.4	0.000	1.401	0.0611
1.642	0.56a	10.00s	1,218.3	0.000	2.707	0.2410
1.675	0.84a	15.00s	1,218.4	0.000	3.181	0.5049
1.682	0.90a	16.10s	1,218.4	0.000	3.189	0.5659
1.704	1.13a	20.00s	1,218.4	0.000	3.101	0.7819
1.727	1.43a	25.00s	1,218.4	0.000	2.840	1.0433
1.742	1.73a	30.00s	1,218.4	0.000	2.494	1.2767
1.750	2.03a	35.00s	1,218.4	0.000	2.100	1.4774
1.750	2.33a	40.00s	1,218.4	0.000	1.673	1.6423
1.742	2.62a	45.00s	1,218.4	0.000	1.225	1.7689
1.722	2.90a	50.00s	1,218.4	0.000	0.762	1.8557
1.691	3.15a	55.00s	1,218.4	0.000	0.290	1.9017
1.664	3.30a	58.05s	1,218.4	0.000	0.000	1.9094
1.645	3.38a	60.00s	1,218.4	0.000	-0.186	1.9062

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

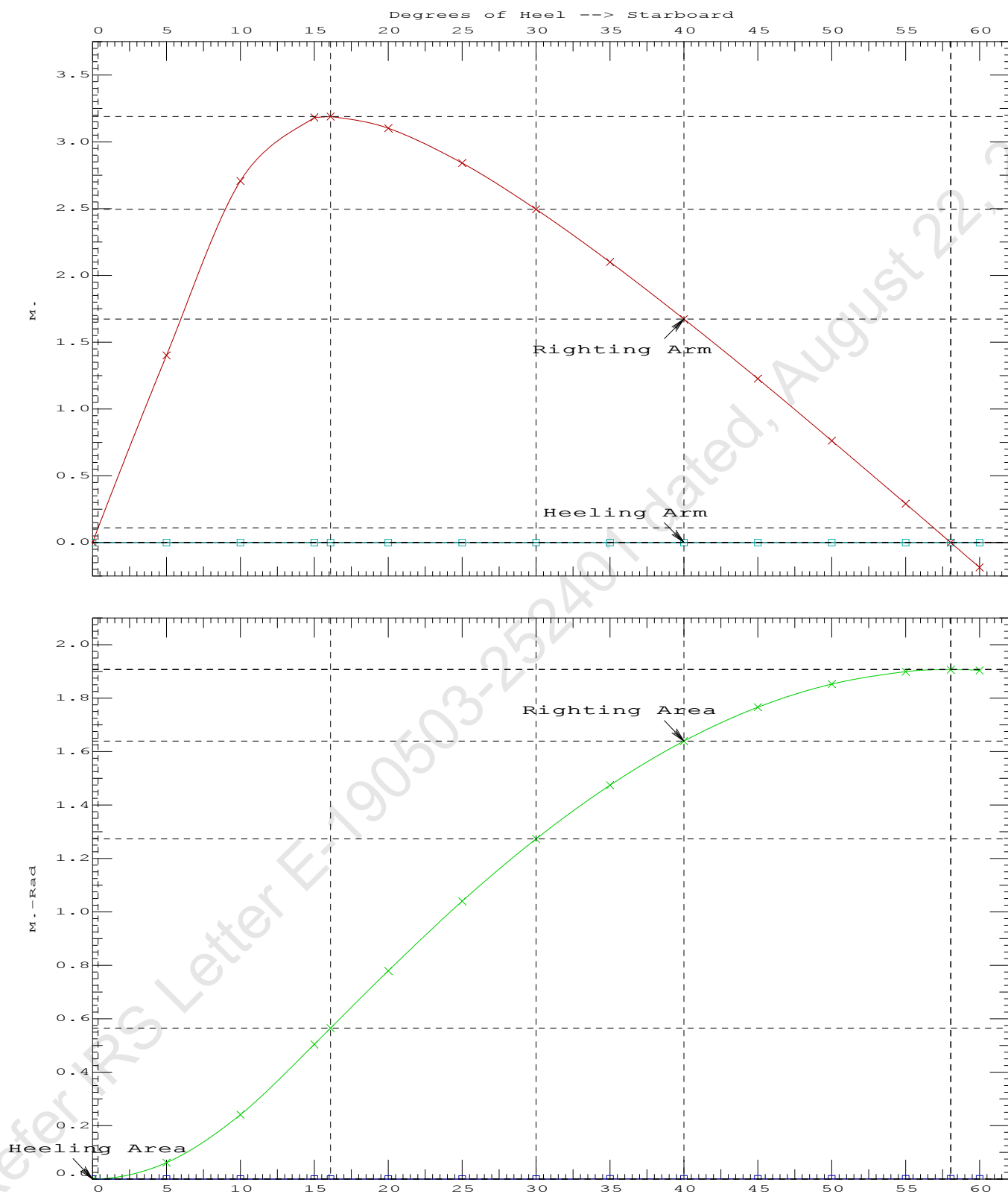
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM	STABILITY CRITERIA FOR CRANE OPERAT	Min/Max	Attained
(1)	Absolute Angle at Equilibrium	< 15.00 deg	0.00 P
(2)	Rise in Abs. RA from Equilibrium to MaxRA	> 66.7 %	388571 P
(3)	Abs Ratio from Equilibrium to RAzero or Flood	> 1.667	2297.24 P

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GHS 17.22

PONTOON

Condition 07 : Operating Condition of 350T CraneCrane Lifting 40.0T @ 35m Radius Aft

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PONTOON

Condition 07 : Operating Condition of 350T Crane**Crane Lifting 40.0T @ 35m Radius Aft**

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.243 @ 50.00f, 1.469 @ 25.00f, 1.695 @ 0.00

Trim: Aft 0.452/50.000, Heel: 0.00 deg.

Part-----			Weight (MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			350.00	11.500f	0.000	6.325	
Crane Load			40.00	23.500a	0.000	21.478	
Total Fixed----->			795.97	16.627f	0.000	5.391	
	Load-----	SpGr-----	Weight (MT)----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			422.40	37.208f	0.000	1.596	470.09
Total Weight----->			1,218.37	23.762f	0.000	4.075	
			Displ (MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,218.37	23.732f	0.001s	0.750	

	Righting Arms:			0.000	0.001s		
	External Arms:			0.000	0.001		
	Residual Righting Arms:			0.000	0.000s		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		861.6	24.569f	0.031s	138.11	19.142
			MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
			8.83		32.84	134.78	15.816
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

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PONTOON

After hook load drop

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.490 @ 50.00f, 1.425 @ 25.00f, 1.360 @ 0.00

Trim: Fwd 0.130/50.000, Heel: 0.00 deg.

Part-----			Weight (MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			350.00	11.500f	0.000	6.325	
Total Fixed----->			755.97	18.750f	0.000	4.539	
	Load----	SpGr-----	Weight (MT)----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			422.40	37.208f	0.000	1.596	470.09
Total Weight----->			1,178.37	25.366f	0.000	3.484	
			Displ (MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,178.37	25.374f	0.002s	0.721	

	Righting Arms:			0.000	0.002s		
	External Arms:			0.000	0.001		
	Residual Righting Arms:			0.000	0.001s		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		857.9	25.250f	0.033s	141.00	19.758
			MT/cm				
			8.79				

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: INFINITY; t3: 30.0; t: 15.00
limmarg: INFINITY; t3: 15.0; t: 7.50
limmarg: INFINITY; t3: 7.5; t: 3.75
limmarg: INFINITY; t3: 3.8; t: 1.88
limmarg: INFINITY; t3: 1.9; t: 0.94
limmarg: INFINITY; t3: 1.0; t: 0.47
limmarg: INFINITY; t3: 0.5; t: 0.23
limmarg: INFINITY; t3: 0.3; t: 0.12
limmarg: INFINITY; t3: 0.2; t: 0.06
limmarg: INFINITY; t3: 0.1; t: 0.03
limmarg: INFINITY; t3: 0.1; t: 0.01

Theta3 is 0.07 degrees, as demonstrated below.

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PONTOON

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.750f TCG = 0.000 VCG = 4.539

Origin Depth	Degrees of Trim	Displacement Heel	Weight (MT)	Residual Arms in Trim	Res. in Heel	Res. Area
1.348	0.15f	0.00	1,178.4	0.000	0.001	0.0000
1.348	0.15f	0.07s	1,178.4	0.000	0.023	0.0000
1.333	0.15f	5.00s	1,178.4	0.000	1.537	0.0672
1.280	0.16f	10.00s	1,178.4	0.000	2.983	0.2651
1.136	0.24f	15.00s	1,178.4	0.000	3.574	0.5573
1.054	0.28f	17.65s	1,178.2	0.000	3.614	0.7241
0.979	0.32f	20.00s	1,178.4	0.000	3.582	0.8719
0.814	0.40f	25.00s	1,178.4	0.000	3.400	1.1789
0.641	0.49f	30.00s	1,178.4	0.000	3.128	1.4644

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 755.97 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 0.1 > 1.000 LARGE

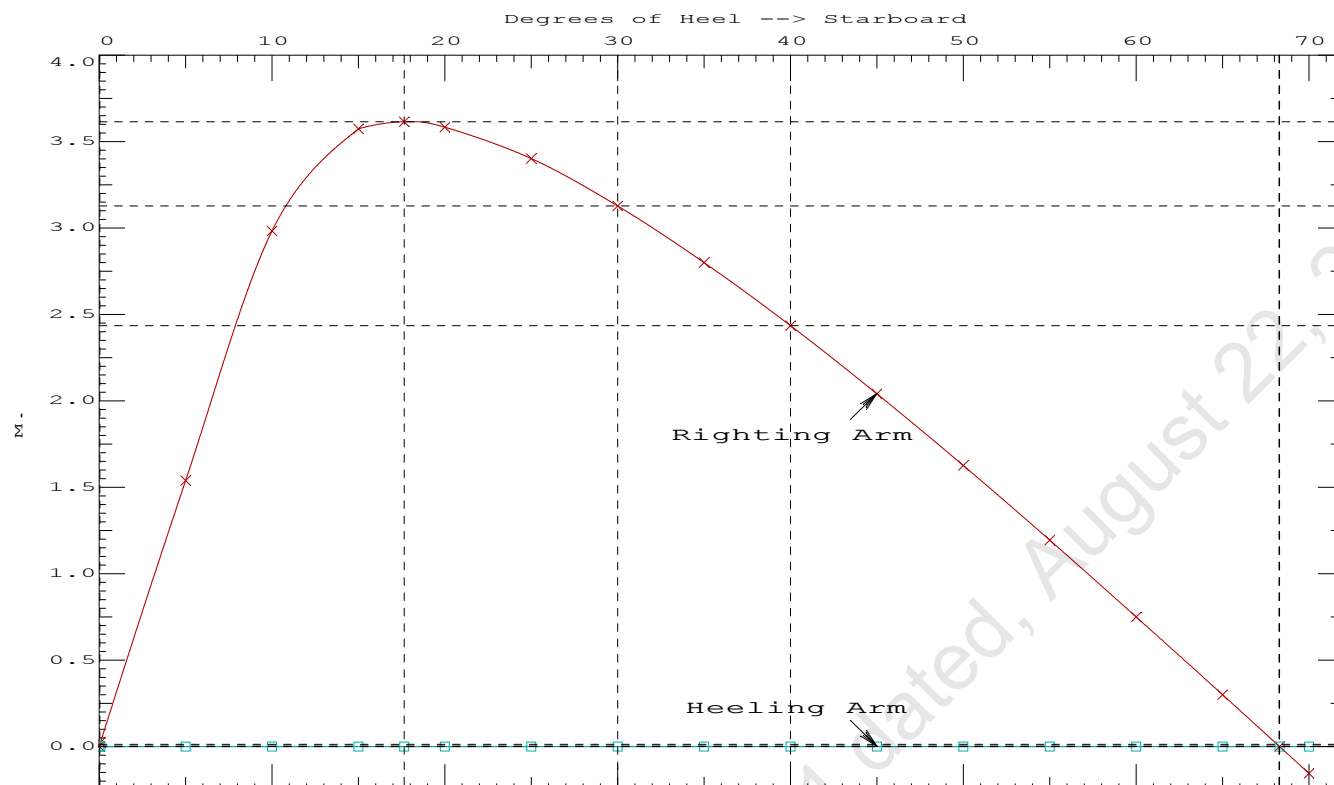
PONTOON

Fixed CG: LCG = 18.750f TCG = 0.000 VCG = 4.539

Note: The Center of Gravity shown above is for the Fixed Weight of 755.97 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

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LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAZero    >    1.000          LARGE
(2) Angle from abs 0.1 deg to RAZero                  >    20.00   deg    68.22 P
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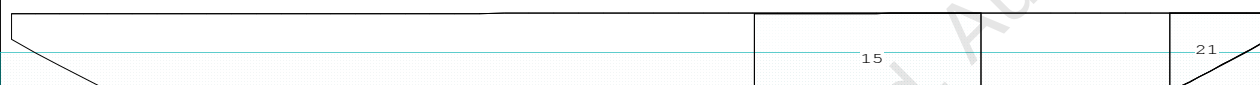
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PONTOON

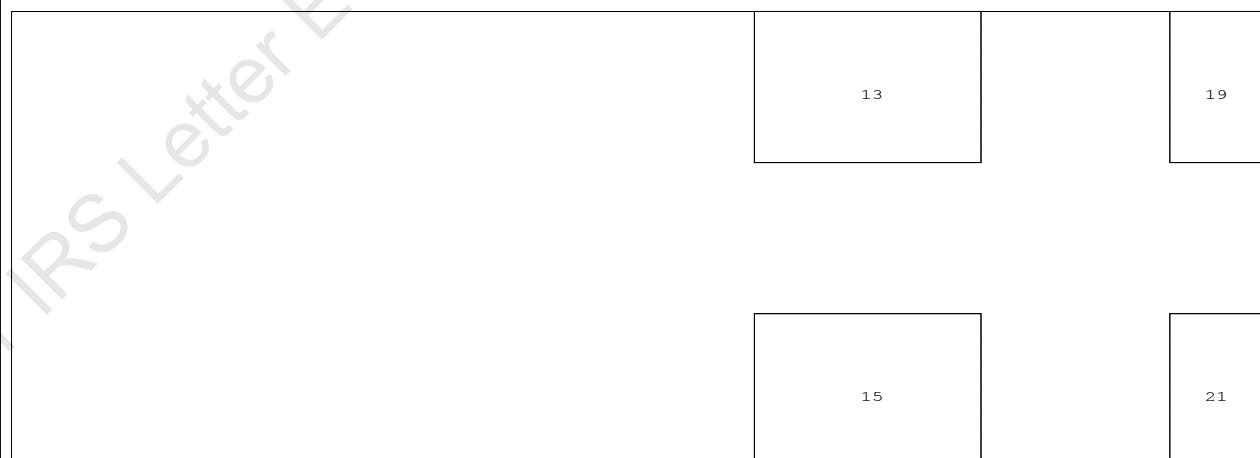
Condition 08 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Stbd

Condition Graphic - Draft: 1.454 @ 50.000f, 1.487 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

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PONTOON

Condition 08 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Stbd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.454 @ 50.00f, 1.471 @ 25.00f, 1.487 @ 0.00

Trim: Aft 0.032/50.000, Heel: zero

Part-----			Weight (MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			350.00	11.500f	0.000	6.325	
Crane Load			40.00	11.500f	0.000	21.478	
Total Fixed----->			795.97	18.385f	0.000	5.391	
	Load----	SpGr-----	Weight (MT)----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			422.40	37.208f	0.000	1.596	470.09
Total Weight----->			1,218.37	24.911f	0.000	4.075	
			Displ (MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,218.37	24.909f	0.000	0.744	

Righting Arms: 0.000 0.000

Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025	866.3	24.984f	0.000	140.38	19.314
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
		8.88	33.40	137.05	15.982	

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

412.1 Cu.M. (16%) SALT WATER

350.00 MT Crane

40.00 MT Crane Load

DRAFT AT FP (M) = 1.454
DRAFT AT MS (M) = 1.471
DRAFT AT AP (M) = 1.487
DRAFT AT FWD MARK (M) = 1.457
DRAFT AT AFT MARK (M) = 1.484

HYDROSTATIC PROPERTIES

Trim: Aft 0.032/50.000, No Heel, VCG = 4.075

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/
Draft----	Weight (MT)----	LCB-----	VCB-----	cm-----
1.471	1,218.37	24.909f	0.744	8.88
				24.984f
				33.40
				137.05
				15.982

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 470.1 m.-MT

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PONTOON

Condition 08 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Stbd

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 1.454 @ 50.00f, 1.471 @ 25.00f, 1.487 @ 0.00

Trim: Aft 0.032/50.000, Heel: zero

Part-----	LPA*Sf-----	HCP-----	Arm----	Pressure-----	Moment
HULL	77.0	0.780	1.499	0.05500	6.35
CRANE	57.0	4.032	4.751	0.05500	14.89
Total wind heeling moment to starboard----->					21.25
Distances in METERS.-----Pressure in MT/Sqm.-----					Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.911f TCG = 0.000 VCG = 4.075

Free Surface Adjustment: 0.386

Adjusted CG: LCG = 24.911f TCG = 0.000 VCG = 4.461

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---Trim---Heel---	Weight (MT)---	in Trim--in Heel---	Area----	Angle
1.475 0.04a 0.00	1,218.4	0.000 -0.017	0.0000	4.948
1.475 0.04a 0.05s	1,218.4	0.000 0.000	-0.0000	4.901
1.461 0.04a 4.95s	1,218.4	0.000 1.371	0.0586	0.000
1.460 0.04a 5.05s	1,218.4	0.000 1.398	0.0610	-0.099
1.415 0.04a 10.05s	1,218.4	0.000 2.724	0.2414	-5.099
1.335 0.06a 15.05s	1,218.4	0.000 3.201	0.5071	-10.099
1.317 0.06a 16.10s	1,218.4	0.000 3.209	0.5662	-11.154
1.246 0.08a 20.05s	1,218.4	0.000 3.119	0.7858	-15.099
1.148 0.10a 25.05s	1,218.4	0.000 2.857	1.0488	-20.099
1.042 0.12a 30.05s	1,218.4	0.000 2.511	1.2837	-25.099
0.928 0.15a 35.05s	1,218.4	0.000 2.116	1.4859	-30.099
0.807 0.17a 40.05s	1,218.4	0.000 1.689	1.6522	-35.099
0.681 0.19a 45.05s	1,218.4	0.000 1.239	1.7802	-40.099
0.550 0.21a 50.05s	1,218.4	0.000 0.774	1.8681	-45.099
0.414 0.23a 55.05s	1,218.4	0.000 0.299	1.9151	-50.099
0.328 0.24a 58.17s	1,218.4	0.000 0.000	1.9232	-53.219
0.275 0.24a 60.05s	1,218.4	0.000 -0.181	1.9202	-55.099
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.				

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.25 (constant)

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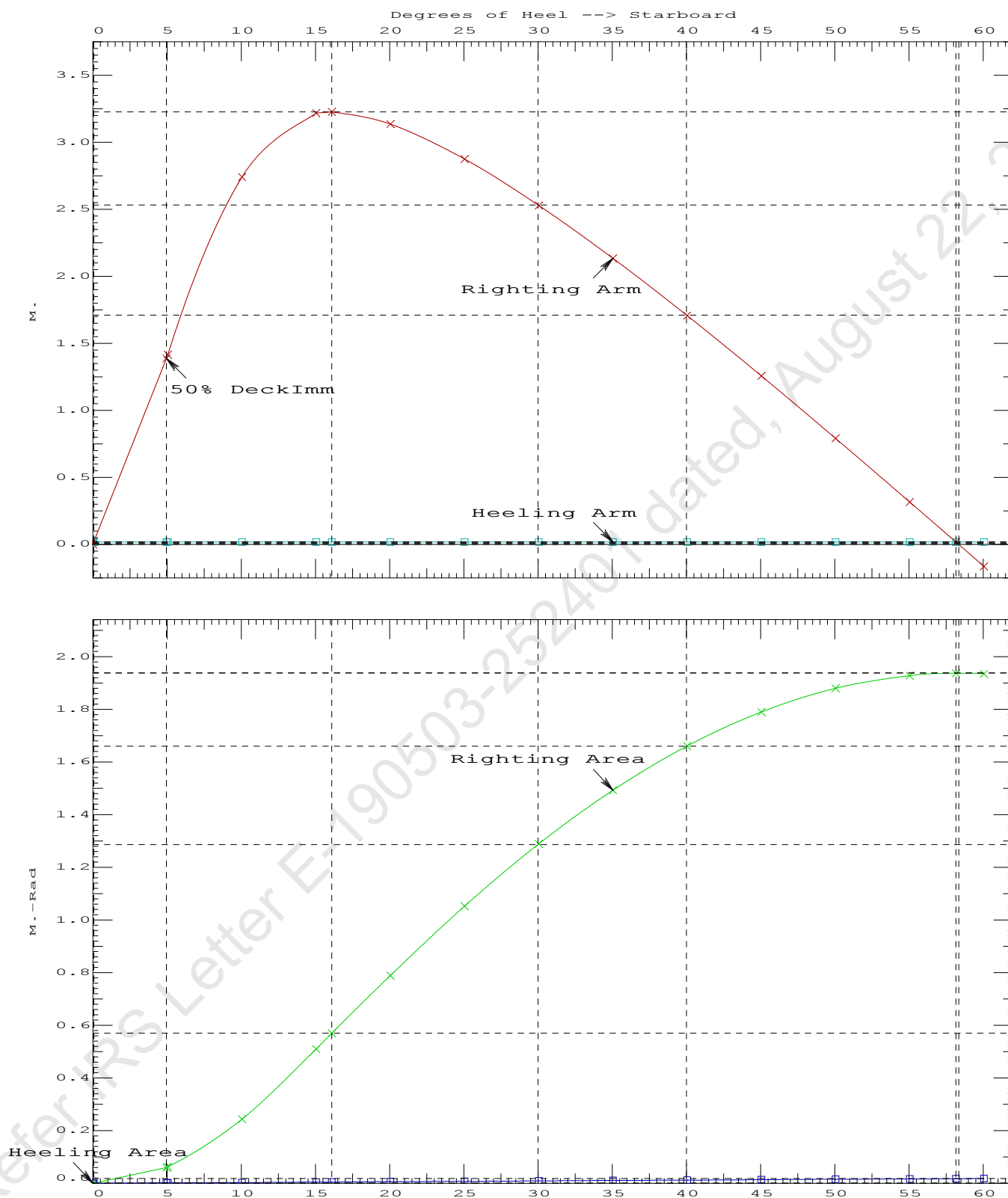
PONTOON**Condition 08 : Operating Condition of 350T Crane**
Crane Lifting 40.0T @ 35m Radius Stbd

LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max-----Attained
(1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad 0.5662 P
(2) Angle from Equilibrium to RAzero or Flood > 20.00 deg 58.12 P
(3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg 4.90 P

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PONTOON

Condition 08 : Operating Condition of 350T CraneCrane Lifting 40.0T @ 35m Radius Stbd

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PONTOON

Condition 08 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.911f TCG = 0.000 VCG = 4.075
Free Surface Adjustment: 0.386
Adjusted CG: LCG = 24.911f TCG = 0.027p VCG = 4.460

Origin Depth	Degrees of Trim	Displacement Heel	Weight (MT)	Residual Arms in Trim	Res. in Heel	Area
1.465	0.04a	3.99s	1,218.4	0.000	0.000	0.0000
1.428	0.04a	8.99s	1,218.4	0.000	1.357	0.0592
1.353	0.06a	13.99s	1,218.4	0.000	2.067	0.2179
1.315	0.07a	16.24s	1,218.4	0.000	2.103	0.3001
1.266	0.08a	18.99s	1,218.4	0.000	2.051	0.4007
1.169	0.10a	23.99s	1,218.4	0.000	1.814	0.5722
1.065	0.12a	28.99s	1,218.4	0.000	1.481	0.7167
0.952	0.14a	33.99s	1,218.4	0.000	1.094	0.8294
0.833	0.16a	38.99s	1,218.4	0.000	0.671	0.9066
0.708	0.18a	43.99s	1,218.4	0.000	0.224	0.9459
0.645	0.19a	46.42s	1,218.4	0.000	0.000	0.9506
0.578	0.20a	48.99s	1,218.4	0.000	-0.240	0.9453
0.443	0.22a	53.99s	1,218.4	0.000	-0.715	0.9037
0.305	0.24a	58.99s	1,218.4	0.000	-1.197	0.8203
0.164	0.26a	63.99s	1,218.4	0.000	-1.680	0.6948

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1400.00

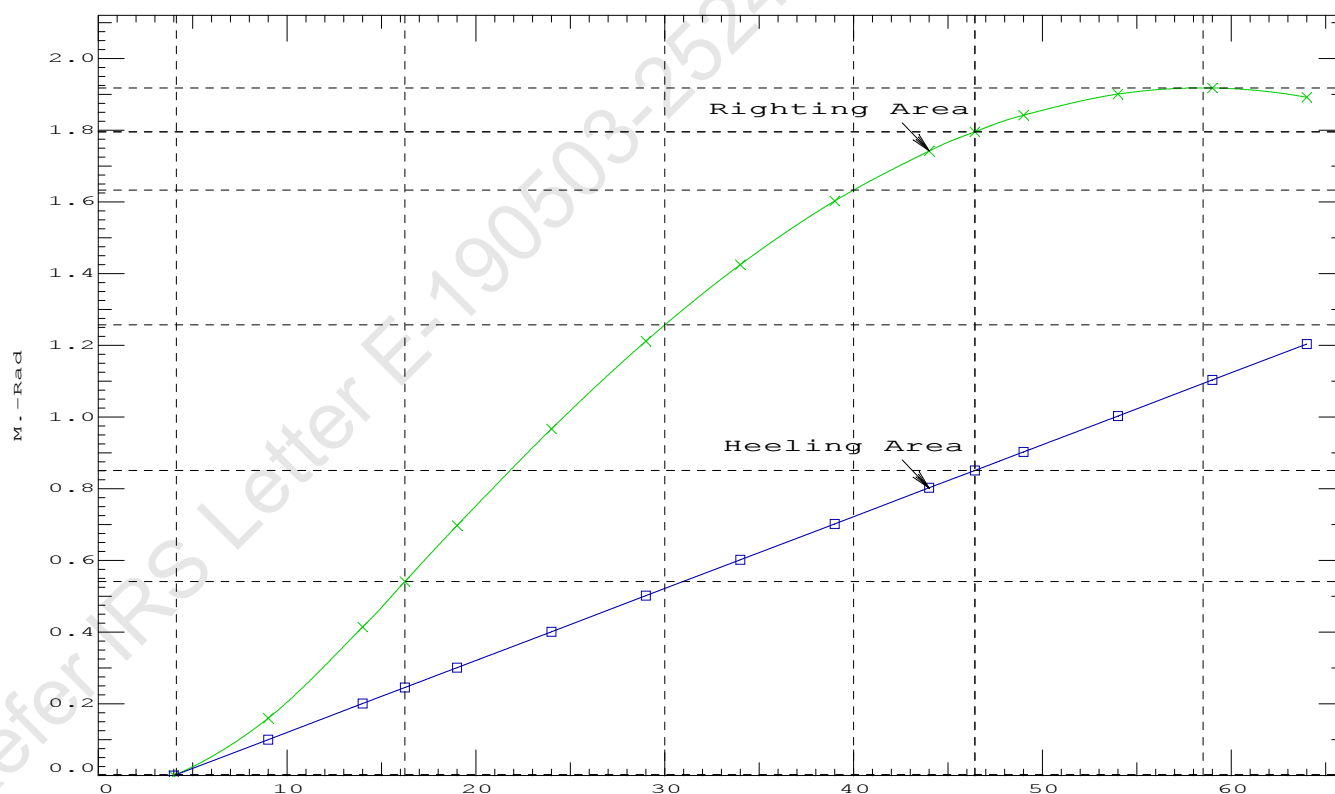
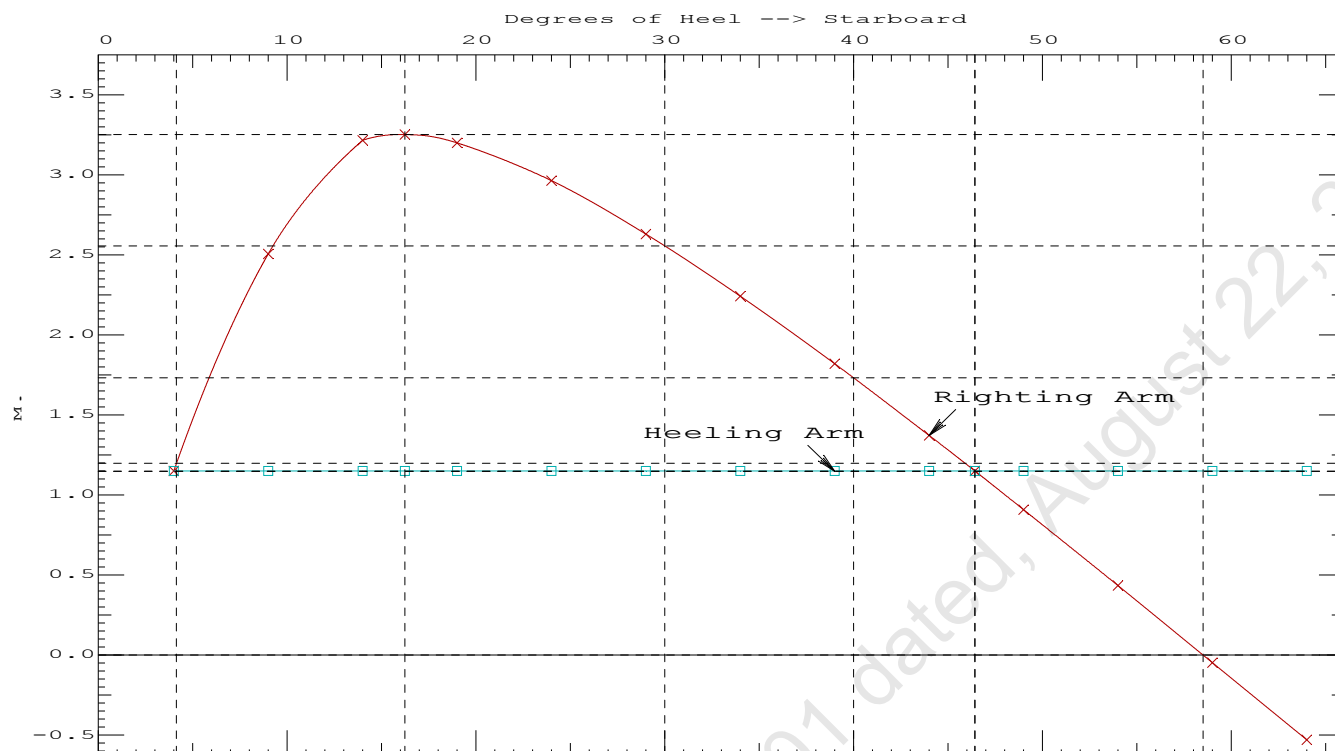
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM	STABILITY CRITERIA FOR CRANE OPERAT	Min/Max	Attained
(1)	Absolute Angle at Equilibrium	< 15.00 deg	3.99 P
(2)	Rise in Abs. RA from Equilibrium to MaxRA	> 66.7 %	183.0 P
(3)	Abs Ratio from Equilibrium to RAzero or Flood	> 1.667	2.117 P

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PONTOON

Condition 08 : Operating Condition of 350T CraneCrane Lifting 40.0T @ 35m Radius Stbd

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PONTOON

Condition 08 : Operating Condition of 350T CraneCrane Lifting 40.0T @ 35m Radius Stbd

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.448 @ 50.00f, 1.464 @ 25.00f, 1.480 @ 0.00

Trim: Aft 0.033/50.000, Heel: Stbd 3.99 deg.

Part			Weight (MT)	LCG	TCG	VCG	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			350.00	11.500f	0.000	6.325	
Crane Load			40.00	11.500f	0.000	21.478	
Total Fixed	----->		795.97	18.385f	0.000	5.391	
	Load	SpGr	Weight (MT)	LCG	TCG	VCG	FSM
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks	----->		422.40	37.208f	0.000	1.596	470.09
Total Weight	----->		1,218.37	24.911f	0.000	4.075	
			Displ (MT)	LCB	TCB	VCB	
HULL	1.025		1,218.37	24.909f	1.381s	0.792	

	Righting Arms:			0.000	1.149s		
	External Arms:			0.000	1.149s		
	Residual Righting Arms:			0.000	0.000s		
Part	SpGr		WPA	LCF	TCF	BML	BMT
Total Waterplane	----> 1.025		865.9	24.970f	0.162s	139.89	19.352
			MT/cm				
			8.88				

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 3.99

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PONTOON

After hook load drop

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.490 @ 50.00f, 1.425 @ 25.00f, 1.360 @ 0.00

Trim: Fwd 0.130/50.000, Heel: 0.00 deg.

Part-----			Weight (MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			350.00	11.500f	0.000	6.325	
Total Fixed----->			755.97	18.750f	0.000	4.539	
	Load----	SpGr-----	Weight (MT)----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			422.40	37.208f	0.000	1.596	470.09
Total Weight----->			1,178.37	25.366f	0.000	3.484	
			Displ (MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,178.37	25.374f	0.002s	0.721	

	Righting Arms:			0.000	0.002s		
	External Arms:			0.000	0.001		
	Residual Righting Arms:			0.000	0.001s		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->		1.025	857.9	25.250f	0.033s	141.00	19.758
			MT/cm				
			8.79				

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: 3316.38; t3: 30.0; t: 15.0
limmarg: 1200.05; t3: 15.0; t: 7.5
limmarg: 251.36; t3: 7.5; t: 3.8
limmarg: -14.19; t3: 3.7; t: 1.9
limmarg: 96.25; t3: 5.6; t: 0.9
limmarg: 38.28; t3: 4.7; t: 0.5
limmarg: 10.50; t3: 4.2; t: 0.2
limmarg: 0.25; t3: 4.0; t: 0.1
limmarg: -4.69; t3: 3.9; t: 0.1
limmarg: 0.25; t3: 4.0; t: 0.0
limmarg: 0.25; t3: 4.0; t: 0.0

Theta3 is 4.00 degrees, as demonstrated below.

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PONTOON

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.750f TCG = 0.000 VCG = 4.539

Origin	Degrees of	Displacement	Residual Arms	Res.
Depth---Trim---Heel---Weight (MT)---in Trim--in Heel---> Area				
1.339	0.15f	3.99s	1,178.3	0.000 -1.230 0.0000
1.348	0.15f	0.00	1,178.4	0.000 -0.001 -0.0429
1.339	0.15f	4.00p	1,178.4	0.000 1.233 0.0001
1.333	0.15f	5.00p	1,178.4	0.000 1.539 0.0243
1.280	0.16f	10.00p	1,178.4	0.000 2.984 0.2221
1.136	0.24f	15.00p	1,178.4	0.000 3.576 0.5145
1.054	0.28f	17.65p	1,178.2	0.000 3.616 0.6815
0.979	0.32f	20.00p	1,178.4	0.000 3.584 0.8292
0.814	0.40f	25.00p	1,178.4	0.000 3.402 1.1364
0.641	0.49f	30.00p	1,178.4	0.000 3.129 1.4220

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 755.97 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs -4 deg to abs 4 > 1.000 1.003 P

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PONTOON

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.750f TCG = 0.000 VCG = 4.539

Origin	Degrees of	Displacement	Residual Arms	Res.
Depth---Trim---Heel---Weight (MT)---in Trim--in Heel---> Area				
1.339 0.15f 3.99s 1,178.3	0.000	-1.230	0.0000	
1.348 0.15f 0.00 1,178.4	0.000	-0.001	-0.0429	
1.339 0.15f 4.00p 1,178.4	0.000	1.233	0.0001	
1.333 0.15f 5.00p 1,178.4	0.000	1.539	0.0243	
1.280 0.16f 10.00p 1,178.4	0.000	2.984	0.2221	
1.136 0.24f 15.00p 1,178.4	0.000	3.576	0.5145	
1.054 0.28f 17.65p 1,178.2	0.000	3.616	0.6815	
0.979 0.32f 20.00p 1,178.4	0.000	3.584	0.8292	
0.814 0.40f 25.00p 1,178.4	0.000	3.402	1.1364	
0.641 0.49f 30.00p 1,178.4	0.000	3.129	1.4220	
0.461 0.57f 35.00p 1,178.4	0.000	2.802	1.6812	
0.276 0.65f 40.00p 1,178.4	0.000	2.436	1.9100	
0.088 0.73f 45.00p 1,178.4	0.000	2.043	2.1057	
-0.102 0.81f 50.00p 1,178.4	0.000	1.627	2.2660	
-0.291 0.88f 55.00p 1,178.4	0.000	1.195	2.3892	
-0.479 0.95f 60.00p 1,178.4	0.000	0.752	2.4743	
-0.662 1.01f 65.00p 1,178.4	0.000	0.301	2.5203	
-0.780 1.04f 68.31p 1,178.4	0.000	0.000	2.5289	
-0.839 1.05f 70.00p 1,178.4	0.000	-0.154	2.5267	

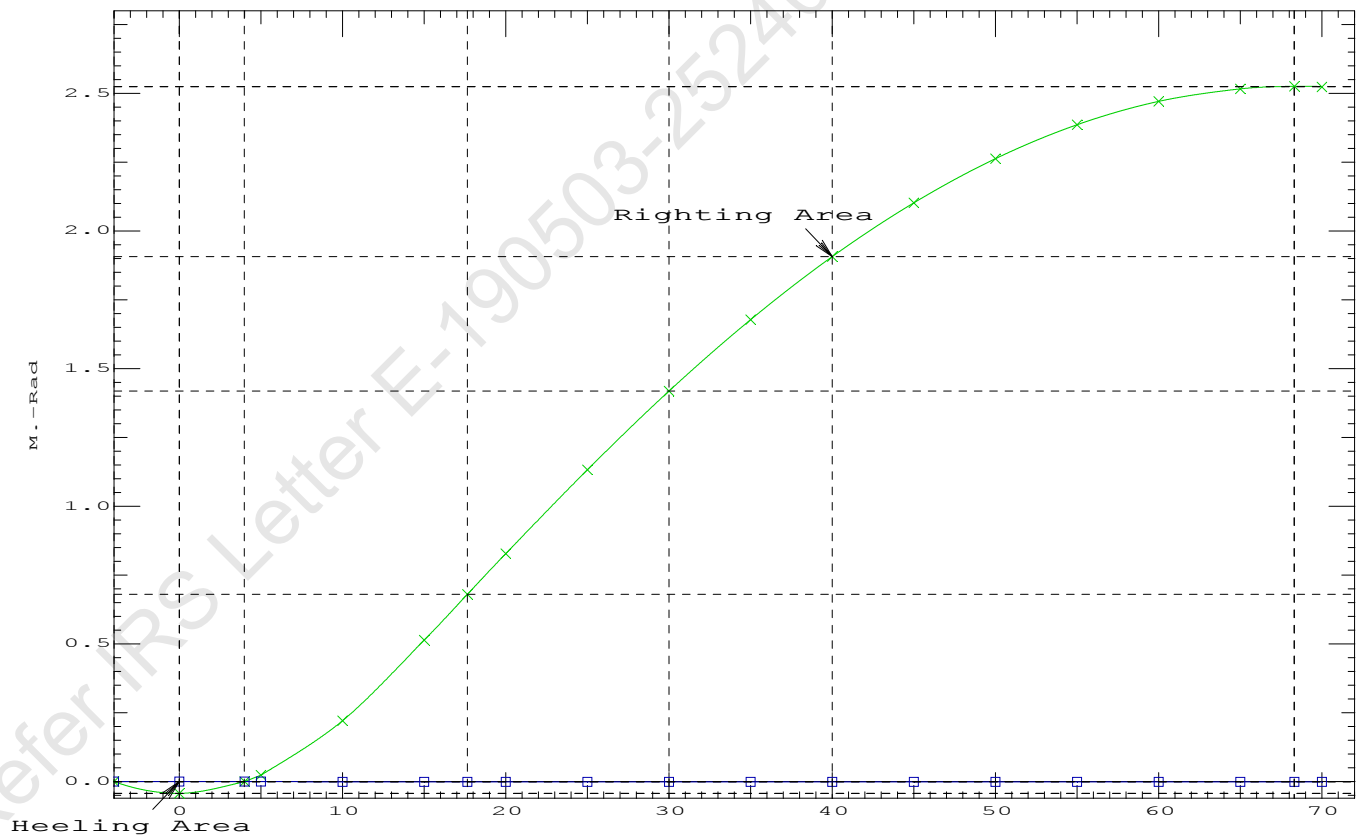
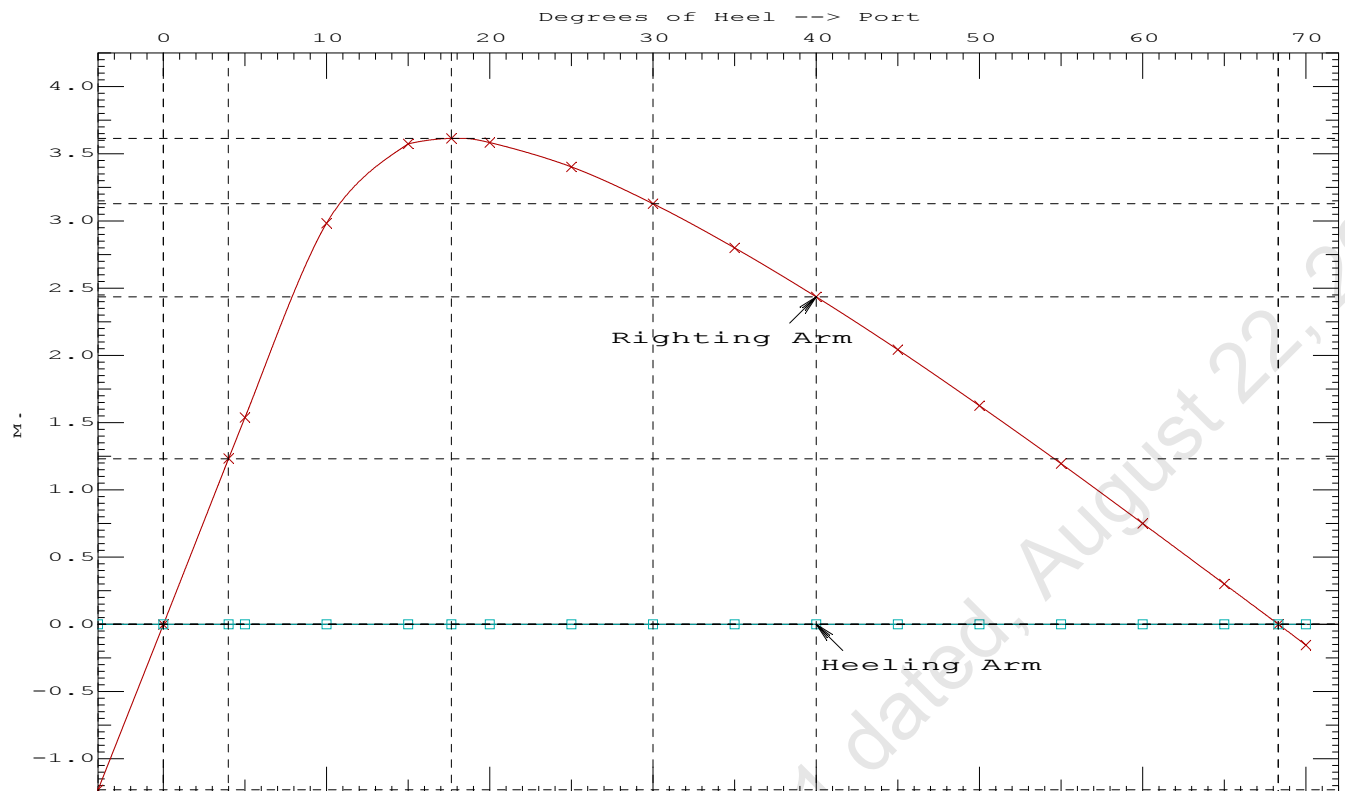
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 755.97 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs -4 deg to RAzero > 1.000 59.973 P
(2) Angle from abs 4 deg to RAzero > 20.00 deg 64.31 P

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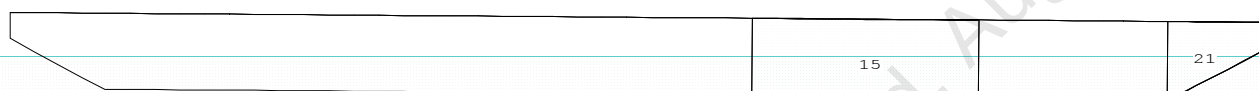
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PONTOON

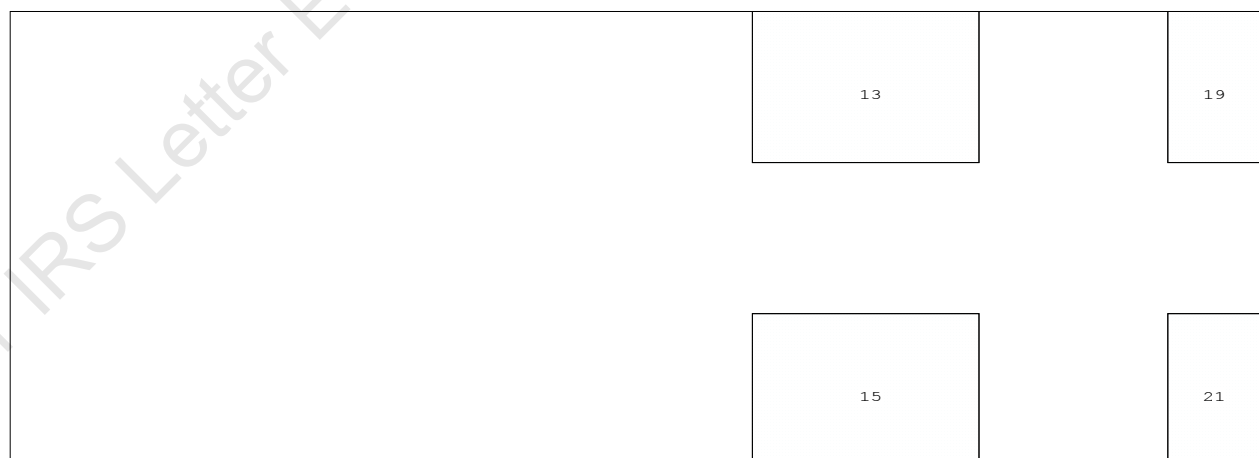
Condition 09 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Fwd

Condition Graphic - Draft: 1.663 @ 50.000f, 1.276 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

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PONTOON

Condition 09 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Fwd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.663 @ 50.00f, 1.469 @ 25.00f, 1.276 @ 0.00

Trim: Fwd 0.387/50.000, Heel: zero

Part-----			Weight (MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			350.00	11.500f	0.000	6.325	
Crane Load			40.00	46.500f	0.000	21.478	
Total Fixed----->			795.97	20.144f	0.000	5.391	
	Load----	SpGr-----	Weight (MT)----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			422.40	37.208f	0.000	1.596	470.09
Total Weight----->			1,218.37	26.060f	0.000	4.075	
			Displ (MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,218.28	26.086f	0.000	0.748	

Righting Arms: 0.000 0.000

Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025	865.7	25.336f	0.000	140.11	19.280
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
		8.87	33.33	136.78	15.953	

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

412.1 Cu.M. (16%) SALT WATER

350.00 MT Crane

40.00 MT Crane Load

DRAFT AT FP (M) = 1.663
DRAFT AT MS (M) = 1.469
DRAFT AT AP (M) = 1.276
DRAFT AT FWD MARK (M) = 1.632
DRAFT AT AFT MARK (M) = 1.307

HYDROSTATIC PROPERTIES

Trim: Fwd 0.387/50.000, No Heel, VCG = 4.075

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/
Draft----	Weight (MT)----	LCB-----	VCB-----	cm-----
1.472	1,218.28	26.086f	0.748	8.87
				25.336f
				33.33
				136.78
				15.953

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 470.1 m.-MT

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PONTOON

Condition 09 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Fwd

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 1.663 @ 50.00f, 1.469 @ 25.00f, 1.276 @ 0.00

Trim: Fwd 0.387/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	77.0	0.786	1.507	0.05500	6.38
CRANE	57.0	4.158	4.879	0.05500	15.30
Total wind heeling moment to starboard-->					21.67
Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT					

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 26.060f TCG = 0.000 VCG = 4.075

Free Surface Adjustment: 0.386

Adjusted CG: LCG = 26.057f TCG = 0.000 VCG = 4.461

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---Trim---Heel---	Weight (MT)---	in Trim--in Heel---	Area---Angle	
1.263 0.44f 0.00	1,218.3	0.000 -0.017	0.0000	4.343
1.263 0.44f 0.06s	1,218.4	0.000 0.000	-0.0000	4.284
1.252 0.44f 4.34s	1,218.4	0.000 1.201	0.0449	0.000
1.248 0.45f 5.06s	1,218.4	0.000 1.400	0.0611	-0.716
1.185 0.48f 10.06s	1,218.4	0.000 2.708	0.2411	-5.716
0.994 0.72f 15.06s	1,218.4	0.000 3.175	0.5047	-10.716
0.953 0.77f 16.09s	1,218.4	0.000 3.182	0.5621	-11.750
0.787 0.97f 20.06s	1,218.4	0.000 3.092	0.7811	-15.716
0.568 1.23f 25.06s	1,218.4	0.000 2.829	1.0416	-20.716
0.340 1.48f 30.06s	1,218.4	0.000 2.483	1.2740	-25.716
0.103 1.74f 35.06s	1,218.4	0.000 2.088	1.4738	-30.716
-0.138 2.00f 40.06s	1,218.4	0.000 1.661	1.6377	-35.716
-0.382 2.25f 45.06s	1,218.4	0.000 1.212	1.7632	-40.716
-0.626 2.48f 50.06s	1,218.4	0.000 0.749	1.8489	-45.716
-0.865 2.70f 55.06s	1,218.4	0.000 0.276	1.8937	-50.716
-1.000 2.82f 57.95s	1,218.4	0.000 0.000	1.9007	-53.608
-1.097 2.90f 60.06s	1,218.4	0.000 -0.201	1.8969	-55.716
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.				

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.67 (constant)

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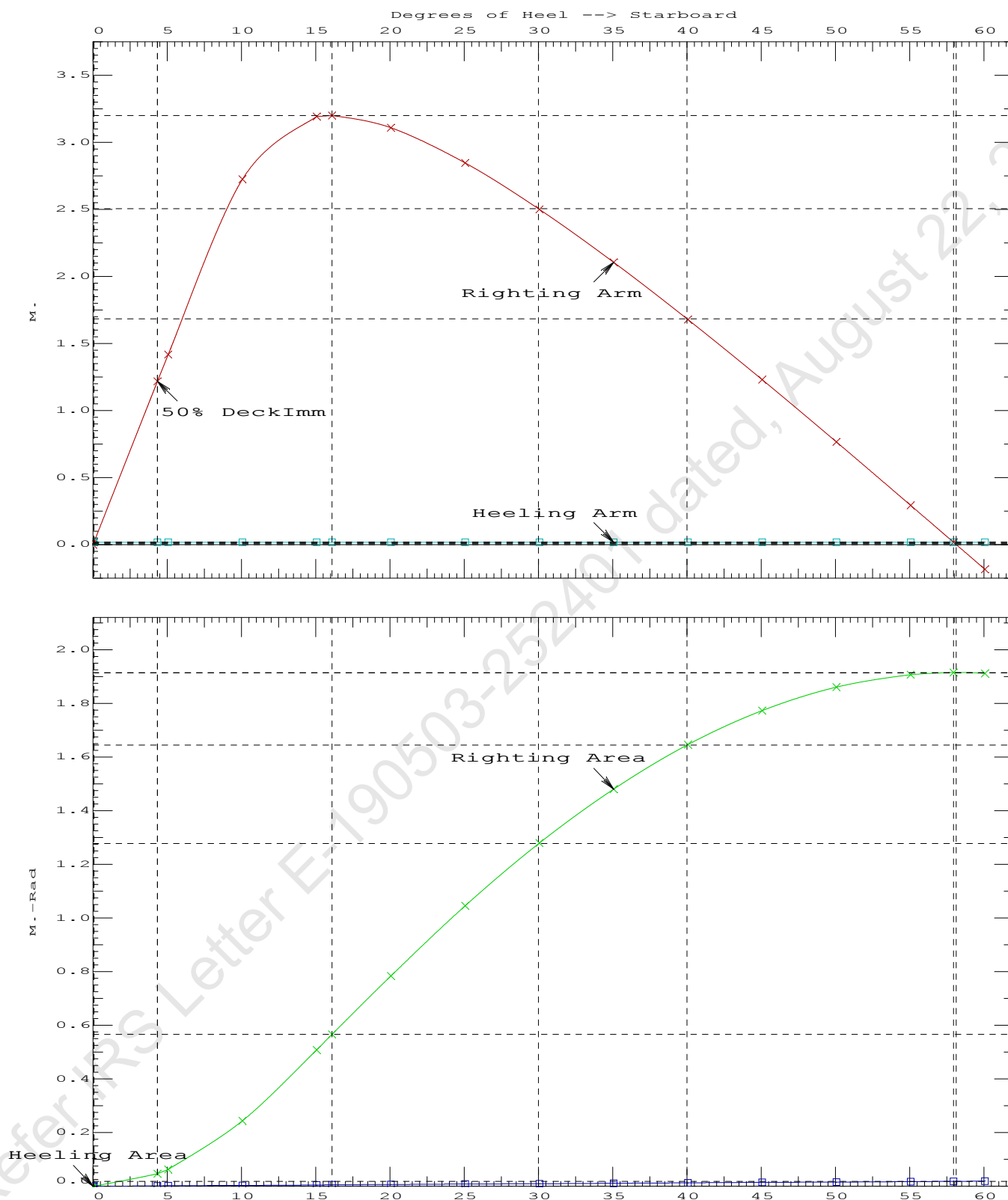
PONTOON**Condition 09 : Operating Condition of 350T Crane**
Crane Lifting 40.0T @ 35m Radius Fwd

LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max-----Attained
(1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad 0.5621 P
(2) Angle from Equilibrium to RAzero or Flood > 20.00 deg 57.89 P
(3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg 4.28 P

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PONTOON

Condition 09 : Operating Condition of 350T CraneCrane Lifting 40.0T @ 35m Radius Fwd

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PONTOON

Condition 09 : Operating Condition of 350T Crane
Crane Lifting 40.0T @ 35m Radius Fwd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 26.060f TCG = 0.000 VCG = 4.075
Free Surface Adjustment: 0.386
Adjusted CG: LCG = 26.057f TCG = 0.000 VCG = 4.461

Origin	Degrees of	Displacement	Residual Arms	Res.
Depth---Trim---Heel---	Weight (MT)---	in Trim--in Heel---	Area	
1.263 0.44f 0.00	1,218.4	0.000 0.001	0.0000	
1.248 0.45f 5.00s	1,218.4	0.000 1.401	0.0612	
1.187 0.48f 10.00s	1,218.4	0.000 2.712	0.2413	
0.997 0.72f 15.00s	1,218.4	0.000 3.190	0.5059	
0.952 0.77f 16.10s	1,218.4	0.000 3.199	0.5673	
0.789 0.97f 20.00s	1,218.4	0.000 3.111	0.7838	
0.571 1.22f 25.00s	1,218.4	0.000 2.850	1.0461	
0.342 1.48f 30.00s	1,218.4	0.000 2.504	1.2803	
0.106 1.74f 35.00s	1,218.4	0.000 2.109	1.4819	
-0.135 1.99f 40.00s	1,218.4	0.000 1.683	1.6476	
-0.380 2.24f 45.00s	1,218.4	0.000 1.234	1.7751	
-0.623 2.48f 50.00s	1,218.4	0.000 0.771	1.8627	
-0.862 2.70f 55.00s	1,218.4	0.000 0.298	1.9094	
-1.008 2.83f 58.13s	1,218.4	0.000 0.000	1.9175	
-1.094 2.90f 60.00s	1,218.4	0.000 -0.179	1.9146	

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

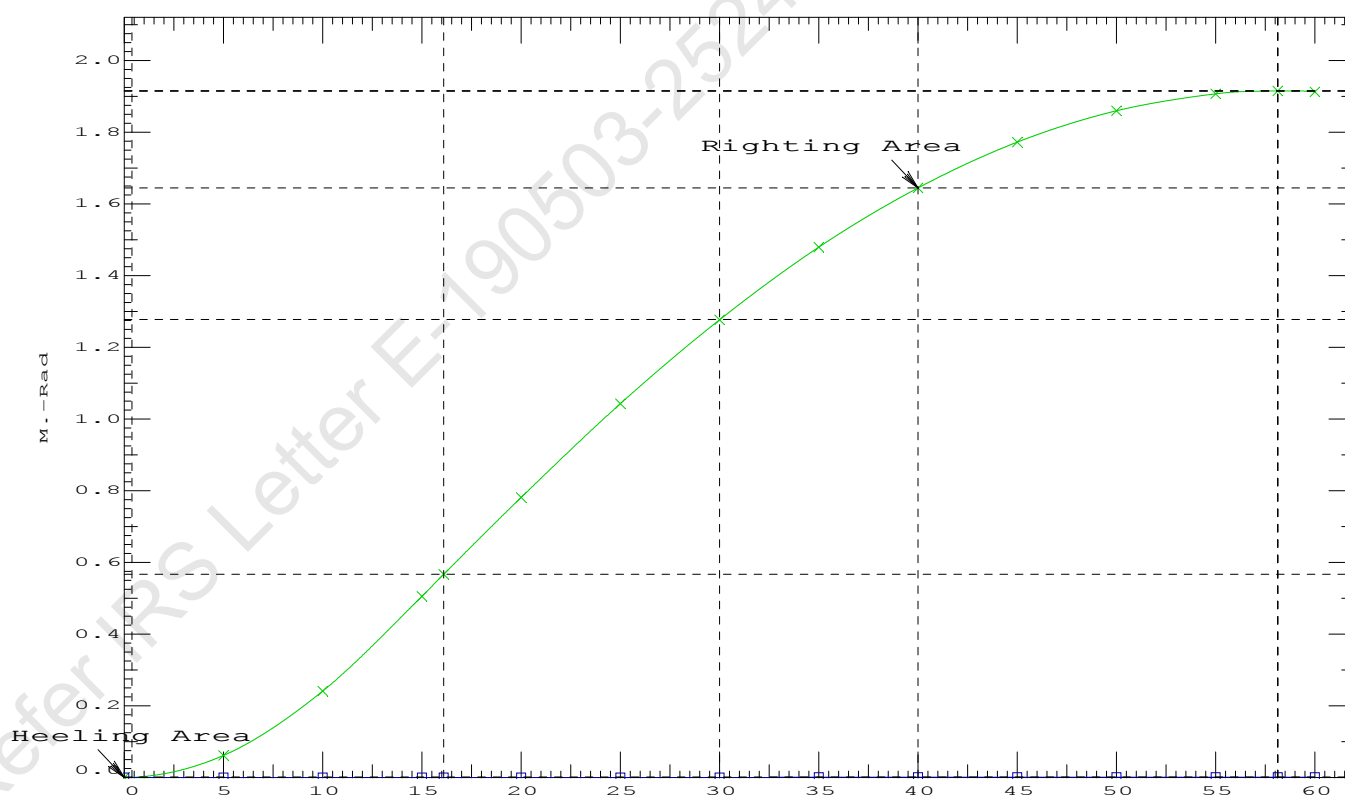
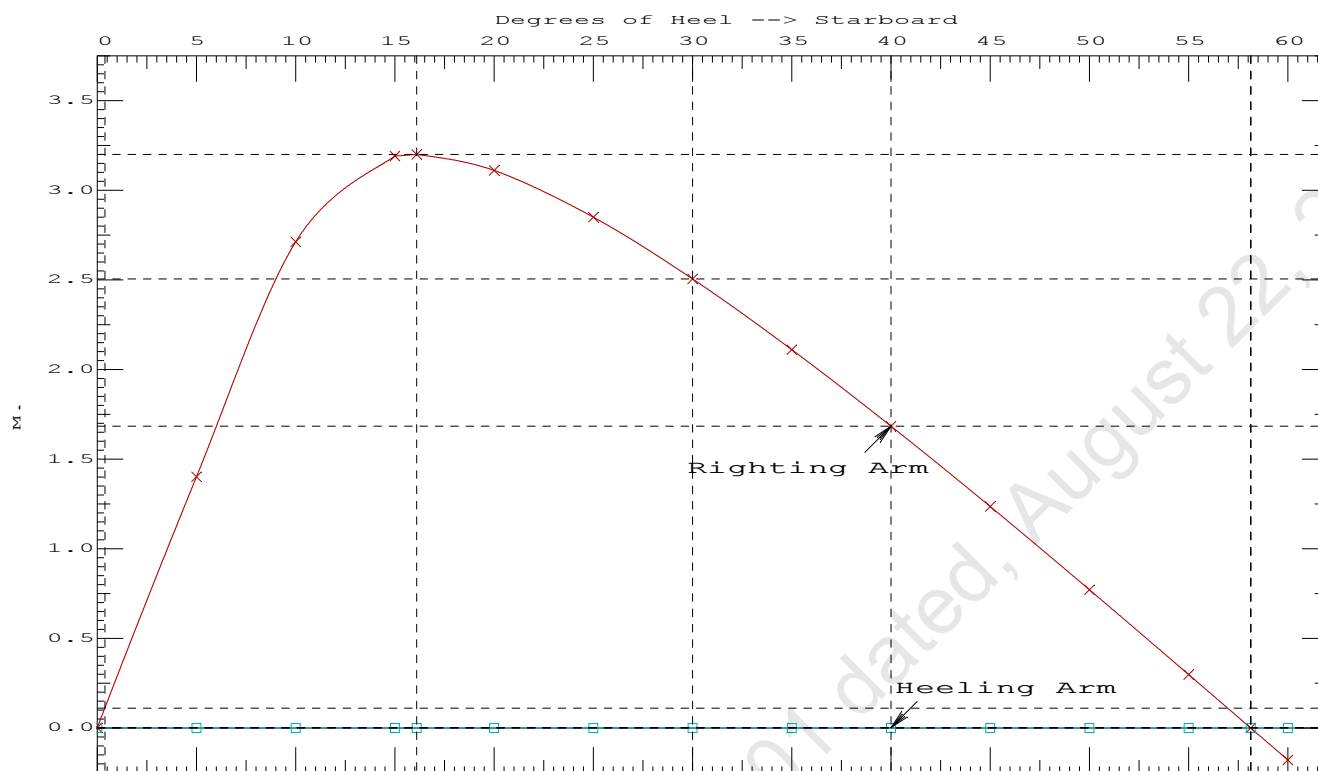
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	0.00 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	389735 P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667	2303.93 P

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PONTOON

Condition 09 : Operating Condition of 350T CraneCrane Lifting 40.0T @ 35m Radius Fwd

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Condition 09 : Operating Condition of 350T Crane**Crane Lifting 40.0T @ 35m Radius Fwd**

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.663 @ 50.00f, 1.469 @ 25.00f, 1.276 @ 0.00

Trim: Fwd 0.387/50.000, Heel: 0.00 deg.

Part-----			Weight (MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			350.00	11.500f	0.000	6.325	
Crane Load			40.00	46.500f	0.000	21.478	
Total Fixed----->			795.97	20.144f	0.000	5.391	
	Load-----	SpGr-----	Weight (MT)----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			422.40	37.208f	0.000	1.596	470.09
Total Weight----->			1,218.37	26.060f	0.000	4.075	
			Displ (MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,218.37	26.086f	0.001s	0.748	

	Righting Arms:			0.000	0.001s		
	External Arms:			0.000	0.001		
	Residual Righting Arms:			0.000	0.001s		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		861.4	25.400f	0.032s	138.02	19.145
			MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
			8.83		32.82	134.69	15.818
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

PONTOON

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.490 @ 50.00f, 1.425 @ 25.00f, 1.360 @ 0.00
Trim: Fwd 0.130/50.000, Heel: 0.00 deg.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

```
limmarg: INFINITY; t3: 30; t: 15.00
limmarg: INFINITY; t3: 15; t: 7.50
limmarg: INFINITY; t3: 8; t: 3.75
limmarg: INFINITY; t3: 4; t: 1.88
limmarg: INFINITY; t3: 2; t: 0.94
limmarg: INFINITY; t3: 1; t: 0.47
limmarg: INFINITY; t3: 1; t: 0.23
limmarg: INFINITY; t3: 1; t: 0.12
limmarg: INFINITY; t3: 1; t: 0.06
limmarg: INFINITY; t3: 1; t: 0.03
limmarg: INFINITY; t3: 1; t: 0.01
```

Theta3 is 0.97 degrees, as demonstrated below.

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PONTOON**RESIDUAL RIGHTING ARMS vs HEEL ANGLE**

Fixed CG: LCG = 18.750f TCG = 0.000 VCG = 4.539

Origin Depth	Degrees of Trim	Displacement Heel	Weight (MT)	Residual Arms in Trim	Res. in Heel	Area
1.348	0.15f	0.00	1,178.4	0.000	0.001	0.0000
1.345	0.15f	0.97s	1,178.4	0.000	0.300	0.0025
1.333	0.15f	5.00s	1,178.4	0.000	1.537	0.0672
1.280	0.16f	10.00s	1,178.4	0.000	2.983	0.2651
1.136	0.24f	15.00s	1,178.4	0.000	3.574	0.5574
1.054	0.28f	17.65s	1,178.2	0.000	3.614	0.7241
0.979	0.32f	20.00s	1,178.4	0.000	3.582	0.8719
0.814	0.40f	25.00s	1,178.4	0.000	3.400	1.1789
0.641	0.49f	30.00s	1,178.4	0.000	3.128	1.4644

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 755.97 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 1 > 1.000 LARGE

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PONTOON

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.750f TCG = 0.000 VCG = 4.539

Origin Depth	Degrees of Trim	Displacement Heel	Weight (MT)	Residual Arms in Trim	Res. in Heel	Area
1.348	0.15f	0.00	1,178.4	0.000	0.001	0.0000
1.345	0.15f	0.97s	1,178.4	0.000	0.300	0.0025
1.333	0.15f	5.00s	1,178.4	0.000	1.537	0.0672
1.280	0.16f	10.00s	1,178.4	0.000	2.983	0.2651
1.136	0.24f	15.00s	1,178.4	0.000	3.574	0.5574
1.054	0.28f	17.65s	1,178.2	0.000	3.614	0.7241
0.979	0.32f	20.00s	1,178.4	0.000	3.582	0.8719
0.814	0.40f	25.00s	1,178.4	0.000	3.400	1.1789
0.641	0.49f	30.00s	1,178.4	0.000	3.128	1.4644
0.461	0.57f	35.00s	1,178.4	0.000	2.800	1.7234
0.276	0.65f	40.00s	1,178.4	0.000	2.435	1.9521
0.088	0.73f	45.00s	1,178.4	0.000	2.041	2.1476
-0.102	0.81f	50.00s	1,178.4	0.000	1.625	2.3077
-0.291	0.88f	55.00s	1,178.4	0.000	1.194	2.4308
-0.479	0.95f	60.00s	1,178.4	0.000	0.750	2.5157
-0.662	1.01f	65.00s	1,178.4	0.000	0.299	2.5615
-0.779	1.04f	68.29s	1,178.4	0.000	0.000	2.5701
-0.839	1.05f	70.00s	1,178.4	0.000	-0.156	2.5678

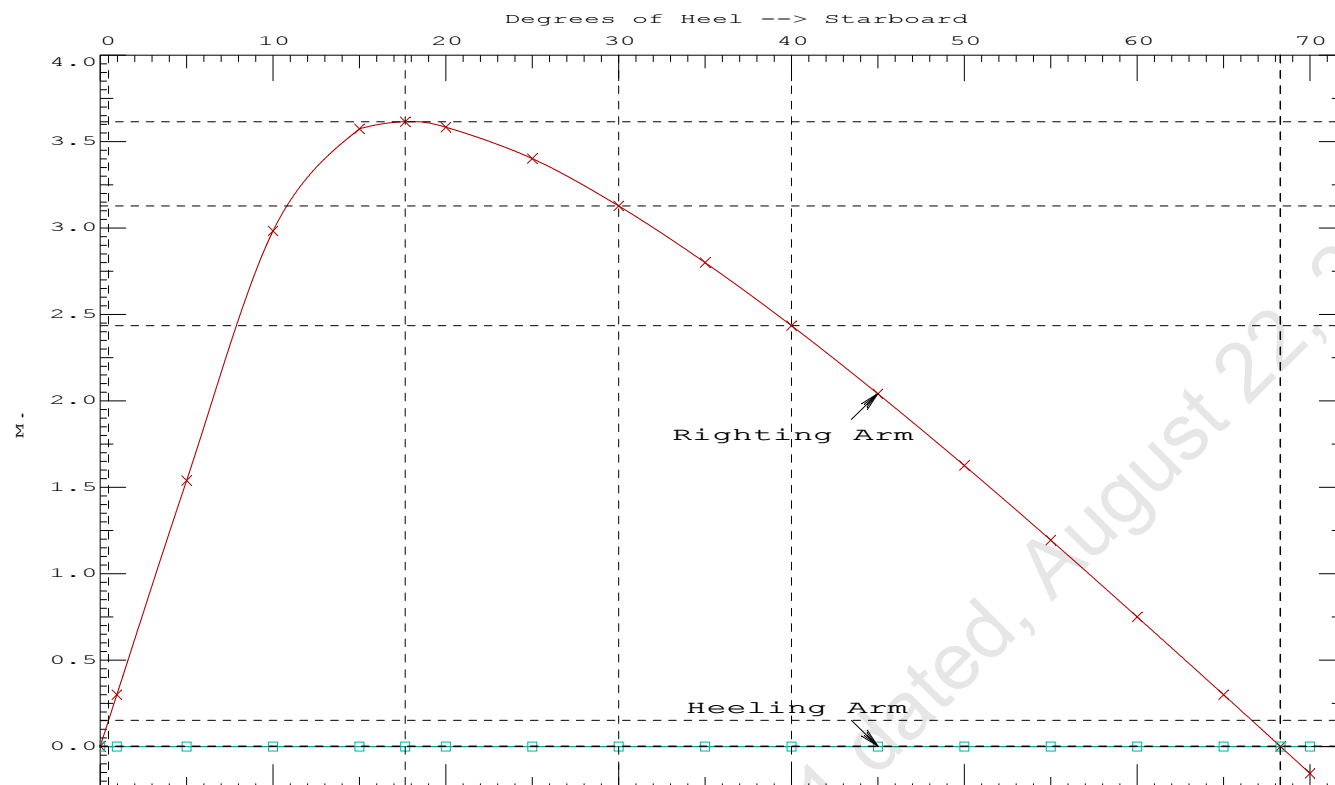
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 755.97 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAzero > 1.000 LARGE
(2) Angle from abs 1 deg to RAzero > 20.00 deg 67.32 P

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GHS 17.22 PONTOON



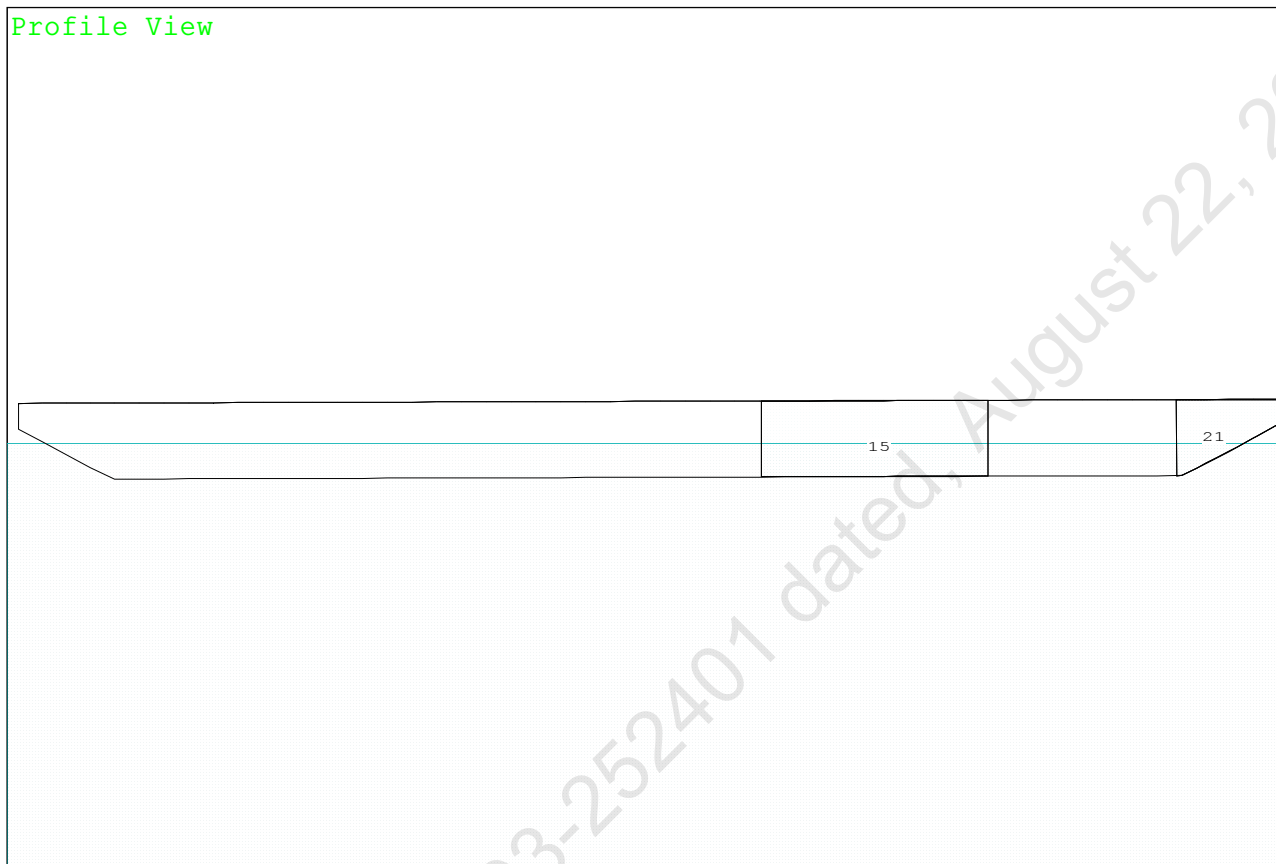
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PONTOON

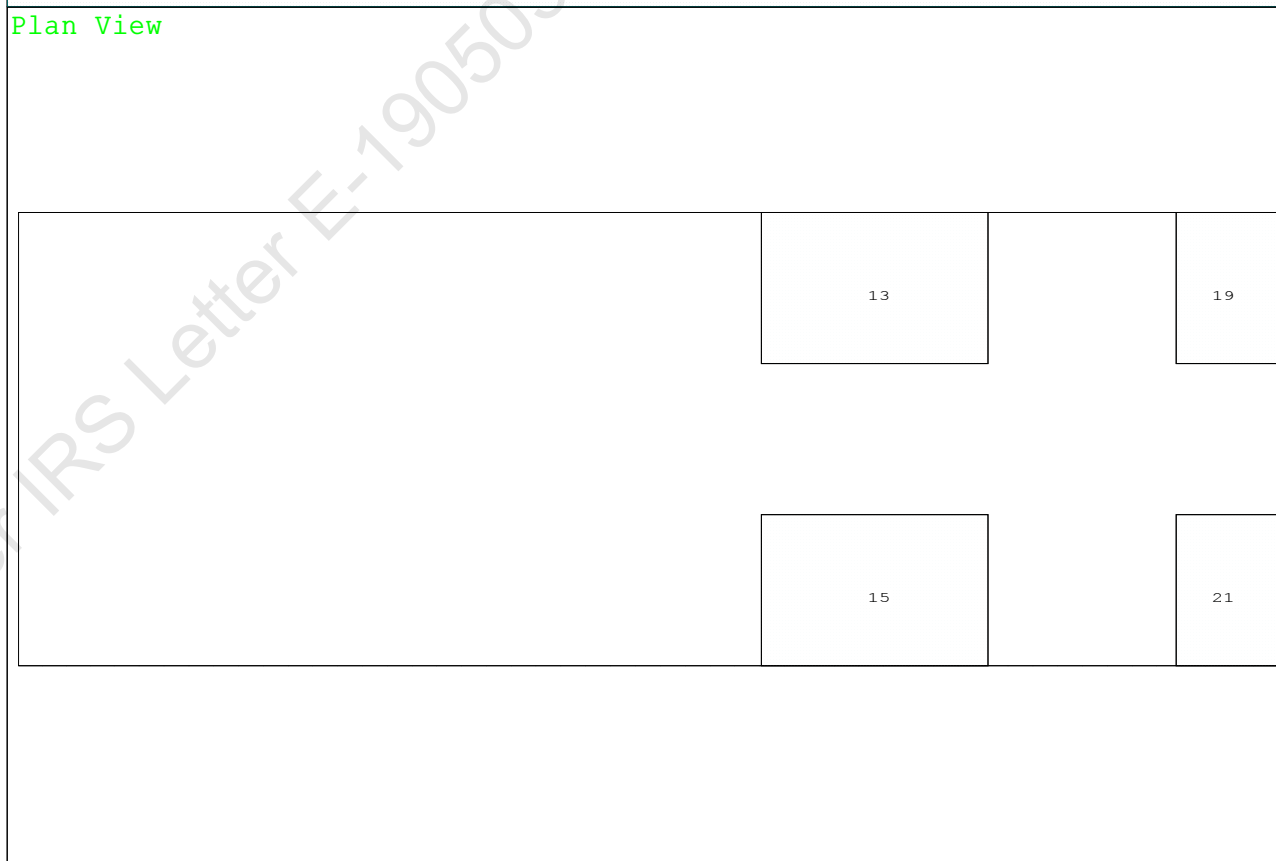
Condition 10 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Aft

Condition Graphic - Draft: 1.273 @ 50.000f, 1.434 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

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PONTOON

Condition 10 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Aft

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.273 @ 50.00f, 1.353 @ 25.00f, 1.434 @ 0.00

Trim: Aft 0.161/50.000, Heel: zero

Part-----			Weight (MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Crane Load			36.60	10.500a	0.000	76.977	
Total Fixed----->			692.57	16.807f	0.000	7.704	
	Load----	SpGr----	Weight (MT)----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			422.40	37.208f	0.000	1.596	470.09
Total Weight----->			1,114.97	24.536f	0.000	5.390	
			Displ (MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,114.94	24.521f	0.000	0.684	

Righting Arms: 0.000 0.000

Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025	858.4	24.848f	0.000	149.26	20.905
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
		8.80	32.23	144.56	16.199	

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

412.1 Cu.M. (16%) SALT WATER

250.00 MT Crane

36.60 MT Crane Load

DRAFT AT FP (M) = 1.273
DRAFT AT MS (M) = 1.353
DRAFT AT AP (M) = 1.434
DRAFT AT FWD MARK (M) = 1.286
DRAFT AT AFT MARK (M) = 1.421

HYDROSTATIC PROPERTIES

Trim: Aft 0.161/50.000, No Heel, VCG = 5.390

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/
Draft----	Weight (MT)----	LCB-----	VCB-----	cm trim-----
1.354	1,114.94	24.521f	0.684	8.80
				24.848f
				32.23
				144.56
				16.199

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 470.1 m.-MT

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PONTOON

Condition 10 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Aft

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 1.273 @ 50.00f, 1.353 @ 25.00f, 1.434 @ 0.00

Trim: Aft 0.161/50.000, Heel: zero

Part-----	LPA*Sf-----	HCP-----	Arm----	Pressure-----	Moment
HULL	82.6	0.842	1.505	0.05500	6.84
CRANE	57.0	4.111	4.774	0.05500	14.97
Total wind heeling moment to starboard----->					21.80
Distances in METERS.-----					Pressure in MT/Sqm.-----
					Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.536f TCG = 0.000 VCG = 5.390

Free Surface Adjustment: 0.422

Adjusted CG: LCG = 24.537f TCG = 0.000 VCG = 5.812

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---Trim---Heel---	Weight (MT)---	in Trim--in Heel---	Area----	Angle
1.422 0.18a 0.00	1,115.0	0.000 -0.019	0.0000	5.136
1.422 0.18a 0.05s	1,115.0	0.000 0.000	-0.0000	5.084
1.407 0.18a 5.05s	1,115.0	0.000 1.419	0.0619	0.084
1.407 0.18a 5.14s	1,115.0	0.000 1.442	0.0640	0.000
1.359 0.20a 10.05s	1,115.0	0.000 2.707	0.2429	-4.916
1.254 0.29a 15.05s	1,115.0	0.000 3.119	0.5044	-9.916
1.142 0.39a 20.05s	1,115.0	0.000 2.942	0.7732	-14.916
1.024 0.50a 25.05s	1,115.0	0.000 2.572	1.0151	-19.916
0.900 0.61a 30.05s	1,115.0	0.000 2.114	1.2202	-24.916
0.772 0.71a 35.05s	1,115.0	0.000 1.607	1.3829	-29.916
0.640 0.82a 40.05s	1,115.0	0.000 1.071	1.5000	-34.916
0.504 0.92a 45.05s	1,115.0	0.000 0.516	1.5694	-39.916
0.378 1.01a 49.62s	1,115.0	0.000 0.000	1.5900	-44.483
0.365 1.02a 50.05s	1,115.0	0.000 -0.049	1.5898	-44.916
0.224 1.11a 55.05s	1,115.0	0.000 -0.618	1.5607	-49.916
0.079 1.19a 60.05s	1,115.0	0.000 -1.186	1.4820	-54.916
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.				

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.80 (constant)

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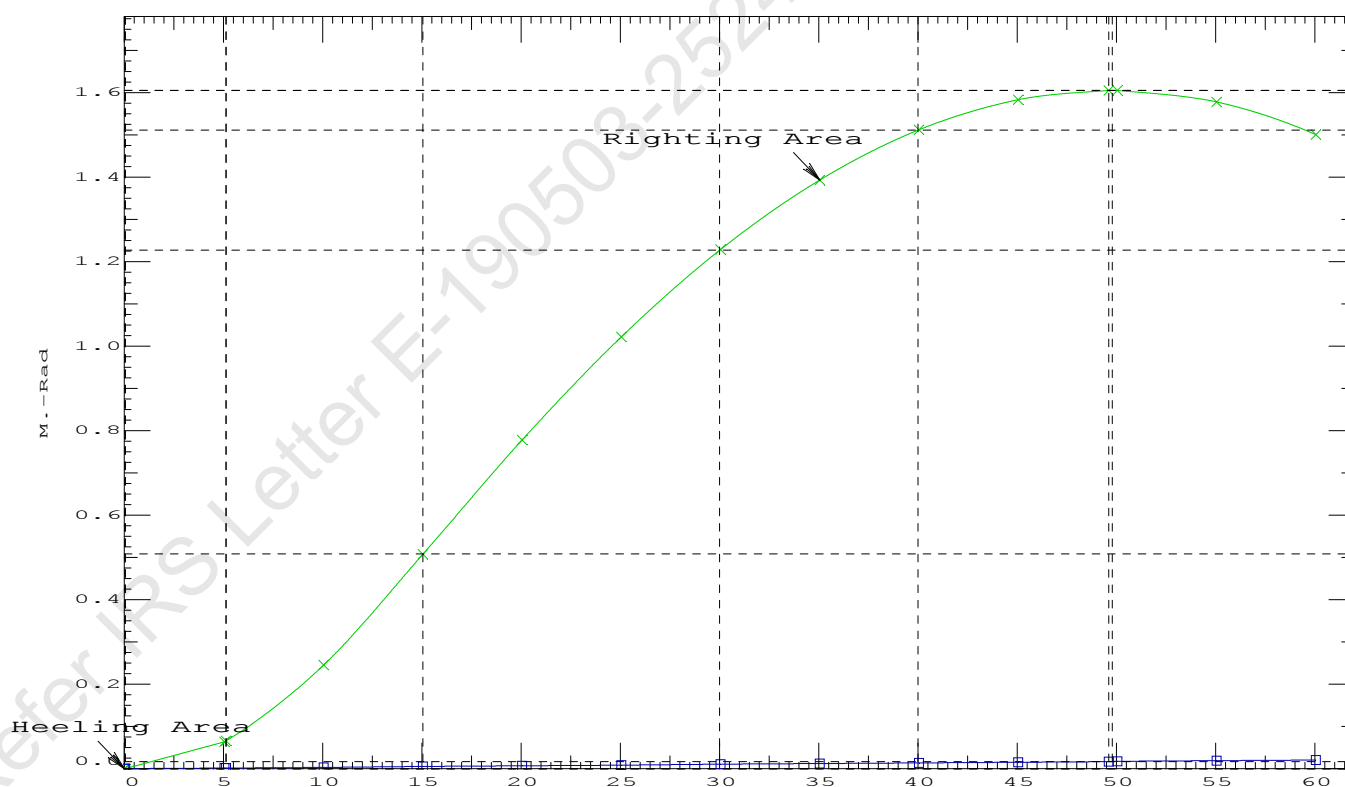
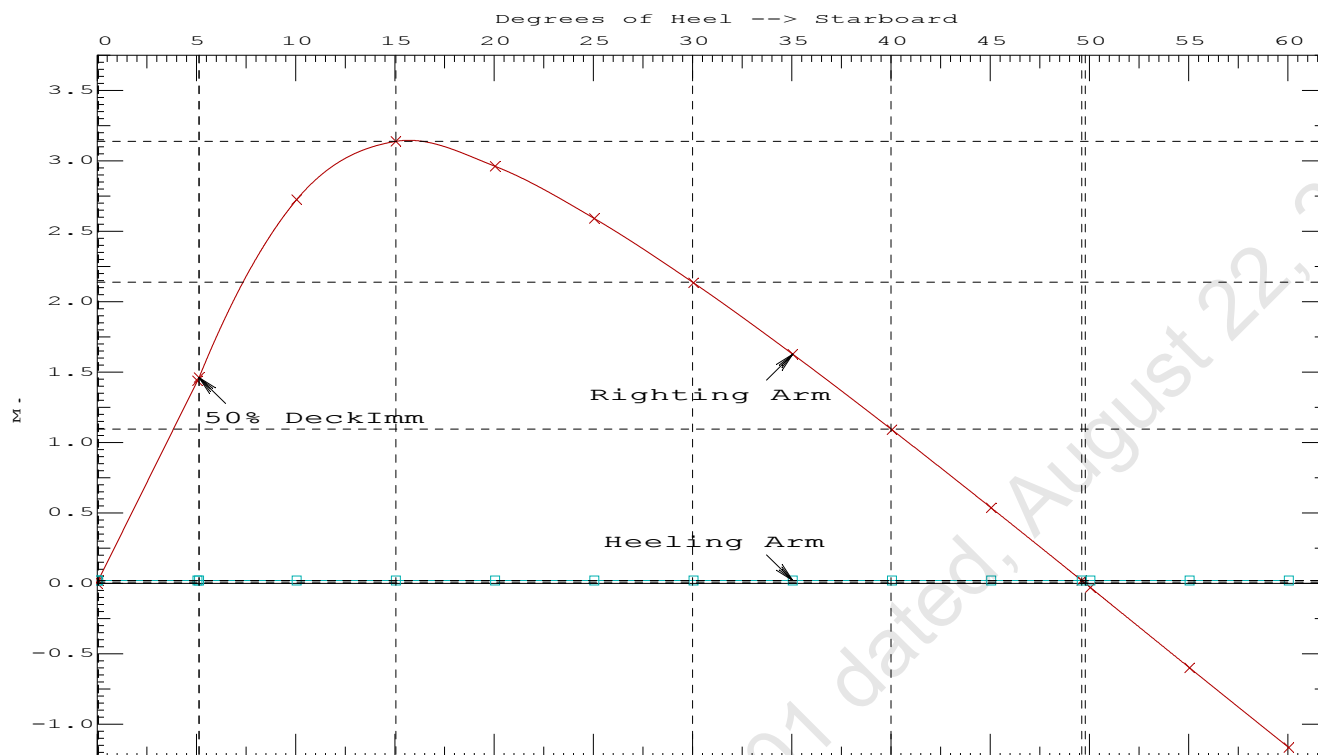
PONTOON**Condition 10 : Operating Condition of 250T Crane**
Crane Lifting 36.6T @ 18m Radius Aft

LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max-----Attained
(1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad 0.5044 P
(2) Angle from Equilibrium to RAzero or Flood > 20.00 deg 49.57 P
(3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg 5.08 P

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PONTOON

Condition 10 : Operating Condition of 250T CraneCrane Lifting 36.6T @ 18m Radius Aft

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PONTOON

Condition 10 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Aft

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.536f TCG = 0.000 VCG = 5.390
Free Surface Adjustment: 0.422
Adjusted CG: LCG = 24.537f TCG = 0.000 VCG = 5.812

Origin	Degrees of	Displacement	Residual Arms	Res.
Depth---	Trim----	Heel----	Weight (MT)----	in Trim--in Heel----> Area
1.422	0.18a	0.00	1,115.0	0.000 0.000 0.0000
1.407	0.18a	5.00s	1,115.0	0.000 1.419 0.0619
1.360	0.20a	10.00s	1,115.0	0.000 2.711 0.2431
1.255	0.29a	15.00s	1,115.0	0.000 3.133 0.5055
1.143	0.39a	20.00s	1,115.0	0.000 2.959 0.7757
1.025	0.50a	25.00s	1,115.0	0.000 2.591 1.0193
0.902	0.61a	30.00s	1,115.0	0.000 2.133 1.2261
0.773	0.71a	35.00s	1,115.0	0.000 1.627 1.3905
0.641	0.82a	40.00s	1,115.0	0.000 1.091 1.5094
0.506	0.92a	45.00s	1,115.0	0.000 0.537 1.5805
0.374	1.01a	49.75s	1,115.0	0.000 0.000 1.6028
0.367	1.02a	50.00s	1,115.0	0.000 -0.029 1.6028
0.225	1.11a	55.00s	1,115.0	0.000 -0.598 1.5755
0.080	1.19a	60.00s	1,115.0	0.000 -1.165 1.4985

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

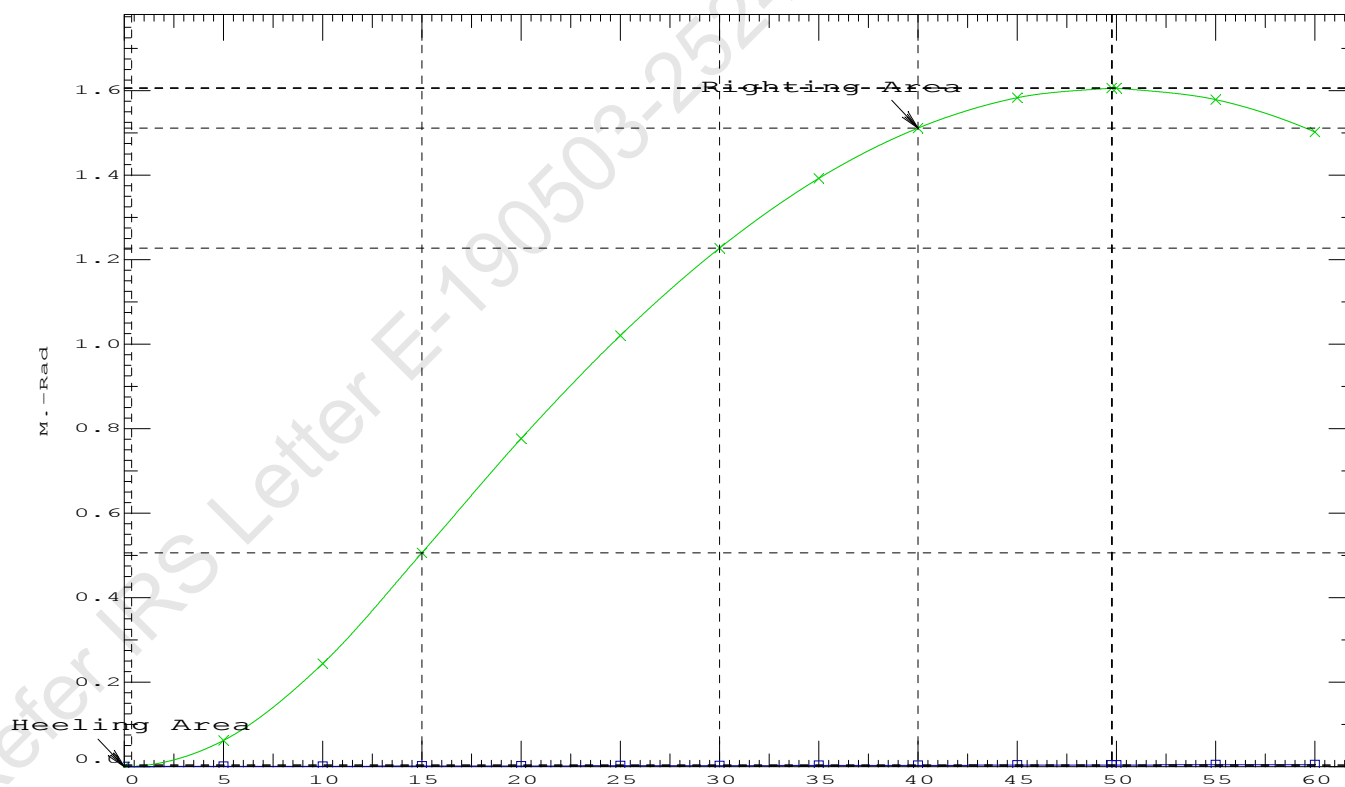
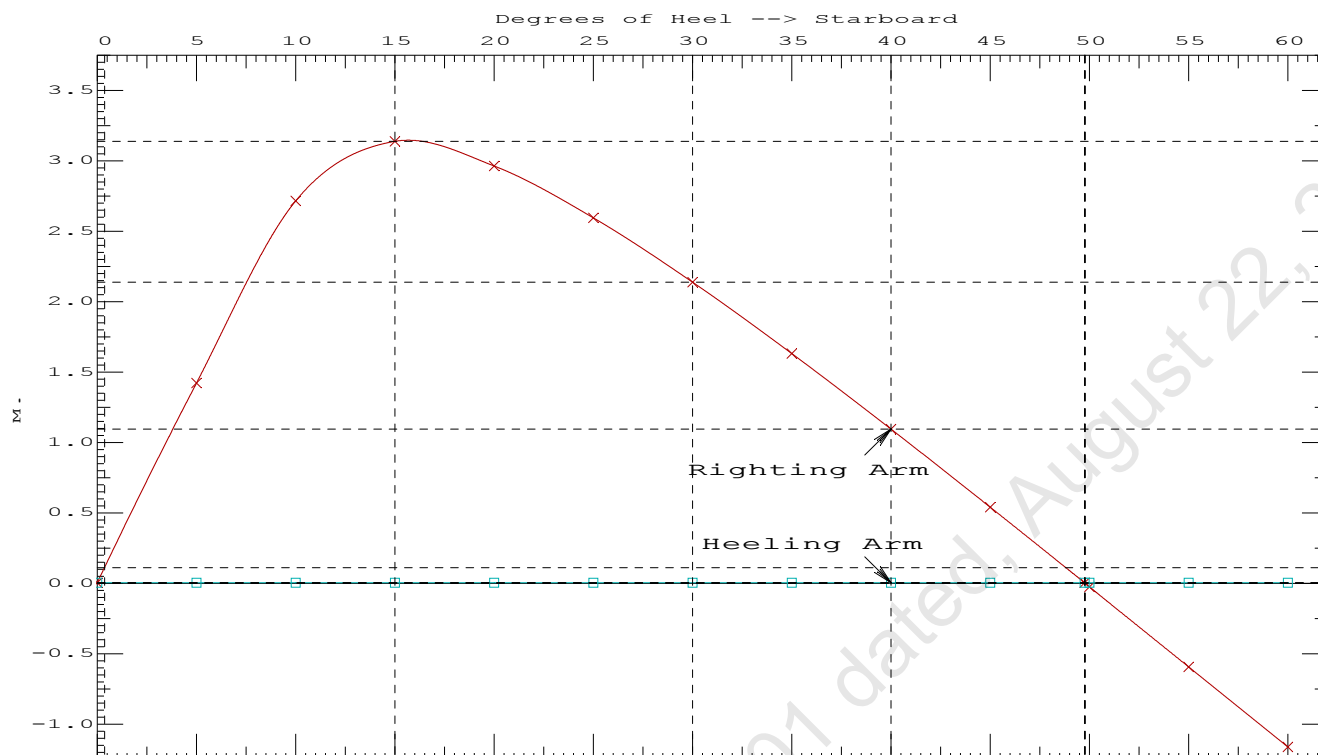
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	0.00 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	69871.1 P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667	412.647 P

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PONTOON

Condition 10 : Operating Condition of 250T CraneCrane Lifting 36.6T @ 18m Radius Aft

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GHS 17.22 PONTOON

Condition 10 : Operating Condition of 250T Crane**Crane Lifting 36.6T @ 18m Radius Aft**

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.273 @ 50.00f, 1.353 @ 25.00f, 1.434 @ 0.00

Trim: Aft 0.161/50.000, Heel: 0.00 deg.

Part-----			Weight (MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Crane Load			36.60	10.500a	0.000	76.977	
Total Fixed----->			692.57	16.807f	0.000	7.704	
	Load-----	SpGr-----	Weight (MT)----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			422.40	37.208f	0.000	1.596	470.09
Total Weight----->			1,114.97	24.536f	0.000	5.390	
			Displ (MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,114.97	24.521f	0.005s	0.684	

	Righting Arms:			0.000	0.005s		
	External Arms:			0.000	0.004s		
	Residual Righting Arms:			0.000	0.000s		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		848.4	24.952f	0.064s	144.12	20.655
			MT/cm				
			8.70				

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

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PONTOON

After hook load drop

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.432 @ 50.00f, 1.311 @ 25.00f, 1.191 @ 0.00

Trim: Fwd 0.241/50.000, Heel: 0.00 deg.

Part-----			Weight (MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Total Fixed----->			655.97	18.330f	0.000	3.838	
	Load----	SpGr-----	Weight (MT)----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			422.40	37.208f	0.000	1.596	470.09
Total Weight----->			1,078.37	25.725f	0.000	2.960	
			Displ (MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,078.37	25.736f	0.004s	0.663	

	Righting Arms:			0.000	0.004s		
	External Arms:			0.000	0.001s		
	Residual Righting Arms:			0.000	0.003s		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->		1.025	847.0	25.085f	0.055s	148.28	21.310
			MT/cm				
			8.68				

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: INFINITY; t3: 30.0; t: 15.00
limmarg: INFINITY; t3: 15.0; t: 7.50
limmarg: INFINITY; t3: 7.5; t: 3.75
limmarg: INFINITY; t3: 3.8; t: 1.88
limmarg: INFINITY; t3: 1.9; t: 0.94
limmarg: INFINITY; t3: 1.0; t: 0.47
limmarg: INFINITY; t3: 0.5; t: 0.23
limmarg: INFINITY; t3: 0.3; t: 0.12
limmarg: INFINITY; t3: 0.2; t: 0.06
limmarg: INFINITY; t3: 0.1; t: 0.03
limmarg: INFINITY; t3: 0.1; t: 0.01

Theta3 is 0.07 degrees, as demonstrated below.

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PONTOON

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.330f TCG = 0.000 VCG = 3.838

Origin Depth	Degrees of Trim	Heel	Displacement Weight (MT)	Residual Arms in Trim	in Heel	Res. Area
1.179	0.28f	0.00	1,078.2	0.000	-0.003	0.0000
1.179	0.28f	0.00	1,078.2	0.000	0.001	-0.0000
1.179	0.28f	0.07p	1,078.4	0.000	0.029	0.0000
1.165	0.28f	5.00p	1,078.4	0.000	1.727	0.0756
1.094	0.30f	10.00p	1,078.3	0.000	3.290	0.2956
0.877	0.43f	15.00p	1,078.4	0.000	3.965	0.6186
0.699	0.53f	18.73p	1,078.4	0.000	4.045	0.8815
0.637	0.57f	20.00p	1,078.4	0.000	4.036	0.9709
0.389	0.72f	25.00p	1,078.4	0.000	3.898	1.3196
0.135	0.87f	30.00p	1,078.4	0.000	3.659	1.6501

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 655.97 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 0.1 > 1.000 LARGE

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PONTOON

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.330f TCG = 0.000 VCG = 3.838

Origin	Degrees of	Displacement	Residual Arms	Res.
Depth---Trim---Heel---	Weight (MT)---	in Trim--in Heel---	Area	
1.179 0.28f 0.00	1,078.2	0.000 -0.003	0.0000	
1.179 0.28f 0.00	1,078.2	0.000 0.001	-0.0000	
1.179 0.28f 0.07p	1,078.4	0.000 0.029	0.0000	
1.165 0.28f 5.00p	1,078.4	0.000 1.727	0.0756	
1.094 0.30f 10.00p	1,078.3	0.000 3.290	0.2956	
0.877 0.43f 15.00p	1,078.4	0.000 3.965	0.6186	
0.699 0.53f 18.73p	1,078.4	0.000 4.045	0.8815	
0.637 0.57f 20.00p	1,078.4	0.000 4.036	0.9709	
0.389 0.72f 25.00p	1,078.4	0.000 3.898	1.3196	
0.135 0.87f 30.00p	1,078.4	0.000 3.659	1.6501	
-0.122 1.01f 35.00p	1,078.4	0.000 3.359	1.9568	
-0.382 1.16f 40.00p	1,078.4	0.000 3.015	2.2352	
-0.640 1.30f 45.00p	1,078.4	0.000 2.637	2.4821	
-0.896 1.44f 50.00p	1,078.4	0.000 2.233	2.6948	
-1.146 1.57f 55.00p	1,078.4	0.000 1.807	2.8713	
-1.388 1.68f 60.00p	1,078.4	0.000 1.365	3.0098	
-1.618 1.79f 65.00p	1,078.4	0.000 0.911	3.1092	
-1.835 1.88f 70.00p	1,078.4	0.000 0.448	3.1686	
-2.028 1.94f 74.80p	1,078.4	0.000 0.000	3.1873	
-2.036 1.94f 75.00p	1,078.4	0.000 -0.019	3.1873	

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

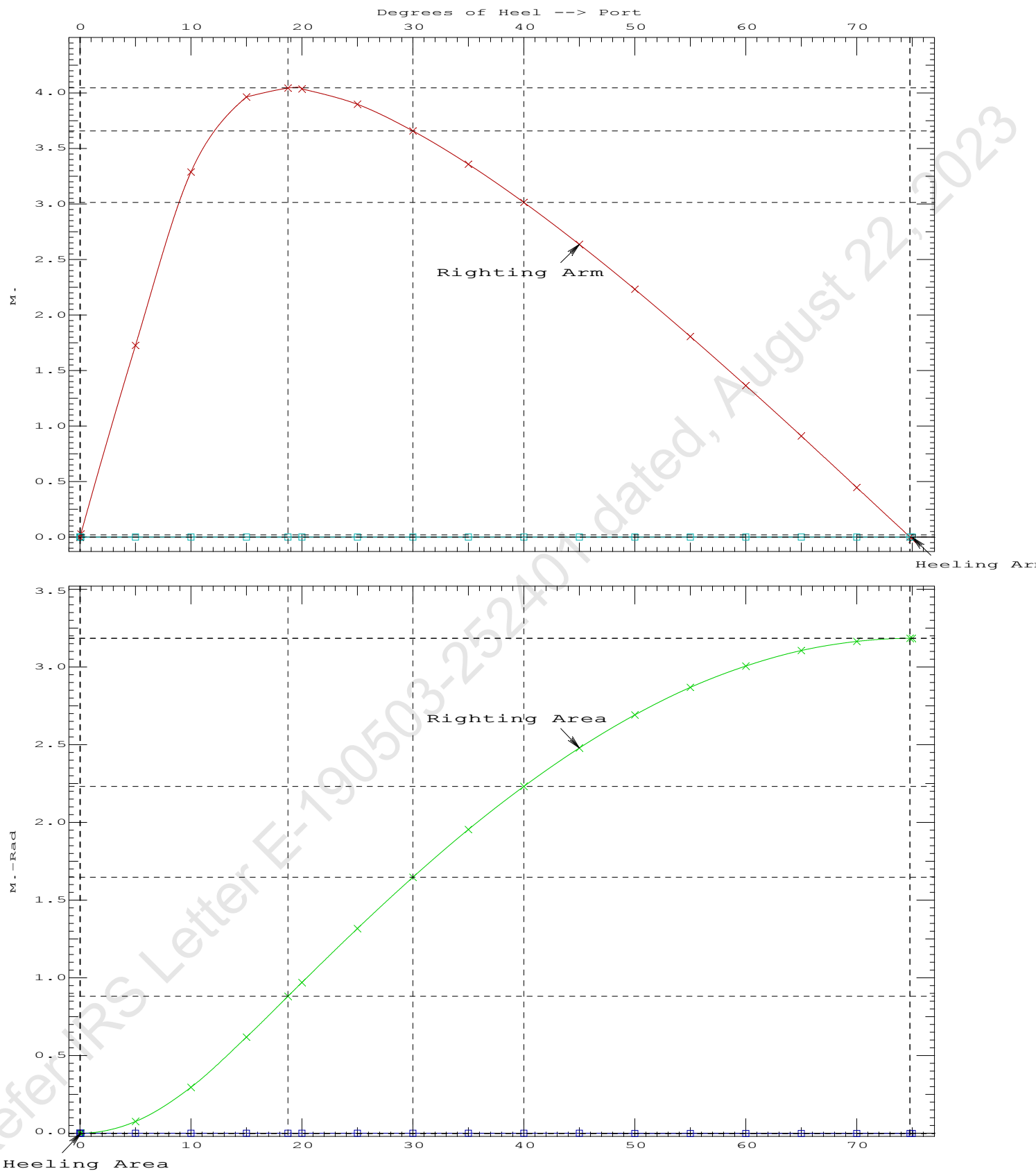
Note: The Center of Gravity shown above is for the Fixed Weight of 655.97 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAZero > 1.000 LARGE
(2) Angle from abs 0.1 deg to RAZero > 20.00 deg 74.73 P

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PONTOON



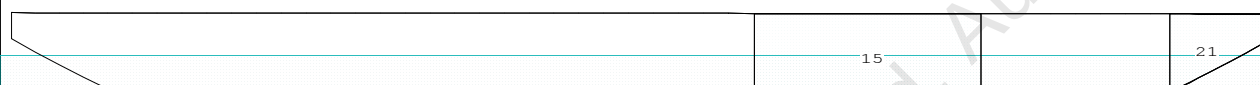
24-06-21 15:56:55 Cybermarine Technologies PTE. Ltd.
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PONTOON

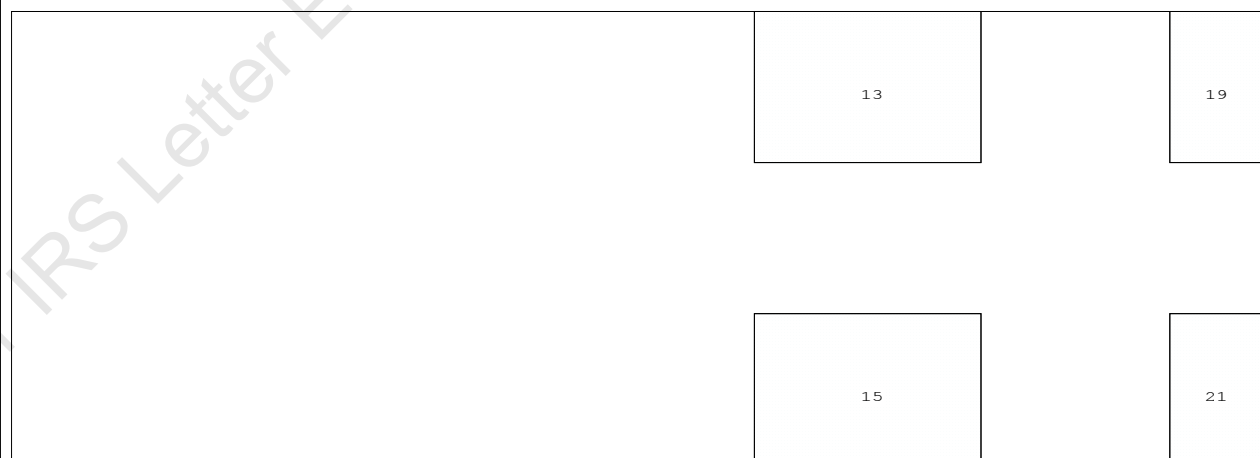
Condition 11 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Stbd

Condition Graphic - Draft: 1.376 @ 50.000f, 1.332 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

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Condition 11 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Stbd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.376 @ 50.00f, 1.354 @ 25.00f, 1.332 @ 0.00

Trim: Fwd 0.044/50.000, Heel: zero

Part-----			Weight (MT)-----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Crane Load			36.60	7.500f	0.000	76.977	
Total Fixed----->			692.57	17.758f	0.000	7.704	
	Load-----	SpGr-----	Weight (MT)-----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			422.40	37.208f	0.000	1.596	470.09
Total Weight----->			1,114.97	25.127f	0.000	5.390	
			Displ (MT)-----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,114.93	25.132f	0.000	0.683	

Righting Arms: 0.001f 0.000

Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----->	1.025	858.4	25.042f	0.000	149.22	20.900
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
		8.80	32.23	144.52	16.193	

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

412.1 Cu.M. (16%) SALT WATER

250.00 MT Crane

36.60 MT Crane Load

DRAFT AT FP (M) = 1.376
DRAFT AT MS (M) = 1.354
DRAFT AT AP (M) = 1.332
DRAFT AT FWD MARK (M) = 1.372
DRAFT AT AFT MARK (M) = 1.335

HYDROSTATIC PROPERTIES

Trim: Fwd 0.044/50.000, No Heel, VCG = 5.390

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/				
Draft----	Weight (MT)-----	LCB-----	VCB-----	cm-----	LCF-----	cm trim-----	GML-----	GMT-----
1.354	1,114.93	25.132f	0.683	8.80	25.042f	32.23	144.52	16.193

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 470.1 m.-MT

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PONTOON

Condition 11 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Stbd

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 1.376 @ 50.00f, 1.354 @ 25.00f, 1.332 @ 0.00

Trim: Fwd 0.044/50.000, Heel: zero

Part-----	LPA*Sf-----	HCP-----	Arm----	Pressure-----	Moment
HULL	82.6	0.841	1.504	0.05500	6.83
CRANE	57.0	4.172	4.834	0.05500	15.16
Total wind heeling moment to starboard----->					21.99
Distances in METERS.-----Pressure in MT/Sqm.-----					Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.127f TCG = 0.000 VCG = 5.390

Free Surface Adjustment: 0.422

Adjusted CG: LCG = 25.126f TCG = 0.000 VCG = 5.812

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---Trim---Heel---	Weight (MT)---	in Trim--in Heel---	Area----	Angle
1.320 0.05f 0.00	1,114.9	0.000 -0.019	0.0000	5.367
1.319 0.05f 0.05s	1,115.0	0.000 0.000	-0.0000	5.312
1.305 0.05f 5.05s	1,115.0	0.000 1.419	0.0619	0.312
1.303 0.05f 5.37s	1,115.0	0.000 1.507	0.0699	0.000
1.248 0.05f 10.05s	1,115.0	0.000 2.709	0.2432	-4.688
1.092 0.08f 15.05s	1,115.2	0.000 3.122	0.5050	-9.688
0.923 0.11f 20.05s	1,115.0	0.000 2.946	0.7741	-14.688
0.746 0.14f 25.05s	1,115.0	0.000 2.576	1.0164	-19.688
0.563 0.17f 30.05s	1,115.0	0.000 2.118	1.2219	-24.688
0.376 0.19f 35.05s	1,115.0	0.000 1.612	1.3850	-29.688
0.184 0.22f 40.05s	1,115.0	0.000 1.075	1.5024	-34.688
-0.008 0.25f 45.05s	1,115.0	0.000 0.520	1.5721	-39.688
-0.186 0.28f 49.65s	1,115.0	0.000 0.000	1.5931	-44.287
-0.201 0.28f 50.05s	1,115.0	0.000 -0.046	1.5929	-44.688
-0.393 0.30f 55.05s	1,115.0	0.000 -0.615	1.5641	-49.688
-0.581 0.32f 60.05s	1,115.0	0.000 -1.183	1.4856	-54.688
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.				

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 21.99 (constant)

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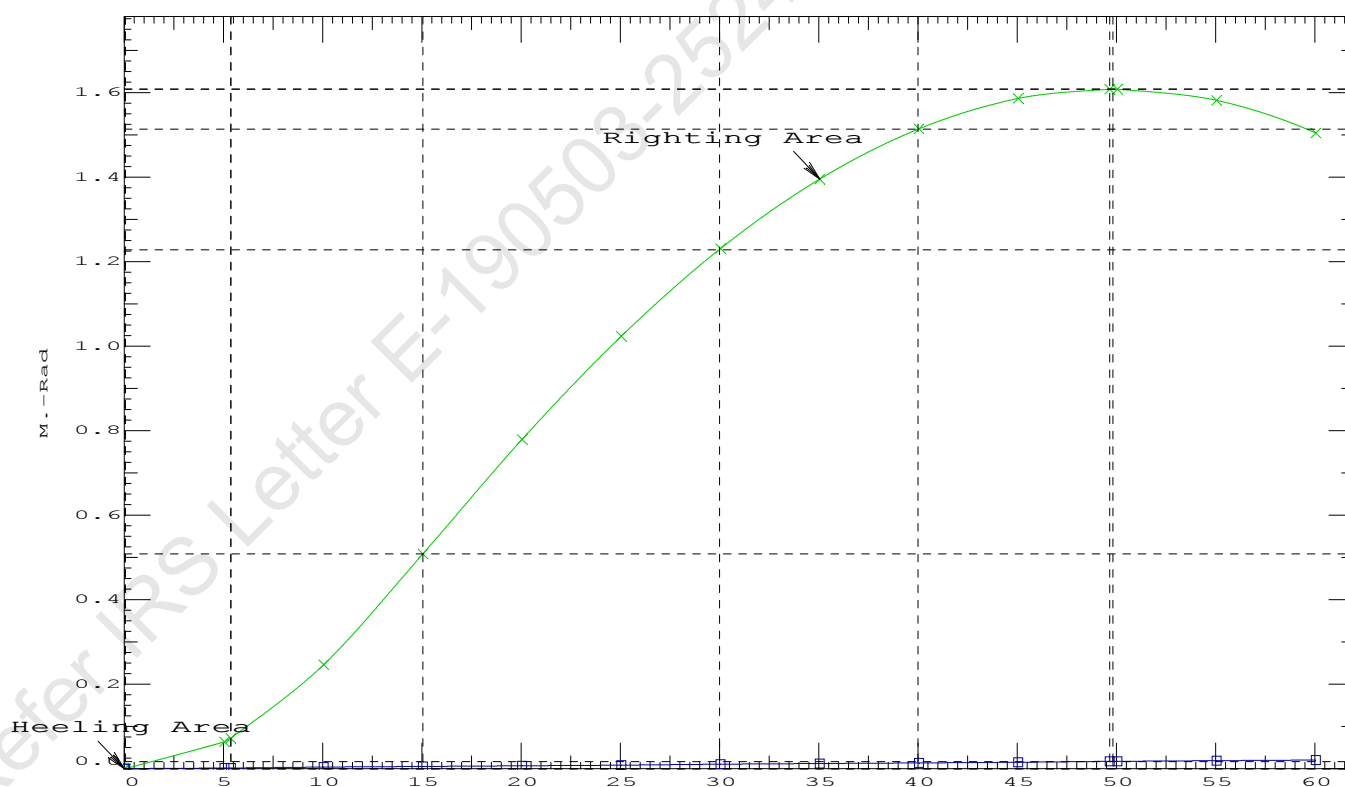
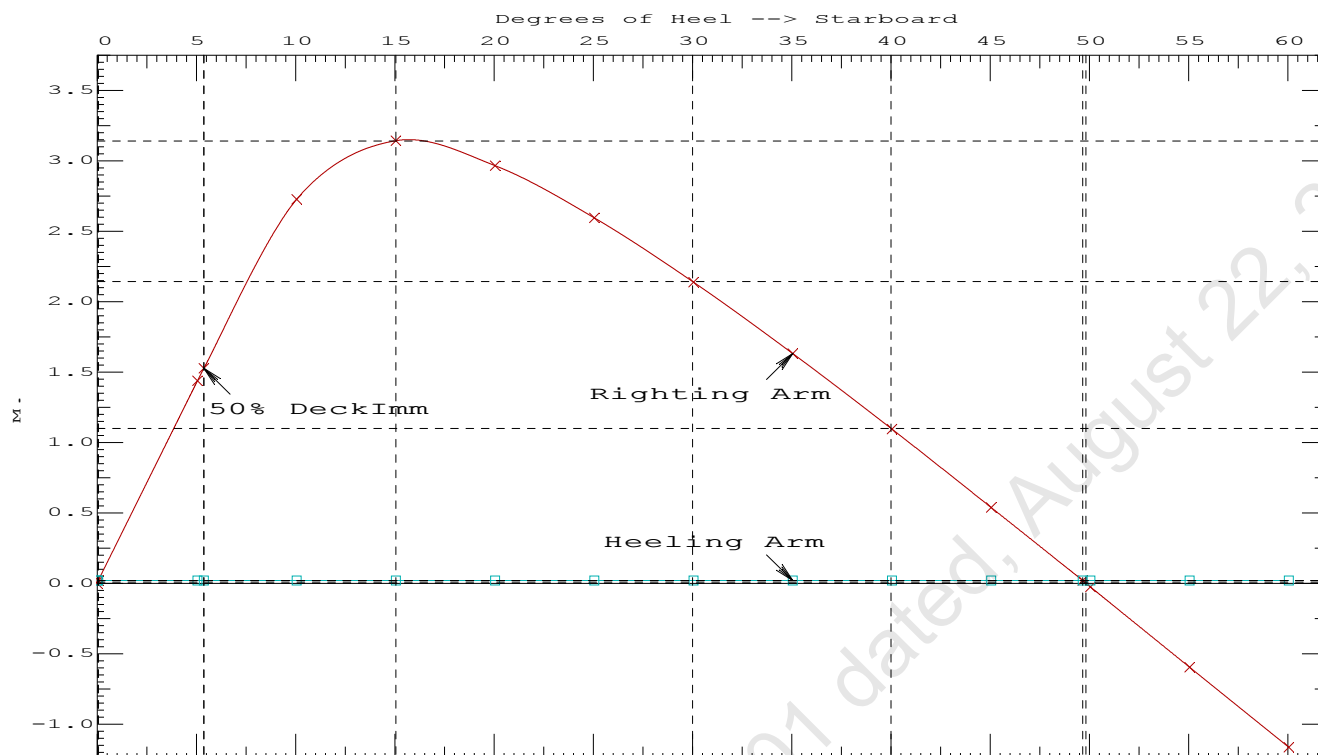
PONTOON**Condition 11 : Operating Condition of 250T Crane**
Crane Lifting 36.6T @ 18m Radius Stbd

LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max-----Attained
(1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad 0.5050 P
(2) Angle from Equilibrium to RAzero or Flood > 20.00 deg 49.60 P
(3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg 5.31 P

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PONTOON

Condition 11 : Operating Condition of 250T CraneCrane Lifting 36.6T @ 18m Radius Stbd

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PONTOON

Condition 11 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.127f TCG = 0.000 VCG = 5.390
Free Surface Adjustment: 0.422
Adjusted CG: LCG = 25.126f TCG = 0.015p VCG = 5.811

Origin	Degrees of		Displacement	Residual Arms		Res.
Depth---	Trim---	Heel---	Weight (MT)---	in Trim--	in Heel---	Area
1.316	0.05f	2.02s	1,115.0	0.000	0.000	0.0000
1.291	0.05f	7.02s	1,115.1	0.000	1.407	0.0614
1.190	0.06f	12.02s	1,115.0	0.000	2.442	0.2320
1.083	0.08f	15.33s	1,115.2	0.000	2.564	0.3786
1.026	0.09f	17.02s	1,115.0	0.000	2.532	0.4539
0.854	0.12f	22.02s	1,115.0	0.000	2.257	0.6670
0.675	0.15f	27.02s	1,115.0	0.000	1.845	0.8470
0.490	0.18f	32.02s	1,115.0	0.000	1.365	0.9875
0.301	0.21f	37.02s	1,115.0	0.000	0.844	1.0842
0.109	0.24f	42.02s	1,115.0	0.000	0.298	1.1342
0.006	0.25f	44.69s	1,115.0	0.000	0.000	1.1412
-0.084	0.26f	47.02s	1,115.0	0.000	-0.263	1.1358
-0.277	0.29f	52.02s	1,115.0	0.000	-0.832	1.0882
-0.467	0.31f	57.02s	1,115.0	0.000	-1.402	0.9907
-0.654	0.33f	62.02s	1,115.0	0.000	-1.969	0.8436

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 658.80

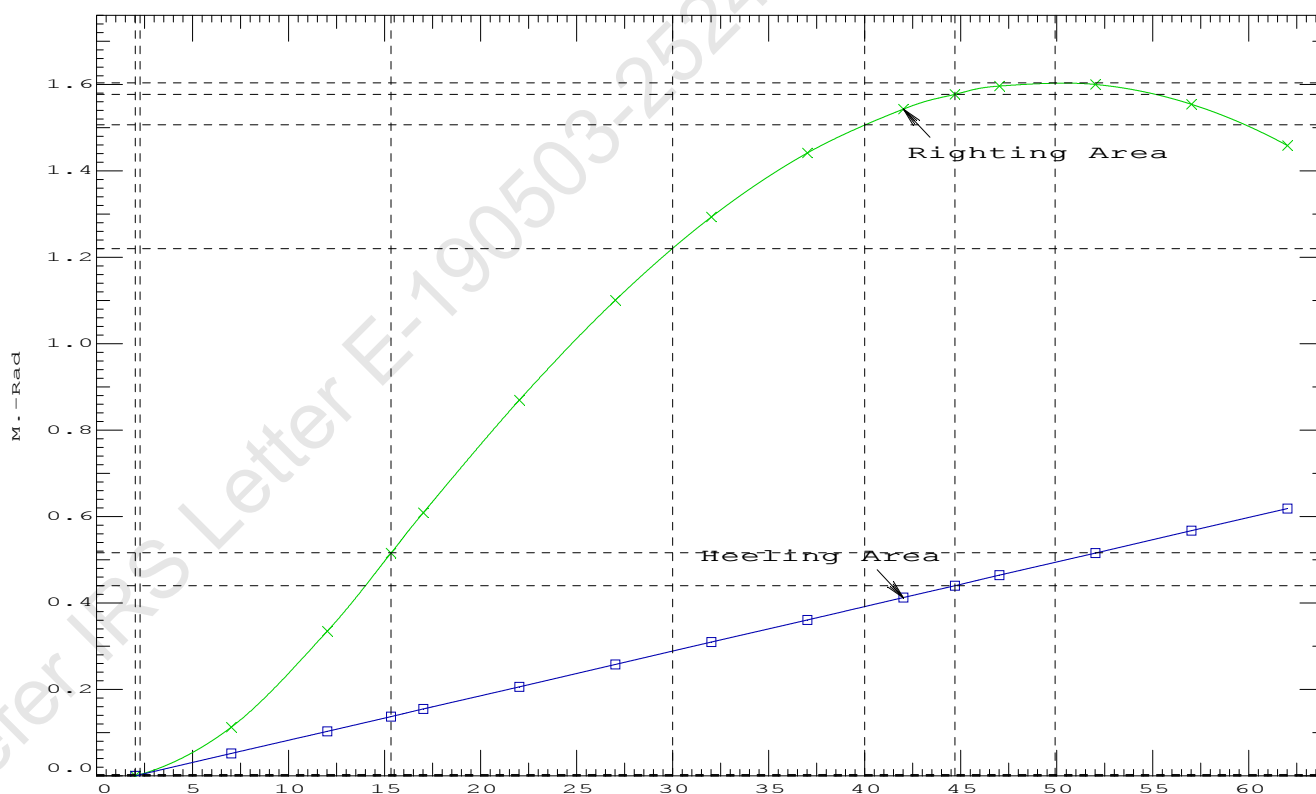
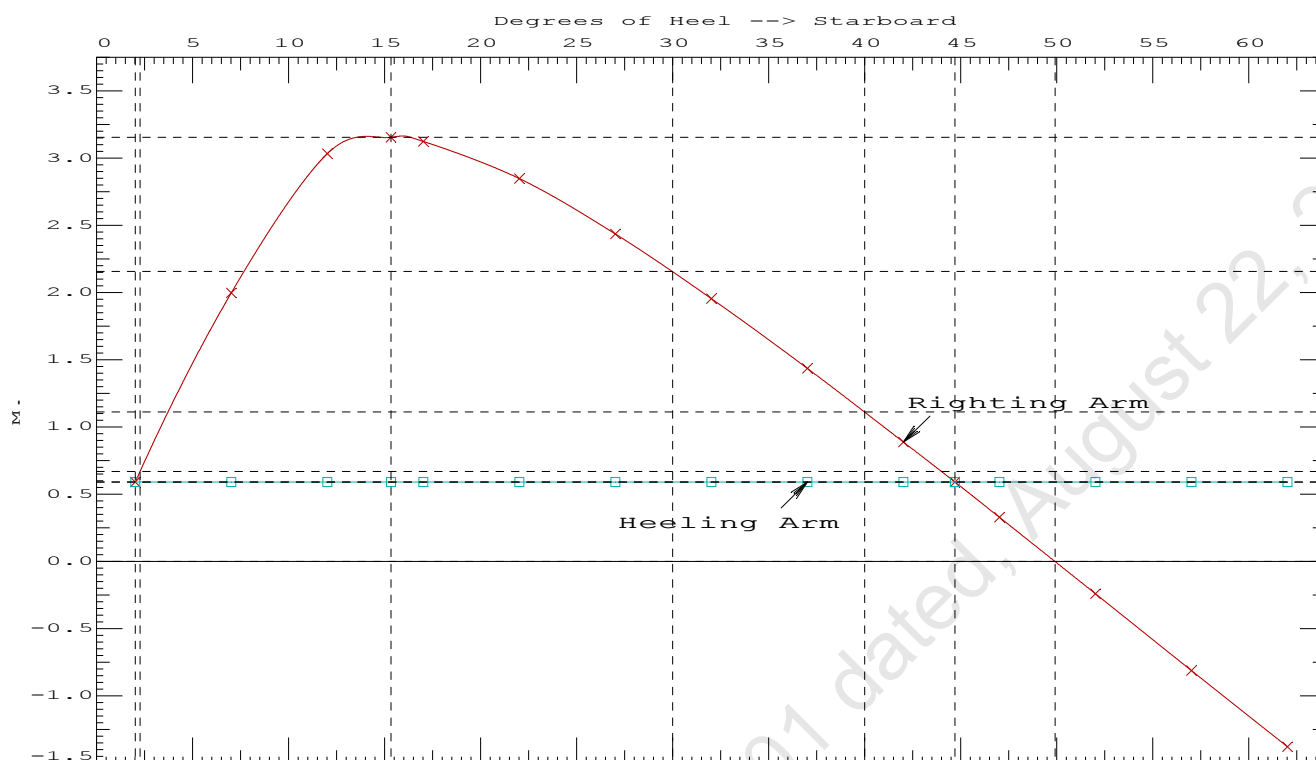
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	2.02 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	434.0 P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667	3.593 P

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PONTOON

Condition 11 : Operating Condition of 250T CraneCrane Lifting 36.6T @ 18m Radius Stbd

PONTOON

Crane Lifting 36.6T @ 18m Radius Stbd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.373 @ 50.00f, 1.351 @ 25.00f, 1.329 @ 0.00

Trim: Fwd 0.044/50.000, Heel: Stbd 2.02 deg.

Part			Weight (MT)	LCG	TCG	VCG	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Crane Load			36.60	7.500f	0.000	76.977	
Total Fixed	----->		692.57	17.758f	0.000	7.704	
	Load	SpGr	Weight (MT)	LCG	TCG	VCG	FSM
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks	----->		422.40	37.208f	0.000	1.596	470.09
Total Weight	----->		1,114.97	25.127f	0.000	5.390	
			Displ (MT)	LCB	TCB	VCB	
HULL		1.025	1,114.97	25.131f	0.757s	0.696	

Righting Arms:	0.000	0.591s
External Arms:	0.000	0.591s
Residual Righting Arms:	0.000	0.000s

Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----	1.025	856.2	25.029f	0.097s	148.04	20.840
		MT/cm				
		8.78				

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 2.02

PONTOON

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.432 @ 50.00f, 1.311 @ 25.00f, 1.191 @ 0.00
Trim: Fwd 0.241/50.000, Heel: 0.00 deg.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

```
limmarg: 13212.80; t3: 30.00; t: 15.00
limmarg: 4890.65; t3: 15.00; t: 7.35
limmarg: 1318.83; t3: 7.65; t: 3.60
limmarg: 300.00; t3: 4.05; t: 1.76
limmarg: 28.25; t3: 2.29; t: 0.86
limmarg: -49.85; t3: 1.43; t: 0.42
limmarg: -16.13; t3: 1.85; t: 0.21
limmarg: 3.88; t3: 2.06; t: 0.10
limmarg: -5.92; t3: 1.96; t: 0.05
limmarg: -1.08; t3: 2.01; t: 0.02
limmarg: 0.89; t3: 2.03; t: 0.01
```

Theta3 is 2.03 degrees, as demonstrated below.

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PONTOON

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.330f TCG = 0.000 VCG = 3.838

Origin Depth	Degrees of Trim	Displacement Heel	Weight (MT)	Residual Arms in Trim	Res. in Heel	Area
1.175	0.28f	2.02s	1,078.3	0.000	-0.700	0.0000
1.179	0.28f	0.00	1,078.4	0.000	-0.003	-0.0124
1.179	0.28f	0.00	1,078.3	0.000	0.001	-0.0124
1.175	0.28f	2.03p	1,078.4	0.000	0.705	0.0001
1.165	0.28f	5.00p	1,078.4	0.000	1.727	0.0632
1.094	0.30f	10.00p	1,078.3	0.000	3.290	0.2831
0.877	0.43f	15.00p	1,078.4	0.000	3.965	0.6061
0.699	0.53f	18.73p	1,078.4	0.000	4.045	0.8689
0.637	0.57f	20.00p	1,078.4	0.000	4.036	0.9583
0.389	0.72f	25.00p	1,078.4	0.000	3.898	1.3070
0.135	0.87f	30.00p	1,078.4	0.000	3.659	1.6375

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 655.97 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs -2 deg to abs 2 > 1.000 1.009 P

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PONTOON

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.330f TCG = 0.000 VCG = 3.838

Origin Depth	Degrees of Trim	Displacement Heel	Residual Arms Weight (MT)	Res. in Trim	Res. in Heel	Area
1.175	0.28f	2.02s	1,078.3	0.000	-0.700	0.0000
1.179	0.28f	0.00	1,078.4	0.000	-0.003	-0.0124
1.179	0.28f	0.00	1,078.3	0.000	0.001	-0.0124
1.175	0.28f	2.03p	1,078.4	0.000	0.705	0.0001
1.165	0.28f	5.00p	1,078.4	0.000	1.727	0.0632
1.094	0.30f	10.00p	1,078.3	0.000	3.290	0.2831
0.877	0.43f	15.00p	1,078.4	0.000	3.965	0.6061
0.699	0.53f	18.73p	1,078.4	0.000	4.045	0.8689
0.637	0.57f	20.00p	1,078.4	0.000	4.036	0.9583
0.389	0.72f	25.00p	1,078.4	0.000	3.898	1.3070
0.135	0.87f	30.00p	1,078.4	0.000	3.659	1.6375
-0.122	1.01f	35.00p	1,078.4	0.000	3.359	1.9442
-0.382	1.16f	40.00p	1,078.4	0.000	3.015	2.2226
-0.640	1.30f	45.00p	1,078.4	0.000	2.637	2.4695
-0.896	1.44f	50.00p	1,078.4	0.000	2.233	2.6822
-1.146	1.57f	55.00p	1,078.4	0.000	1.807	2.8587
-1.388	1.68f	60.00p	1,078.4	0.000	1.365	2.9973
-1.618	1.79f	65.00p	1,078.4	0.000	0.911	3.0967
-1.835	1.88f	70.00p	1,078.4	0.000	0.448	3.1560
-2.028	1.94f	74.80p	1,078.4	0.000	0.000	3.1748
-2.036	1.94f	75.00p	1,078.4	0.000	-0.019	3.1747

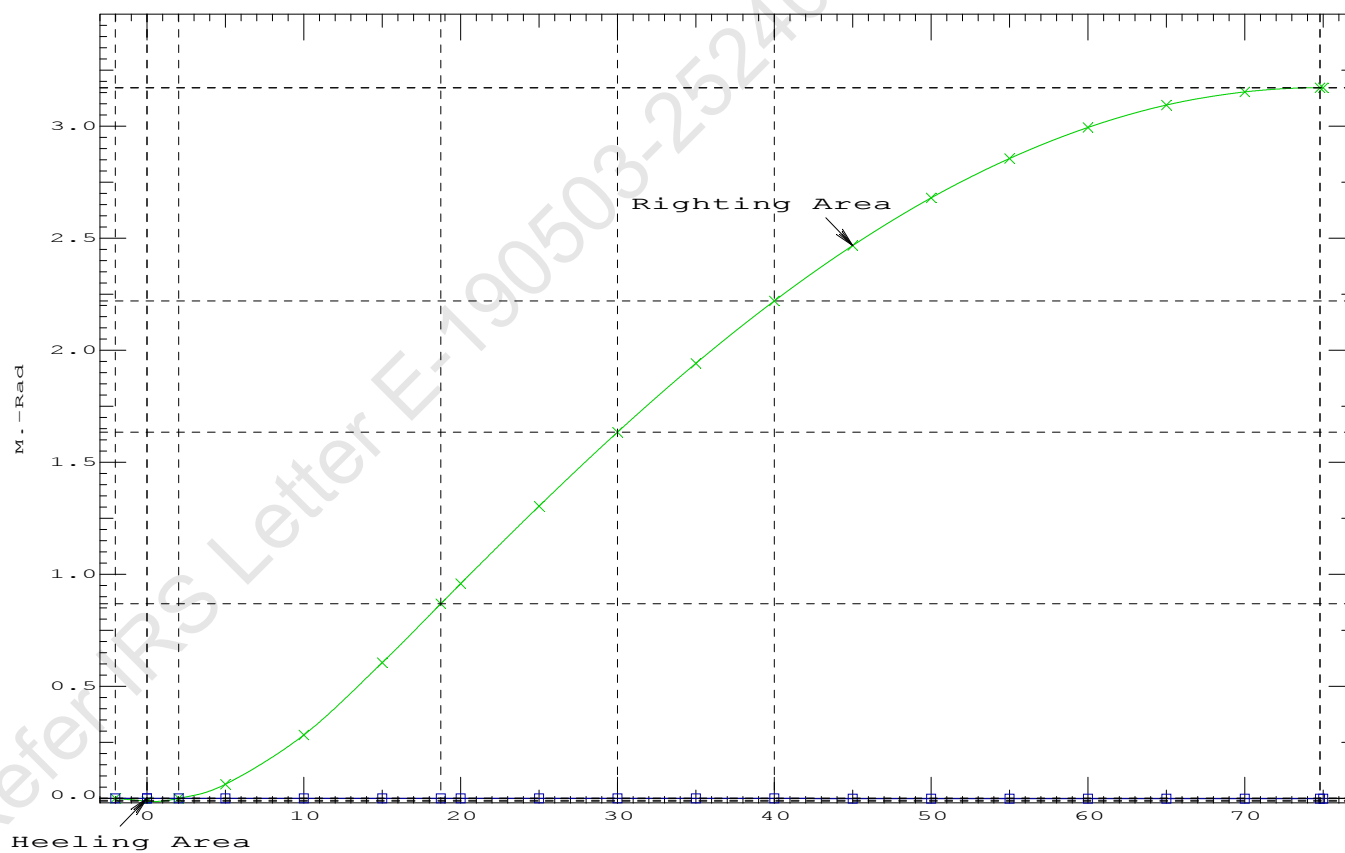
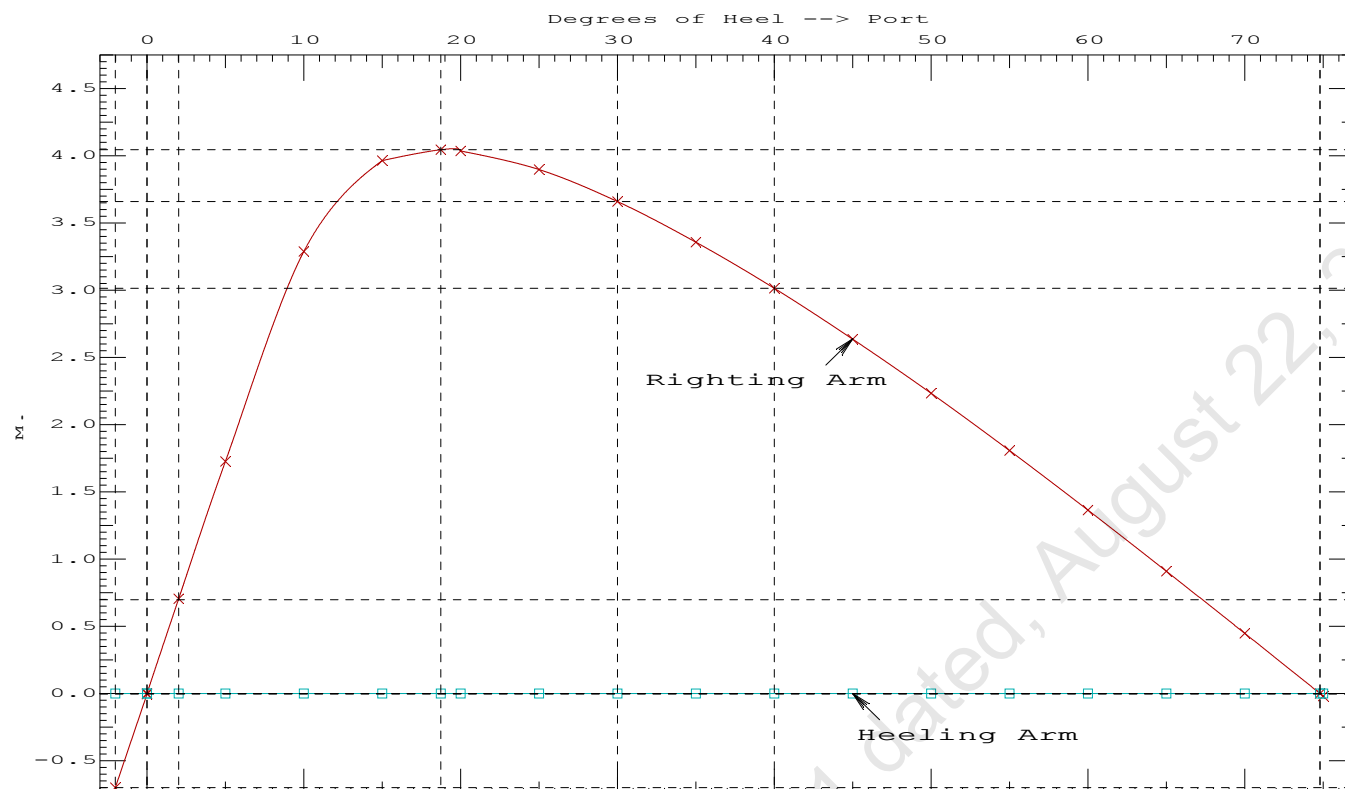
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 655.97 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs -2 deg to RAzero > 1.000 257.146 P
(2) Angle from abs 2 deg to RAzero > 20.00 deg 72.77 P

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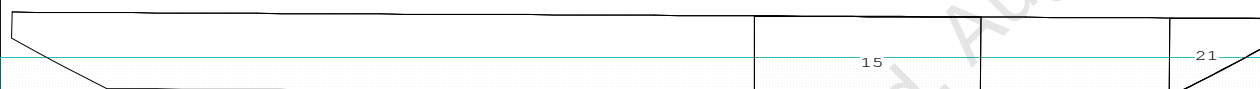
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PONTOON

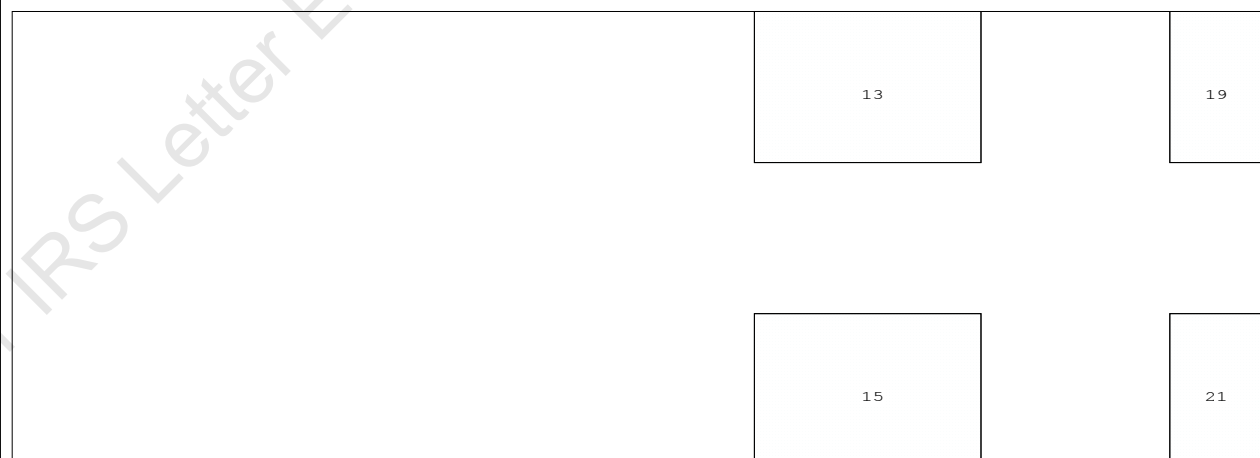
Condition 12 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Fwd

Condition Graphic - Draft: 1.477 @ 50.000f, 1.229 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

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PONTOON

Condition 12 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Fwd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.477 @ 50.00f, 1.353 @ 25.00f, 1.229 @ 0.00

Trim: Fwd 0.248/50.000, Heel: zero

Part-----			Weight (MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Crane Load			36.60	25.500f	0.000	76.977	
Total Fixed----->			692.57	18.709f	0.000	7.704	
	Load-----	SpGr-----	Weight (MT)---- <td>LCG-----</td> <td>TCG-----</td> <td>VCG-----</td> <td>FSM-----</td>	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			422.40	37.208f	0.000	1.596	470.09
Total Weight----->			1,114.97	25.718f	0.000	5.390	
			Displ (MT)---- <td>LCB-----</td> <td>TCB-----</td> <td>VCB-----</td> <td></td>	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,114.92	25.741f	0.000	0.685	

Righting Arms: 0.000 0.000

Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025	858.1	25.221f	0.000	149.07	20.877
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
		8.80	32.19	144.36	16.172	

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

412.1 Cu.M. (16%) SALT WATER

250.00 MT Crane

36.60 MT Crane Load

DRAFT AT FP (M) = 1.477
DRAFT AT MS (M) = 1.353
DRAFT AT AP (M) = 1.229
DRAFT AT FWD MARK (M) = 1.457
DRAFT AT AFT MARK (M) = 1.249

HYDROSTATIC PROPERTIES

Trim: Fwd 0.248/50.000, No Heel, VCG = 5.390

LCF Displacement Buoyancy-Ctr. Weight/ Moment/
Draft---Weight (MT)----LCB-----VCB-----cm-----LCF---cm trim----GML-----GMT
1.354 1,114.92 25.741f 0.685 8.80 25.221f 32.19 144.36 16.172
Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 470.1 m.-MT

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PONTOON

Condition 12 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Fwd

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 1.477 @ 50.00f, 1.353 @ 25.00f, 1.229 @ 0.00

Trim: Fwd 0.248/50.000, Heel: zero

Part	LPA*SF	HCP	Arm	Pressure	Moment
HULL	82.6	0.843	1.506	0.05500	6.84
CRANE	57.0	4.233	4.897	0.05500	15.35
Total wind heeling moment to starboard-->					22.19
Distances in METERS.-----Pressure in MT/Sqm.-----Moment in m.-MT					

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.718f TCG = 0.000 VCG = 5.390

Free Surface Adjustment: 0.422

Adjusted CG: LCG = 25.715f TCG = 0.000 VCG = 5.812

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---Trim---Heel---	Weight (MT)---	in Trim--in Heel---	Area---Angle	
1.217 0.28f 0.00	1,115.0	0.000 -0.019	0.0000	4.974
1.217 0.28f 0.06s	1,115.0	0.000 0.000	-0.0000	4.915
1.203 0.28f 4.97s	1,115.0	0.000 1.396	0.0599	0.000
1.202 0.29f 5.06s	1,115.0	0.000 1.420	0.0619	-0.085
1.135 0.31f 10.06s	1,115.0	0.000 2.705	0.2429	-5.085
0.930 0.45f 15.06s	1,115.2	0.000 3.111	0.5039	-10.085
0.704 0.61f 20.06s	1,115.0	0.000 2.934	0.7719	-15.085
0.469 0.77f 25.06s	1,115.0	0.000 2.564	1.0133	-20.085
0.226 0.94f 30.06s	1,115.0	0.000 2.106	1.2177	-25.085
-0.021 1.10f 35.06s	1,115.0	0.000 1.599	1.3797	-30.085
-0.271 1.27f 40.06s	1,115.0	0.000 1.063	1.4961	-35.085
-0.521 1.43f 45.06s	1,115.0	0.000 0.509	1.5648	-40.085
-0.744 1.56f 49.56s	1,115.0	0.000 0.000	1.5849	-44.590
-0.768 1.58f 50.06s	1,115.0	0.000 -0.056	1.5846	-45.085
-1.009 1.71f 55.06s	1,115.0	0.000 -0.625	1.5549	-50.085
-1.240 1.84f 60.06s	1,115.0	0.000 -1.192	1.4757	-55.085
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.				

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 22.19 (constant)

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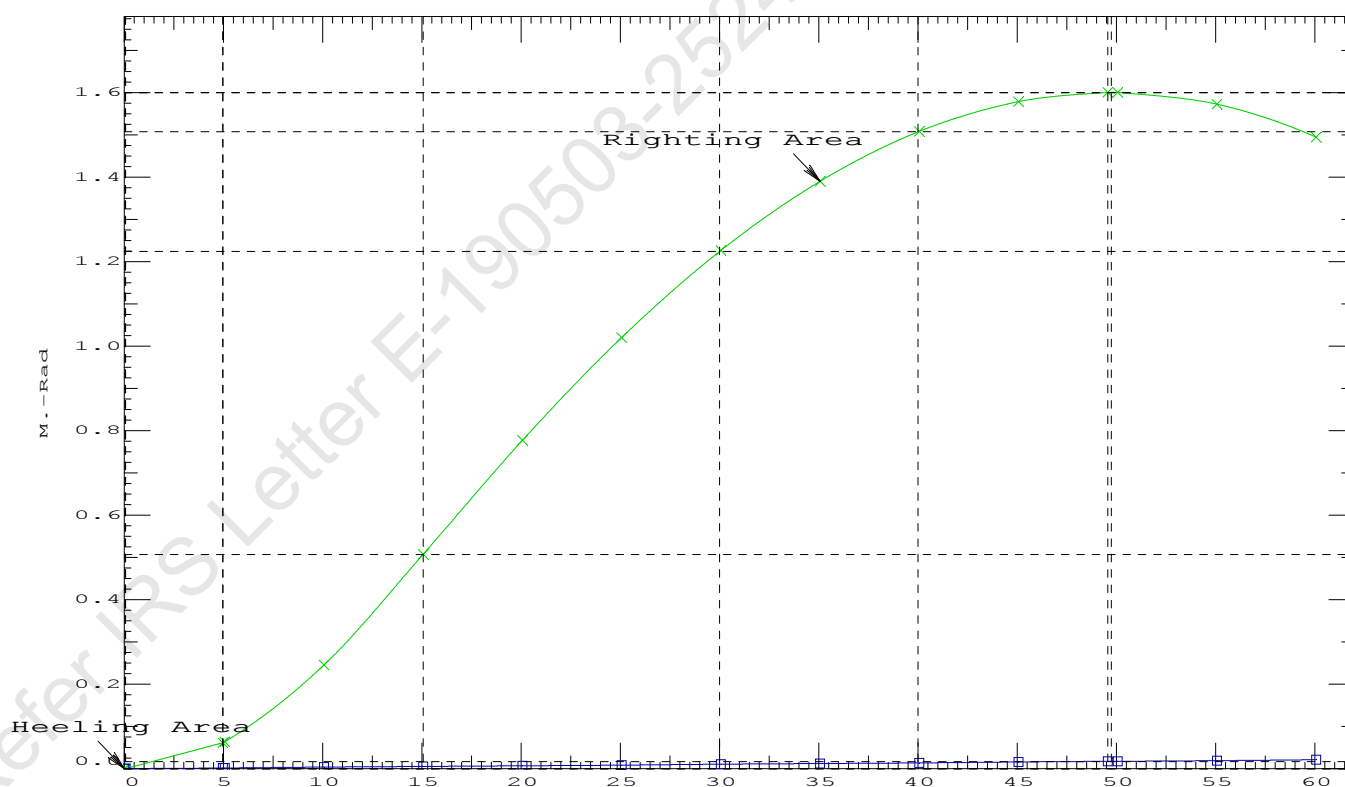
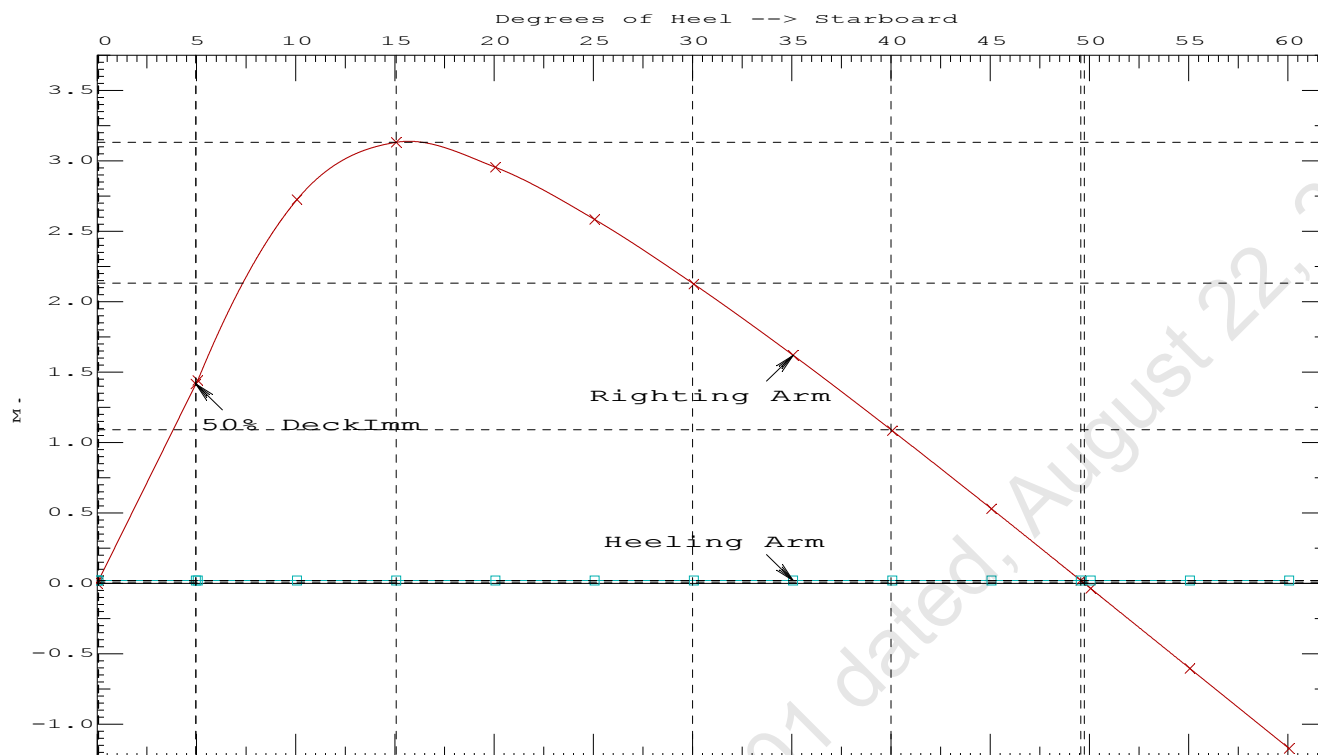
PONTOON**Condition 12 : Operating Condition of 250T Crane**
Crane Lifting 36.6T @ 18m Radius Fwd

LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max-----Attained
(1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad 0.5039 P
(2) Angle from Equilibrium to RAzero or Flood > 20.00 deg 49.50 P
(3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg 4.91 P

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PONTOON

Condition 12 : Operating Condition of 250T CraneCrane Lifting 36.6T @ 18m Radius Fwd

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PONTOON

Condition 12 : Operating Condition of 250T Crane
Crane Lifting 36.6T @ 18m Radius Fwd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.718f TCG = 0.000 VCG = 5.390
Free Surface Adjustment: 0.422
Adjusted CG: LCG = 25.715f TCG = 0.000 VCG = 5.812

Origin	Degrees of		Displacement	Residual Arms		Res.
Depth---	Trim---	Heel---	Weight (MT)---	in Trim--	in Heel---	Area
1.217	0.28f	0.01s	1,115.0	0.000	0.000	0.0000
1.202	0.28f	5.01s	1,114.9	0.000	1.422	0.0620
1.136	0.31f	10.01s	1,115.0	0.000	2.710	0.2433
0.932	0.45f	15.01s	1,115.0	0.000	3.127	0.5053
0.706	0.61f	20.01s	1,115.0	0.000	2.952	0.7748
0.471	0.77f	25.01s	1,115.0	0.000	2.583	1.0178
0.229	0.94f	30.01s	1,115.0	0.000	2.126	1.2239
-0.019	1.10f	35.01s	1,115.0	0.000	1.620	1.3877
-0.269	1.27f	40.01s	1,115.0	0.000	1.084	1.5058
-0.519	1.42f	45.01s	1,115.0	0.000	0.529	1.5763
-0.750	1.57f	49.70s	1,115.0	0.000	0.000	1.5980
-0.766	1.58f	50.01s	1,115.0	0.000	-0.036	1.5979
-1.006	1.71f	55.01s	1,115.0	0.000	-0.604	1.5700
-1.238	1.83f	60.01s	1,115.0	0.000	-1.171	1.4926

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 470.1 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

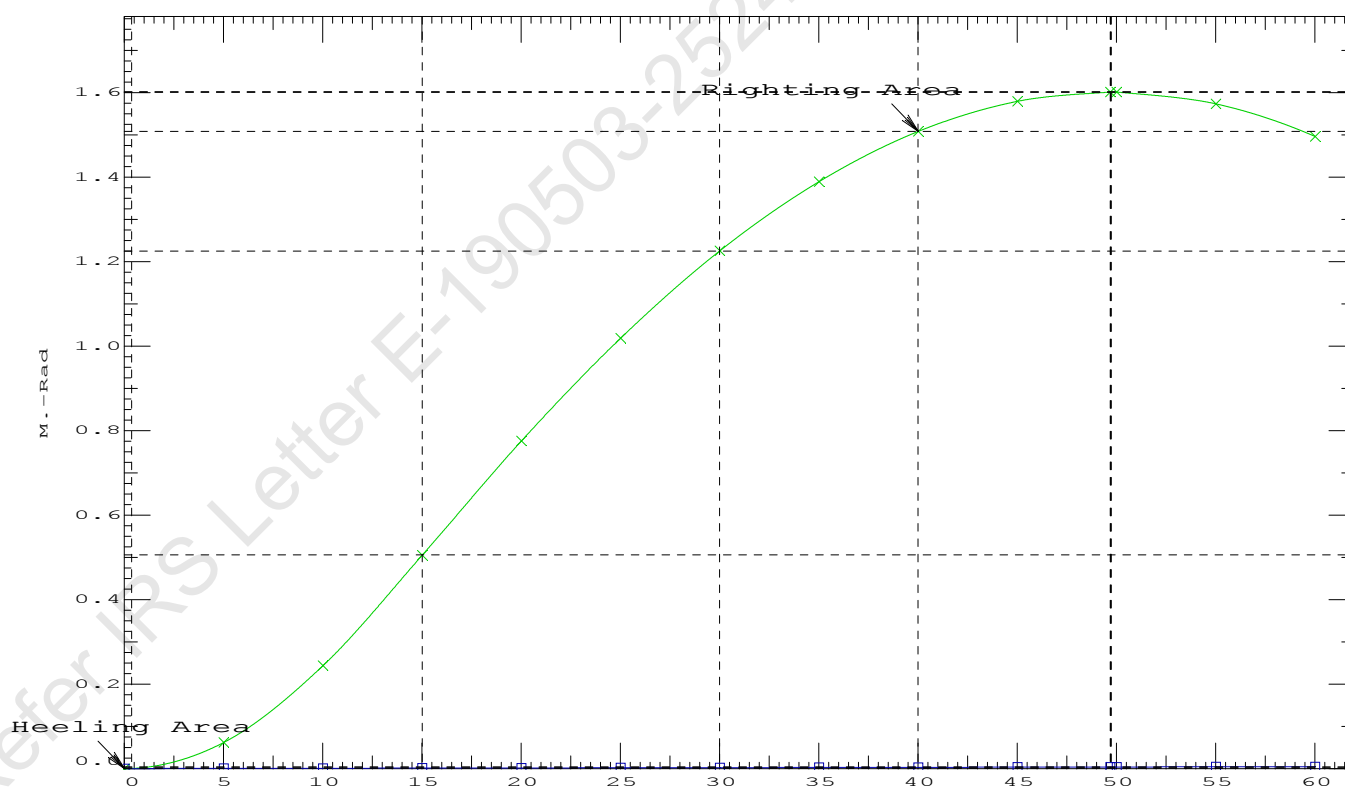
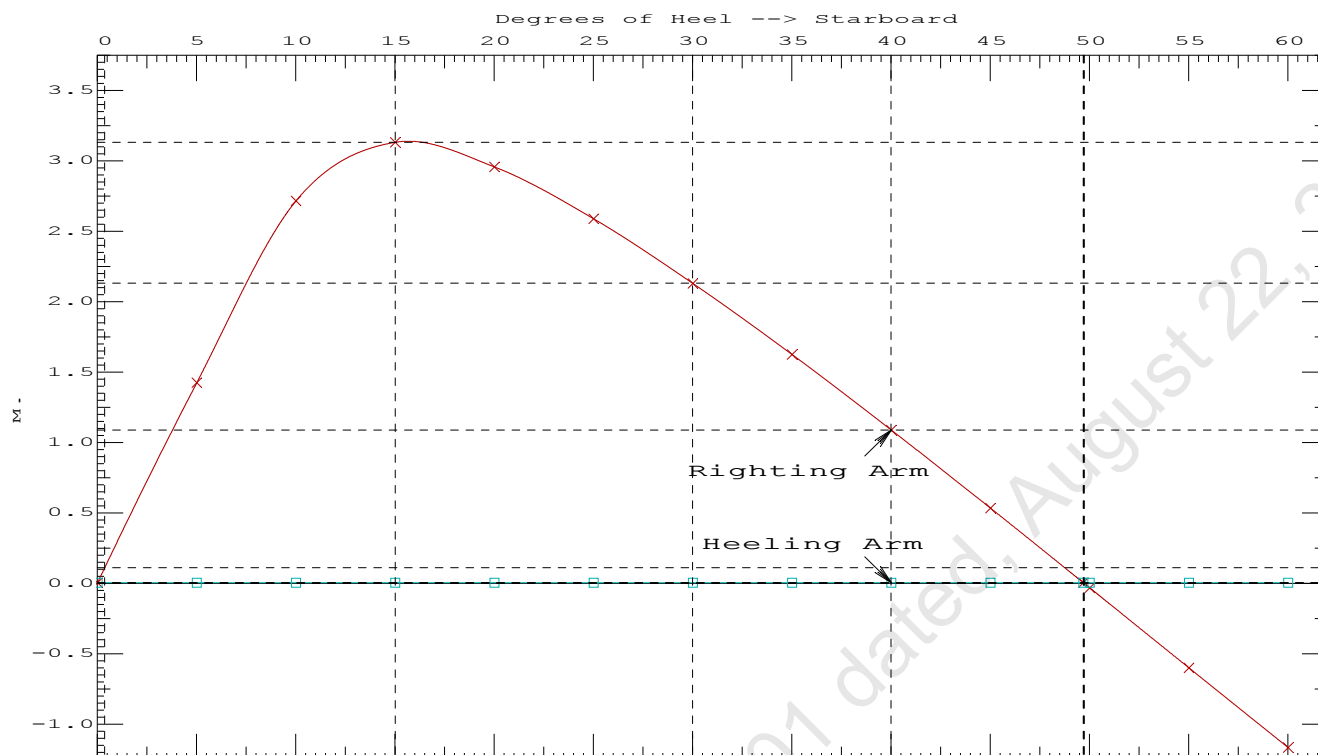
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	0.01 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	69727.5 P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667	411.926 P

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PONTOON

Condition 12 : Operating Condition of 250T CraneCrane Lifting 36.6T @ 18m Radius Fwd

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PONTOON

Condition 12 : Operating Condition of 250T Crane**Crane Lifting 36.6T @ 18m Radius Fwd**

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.477 @ 50.00f, 1.353 @ 25.00f, 1.229 @ 0.00

Trim: Fwd 0.248/50.000, Heel: Stbd 0.01 deg.

Part-----			Weight (MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			250.00	7.500f	0.000	5.200	
Crane Load			36.60	25.500f	0.000	76.977	
Total Fixed----->			692.57	18.709f	0.000	7.704	
	Load-----	SpGr-----	Weight (MT)----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
07_WBTK.P	1.000	1.025	49.60	47.660f	5.971p	1.880	72.32*
07_WBTK.S	1.000	1.025	49.60	47.660f	5.971s	1.880	72.32*
Total Tanks----->			422.40	37.208f	0.000	1.596	470.09
Total Weight----->			1,114.97	25.718f	0.000	5.390	
			Displ (MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,114.97	25.741f	0.005s	0.685	

	Righting Arms:			0.000	0.004s		
	External Arms:			0.000	0.004s		
	Residual Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		855.8	25.287f	0.018s	147.87	20.797
			MT/cm				
			8.77				

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.01

PONTOON

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.432 @ 50.00f, 1.311 @ 25.00f, 1.191 @ 0.00
Trim: Fwd 0.241/50.000, Heel: 0.00 deg.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

```
limmarg: 187583056.00; t3: 30; t: 15.00
limmarg: 70316040.00; t3: 15; t: 7.50
limmarg: 19217014.00; t3: 8; t: 3.75
limmarg: 6199934.00; t3: 4; t: 1.88
limmarg: 1549649.88; t3: 2; t: 0.94
limmarg: 388918.69; t3: 1; t: 0.47
limmarg: 98306.09; t3: 1; t: 0.23
limmarg: 205963.45; t3: 1; t: 0.12
limmarg: 268537.13; t3: 1; t: 0.06
limmarg: 306181.31; t3: 1; t: 0.03
limmarg: 325938.19; t3: 1; t: 0.01
```

Theta3 is 0.97 degrees, as demonstrated below.

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PONTOON

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.330f TCG = 0.000 VCG = 3.838

Origin Depth	Degrees of Trim	Displacement Heel	Weight (MT)	Residual Arms in Trim	Res. in Heel	Area
1.179	0.28f	0.01s	1,078.3	0.000	-0.007	0.0000
1.179	0.28f	0.00	1,078.3	0.000	-0.003	-0.0000
1.179	0.28f	0.00	1,078.3	0.000	0.001	-0.0000
1.177	0.28f	0.97p	1,078.4	0.000	0.338	0.0029
1.165	0.28f	5.00p	1,078.3	0.000	1.727	0.0755
1.094	0.30f	10.00p	1,078.3	0.000	3.290	0.2956
0.877	0.43f	15.00p	1,078.4	0.000	3.965	0.6186
0.699	0.53f	18.73p	1,078.4	0.000	4.045	0.8815
0.637	0.57f	20.00p	1,078.4	0.000	4.036	0.9709
0.389	0.72f	25.00p	1,078.4	0.000	3.898	1.3196
0.135	0.87f	30.00p	1,078.4	0.000	3.659	1.6500

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 655.97 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 1 > 1.000 3260.38 P

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PONTOON

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 18.330f TCG = 0.000 VCG = 3.838

Origin	Degrees of	Displacement	Residual Arms	Res.
Depth---Trim---Heel---	Weight (MT)---	in Trim--in Heel---	Area	
1.179 0.28f 0.01s	1,078.3	0.000 -0.007	0.0000	
1.179 0.28f 0.00	1,078.3	0.000 -0.003	-0.0000	
1.179 0.28f 0.00	1,078.3	0.000 0.001	-0.0000	
1.177 0.28f 0.97p	1,078.4	0.000 0.338	0.0029	
1.165 0.28f 5.00p	1,078.3	0.000 1.727	0.0755	
1.094 0.30f 10.00p	1,078.3	0.000 3.290	0.2956	
0.877 0.43f 15.00p	1,078.4	0.000 3.965	0.6186	
0.699 0.53f 18.73p	1,078.4	0.000 4.045	0.8815	
0.637 0.57f 20.00p	1,078.4	0.000 4.036	0.9709	
0.389 0.72f 25.00p	1,078.4	0.000 3.898	1.3196	
0.135 0.87f 30.00p	1,078.4	0.000 3.659	1.6500	
-0.122 1.01f 35.00p	1,078.4	0.000 3.359	1.9567	
-0.382 1.16f 40.00p	1,078.4	0.000 3.015	2.2352	
-0.640 1.30f 45.00p	1,078.4	0.000 2.637	2.4821	
-0.896 1.44f 50.00p	1,078.4	0.000 2.233	2.6948	
-1.146 1.57f 55.00p	1,078.4	0.000 1.807	2.8712	
-1.388 1.68f 60.00p	1,078.4	0.000 1.365	3.0098	
-1.618 1.79f 65.00p	1,078.4	0.000 0.911	3.1092	
-1.835 1.88f 70.00p	1,078.4	0.000 0.448	3.1685	
-2.028 1.94f 74.80p	1,078.4	0.000 0.000	3.1873	
-2.036 1.94f 75.00p	1,078.4	0.000 -0.019	3.1873	

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

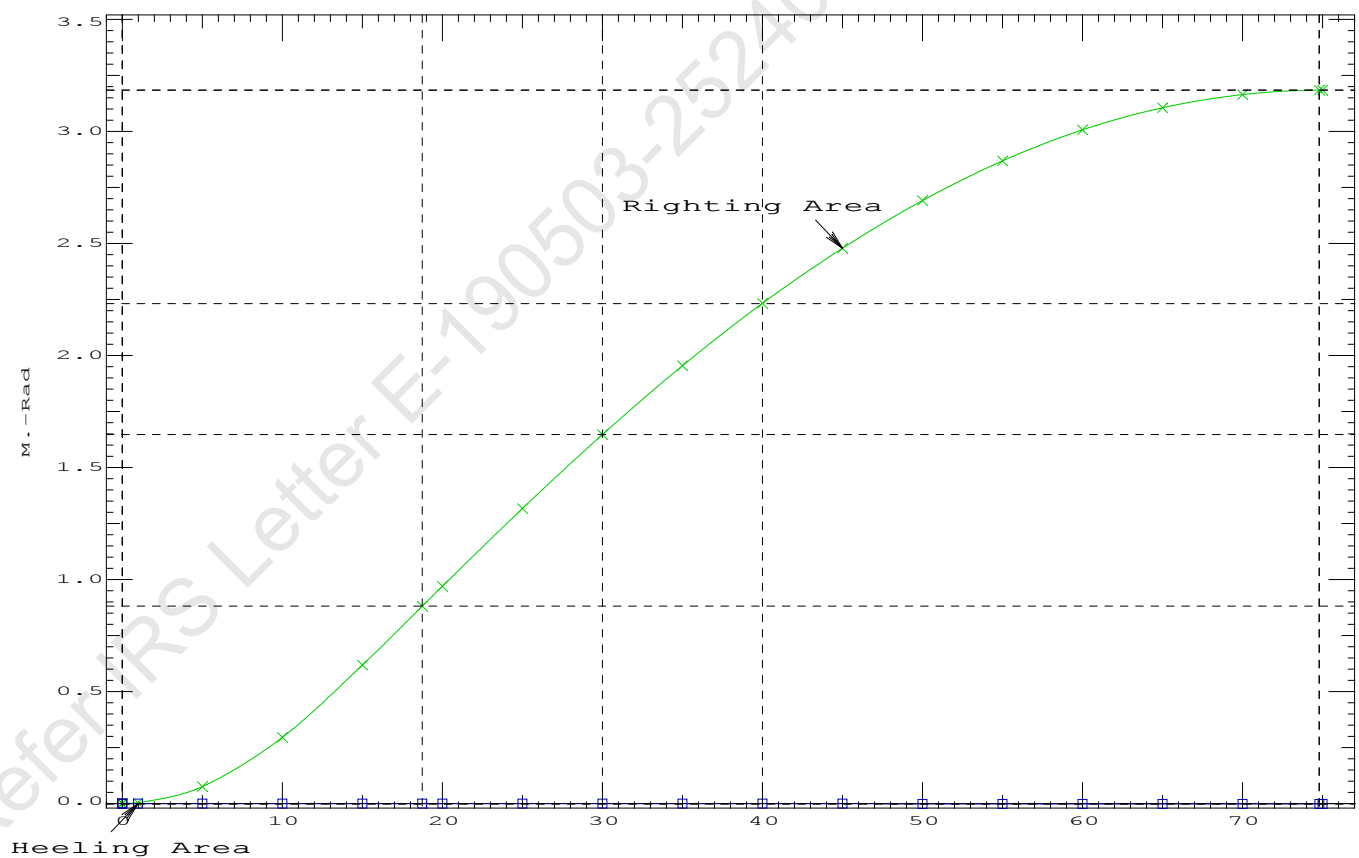
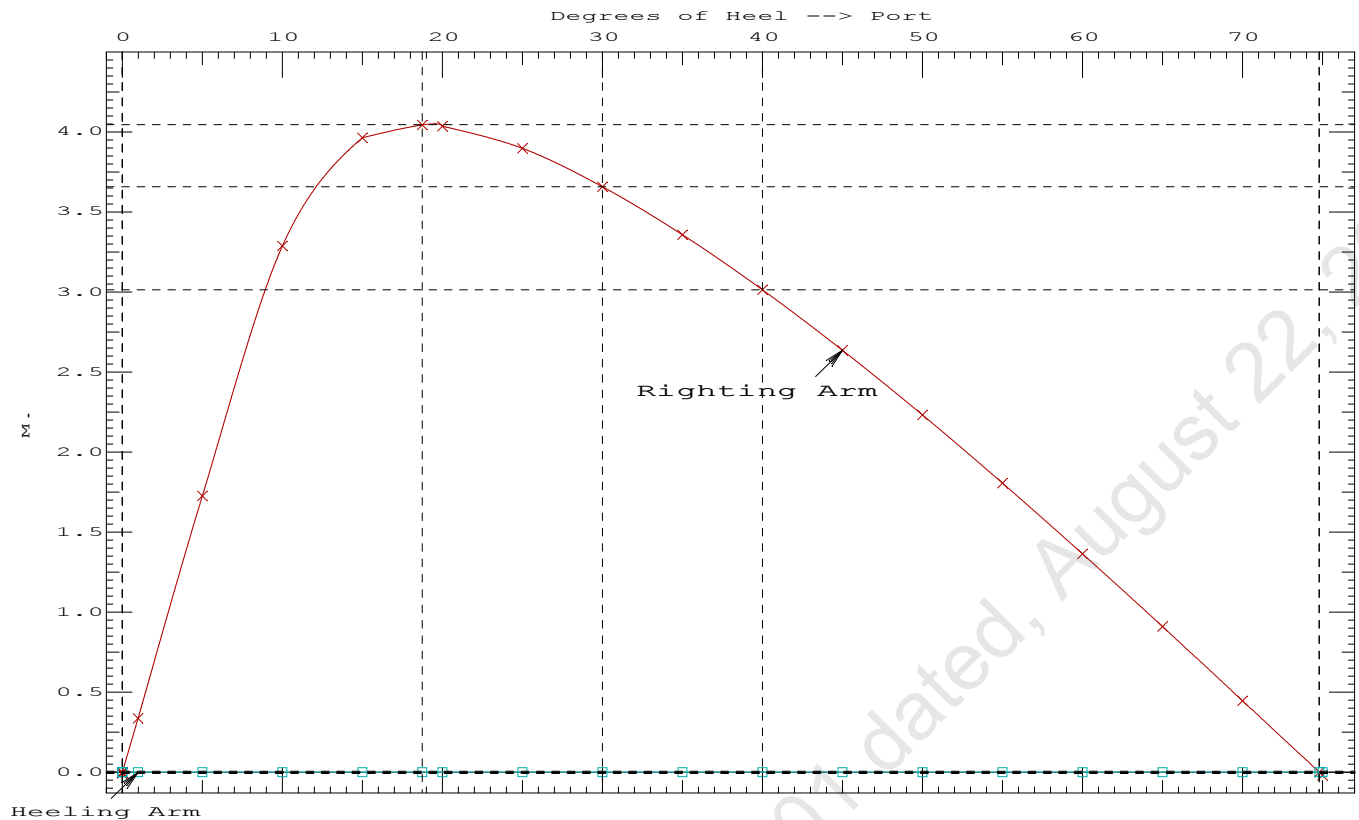
Note: The Center of Gravity shown above is for the Fixed Weight of 655.97 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAzero > 1.000 3623723 P
(2) Angle from abs 1 deg to RAzero > 20.00 deg 73.83 P

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PONTOON



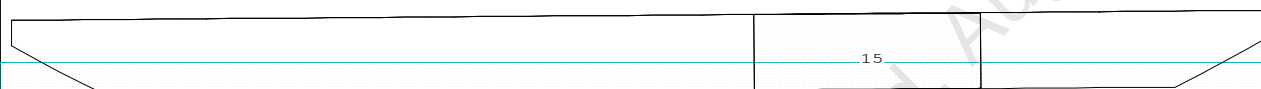
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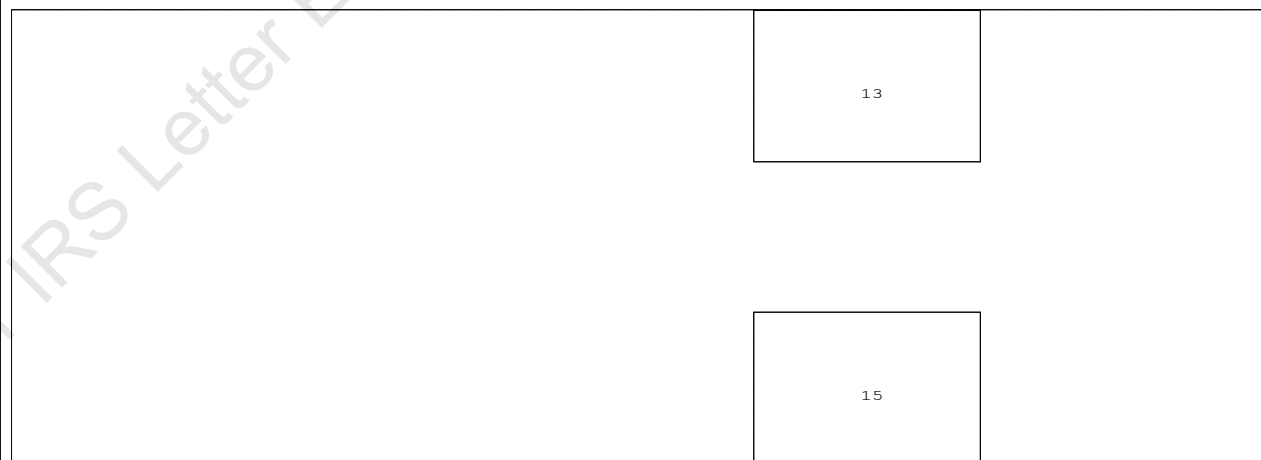
Condition 13 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Aft

Condition Graphic - Draft: 0.942 @ 50.000f, 1.340 @ 0.000 Heel: zero

Profile View



Plan View



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PONTOON

Condition 13 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Aft

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 0.942 @ 50.00f, 1.141 @ 25.00f, 1.340 @ 0.00

Trim: Aft 0.398/50.000, Heel: zero

Part-----	Weight (MT)	LCG	TCG	VCG			
LIGHT SHIP	405.97	25.000f	0.000	3.000			
Crane	150.00	7.500f	0.000	5.200			
Crane Load	51.30	4.500a	0.000	33.218			
Total Fixed----->	607.27	18.185f	0.000	6.096			
Load-----	SpGr-----	Weight (MT)	LCG	TCG	VCG	FSM	
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
Total Tanks----->			323.20	34.000f	0.000	1.509	325.46
Total Weight----->			930.47	23.679f	0.000	4.503	
			Displ (MT)	LCB	TCB	VCB	
HULL	1.025		930.47	23.646f	0.000	0.579	

Righting Arms:			0.001	0.000			
Part-----	SpGr-----	WPA	LCF	TCF	BML	BMT	
Total Waterplane---->	1.025	844.5	24.662f	0.000	170.28	24.799	
		MT/cm	m.	MT/cm	GML	GMT	
		8.66		30.96	166.36	20.875	
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

315.3 Cu.M. (13%) SALT WATER

150.00 MT Crane

51.30 MT Crane Load

DRAFT AT FP (M) = 0.942
DRAFT AT MS (M) = 1.141
DRAFT AT AP (M) = 1.340
DRAFT AT FWD MARK (M) = 0.973
DRAFT AT AFT MARK (M) = 1.308

HYDROSTATIC PROPERTIES

Trim: Aft 0.398/50.000, No Heel, VCG = 4.503

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/
Draft---	Weight (MT)	LCB	VCB	cm trim---
1.143	930.47	23.646f	0.579	8.66
				24.662f
				30.96
				166.36
				20.875

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 325.5 m.-MT

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PONTOON

Condition 13 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Aft

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 0.942 @ 50.00f, 1.141 @ 25.00f, 1.340 @ 0.00

Trim: Aft 0.398/50.000, Heel: zero

Part-----	LPA*Sf-----	HCP-----	Arm----	Pressure-----	Moment
HULL	92.6	0.954	1.520	0.05500	7.74
CRANE	57.0	4.252	4.817	0.05500	15.10
Total wind heeling moment to starboard----->					22.84
Distances in METERS.-----Pressure in MT/Sqm.-----					Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 23.679f TCG = 0.000 VCG = 4.503

Free Surface Adjustment: 0.350

Adjusted CG: LCG = 23.681f TCG = 0.000 VCG = 4.853

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---Trim---Heel---	Weight (MT)---	in Trim--in Heel---	Area----	Angle
1.328 0.46a 0.00	930.46	0.000 -0.024	0.0000	5.509
1.328 0.46a 0.05s	930.47	0.000 0.000	-0.0000	5.457
1.313 0.46a 5.05s	930.46	0.000 1.817	0.0793	0.457
1.310 0.46a 5.51s	930.41	0.000 1.978	0.0944	0.000
1.259 0.53a 10.05s	930.33	0.000 3.277	0.3048	-4.543
1.099 0.68a 15.05s	930.47	0.000 3.792	0.6187	-9.543
1.044 0.76a 16.79s	930.47	0.000 3.816	0.7340	-11.277
0.940 0.92a 20.05s	930.47	0.000 3.740	0.9504	-14.543
0.779 1.15a 25.05s	930.47	0.000 3.456	1.2669	-19.543
0.617 1.39a 30.05s	930.47	0.000 3.063	1.5521	-24.543
0.454 1.63a 35.05s	930.47	0.000 2.608	1.8000	-29.543
0.292 1.87a 40.05s	930.47	0.000 2.112	2.0063	-34.543
0.130 2.10a 45.05s	930.47	0.000 1.588	2.1679	-39.543
-0.030 2.31a 50.05s	930.47	0.000 1.044	2.2829	-44.543
-0.189 2.51a 55.05s	930.47	0.000 0.487	2.3498	-49.543
-0.327 2.67a 59.38s	930.47	0.000 0.000	2.3682	-53.869
-0.348 2.69a 60.05s	930.47	0.000 -0.076	2.3678	-54.543
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.				

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 325.5 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 22.84 (constant)

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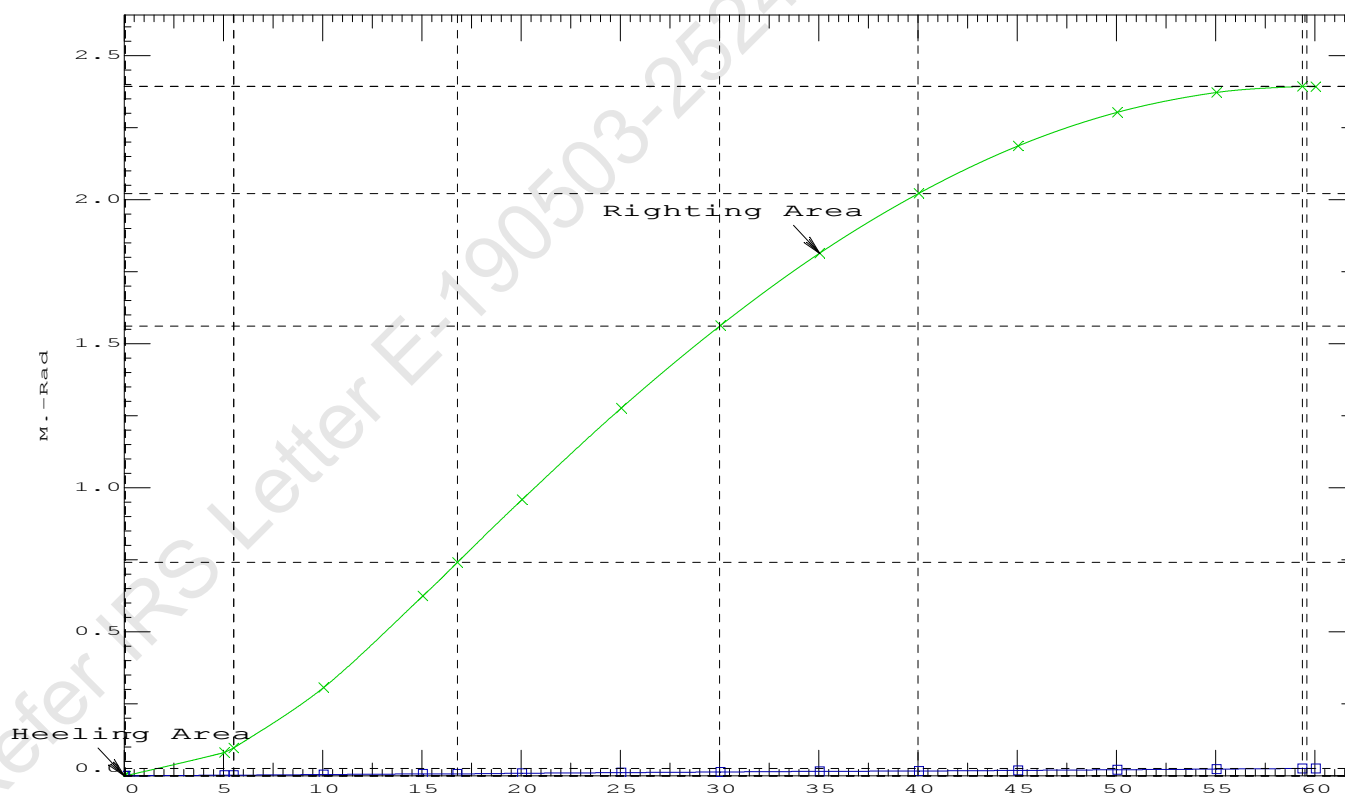
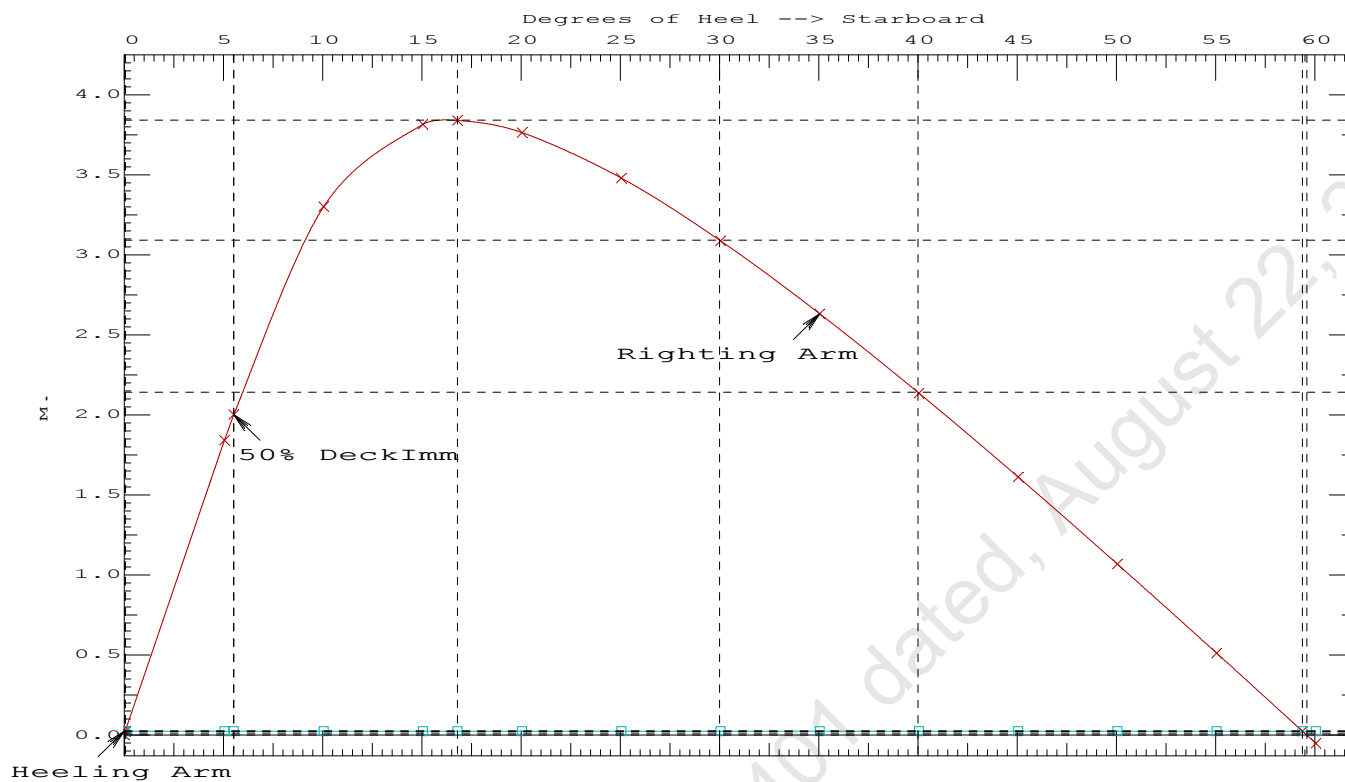
PONTOON**Condition 13 : Operating Condition of 150T Crane**
Crane Lifting 51.3T @ 12m Radius Aft

LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max-----Attained
(1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad 0.7340 P
(2) Angle from Equilibrium to RAzero or Flood > 20.00 deg 59.33 P
(3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg 5.46 P

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PONTOON

Condition 13 : Operating Condition of 150T CraneCrane Lifting 51.3T @ 12m Radius Aft

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PONTOON

Condition 13 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Aft

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 23.679f TCG = 0.000 VCG = 4.503
Free Surface Adjustment: 0.350
Adjusted CG: LCG = 23.681f TCG = 0.000 VCG = 4.853

Origin	Degrees of		Displacement	Residual Arms		Res.
Depth---	Trim---	Heel---	Weight (MT)---	in Trim--	in Heel---	Area
1.328	0.46a	0.00	930.46	0.000	0.002	0.0000
1.313	0.46a	5.00s	930.46	0.000	1.818	0.0794
1.260	0.53a	10.00s	930.33	0.000	3.286	0.3046
1.100	0.68a	15.00s	930.47	0.000	3.809	0.6199
1.044	0.76a	16.79s	930.47	0.000	3.835	0.7394
0.942	0.91a	20.00s	930.47	0.000	3.761	0.9533
0.781	1.15a	25.00s	930.47	0.000	3.478	1.2717
0.618	1.39a	30.00s	930.47	0.000	3.086	1.5589
0.456	1.63a	35.00s	930.47	0.000	2.632	1.8089
0.293	1.87a	40.00s	930.47	0.000	2.136	2.0172
0.132	2.10a	45.00s	930.47	0.000	1.612	2.1810
-0.028	2.31a	50.00s	930.47	0.000	1.069	2.2981
-0.188	2.51a	55.00s	930.47	0.000	0.512	2.3671
-0.332	2.68a	59.55s	930.47	0.000	0.000	2.3875
-0.347	2.69a	60.00s	930.47	0.000	-0.051	2.3873

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 325.5 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 5.00

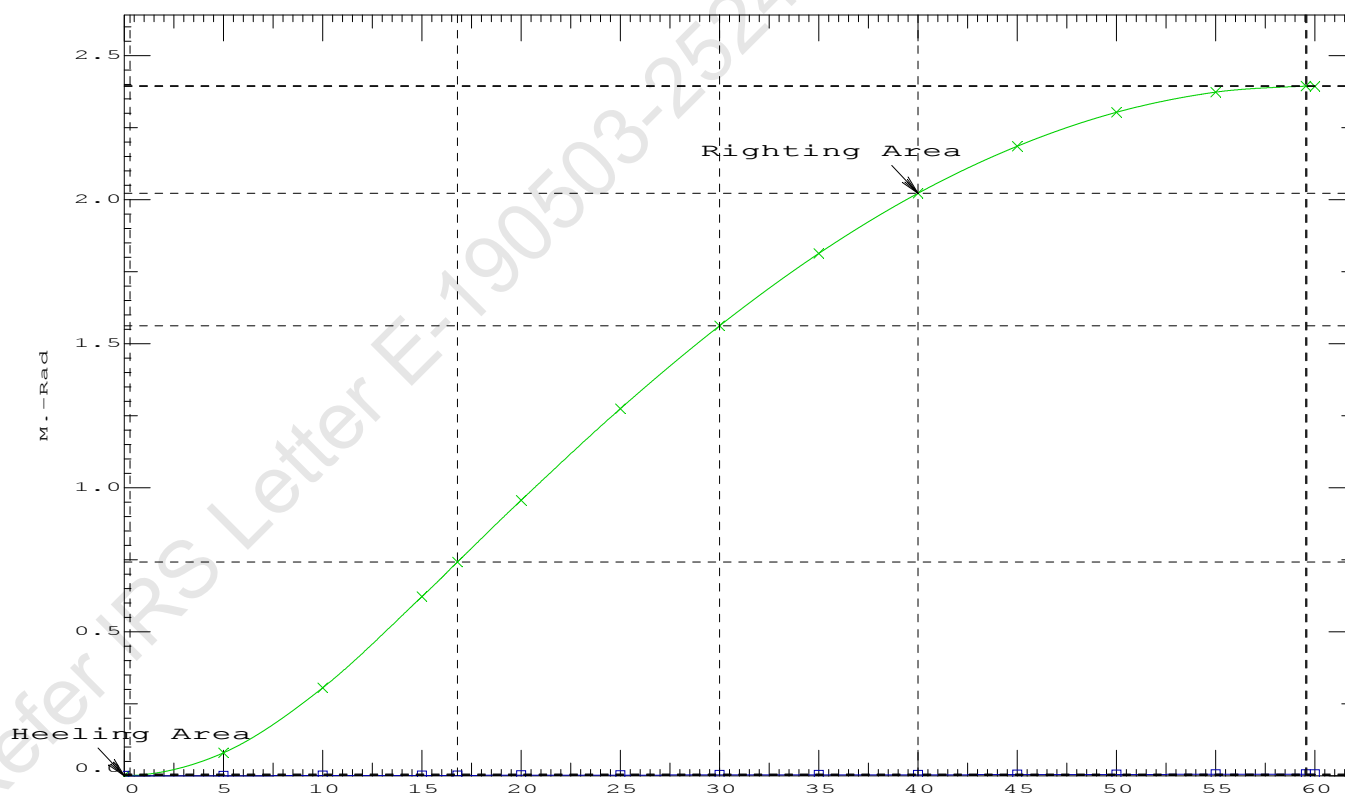
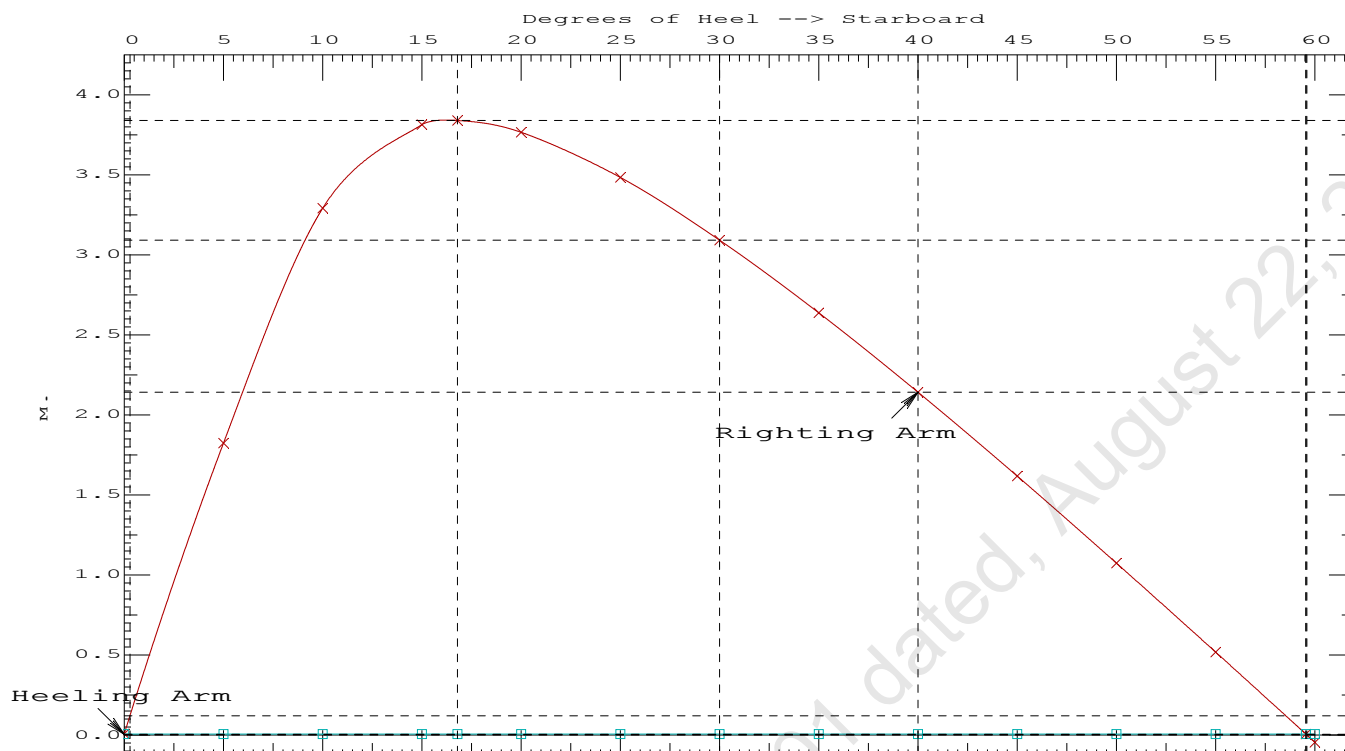
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	0.00 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	71366.5 P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667	428.504 P

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PONTOON

Condition 13 : Operating Condition of 150T CraneCrane Lifting 51.3T @ 12m Radius Aft

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PONTOON**Condition 14 : Operating Condition of 150T Crane**

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 0.942 @ 50.00f, 1.141 @ 25.00f, 1.340 @ 0.00

Trim: Aft 0.398/50.000, Heel: 0.00 deg.

Part-----			Weight (MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			150.00	7.500f	0.000	5.200	
Crane Load			51.30	4.500a	0.000	33.218	
Total Fixed----->			607.27	18.185f	0.000	6.096	
	Load-----	SpGr-----	Weight (MT)----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
Total Tanks----->			323.20	34.000f	0.000	1.509	325.46
Total Weight----->			930.47	23.679f	0.000	4.503	
			Displ (MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	930.46	23.647f	0.007s	0.579	

	Righting Arms:			0.000	0.007s		
	External Arms:			0.000	0.001s		
	Residual Righting Arms:			0.000	0.006s		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		832.2	24.559f	0.075s	163.01	24.461
			MT/cm				
			8.53				
Distances in METERS.-----				-----Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

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PONTOON

After hook load drop
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.129 @ 50.00f, 1.083 @ 25.00f, 1.036 @ 0.00
Trim: Fwd 0.092/50.000, Heel: 0.00 deg.

Part	Weight (MT)	LCG	TCG	VCG
LIGHT SHIP	405.97	25.000f	0.000	3.000
Crane	150.00	7.500f	0.000	5.200
Total Fixed	555.97	20.279f	0.000	3.594

Load	SpGr	Weight (MT)	LCG	TCG	VCG	FSM
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509
Total Tanks			323.20	34.000f	0.000	1.509
Total Weight			879.17	25.323f	0.000	2.827

HULL	Displ (MT)	LCB	TCB	VCB
1.025	879.17	25.327f	0.001s	0.543

Righting Arms:	0.000	0.001s
External Arms:	0.000	0.001s
Residual Righting Arms:	0.000	0.000

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane	1.025	838.7	25.061f	0.009s	176.53	25.992

MT/cm	m.	MT/cm	GML	GMT
8.60		30.64	174.24	23.707

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: INFINITY; t3: 30.0; t: 15.00
limmarg: INFINITY; t3: 15.0; t: 7.50
limmarg: INFINITY; t3: 7.5; t: 3.75
limmarg: INFINITY; t3: 3.8; t: 1.88
limmarg: INFINITY; t3: 1.9; t: 0.94
limmarg: INFINITY; t3: 1.0; t: 0.47
limmarg: INFINITY; t3: 0.5; t: 0.23
limmarg: INFINITY; t3: 0.3; t: 0.12
limmarg: INFINITY; t3: 0.2; t: 0.06
limmarg: INFINITY; t3: 0.1; t: 0.03
limmarg: INFINITY; t3: 0.1; t: 0.01

Theta3 is 0.07 degrees, as demonstrated below.

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 20.279f TCG = 0.000 VCG = 3.594

Origin	Degrees of	Displacement	Residual Arms	Res.
Depth	Trim	Heel	Weight (MT)	in Trim--in Heel--> Area
1.024	0.11f	0.00	879.17	0.000 0.000 0.0000
1.024	0.11f	0.07s	879.16	0.000 0.029 0.0000
1.009	0.11f	5.00s	879.18	0.000 2.110 0.0921
0.910	0.12f	10.00s	879.16	0.000 3.781 0.3523
0.646	0.15f	15.00s	879.17	0.000 4.482 0.7199
0.364	0.20f	19.52s	879.17	0.000 4.622 1.0817
0.334	0.21f	20.00s	879.17	0.000 4.621 1.1208

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0.018	0.26f	25.00s	879.17	0.000	4.506	1.5216
-0.299	0.31f	30.00s	879.17	0.000	4.269	1.9053

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 555.97 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 0.1 > 1.000 LARGE

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PONTOON

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

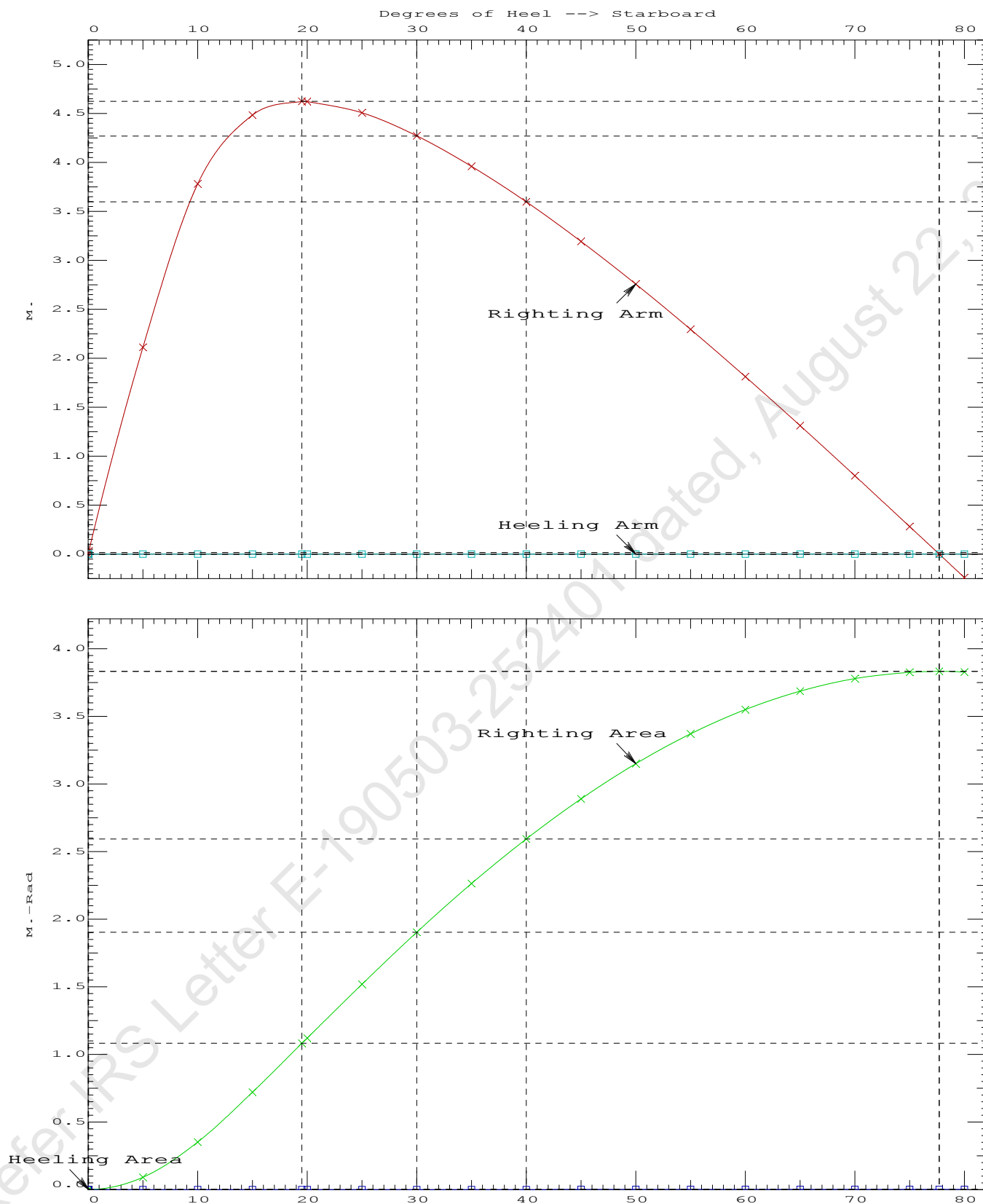
Fixed CG: LCG = 20.279f TCG = 0.000 VCG = 3.594

Origin	Degrees of		Displacement	Residual Arms		Res.
Depth---	Trim---	Heel---	Weight (MT)---	in Trim--	in Heel---	Area
1.024	0.11f	0.00	879.17	0.000	0.000	0.0000
1.024	0.11f	0.07s	879.16	0.000	0.029	0.0000
1.009	0.11f	5.00s	879.18	0.000	2.110	0.0921
0.910	0.12f	10.00s	879.16	0.000	3.781	0.3523
0.646	0.15f	15.00s	879.17	0.000	4.482	0.7199
0.364	0.20f	19.52s	879.17	0.000	4.622	1.0817
0.334	0.21f	20.00s	879.17	0.000	4.621	1.1208
0.018	0.26f	25.00s	879.17	0.000	4.506	1.5216
-0.299	0.31f	30.00s	879.17	0.000	4.269	1.9053
-0.614	0.36f	35.00s	879.17	0.000	3.959	2.2649
-0.926	0.41f	40.00s	879.17	0.000	3.596	2.5949
-1.231	0.46f	45.00s	879.17	0.000	3.193	2.8914
-1.528	0.51f	50.00s	879.17	0.000	2.757	3.1512
-1.814	0.56f	55.00s	879.17	0.000	2.294	3.3718
-2.085	0.60f	60.00s	879.17	0.000	1.811	3.5511
-2.341	0.63f	65.00s	879.17	0.000	1.311	3.6874
-2.579	0.67f	70.00s	879.17	0.000	0.800	3.7796
-2.796	0.69f	75.00s	879.17	0.000	0.281	3.8268
-2.904	0.70f	77.69s	879.17	0.000	0.000	3.8334
-2.991	0.71f	80.00s	879.17	0.000	-0.241	3.8286
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.						

Note: The Center of Gravity shown above is for the Fixed Weight of 555.97 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAzero > 1.000 LARGE
(2) Angle from abs 0.1 deg to RAzero > 20.00 deg 77.62 P

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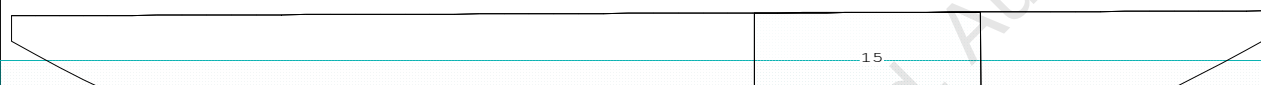
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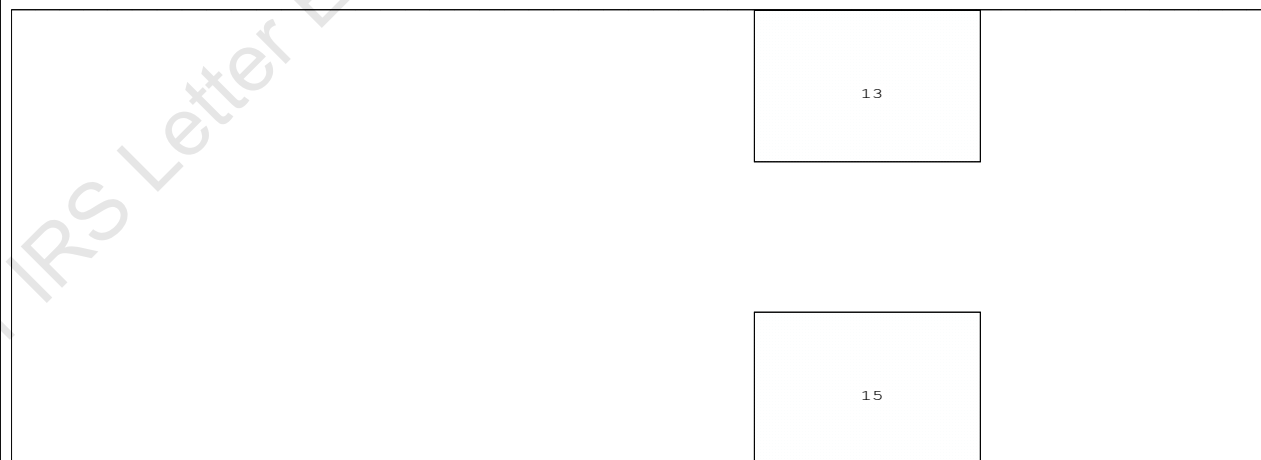
Condition 14 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Stbd

Condition Graphic - Draft: 1.042 @ 50.000f, 1.241 @ 0.000 Heel: zero

Profile View



Plan View



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PONTOON

Condition 14 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Stbd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.042 @ 50.00f, 1.142 @ 25.00f, 1.241 @ 0.00

Trim: Aft 0.199/50.000, Heel: zero

Part-----	Weight (MT)	LCG	TCG	VCG		
LIGHT SHIP	405.97	25.000f	0.000	3.000		
Crane	150.00	7.500f	0.000	5.200		
Crane Load	51.30	7.500f	0.000	33.218		
Total Fixed----->	607.27	19.199f	0.000	6.096		
Load-----	SpGr	Weight (MT)	LCG	TCG	VCG	FSM
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509
Total Tanks----->			323.20	34.000f	0.000	1.509
Total Weight----->			930.47	24.340f	0.000	4.503
		Displ (MT)	LCB	TCB	VCB	
HULL	1.025	930.43	24.325f	0.000	0.575	

Righting Arms:			0.000	0.000		
Part-----	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	843.8	24.826f	0.000	169.86	24.743
		MT/cm	m.	MT/cm	GML	GMT
		8.65		30.88	165.93	20.815
Distances in METERS.-----			Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

315.3 Cu.M. (13%) SALT WATER

150.00 MT Crane

51.30 MT Crane Load

DRAFT AT FP (M) = 1.042
DRAFT AT MS (M) = 1.142
DRAFT AT AP (M) = 1.241
DRAFT AT FWD MARK (M) = 1.058
DRAFT AT AFT MARK (M) = 1.225

HYDROSTATIC PROPERTIES

Trim: Aft 0.199/50.000, No Heel, VCG = 4.503

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/				
Draft---	Weight (MT)	LCB	VCB	cm	LCF	cm trim	GML	GMT
1.142	930.43	24.325f	0.575	8.65	24.826f	30.88	165.93	20.815
Distances in METERS.-----				Specific Gravity = 1.025.-----				
				Moment in m.-MT.				
				Trim is per 50.00m.				

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 325.5 m.-MT

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PONTOON

Condition 14 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Stbd

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 1.042 @ 50.00f, 1.142 @ 25.00f, 1.241 @ 0.00

Trim: Aft 0.199/50.000, Heel: zero

Part-----	LPA*Sf-----	HCP-----	Arm----	Pressure-----	Moment
HULL	92.6	0.952	1.513	0.05500	7.70
CRANE	57.0	4.311	4.872	0.05500	15.27
Total wind heeling moment to starboard----->					22.97
Distances in METERS.-----Pressure in MT/Sqm.-----					Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.340f TCG = 0.000 VCG = 4.503

Free Surface Adjustment: 0.350

Adjusted CG: LCG = 24.342f TCG = 0.000 VCG = 4.853

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---Trim---Heel---	Weight (MT)---	in Trim--in Heel---	Area----	Angle
1.229 0.23a 0.00	930.46	0.000 -0.024	0.0000	5.973
1.229 0.23a 0.06s	930.47	0.000 0.000	-0.0000	5.909
1.214 0.23a 5.06s	930.47	0.000 1.823	0.0795	0.909
1.208 0.23a 5.97s	930.44	0.000 2.141	0.1110	0.000
1.147 0.26a 10.06s	930.44	0.000 3.290	0.3065	-4.091
0.949 0.34a 15.06s	930.47	0.000 3.815	0.6219	-9.090
0.877 0.38a 16.79s	930.47	0.000 3.839	0.7375	-10.818
0.740 0.46a 20.06s	930.47	0.000 3.763	0.9557	-14.090
0.527 0.58a 25.06s	930.47	0.000 3.478	1.2742	-19.090
0.312 0.70a 30.06s	930.47	0.000 3.086	1.5614	-24.091
0.097 0.82a 35.06s	930.47	0.000 2.630	1.8113	-29.091
-0.117 0.93a 40.06s	930.47	0.000 2.134	2.0194	-34.091
-0.328 1.05a 45.06s	930.47	0.000 1.609	2.1829	-39.091
-0.536 1.16a 50.06s	930.47	0.000 1.063	2.2996	-44.091
-0.739 1.26a 55.06s	930.47	0.000 0.505	2.3682	-49.091
-0.916 1.34a 59.53s	930.47	0.000 0.000	2.3878	-53.552
-0.937 1.35a 60.06s	930.47	0.000 -0.061	2.3876	-54.091
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.				

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 325.5 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 22.97 (constant)

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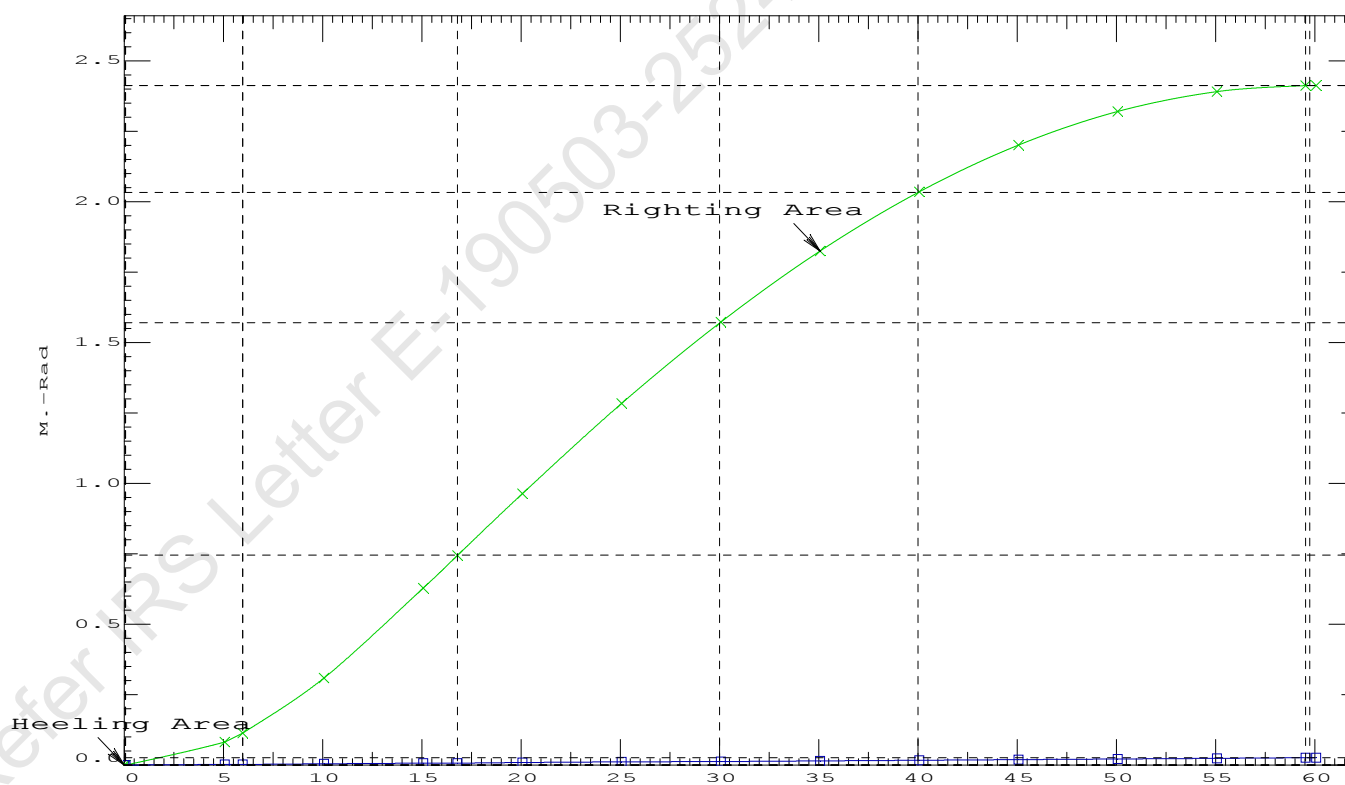
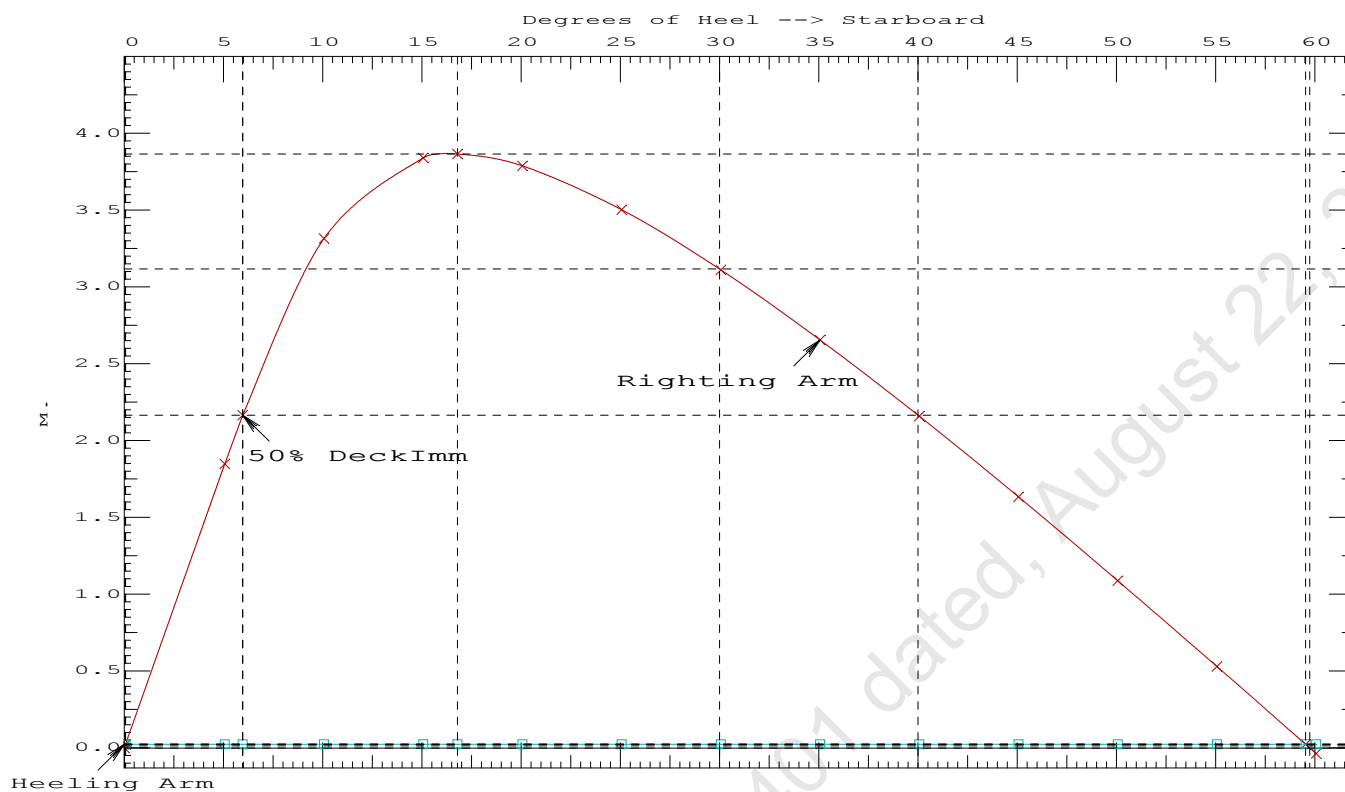
PONTOON**Condition 14 : Operating Condition of 150T Crane**
Crane Lifting 51.3T @ 12m Radius Stbd

LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max-----Attained
(1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad 0.7375 P
(2) Angle from Equilibrium to RAzero or Flood > 20.00 deg 59.46 P
(3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg 5.91 P

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PONTOON

Condition 14 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Stbd



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PONTOON

Condition 14 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.340f TCG = 0.000 VCG = 4.503
Free Surface Adjustment: 0.350
Adjusted CG: LCG = 24.342f TCG = 0.011p VCG = 4.853

Origin	Degrees of		Displacement	Residual Arms		Res.
Depth---	Trim---	Heel---	Weight (MT)---	in Trim--	in Heel---	Area
1.225	0.23a	1.77s	930.47	0.000	0.000	0.0000
1.202	0.23a	6.77s	930.47	0.000	1.786	0.0779
1.087	0.28a	11.77s	930.45	0.000	2.926	0.2882
0.878	0.38a	16.77s	930.47	0.000	3.212	0.5610
0.667	0.50a	21.77s	930.47	0.000	3.056	0.8377
0.454	0.62a	26.77s	930.47	0.000	2.726	1.0913
0.239	0.74a	31.77s	930.47	0.000	2.308	1.3116
0.024	0.86a	36.77s	930.47	0.000	1.837	1.4928
-0.189	0.97a	41.77s	930.47	0.000	1.329	1.6312
-0.399	1.09a	46.77s	930.47	0.000	0.795	1.7240
-0.605	1.19a	51.77s	930.47	0.000	0.244	1.7695
-0.694	1.24a	53.94s	930.47	0.000	0.000	1.7741
-0.807	1.29a	56.77s	930.47	0.000	-0.319	1.7663
-1.003	1.38a	61.77s	930.47	0.000	-0.886	1.7138

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 325.5 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 615.60

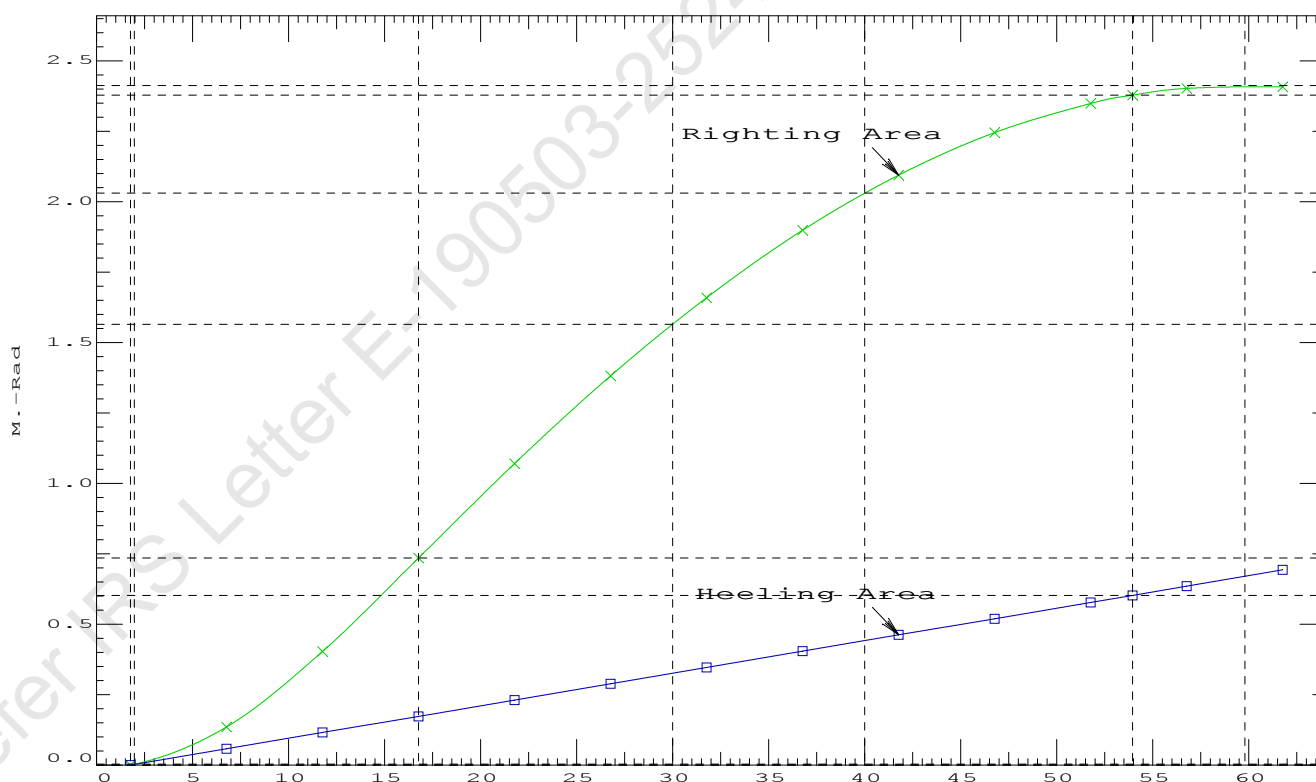
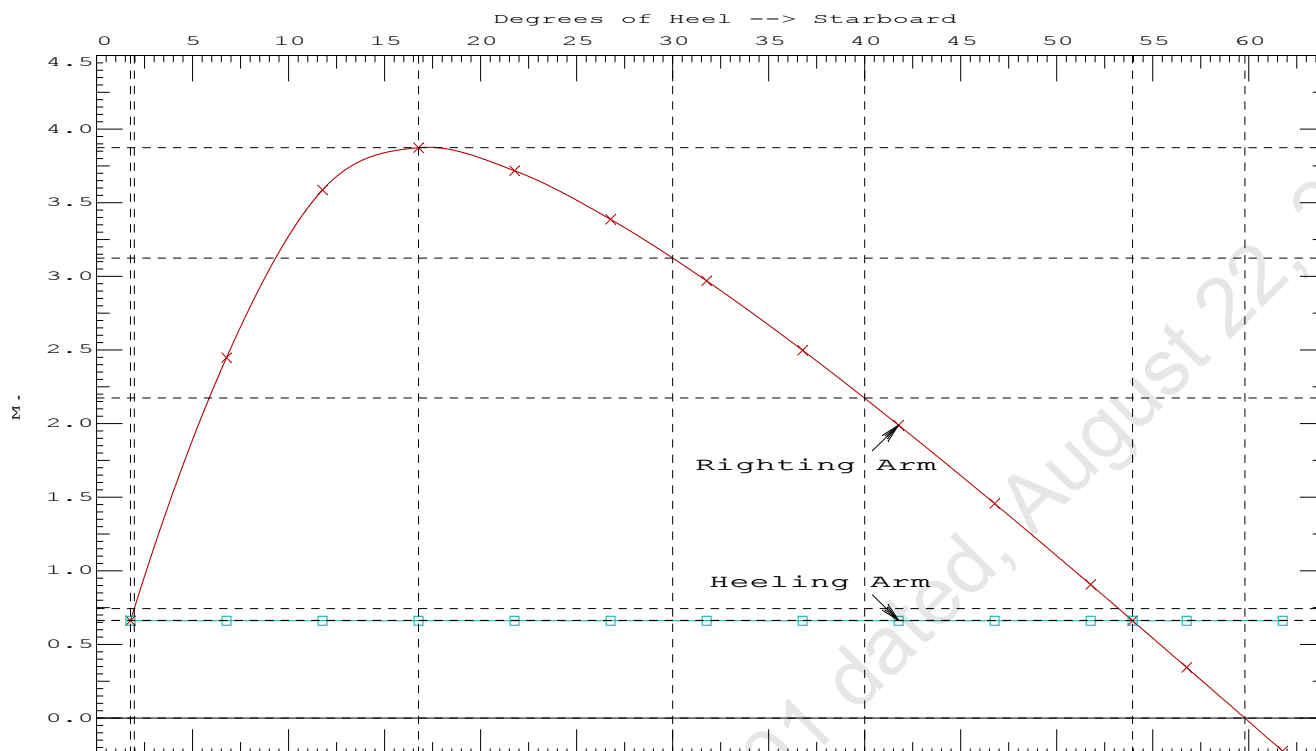
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	1.77 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	485.6 P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667	3.945 P

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PONTOON

Condition 14 : Operating Condition of 150T CraneCrane Lifting 51.3T @ 12m Radius Stbd

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Condition 14 : Operating Condition of 150T Crane**Crane Lifting 51.3T @ 12m Radius Stbd**

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.041 @ 50.00f, 1.140 @ 25.00f, 1.238 @ 0.00

Trim: Aft 0.196/50.000, Heel: Stbd 1.77 deg.

Part-----			Weight (MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			150.00	7.500f	0.000	5.200	
Crane Load			51.30	7.500f	0.000	33.218	
Total Fixed----->			607.27	19.199f	0.000	6.096	
	Load-----	SpGr-----	Weight (MT)----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
Total Tanks----->			323.20	34.000f	0.000	1.509	325.46
Total Weight----->			930.47	24.340f	0.000	4.503	
			Displ (MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	930.58	24.325f	0.783s	0.587	

	Righting Arms:			0.001	0.662s		
	External Arms:			0.000	0.662s		
	Residual Righting Arms:			0.001	0.000s		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		843.5	24.821f	0.103s	169.63	24.747
			MT/cm				
			8.65				

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 1.77

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PONTOON

After hook load drop
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.129 @ 50.00f, 1.083 @ 25.00f, 1.036 @ 0.00
Trim: Fwd 0.092/50.000, Heel: 0.00 deg.

Part-----	Weight (MT)----	LCG-----	TCG-----	VCG-----			
LIGHT SHIP	405.97	25.000f	0.000	3.000			
Crane	150.00	7.500f	0.000	5.200			
Total Fixed----->	555.97	20.279f	0.000	3.594			
Load-----	SpGr-----	Weight (MT)----	LCG-----	TCG-----	VCG-----	FSM-----	
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
Total Tanks----->			323.20	34.000f	0.000	1.509	325.46
Total Weight----->			879.17	25.323f	0.000	2.827	
			Displ (MT)----	LCB-----	TCB-----	VCB-----	
HULL	1.025		879.17	25.327f	0.001s	0.543	

Righting Arms:			0.000	0.001s			
External Arms:			0.000	0.001s			
Residual Righting Arms:			0.000	0.000			
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----	
Total Waterplane---->	1.025	838.7	25.061f	0.009s	176.53	25.992	
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----		
		8.60	30.64	174.24	23.708		
Distances in METERS.-----							
Moments in m.-MT.							

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: 16332.01; t3: 30.00; t: 15.00
limmarg: 6109.00; t3: 15.00; t: 7.50
limmarg: 1677.12; t3: 7.50; t: 3.75
limmarg: 350.02; t3: 3.75; t: 1.88
limmarg: 12.55; t3: 1.87; t: 0.94
limmarg: -71.97; t3: 0.93; t: 0.47
limmarg: -36.73; t3: 1.40; t: 0.23
limmarg: -14.37; t3: 1.63; t: 0.12
limmarg: -1.37; t3: 1.75; t: 0.06
limmarg: 5.47; t3: 1.81; t: 0.03
limmarg: 2.02; t3: 1.78; t: 0.01

Theta3 is 1.78 degrees, as demonstrated below.

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 20.279f TCG = 0.000 VCG = 3.594

Origin	Degrees of	Displacement	Residual Arms	Res.
Depth---Trim---	Heel---	Weight (MT)---	in Trim--in Heel---	Area
1.022 0.11f	1.77s	879.16	0.000 -0.753	0.0000
1.024 0.11f	0.00	879.17	0.000 0.000	-0.0116
1.022 0.11f	1.78p	879.16	0.000 0.761	0.0002
1.009 0.11f	5.00p	879.18	0.000 2.110	0.0810
0.911 0.12f	10.00p	879.16	0.000 3.782	0.3408
0.647 0.15f	15.00p	879.17	0.000 4.484	0.7085
0.364 0.20f	19.52p	879.17	0.000 4.624	1.0711

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0.334	0.21f	20.00p	879.17	0.000	4.623	1.1096
0.018	0.26f	25.00p	879.17	0.000	4.508	1.5106
-0.299	0.31f	30.00p	879.17	0.000	4.272	1.8946

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 555.97 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Res. Area Ratio from abs -1.8 deg to abs 1.8 > 1.000 1.020 P

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PONTOON

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 20.279f TCG = 0.000 VCG = 3.594

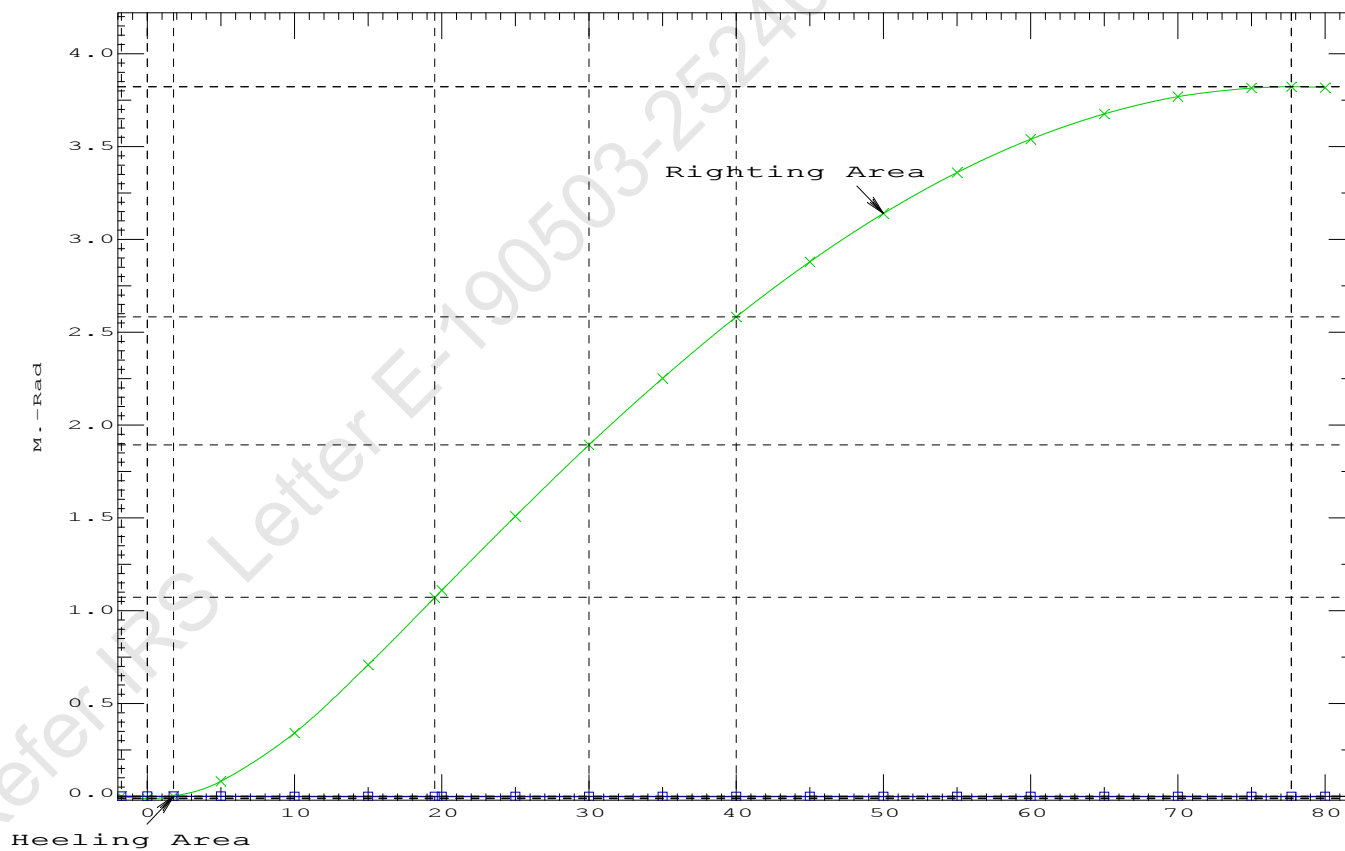
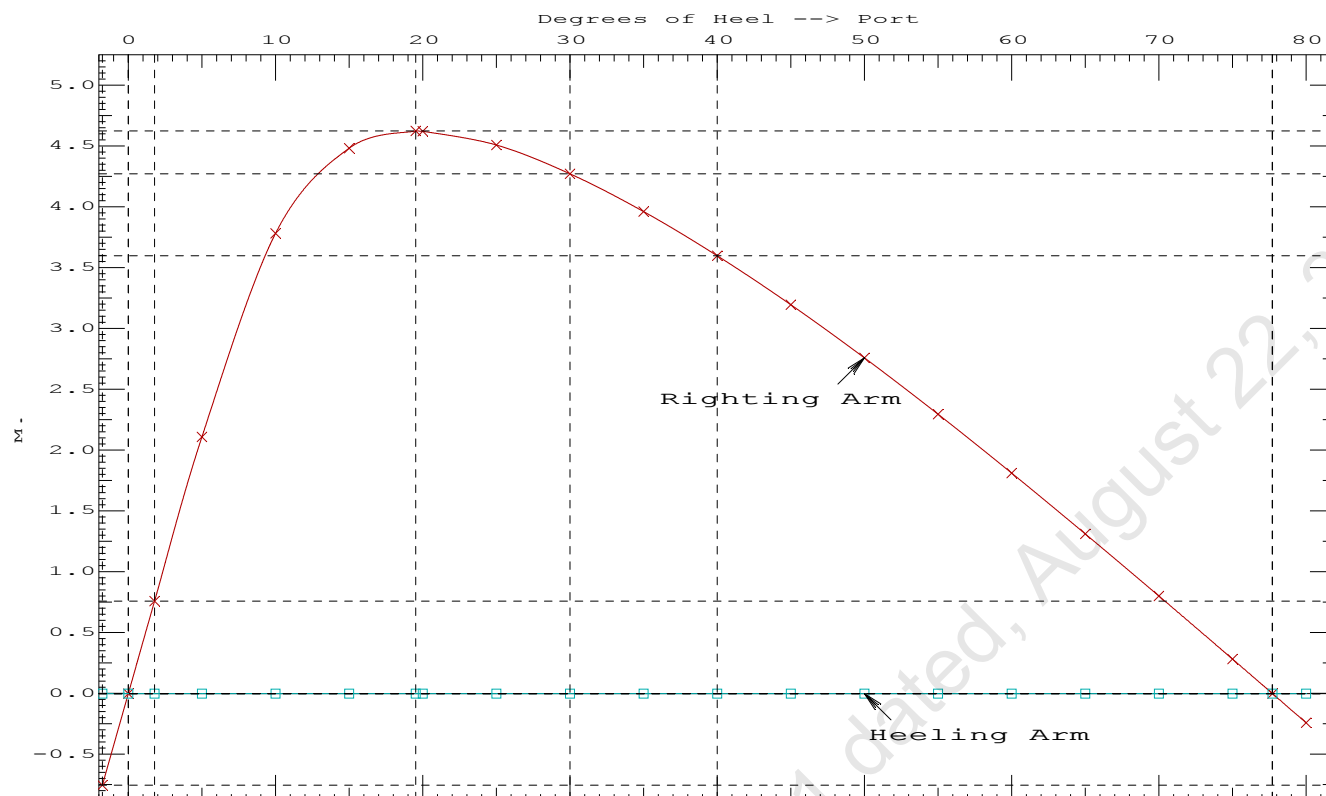
Origin	Degrees of	Displacement	Residual Arms	Res.
Depth---Trim---Heel---Weight (MT)---in Trim--in Heel---> Area				
1.022	0.11f	1.77s	879.16	0.000 -0.753 0.0000
1.024	0.11f	0.00	879.17	0.000 0.000 -0.0116
1.022	0.11f	1.78p	879.16	0.000 0.761 0.0002
1.009	0.11f	5.00p	879.18	0.000 2.110 0.0810
0.911	0.12f	10.00p	879.16	0.000 3.782 0.3408
0.647	0.15f	15.00p	879.17	0.000 4.484 0.7085
0.364	0.20f	19.52p	879.17	0.000 4.624 1.0711
0.334	0.21f	20.00p	879.17	0.000 4.623 1.1096
0.018	0.26f	25.00p	879.17	0.000 4.508 1.5106
-0.299	0.31f	30.00p	879.17	0.000 4.272 1.8946
-0.614	0.36f	35.00p	879.17	0.000 3.961 2.2544
-0.926	0.41f	40.00p	879.17	0.000 3.599 2.5846
-1.231	0.46f	45.00p	879.17	0.000 3.195 2.8813
-1.528	0.51f	50.00p	879.17	0.000 2.759 3.1414
-1.813	0.56f	55.00p	879.17	0.000 2.297 3.3622
-2.085	0.60f	60.00p	879.17	0.000 1.814 3.5418
-2.341	0.63f	65.00p	879.17	0.000 1.314 3.6783
-2.579	0.67f	70.00p	879.17	0.000 0.802 3.7708
-2.796	0.69f	75.00p	879.17	0.000 0.284 3.8182
-2.905	0.70f	77.72p	879.17	0.000 0.000 3.8249
-2.991	0.71f	80.00p	879.16	0.000 -0.238 3.8202

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 555.97 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Res. Area Ratio from abs -1.8 deg to RAzero > 1.000 330.687 P
(2) Angle from abs 1.8 deg to RAzero > 20.00 deg 75.94 P

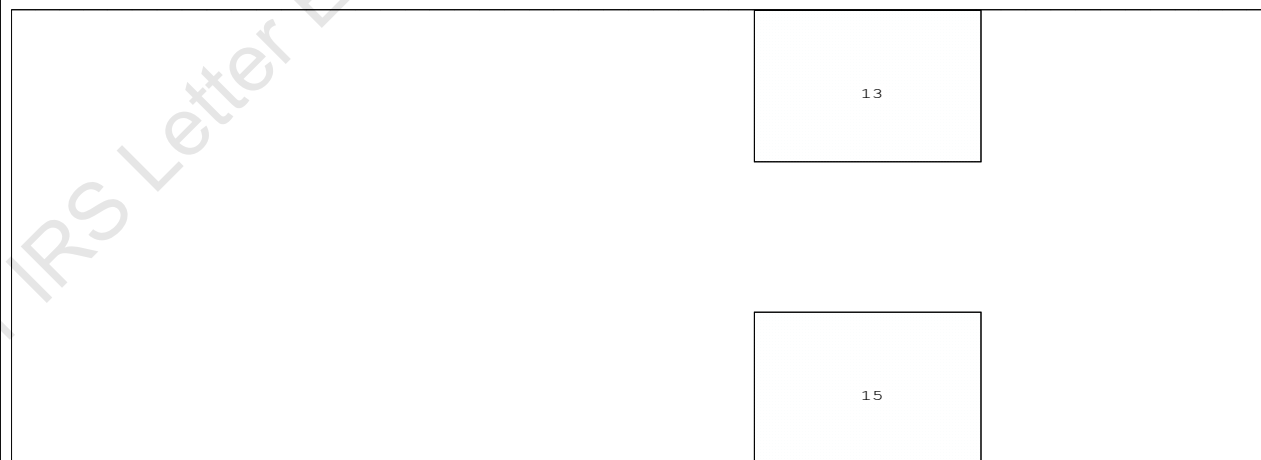
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PONTOON

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PONTOON

Condition 15 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Fwd

Condition Graphic - Draft: 1.142 @ 50.000f, 1.142 @ 0.000 Heel: zero

Profile View**Plan View**

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PONTOON

Condition 15 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Fwd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.142 @ 50.00f, 1.142 @ 25.00f, 1.142 @ 0.00
Trim: 0.000/50.000, Heel: zero

Part-----	Weight (MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP	405.97	25.000f	0.000	3.000	
Crane	150.00	7.500f	0.000	5.200	
Crane Load	51.30	19.500f	0.000	33.218	
Total Fixed----->	607.27	20.213f	0.000	6.096	
Load-----	SpGr-----	Weight (MT)----	LCG-----	TCG-----	VCG-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s
Total Tanks----->			323.20	34.000f	0.000
Total Weight----->			930.47	25.002f	0.000
			Displ (MT)----	LCB-----	TCB-----
HULL	1.025		930.47	25.001f	0.000
					0.573

	Righting Arms:		0.001a	0.000	
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----
Total Waterplane---->	1.025	843.7	25.000f	0.000	169.80
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----
		8.65	30.87	165.87	20.806
Distances in METERS.-----					
-----Moments in m.-MT.					

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

315.3 Cu.M. (13%) SALT WATER

150.00 MT Crane

51.30 MT Crane Load

DRAFT AT FP (M) = 1.142
DRAFT AT MS (M) = 1.142
DRAFT AT AP (M) = 1.142
DRAFT AT FWD MARK (M) = 1.142
DRAFT AT AFT MARK (M) = 1.142

HYDROSTATIC PROPERTIES

Trim: 0.000/50.000, No Heel, VCG = 4.503

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/
Draft----	Weight (MT)----	LCB-----	VCB-----	cm-----
1.142	930.47	25.001f	0.573	8.65
				25.000f
				30.87
				165.87
				20.806
Distances in METERS.-----				
Specific Gravity = 1.025.-----				
Moment in m.-MT.				
Trim is per 50.00m.				

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 325.5 m.-MT

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PONTOON

Condition 15 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Fwd

HEELING MOMENT specification

Lateral Plane Method

Wind pressure toward starboard

USK draft: 1.142 @ 50.00f, 1.142 @ 25.00f, 1.142 @ 0.00

Trim: 0.000/50.000, Heel: zero

Part-----	LPA*SF-----	HCP-----	Arm----	Pressure-----	Moment
HULL	92.6	0.951	1.510	0.05500	7.69
CRANE	57.0	4.370	4.930	0.05500	15.45
Total wind heeling moment to starboard----->					23.14
Distances in METERS.-----Pressure in MT/Sqm.-----					Moment in m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.002f TCG = 0.000 VCG = 4.503

Free Surface Adjustment: 0.350

Adjusted CG: LCG = 25.002f TCG = 0.000 VCG = 4.853

Origin	Degrees of	Displacement	Residual Arms	Res. 50% DeckImm
Depth---Trim---Heel---	Weight (MT)---	in Trim--in Heel---	Area----	Angle
1.130 0.00 0.00	930.41	0.000 -0.024	0.0000	6.497
1.130 0.00 0.07s	930.47	0.000 0.000	-0.0000	6.432
1.115 0.00 5.07s	930.47	0.000 1.824	0.0796	1.432
1.103 0.00 6.50s	930.42	0.000 2.321	0.1313	0.000
1.033 0.00 10.07s	930.55	0.000 3.293	0.3074	-3.568
0.801 0.00 15.07s	930.65	0.000 3.822	0.6233	-8.568
0.706 0.00 16.87s	930.47	0.000 3.847	0.7441	-10.371
0.539 0.00 20.07s	930.47	0.000 3.771	0.9577	-13.568
0.274 0.00 25.07s	930.47	0.000 3.486	1.2768	-18.568
0.007 0.00 30.07s	930.47	0.000 3.093	1.5647	-23.568
-0.261 0.00 35.07s	930.47	0.000 2.638	1.8152	-28.568
-0.525 0.00 40.07s	930.47	0.000 2.141	2.0240	-33.568
-0.787 0.00 45.07s	930.46	0.000 1.616	2.1881	-38.568
-1.042 0.00f 50.07s	930.47	0.000 1.070	2.3055	-43.568
-1.290 0.00f 55.07s	930.47	0.000 0.511	2.3745	-48.568
-1.504 0.00f 59.57s	930.47	0.000 0.000	2.3947	-53.076
-1.527 0.00f 60.07s	930.47	0.000 -0.056	2.3944	-53.568
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.				

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 325.5 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 23.14 (constant)

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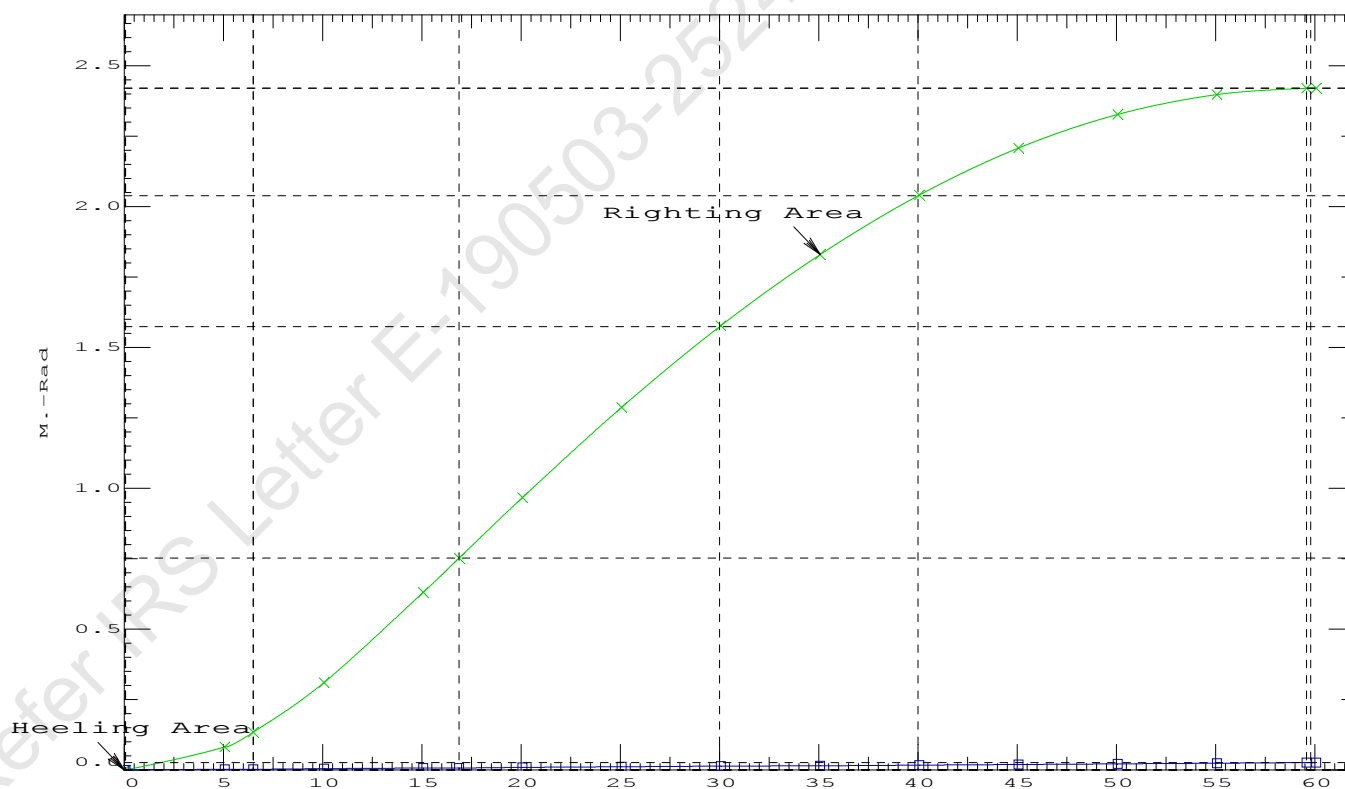
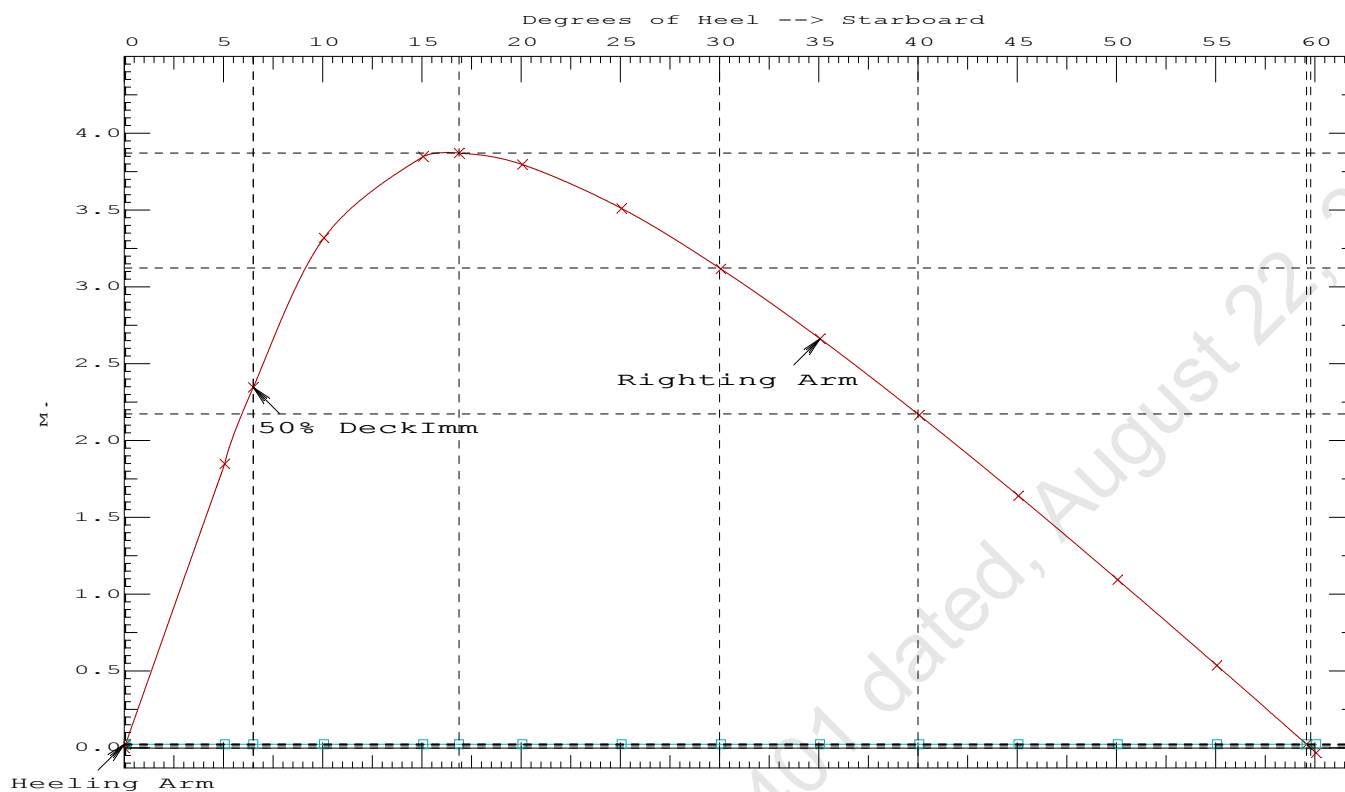
PONTOON**Condition 15 : Operating Condition of 150T Crane**
Crane Lifting 51.3T @ 12m Radius Fwd

LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max-----Attained
(1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad 0.7441 P
(2) Angle from Equilibrium to RAzero or Flood > 20.00 deg 59.51 P
(3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg 6.43 P

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PONTOON

Condition 15 : Operating Condition of 150T CraneCrane Lifting 51.3T @ 12m Radius Fwd

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PONTOON

Condition 15 : Operating Condition of 150T Crane
Crane Lifting 51.3T @ 12m Radius Fwd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.002f TCG = 0.000 VCG = 4.503
Free Surface Adjustment: 0.350
Adjusted CG: LCG = 25.002f TCG = 0.000 VCG = 4.853

Origin	Degrees of	Displacement	Residual Arms	Res.
Depth---Trim---Heel---Weight (MT)---in Trim--in Heel---> Area				
1.130	0.00	0.00	930.47	0.0000
1.115	0.00	5.00s	930.47	0.0796
1.035	0.00	10.00s	930.47	0.3059
0.804	0.00	15.00s	930.65	0.6247
0.707	0.00	16.87s	930.47	0.7510
0.542	0.00	20.00s	930.47	0.9612
0.278	0.00	25.00s	930.47	1.2827
0.010	0.00	30.00s	930.47	1.5731
-0.257	0.00	35.00s	930.47	1.8262
-0.522	0.00	40.00s	930.47	2.0376
-0.784	0.00	45.00s	930.47	2.2043
-1.039	0.00f	50.00s	930.47	2.3243
-1.286	0.00f	55.00s	930.46	2.3961
-1.514	0.00f	59.78s	930.47	2.4187
-1.524	0.00f	60.00s	930.46	2.4186

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 325.5 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

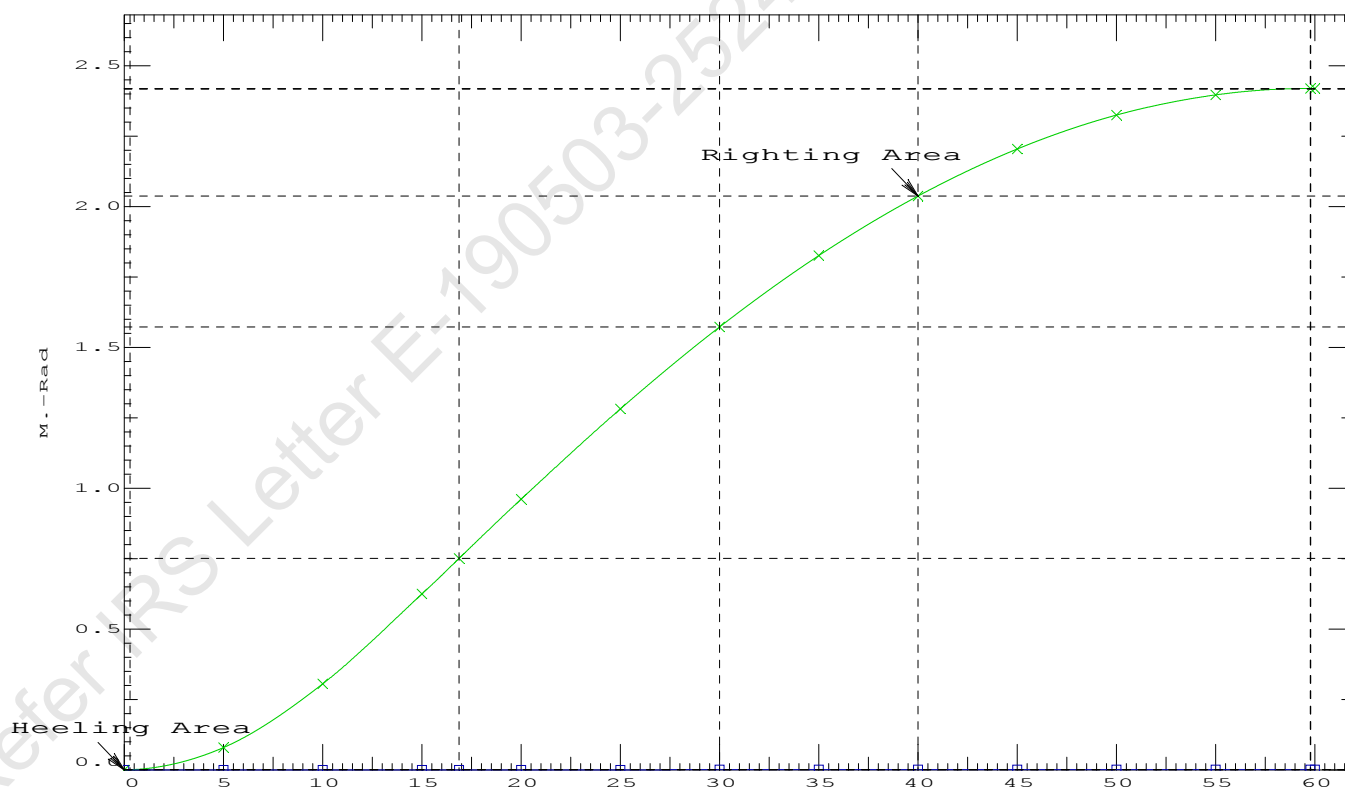
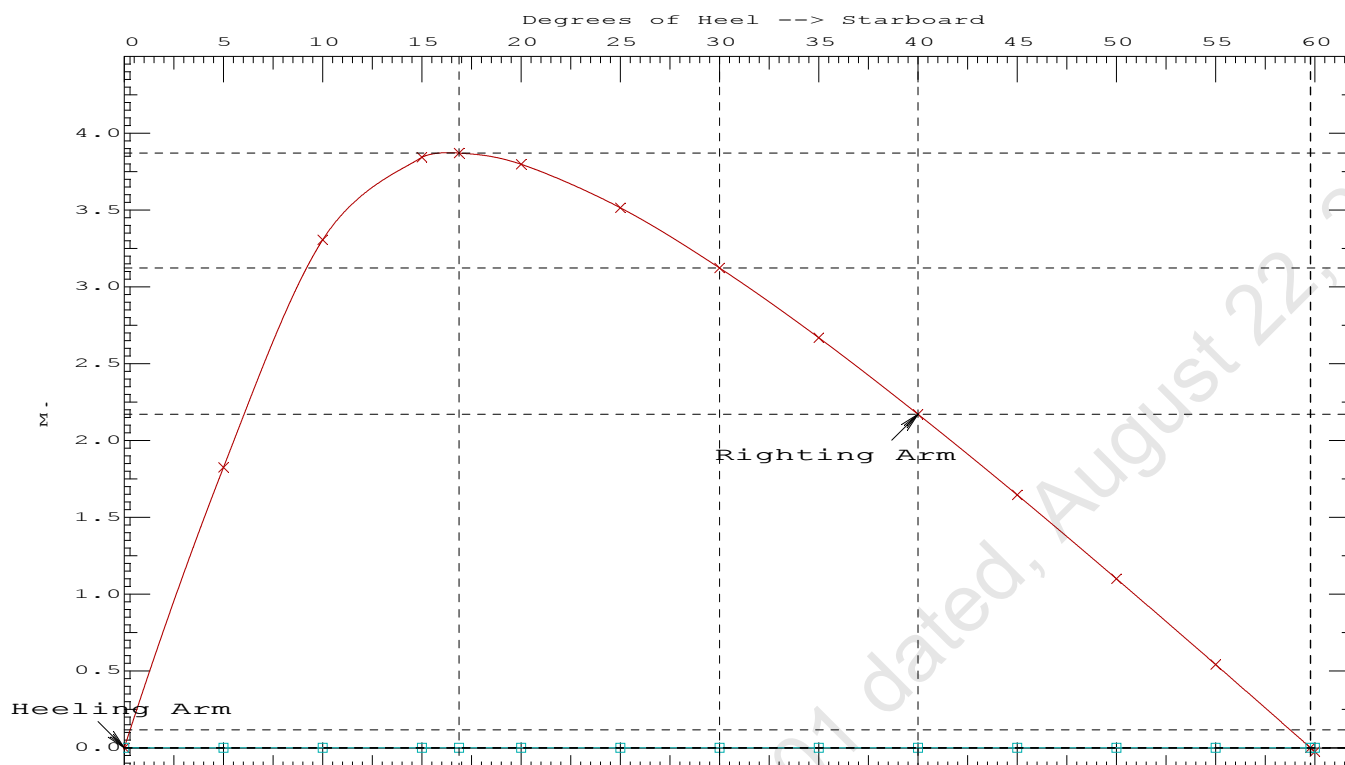
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	0.00 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	360090 P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667	2158.05 P

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PONTOON

Condition 15 : Operating Condition of 150T CraneCrane Lifting 51.3T @ 12m Radius Fwd

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PONTOON

Condition 15 : Operating Condition of 150T Crane**Crane Lifting 51.3T @ 12m Radius Fwd**

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.142 @ 50.00f, 1.142 @ 25.00f, 1.142 @ 0.00

Trim: 0.000/50.000, Heel: 0.00 deg.

Part-----			Weight (MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			405.97	25.000f	0.000	3.000	
Crane			150.00	7.500f	0.000	5.200	
Crane Load			51.30	19.500f	0.000	33.218	
Total Fixed----->			607.27	20.213f	0.000	6.096	
	Load-----	SpGr-----	Weight (MT)----	LCG-----	TCG-----	VCG-----	FSM-----
05_WBTK.P	1.000	1.025	161.60	34.000f	5.980p	1.509	162.73*
05_WBTK.S	1.000	1.025	161.60	34.000f	5.980s	1.509	162.73*
Total Tanks----->			323.20	34.000f	0.000	1.509	325.46
Total Weight----->			930.47	25.002f	0.000	4.503	
			Displ (MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	930.47	25.002f	0.001s	0.573	

	Righting Arms:			0.000	0.001s		
	External Arms:			0.000	0.001s		
	Residual Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		841.6	25.000f	0.018s	168.56	24.627
			MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
			8.63		30.64	164.63	20.698
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

PONTOON

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.129 @ 50.00f, 1.083 @ 25.00f, 1.037 @ 0.00
Trim: Fwd 0.092/50.000, Heel: 0.00 deg.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

```
limmarg: INFINITY; t3: 30; t: 15.00
limmarg: INFINITY; t3: 15; t: 7.50
limmarg: INFINITY; t3: 8; t: 3.75
limmarg: INFINITY; t3: 4; t: 1.88
limmarg: INFINITY; t3: 2; t: 0.94
limmarg: INFINITY; t3: 1; t: 0.47
limmarg: INFINITY; t3: 1; t: 0.23
limmarg: INFINITY; t3: 1; t: 0.12
limmarg: INFINITY; t3: 1; t: 0.06
limmarg: INFINITY; t3: 1; t: 0.03
limmarg: INFINITY; t3: 1; t: 0.01
```

Theta3 is 0.97 degrees, as demonstrated below.

Fixed CG: LCG = 20.279f TCG = 0.000 VCG = 3.594

Origin	Degrees of	Displacement	Residual Arms	Res.		
Depth	Trim	Heel	Weight (MT)	in Trim	in Heel	Area
1.024	0.11f	0.00	879.17	0.000	0.000	0.0000
1.024	0.10f	0.97s	879.16	0.000	0.415	0.0035
1.009	0.11f	5.00s	879.18	0.000	2.110	0.0923
0.910	0.12f	10.00s	879.16	0.000	3.781	0.3526
0.646	0.15f	15.00s	879.17	0.000	4.482	0.7201
0.364	0.20f	19.52s	879.17	0.000	4.622	1.0822
0.334	0.21f	20.00s	879.17	0.000	4.621	1.1210

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0.018	0.26f	25.00s	879.17	0.000	4.505	1.5218
-0.299	0.31f	30.00s	879.17	0.000	4.269	1.9056

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 555.97 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----	STABILITY CRITERION-----	Min/Max-----	Attained
(1) Residual Area Ratio from abs 0 deg to abs 1	>	1.000	LARGE

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PONTOON

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 20.279f TCG = 0.000 VCG = 3.594

Origin	Degrees of	Displacement	Residual Arms	Res.
Depth	Trim	Heel	Weight (MT)	in Trim--in Heel--> Area
1.024	0.11f	0.00	879.17	0.000 0.000 0.0000
1.024	0.10f	0.97s	879.16	0.000 0.415 0.0035
1.009	0.11f	5.00s	879.18	0.000 2.110 0.0923
0.910	0.12f	10.00s	879.16	0.000 3.781 0.3526
0.646	0.15f	15.00s	879.17	0.000 4.482 0.7201
0.364	0.20f	19.52s	879.17	0.000 4.622 1.0822
0.334	0.21f	20.00s	879.17	0.000 4.621 1.1210
0.018	0.26f	25.00s	879.17	0.000 4.505 1.5218
-0.299	0.31f	30.00s	879.17	0.000 4.269 1.9056
-0.614	0.36f	35.00s	879.17	0.000 3.959 2.2651
-0.926	0.41f	40.00s	879.17	0.000 3.596 2.5951
-1.231	0.46f	45.00s	879.17	0.000 3.193 2.8916
-1.528	0.51f	50.00s	879.17	0.000 2.757 3.1515
-1.814	0.56f	55.00s	879.17	0.000 2.294 3.3720
-2.085	0.60f	60.00s	879.17	0.000 1.811 3.5513
-2.341	0.63f	65.00s	879.17	0.000 1.311 3.6877
-2.579	0.67f	70.00s	879.17	0.000 0.800 3.7799
-2.796	0.69f	75.00s	879.17	0.000 0.281 3.8271
-2.904	0.70f	77.69s	879.16	0.000 0.000 3.8337
-2.991	0.71f	80.00s	879.17	0.000 -0.241 3.8288

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

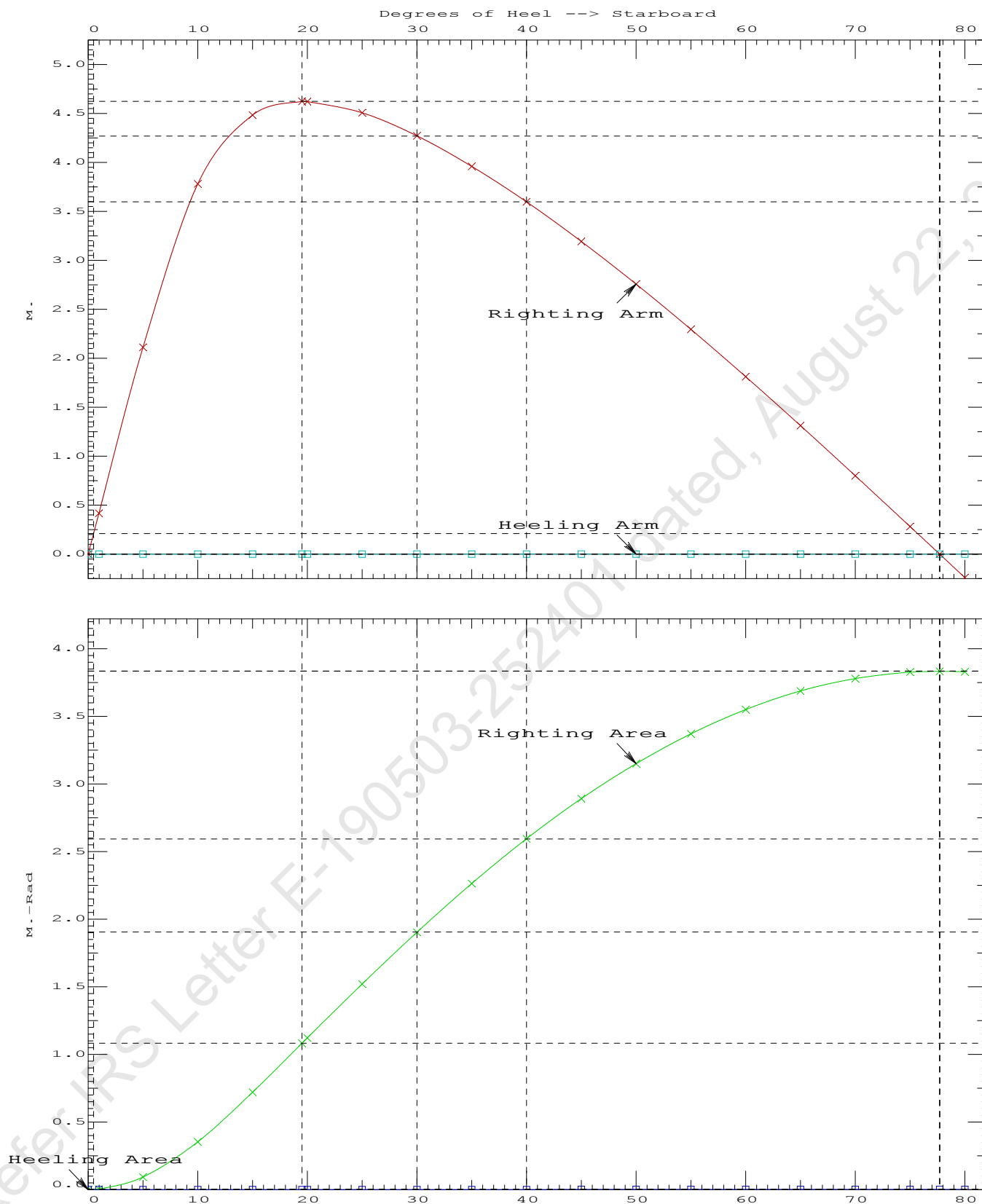
Note: The Center of Gravity shown above is for the Fixed Weight of 555.97 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 1.00

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAZero > 1.000 LARGE
(2) Angle from abs 1 deg to RAZero > 20.00 deg 76.72 P

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LIGHTSHIP PARTICULARS

Lightship Weight	=	405.97	T
VCG	=	3.000	M above Baseline
LCG	=	25.000	M fwd of AP (Fr.0)
TCG	=	0.000	M off centerline

(Reference: Lightship particulars are estimated.)

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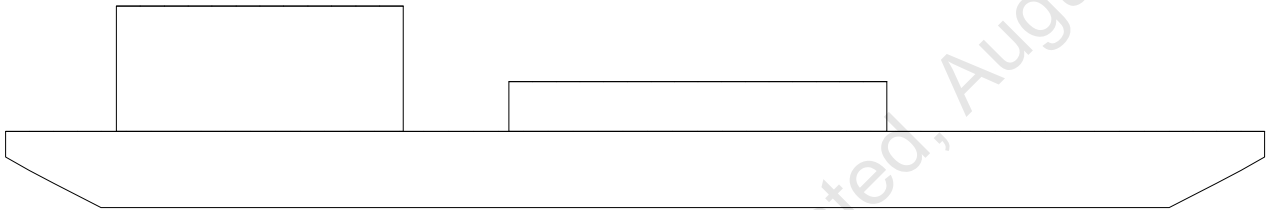
USK draft refers to the line:

0.012 below baseline @ 50.000f and 0.012 below baseline @ 0.000

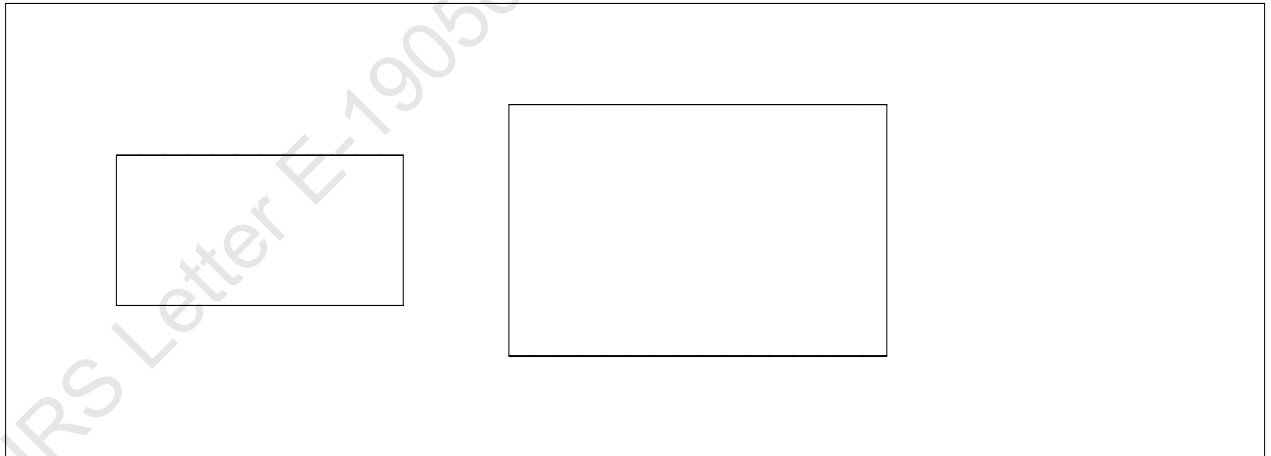
WINDAGE AREA CALCULATION AT DIFFERENT DRAFTS

Condition Graphic

Profile View



Plan View



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LATERAL PLANE STATUS

USK draft: 0.512 @ 50.00f, 0.512 @ 25.00f, 0.512 @ 0.00

Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	22.3	25.000f	-0.253	121.4	25.000f	1.281
CRANE				57.0	10.100f	5.000
DECKCARGO.C				29.4	27.500f	3.500
Total Lateral Plane->	22.3	25.000f	-0.253	207.8	21.267f	2.615
Distances in METERS.						

LATERAL PLANE STATUS

USK draft: 0.612 @ 50.00f, 0.612 @ 25.00f, 0.612 @ 0.00

Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	26.8	25.000f	-0.302	116.9	25.000f	1.229
CRANE				57.0	10.100f	4.900
DECKCARGO.C				29.4	27.500f	3.400
Total Lateral Plane->	26.8	25.000f	-0.302	203.3	21.184f	2.572
Distances in METERS.						

LATERAL PLANE STATUS

USK draft: 0.712 @ 50.00f, 0.712 @ 25.00f, 0.712 @ 0.00

Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	31.3	25.000f	-0.351	112.4	25.000f	1.176
CRANE				57.0	10.100f	4.800
DECKCARGO.C				29.4	27.500f	3.300
Total Lateral Plane->	31.3	25.000f	-0.351	198.8	21.097f	2.529
Distances in METERS.						

LATERAL PLANE STATUS

USK draft: 0.812 @ 50.00f, 0.812 @ 25.00f, 0.812 @ 0.00

Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	35.9	25.000f	-0.400	107.8	25.000f	1.123
CRANE				57.0	10.100f	4.700
DECKCARGO.C				29.4	27.500f	3.200
Total Lateral Plane->	35.9	25.000f	-0.400	194.2	21.006f	2.487
Distances in METERS.						

LATERAL PLANE STATUS

USK draft: 0.912 @ 50.00f, 0.912 @ 25.00f, 0.912 @ 0.00

Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	40.4	25.000f	-0.450	103.3	25.000f	1.070
CRANE				57.0	10.100f	4.600
DECKCARGO.C				29.4	27.500f	3.100
Total Lateral Plane->	40.4	25.000f	-0.450	189.7	20.911f	2.445
Distances in METERS.						

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LATERAL PLANE STATUS

USK draft: 1.012 @ 50.00f, 1.012 @ 25.00f, 1.012 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	45.0	25.000f	-0.497	98.7	25.000f	1.018
CRANE				57.0	10.100f	4.500
DECKCARGO.C				29.4	27.500f	3.000
Total Lateral Plane->	45.0	25.000f	-0.497	185.1	20.809f	2.405
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.112 @ 50.00f, 1.112 @ 25.00f, 1.112 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	49.7	25.000f	-0.545	94.0	25.000f	0.966
CRANE				57.0	10.100f	4.400
DECKCARGO.C				29.4	27.500f	2.900
Total Lateral Plane->	49.7	25.000f	-0.545	180.4	20.699f	2.367
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.212 @ 50.00f, 1.212 @ 25.00f, 1.212 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	54.4	25.000f	-0.594	89.3	25.000f	0.914
CRANE				57.0	10.100f	4.300
DECKCARGO.C				29.4	27.500f	2.800
Total Lateral Plane->	54.4	25.000f	-0.594	175.7	20.584f	2.329
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.312 @ 50.00f, 1.312 @ 25.00f, 1.312 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	59.1	25.000f	-0.642	84.5	25.000f	0.863
CRANE				57.0	10.100f	4.200
DECKCARGO.C				29.4	27.500f	2.700
Total Lateral Plane->	59.1	25.000f	-0.642	170.9	20.462f	2.291
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.412 @ 50.00f, 1.412 @ 25.00f, 1.412 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	63.9	25.000f	-0.691	79.8	25.000f	0.810
CRANE				57.0	10.100f	4.100
DECKCARGO.C				29.4	27.500f	2.600
Total Lateral Plane->	63.9	25.000f	-0.691	166.2	20.333f	2.255
Distances in METERS.-----						

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LATERAL PLANE STATUS

USK draft: 1.512 @ 50.00f, 1.512 @ 25.00f, 1.512 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	68.7	25.000f	-0.739	75.0	25.000f	0.759
CRANE				57.0	10.100f	4.000
DECKCARGO.C				29.4	27.500f	2.500
Total Lateral Plane->	68.7	25.000f	-0.739	161.4	20.194f	2.220
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.612 @ 50.00f, 1.612 @ 25.00f, 1.612 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	73.6	25.000f	-0.786	70.1	25.000f	0.708
CRANE				57.0	10.100f	3.900
DECKCARGO.C				29.4	27.500f	2.400
Total Lateral Plane->	73.6	25.000f	-0.786	156.5	20.044f	2.188
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.712 @ 50.00f, 1.712 @ 25.00f, 1.712 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	78.5	25.000f	-0.834	65.2	25.000f	0.657
CRANE				57.0	10.100f	3.800
DECKCARGO.C				29.4	27.500f	2.300
Total Lateral Plane->	78.5	25.000f	-0.834	151.6	19.883f	2.157
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.812 @ 50.00f, 1.812 @ 25.00f, 1.812 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	83.4	25.000f	-0.882	60.3	25.000f	0.607
CRANE				57.0	10.100f	3.700
DECKCARGO.C				29.4	27.500f	2.200
Total Lateral Plane->	83.4	25.000f	-0.882	146.7	19.712f	2.128
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.912 @ 50.00f, 1.912 @ 25.00f, 1.912 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	88.3	25.000f	-0.930	55.4	25.000f	0.556
CRANE				57.0	10.100f	3.600
DECKCARGO.C				29.4	27.500f	2.100
Total Lateral Plane->	88.3	25.000f	-0.930	141.8	19.529f	2.100
Distances in METERS.-----						

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LATERAL PLANE STATUS

USK draft: 2.012 @ 50.00f, 2.012 @ 25.00f, 2.012 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	93.2	25.000f	-0.978	50.5	25.000f	0.505
CRANE				57.0	10.100f	3.500
DECKCARGO.C				29.4	27.500f	2.000
Total Lateral Plane->	93.2	25.000f	-0.978	136.9	19.333f	2.073
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 2.112 @ 50.00f, 2.112 @ 25.00f, 2.112 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	98.2	25.000f	-1.025	45.5	25.000f	0.455
CRANE				57.0	10.100f	3.400
DECKCARGO.C				29.4	27.500f	1.900
Total Lateral Plane->	98.2	25.000f	-1.025	131.9	19.118f	2.050
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 2.212 @ 50.00f, 2.212 @ 25.00f, 2.212 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	103.2	25.000f	-1.073	40.5	25.000f	0.405
CRANE				57.0	10.100f	3.300
DECKCARGO.C				29.4	27.500f	1.800
Total Lateral Plane->	103.2	25.000f	-1.073	126.9	18.887f	2.029
Distances in METERS.-----						